

Connectivity Bounds in Graphs and Beyond

Erdős Problem 915, from Proof to Search

Louis Vandenbruwaene

Supervisor: Prof. Stijn Cambie
KU Leuven

Thesis presented in
fulfillment of the requirements
for the degree of Master of Science
in Mathematics

Academic year 2026–2027

Open Access Disclaimer

This master's thesis is a student work submitted as part of an academic programme at KU Leuven. The work is made available for examination and archiving in accordance with the applicable KU Leuven regulations and open-access policy for master's theses.

© Copyright by KU Leuven

Without written permission from the supervisors and the author, it is forbidden to reproduce or adapt in any form or by any means any part of this publication. Requests for obtaining the right to reproduce or utilize parts of this publication should be addressed to KU Leuven, Faculty of Science, Celestijnenlaan 200H - box 2100, 3001 Leuven (Heverlee), Telephone +32 16 32 14 01.

Written permission from the supervisor is also required to use the methods, products, schematics, and programs described in this work for industrial or commercial use, and for submitting this publication in scientific contests.

Foreword

This thesis studies how many edges or arcs a network can contain when no pair of vertices is allowed to have too many independent routes between them. The starting point is Erdős Problem 915, an extremal question about edge-disjoint paths in simple graphs. The work is told as a journey. It begins with the variants whose answers can be proved cleanly by hand, watches those proofs grow longer and more delicate as the model changes, and at the point where a single argument no longer suffices it turns to a unified program. That program holds all twelve variants behind one common interface, so a single piece of code serves them all. It measures connectivity exactly with a maximum-flow routine, the standard way to count the independent routes between two points. It runs a prover that sweeps every graph of a small fixed size and, when none beats a target, reports that sweep as a genuine upper-bound proof. And it runs a search that, like hot metal cooling slowly into shape, settles random graphs toward the densest ones the rules allow.

I am grateful to my supervisor, Prof. Stijn Cambie, and my daily advisor for their guidance, corrections, and mathematical feedback throughout the project. One review in particular changed the shape of this thesis: Prof. Cambie put a late draft through a language model different from the one I had been working with, and it came back with a misapplied inequality I had read past many times, together with the counting idea that grew into [Proposition A.22](#) and [Theorem A.24](#). I thank my promotor, Prof. Jan Goedgebeur, whose suggestion of the `nauty` generation pipeline sharply accelerated the directed enumeration. I also thank the people who discussed early versions of the arguments, checked examples, or helped clarify the presentation. Above all I thank my girlfriend Sara, whose patience and encouragement carried me through the long stretches of this work. Any remaining mistakes are my own.

Contribution Statement

This thesis is my own work. I carried out the literature study, recast the twelve variants of the problem in one common language, and designed and wrote the single program that runs through all of them. I built every example, by hand or by computer search, checked each one with that program, wrote the proofs that appear in the thesis, and typeset the whole document. Where a result comes from the literature I say so and cite it. The theorems I lean on but do not claim as mine are those of Mader, Leonard, Sørensen and Thomassen, Menger, Gomory and Hu, and Baranyai, the hypergraph connectivity results of Bérczi, Chandrasekaran, Király and Kulkarni, and the standard probabilistic tools used for the random graphs.

The program has three jobs, and it keeps them separate. It *measures*: given any graph, it computes exactly, through maximum flow, how many independent routes the busiest pair of vertices has. It *proves*: for a fixed number of vertices it sweeps a finite space of graphs, and when nothing in that space beats a target, that sweep is a genuine upper-bound proof rather than a guess. And it *discovers*: a search that cools like a heated metal settling into shape wanders the space of graphs and finds dense ones, which give lower bounds and suggest conjectures but never prove an upper bound by themselves. Every general statement in the thesis rests on a proof by hand or a cited theorem. The computer searches are evidence that points the way, never a replacement for the argument. I used AI language models, including Claude Code, ChatGPT, and Gemini, as tools for literature search, L^AT_EX editing, proofreading, and trying out proof drafts. I checked every mathematical statement and proof in the thesis myself and take full responsibility for the content.

One thing about that use is worth recording, because it is a methodological point and not just a disclosure. A model that has helped write an argument is a poor reviewer of it: it reads the text the way its author does, follows the same intended reading of each sentence, and slides over the same gaps. Handing the finished draft to a *different* model, cold, behaved much more like handing it to a different person. It read each sentence as a standalone claim rather than as a step in an argument it already believed, and that is exactly how it caught a use of Whitney's inequality that runs the wrong way, in a sentence that had survived several careful readings by me and by the model that helped write it. The same pass also proposed a counting argument I had not tried. It was wrong about a good deal else, and every item it raised had to be checked against the source before anything was changed, several of them turning out to be objections to conventions the thesis states rather than to errors. But the pattern is the familiar one from working with people: the value came from the disagreement, not from the agreement, and from the fact that the second reader had no stake in the first reader's phrasing.

The genuinely new results are the following. For hyperedge networks I settle the vertex-disjoint problem completely at the two strictest route limits, $m = 2$ and $m = 3$, for every hyper-

edge size r and every number of vertices n (Theorems A.45 and A.48): the vertex-disjoint and edge-disjoint answers turn out to be equal, both $\lfloor (m-1)(n-1)/(r-1) \rfloor$. The $m = 3$ case rests on a bound for incidence graphs that I prove using Tutte's decomposition of a graph into its three-connected pieces (Lemma A.47). For one-way multigraphs at $m = 3$ I prove an unconditional attachment lemma (Lemma A.34) and an odd-step theorem (Theorem A.36, Remark A.39) that together shrink the whole problem, for every n , down to two finite statements about multigraphs on just seven vertices that a computer can check, one of which the accompanying program has already verified outright, and they do so without assuming the unproved minimum-degree conjecture the earlier route needed. For one-way simple graphs I prove an unconditional quadratic bound (Proposition A.22 and Theorem A.24): under any route limit m , such a graph has at most $\lfloor n^2/4 \rfloor + 4(m-1)(n-1)$ arcs, and deleting a comparable number of arcs always leaves a one-way bipartite graph. This fixes both the leading constant and the order of the second-order term of the standing conjecture (Conjecture 1.14) without any assumption about the shape of the extremal graph, which is what every previous route to that bound needed, and leaves only a constant factor of about eight in one coefficient. The counting idea behind it came out of the external review recorded in the foreword, and I verified it and worked out the sharpening myself. For multigraphs under the alternative counting convention I settle the vertex problem at $m = 2$ and disprove the natural guess, suggested by the data through $m = 4$, that a thickened tree stays optimal for all larger n : a bouquet of thickened cliques sharing a single vertex (Theorem A.56) beats it by a margin growing linearly in n , at a rate that is quadratic rather than linear in m (Section A.8).

Short Summary

Erdős Problem 915, posed by Bollobás and Erdős [1], asks for the maximum number of edges in a graph on n vertices such that no two vertices are connected by m edge-disjoint paths. In modern notation this asks for

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1\}.$$

Mader proved that, for simple graphs and $n \geq m \geq 2$,

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

This thesis revisits this theorem and studies the corresponding extremal problem under three independent changes: the graph model (simple graphs, multigraphs, and hypergraphs), the path-separation notion (edge-disjoint or internally vertex-disjoint), and the presence or absence of directions. Sampled random graphs $G(n, p)$ run alongside throughout as a lens on the typical case rather than as a separate model.

The thesis is organised as a journey from hand proofs to computation. The early variants are settled rigorously: the multigraph value $L_m(n) = (m-1)(n-1)$, the random-graph threshold $p^* = m/n$, the equality of the edge and vertex problems for $m \leq 4$, and the first divergence at $m = 5$, where Sørensen and Thomassen prove $k_5(n) = \lfloor 8n/3 \rfloor - 3$. The directed case forces the turn: the value $\ell_2^{\text{dir}}(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is proved by a long induction, and for $m \geq 3$ the hand proof gives way to a conjecture, after an explicit ten-vertex graph rules out the naive guess. The centrepiece is then a unified program. One multiplicity-matrix representation models almost all twelve variants, an exact maximum-flow checker measures local connectivity, a cut-counting optimisation certifies upper bounds for small sizes, and a search guided by edge sensitivity, run as both simulated annealing and tabu search, discovers extremal graphs. The program rediscovers, each in its own variant, the double star (undirected multigraphs at $m = 2$), the directed hub, the one-directional bipartite digraph, and the augmented-bipartite construction (the directed cases), certifies the directed multigraph values $L_m^{\text{dir}}(n) = 2(n-1)(m-1)$ for $n \leq 6$, and frames the remaining open problems on the directed frontier.

The interplay then pays back in the other direction. The hypergraph vertex problem is settled completely at $m = 2$ and $m = 3$: $k_m^{(r)}(n) = \lfloor (m-1)(n-1)/(r-1) \rfloor$ for every uniformity r , the $m = 3$ case through a new rank bound on incidence graphs proved by way of Tutte's triconnected decomposition. On the directed side, an unconditional attachment lemma and a conditional odd-step theorem reduce the entire directed multigraph problem at $m = 3$, value and extremal characterisation for all n , to two finite, machine-checkable statements about multigraphs on seven vertices, one of which the program has since verified outright, and directed hypergraphs

receive their first bounds, quadratic in n .

Abbreviations and Symbols

GH	Gomory-Hu
MILP	mixed-integer linear program
SA	simulated annealing
SPQR	the triconnected decomposition tree (series, parallel, rigid pieces)
$G = (V, E)$	graph with vertex set V and edge set E
$D = (V, A)$	directed graph with vertex set V and arc set A
n	number of vertices
m	forbidden-connectivity parameter, with all local connectivities at most $m - 1$
r	hyperedge size (every hyperedge spans r vertices)
$\lambda_G(u, v)$	maximum number of edge-disjoint u - v paths
$\lambda_G^{\max}$	maximum local edge-connectivity over all vertex pairs
$\kappa_G(u, v)$	maximum number of internally vertex-disjoint u - v paths
$\kappa_G^{\max}$	maximum local vertex-connectivity over all vertex pairs
$\ell_m(n)$	simple undirected edge-disjoint extremal function
$L_m(n)$	undirected multigraph edge-disjoint extremal function
$k_m(n)$	simple undirected vertex-disjoint extremal function
$\ell_m^{\text{dir}}(n)$	simple directed arc-disjoint extremal function
$L_m^{\text{dir}}(n)$	directed multigraph arc-disjoint extremal function
$k_m^{\text{dir}}(n)$	simple directed vertex-disjoint extremal function
$\ell_m^{(r)}(n), k_m^{(r)}(n)$	r -uniform hypergraph edge- and vertex-disjoint extremal functions
$M^*(n)$	the m -free directed multigraph optimum, $L_m^{\text{dir}}(n) = (m - 1) M^*(n)$
$B_{p,q}$	one-directional complete bipartite digraph, all arcs from a p -part to a q -part
$\mu(u, v)$	multiplicity of an edge or arc between u and v
$\sigma(e)$	edge sensitivity $\lambda^{\max}(G) - \lambda^{\max}(G - e)$
$G(n, p)$	binomial random graph

Contents

Foreword	II
Contribution Statement	III
Short Summary	V
Abbreviations and Symbols	VII
1 The Problem and Its Base Cases	1
1.1 Graphs and routes	1
1.2 Erdős Problem 915	2
1.3 Three axes of variation	4
1.4 Cases with short proofs	6
1.5 The first divergence: vertex-disjoint paths	8
1.6 The directed case and the quadratic phenomenon	10
1.7 The failure of direct proof for $m \geq 3$	13
1.8 The hypergraph variant	15
1.9 Random graphs as intuition	17
1.10 Where this work sits	19
2 Certifying Bounds by Machine	22
2.1 The solver	23
2.2 A single representation for all variants	23
2.3 The connectivity checker built on maximum flow	25
2.3.1 The directed hypergraph checker	31
2.4 Checking every graph of a fixed size	32
2.5 Reaching further: a pruned search for the directed base cases	34
2.6 Generating the directed cases faster	35
2.7 Proving every m at once	36

2.8	What the solver may claim	41
2.9	Reproducibility	42
2.9.1	What a machine-checked claim in this thesis means	42
3	Discovering Bounds by Search	44
3.1	The wall: how enumeration explodes	44
3.2	The temperature search	45
3.3	Temperature and edge sensitivity	47
3.4	Keeping the checker cheap inside the search	49
3.5	Rediscovering the known extremisers	51
3.5.1	Simulated annealing versus tabu search	52
3.6	The trichotomy: proved, certified, conjectured	54
4	Synthesis, Results, and Open Problems	60
4.1	The directed frontier and the single open lemma	60
4.2	Two principal phenomena	64
4.3	Summary of the twelve variants	70
4.3.1	Orientation models for the directed hypergraph	70
4.4	Contributions and limitations of the program	72
4.5	Open problems	73
A	Full Proofs and Computational Transcripts	76
A.1	Menger, Gomory–Hu, and the degree bound	76
A.2	Mader’s theorem in full	79
A.3	The directed arc value at $m = 2$	82
A.4	Structural propositions on the directed frontier	86
A.5	The directed multigraph problem at large n	92
A.6	The hypergraph bounds	101
A.7	The hypergraph vertex problem	104
A.8	The multigraph vertex problem under the other convention	113
A.9	The random threshold	117
A.10	Computational transcripts	119

A.11 How the program checks itself	120
B Supplementary Figures	122
B.1 A gallery of extremal graphs	122
B.2 Search traces across the variants	122
B.3 The variant landscape at $m = 6$ and in three dimensions	122
C The Complete Source Code	127
C.1 The main program	127
C.1.1 How the file opens	127
C.1.2 The objects: graphs, hypergraphs, and the values we already know	131
C.1.3 Measuring, and proving	141
C.1.4 Searching	151
C.1.5 The random model	170
C.1.6 Drawing the figures	175
C.1.7 Walking the whole space	191
C.1.8 The open variants	208
C.1.9 The gallery	218
C.1.10 The self-check	227
C.2 The figure script	233
C.3 The C accelerator	245
C.4 The test suite	248
C.4.1 Shared setup	248
C.4.2 The graph model	248
C.4.3 The closed-form bounds	251
C.4.4 The named constructions	252
C.4.5 The connectivity checker	253
C.4.6 Hypergraphs	257
C.4.7 Edge sensitivity	259
C.4.8 The search	260
C.4.9 The prover	262
C.4.10 The random model	264

C.4.11 The solver	265
-----------------------------	-----

Chapter 1

The Problem and Its Base Cases

A network is useful not only because it connects, but because it connects in more than one way. A single road between two towns is a liability, and a second independent road is insurance. The mathematical content of that intuition is the number of routes between two points that share no segment, and the question this thesis circles is how many connections a network can hold before some pair of points is forced to have too many such routes. This chapter builds the question from nothing, settles the variants whose proofs are short enough to read in an afternoon, and then follows the same question along three axes until the proofs stretch past breaking. By the end the case for a machine has made itself.

1.1 Graphs and routes

To state the question precisely we need only a few ideas. A *graph* G is a finite set of points, called *vertices*, together with a set of links, called *edges*, each joining two of the vertices. We write $V(G)$ for the set of vertices and $E(G)$ for the set of edges, so that $|E(G)|$ is the number of edges, the very quantity this thesis tries to make as large as possible. One pictures the vertices as dots and the edges as lines between them.

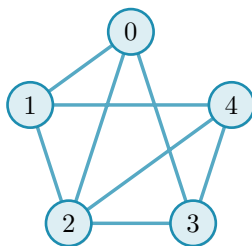


Figure 1.1. A simple graph on five vertices. It is K_5 , the *complete* graph in which every one of the $\binom{5}{2} = 10$ pairs is joined by an edge (the binomial coefficient $\binom{5}{2}$ counts the ways to choose two of the five vertices), with the two edges $\{0, 4\}$ and $\{1, 3\}$ removed, so its edge set $E(G)$ holds eight edges. The five dots are the vertex set $V(G)$ and the lines are the edges. Because at most one line joins any given pair and no edge loops back to its own endpoint, this is a simple graph. The same graph returns in [Figure 1.2](#), once we can count the independent routes between a pair.

A graph is the bare skeleton of a network: the vertices are the towns or the devices, and the edges are the roads or the links between them. We write n for the number of vertices, and the *degree* of a vertex is the number of edges meeting it. Unless we say otherwise our graphs are *simple*, meaning that at most one edge joins any given pair of vertices and its endpoints are distinct.

A *path* from a vertex u to a vertex v is a sequence of distinct vertices that starts at u , ends at v , and joins each consecutive pair by an edge. It is the formal version of a route through the network. Two paths between the same endpoints are *edge-disjoint* if no edge belongs to both, so that the loss of a single edge can break at most one of them. The largest number of pairwise edge-disjoint paths between u and v is their *local edge-connectivity* $\lambda_G(u, v)$, the count of genuinely independent routes the network offers to that pair. The pair with the most independent routes sets the ceiling for the whole graph:

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v),$$

the largest local edge-connectivity anywhere in G . A theorem of Menger [2], recalled in [Appendix A](#), gives this quantity a second face: $\lambda_G(u, v)$ also equals the least number of edges one must remove to cut every route from u to v ([Figure 1.2](#)). Many independent routes between a pair is exactly the same thing as a costly cut between them.

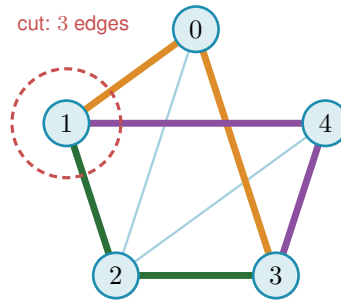


Figure 1.2. Menger’s theorem on the graph of [Figure 1.1](#). The pair 1, 3 carries no edge of its own, yet three routes still join them with no shared edge, 1-0-3 (orange), 1-2-3 (green), and 1-4-3 (purple), each through a different middle vertex, so $\lambda(1, 3) = 3$. The three edges meeting vertex 1 (inside the red dashed ring) are a smallest set whose removal cuts every 1-3 route, and no set of just two edges can cut them all, because the three routes share no edge, so each removed edge breaks at most one route and two removals leave one route intact. The cheapest cut therefore also has size three. The most independent routes a pair can muster always equals the cheapest cut between them, the fact every bound in this thesis leans on.

1.2 Erdős Problem 915

The constraint at the heart of this thesis is a ceiling on connectivity. Requiring $\lambda_G^{\max} \leq m - 1$ says that no pair of vertices is joined by as many as m edge-disjoint paths, or equivalently that every pair can be severed by removing fewer than m edges. The question is how many edges a graph can hold while staying under that ceiling.

Question 1.1 (Erdős Problem 915 [3]). How many edges can a graph on n vertices have if no two of its vertices are joined by m edge-disjoint paths?

Writing the answer as an extremal function,

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1\},$$

the problem asks for $\ell_m(n)$, the most edges possible under the ceiling. Mader settled the simple-graph case completely, proving both halves at once: no graph under the ceiling has more than $\lfloor m(n-1)/2 \rfloor$ edges (the upper bound), and some graph reaches that count (the lower bound). The brackets $\lfloor \cdot \rfloor$ round their content down to the nearest whole number, and their ceiling counterpart $\lceil \cdot \rceil$ rounds up. Both recur throughout, because an edge count is always an integer while the formulas that bound it need not be.

Theorem 1.2 (Mader [4]). *For all $n \geq m \geq 2$,*

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

Mader's own proof is a direct structural induction. The proof given in full in [Appendix A](#) takes a different road, by way of the Gomory–Hu tree, found independently of Mader, together with a characterisation of the graphs that attain equality ([Theorem A.11](#)). This is the road the thesis adopts throughout, because the very same tree returns for the multigraph and hypergraph variants ([Theorem 1.3](#), [Proposition 1.15](#)), so a single argument serves all of them rather than each needing its own. Every graph has such a tree ([Theorem A.2](#)), and we single out the one identity that makes it work, because the same idea reappears when the program of [Chapter 2](#) stores connectivities as a tree (a graph without cycles): a Gomory–Hu tree of G is a weighted tree on the same vertex set in which $\lambda_G(u, v)$ equals the lightest edge on the tree path from u to v . All $\binom{n}{2}$ local connectivities are therefore encoded by $n-1$ numbers, and $\lambda_G^{\max}$ is simply the heaviest edge of the tree.

The whole argument ([Appendix A](#)) is one act of double counting on that tree. The word *charged* below is just accounting: to charge an edge to a tree edge is to assign it there for the count, and the proof works by charging every edge of G to tree edges and then adding the charges up. Each of the $n-1$ tree edges stands for a minimum cut of G , and an original edge of G is charged once for every tree edge lying between its endpoints, so summing the tree-edge weights counts every edge of G at least once and, crucially, counts most of them at least twice. The only edges charged a single time are those joining tree-adjacent vertices, and a *simple* graph has at most $n-1$ of those, one per tree edge. Every other edge is charged twice. Since each tree weight is a local connectivity and so at most $m-1$, the two counts meet at $2|E(G)| - (n-1) \leq (m-1)(n-1)$, which rearranges to the floor. The factor of two, and with it the gap between Mader's $\lfloor m(n-1)/2 \rfloor$ and the larger value $(m-1)(n-1)$ that parallel edges will buy in [Theorem 1.3](#), is exactly the dividend of simplicity, the one place the argument uses that no pair carries a parallel edge.

One example fixes the scale. The complete graph K_n joins every pair, so every pair has $n-1$ routes that share no edge, $\lambda(u, v) = n-1$, while it carries all $\binom{n}{2}$ edges. It therefore meets the ceiling $\lambda^{\max} \leq m-1$ exactly when $m \geq n$, and from that point on the question is settled trivially: the complete graph is feasible, no simple graph on n vertices is denser, so $\ell_m(n) = \binom{n}{2}$ for every $m \geq n$ and raising m further changes nothing. For smaller m the extremal graph must be strictly

sparser than complete, and pinning down exactly how much sparser is the whole problem. That is why the theorem, and the thesis, assume $m \leq n$.

1.3 Three axes of variation

Mader's theorem is the floor we build on, and everything afterwards is a variation on it. There are three independent ways to change the rules, and they generate the landscape this thesis traverses.

The first axis is the *model*, drawn in Figure 1.3. Two independent choices generate it. The first is *arity*: an edge may join exactly two vertices, or a *hyperedge* may join any $r \geq 2$ of them, a route then passing through it between any two of its members. At $r = 2$ a hyperedge is an ordinary edge, so graphs are the $r = 2$ special case and the genuinely new setting begins at $r \geq 3$. The second is *multiplicity*: a *simple* object carries at most one copy of each edge, while a *multi* object allows parallel copies, recorded by a multiplicity $\mu(u, v)$, so a single pair can already carry several independent routes. Because the two choices are independent they form a small lattice. At the top sits the most general object, the *multi-hypergraph*, and lowering one toggle at a time restricts it downward: forbidding repeats gives the simple hypergraph, dropping the arity to two gives the multigraph, and doing both gives the simple graph at the bottom. Every theorem in this thesis is a statement about one node of this lattice, and the three rows of every comparison figure are its simple-graph, multigraph, and hypergraph levels.

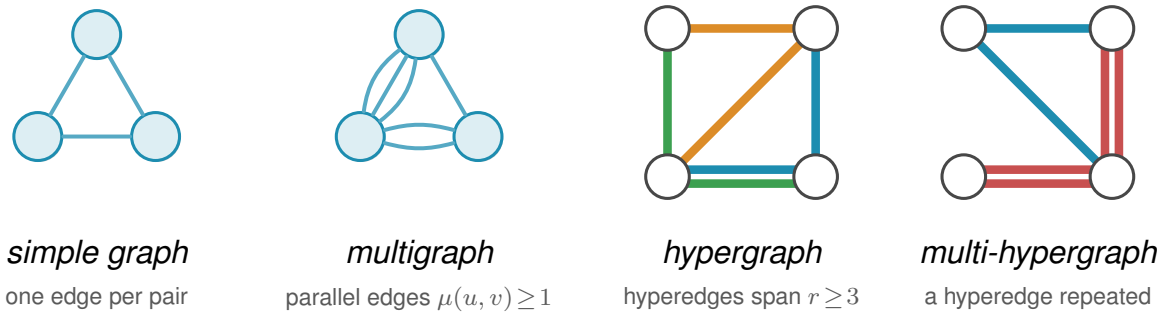


Figure 1.3. The first axis of variation, the *model*, shown for the four object classes this thesis uses. In a *simple graph* each pair of vertices is joined by at most one edge. A *multigraph* permits parallel edges, here a triple and a double, so the multiplicity $\mu(u, v)$ counts how many copies join a pair. A *hypergraph* replaces pairwise edges by hyperedges, each a set of $r \geq 2$ vertices ($r = 2$ gives back ordinary edges, and the depicted examples use $r = 3$), drawn the way a metro map draws a line, threading all of its stations, one colour per hyperedge. Where two hyperedges share a stretch between stations the two lines run side by side, one colour each, exactly as a metro map separates lines along a shared stretch of track. A *multi-hypergraph* lets a hyperedge be repeated, drawn as two parallel lines through the same stations, the hypergraph counterpart of parallel edges. When all hyperedges have the same size r the hypergraph is r -uniform. The same connectivity question, how many independent routes a pair can be forced to share, is asked of all four.

The second axis is the *direction*, a third toggle independent of the other two. Setting it at the top of the model lattice gives the single most general object in the thesis, the *directed multi-hypergraph*, of which every other class is a restriction. Figure 1.4 redraws the four models of Figure 1.3 with arcs in place of edges. An undirected object is the symmetric special case of

a directed one, each edge standing for a matched pair of opposite arcs $u \rightarrow v$ and $v \rightarrow u$, so direction strictly enlarges the family of objects. What it does *not* do is let the directed and undirected problems merge into one. The two opposite arcs offer a round trip the single edge uv does not, the extremal digraphs (directed graphs) are deliberately *lopsided*, with out-degree and in-degree pulled apart in a way no symmetric object can imitate, and the Gomory-Hu tree that settles every undirected case has no directed analogue (Appendix A), so the one-way variants stay distinct in both difficulty and method.

A digraph carries three degree counts at each vertex. The *out-degree* $d^+(v)$ counts the arcs leaving v , and the vertices they reach are its *out-neighbours*. The *in-degree* $d^-(v)$ counts the arcs entering v , and the vertices they come from are its *in-neighbours*. The *total degree* $d(v) = d^+(v) + d^-(v)$ adds the two. A vertex with high out-degree fans arcs outward, one with high in-degree gathers them, and keeping the two balanced is the structural idea behind several of the directed constructions to come. The directed measure itself is easiest to meet on one concrete digraph. Figure 1.5 shows a network with three arc-disjoint routes between the two chosen vertices, which equals the smallest arc-cut that separates them, and the degree bound that limits both, which is exactly how Menger reads in a one-way network.

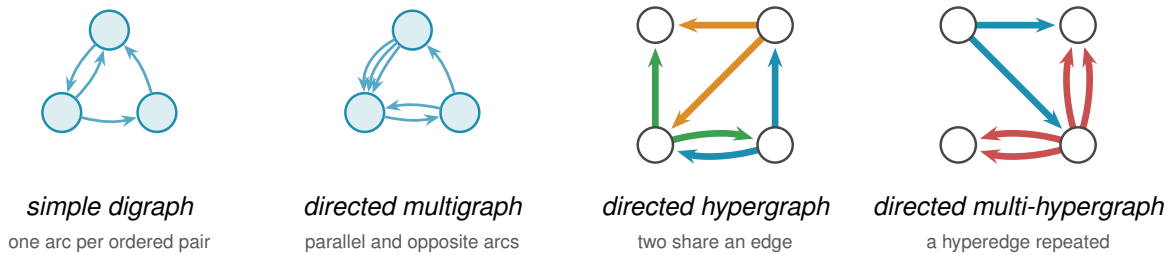


Figure 1.4. The four models of Figure 1.3 redrawn *directed*, the second axis applied to the first. Every edge becomes an arc, and a pair may now carry arcs both ways, the round trip an undirected edge cannot offer. The *simple digraph* still allows at most one arc per ordered pair, so the two-cycle on the top pair is simple. The *directed multigraph* adds parallel arcs, here a triple one way and a bidirected pair. The *directed hypergraph* draws the same three hyperedges as Figure 1.3, in the same colours, each now a metro-styled star, the tail sending one thick coloured arc to each of its heads, the star convention of Figure A.8. Taking each star’s tail to be the middle station of the undirected line keeps every arc on the same edge as before. The blue and green hyperedges still share the bottom edge, now one arc running each way, the directed counterpart of two metro lines sharing a stretch of track. The *directed multi-hypergraph* repeats the red hyperedge, so each of its arcs becomes a parallel pair. This last panel is the directed multi-hypergraph at the top of the model lattice, the most general object in the thesis.

The third axis is the *separation*: routes may be required to avoid sharing an edge, as above, or the stricter *internally vertex-disjoint*, sharing no intermediate vertex, which gives the local vertex-connectivity $\kappa_G(u, v)$ and its maximum $\kappa_G^{\max}$. Whitney’s inequality $\kappa_G(u, v) \leq \lambda_G(u, v)$ [5] records that avoiding shared vertices is at least as demanding as avoiding shared edges, and Figure 1.6 shows the gap on a graph where two routes are edge-disjoint yet forced through one common vertex.

Three models, two directions, two separations: twelve versions of one question. The thesis is the attempt to answer as many of them as honest mathematics allows, and to build a machine

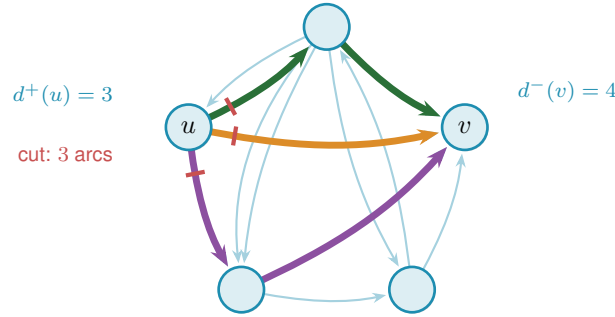


Figure 1.5. A directed version of the K_5 example with multiple arcs. Three arc-disjoint directed routes connect u to v (orange, green, purple). The three red marks cut the arcs leaving u , the first arc of each route, so $\lambda(u, v) = 3$ by Menger's theorem. A directed cut counts only the arcs leaving u 's side, so the lone arc entering u from the top is not part of it. The out-degree $d^+(u) = 3$ and in-degree $d^-(v) = 4$ bound the value from above, and $\lambda(u, v) = \min(3, 4) = 3$ meets the smaller one, at u .

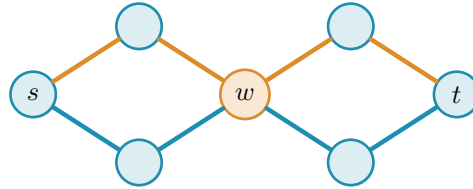


Figure 1.6. The third axis, the *separation*. The two s - t routes drawn here, one orange and one blue, share no edge, so they count as two edge-disjoint paths and $\lambda(s, t) = 2$. Both pass through the single vertex w , however, so they are *not* internally vertex-disjoint, and since every s - t route must cross w we have $\kappa(s, t) = 1$. This is Whitney's inequality $\kappa \leq \lambda$ made concrete: the gap between the two separations is exactly what the vertex-disjoint problem measures, and the rest of the chapter shows it stays shut until $m = 5$.

for the rest. Some answers are short and we prove them here. Some answers are long, and their length is the reason [Chapter 2](#) exists. The rest of this chapter collects the variants whose proofs are short enough to belong in the body, so that the reader meets the easy slope before the climb. [Figure 1.7](#) draws the whole space as one tree, so that the reader has a single map to return to as the chapters visit its branches one at a time.

1.4 Cases with short proofs

Allowing parallel edges removes the parity subtlety, the dependence on whether $m(n-1)$ is even or odd, that produces the floor in Mader's theorem, and the answer becomes exactly linear. Write $L_m(n)$ for the multigraph analogue of $\ell_m(n)$.

Theorem 1.3. *For undirected multigraphs on n vertices, $L_m(n) = (m-1)(n-1)$.*

Proof. For the lower bound, take a spanning tree of K_n , that is, a connected cycle-free subgraph that touches all n vertices and so has exactly $n-1$ edges, and give every tree edge multiplicity $m-1$, meaning $m-1$ parallel copies. The result has $(m-1)(n-1)$ edges. Any two vertices are joined by a unique path in the tree, of some length $k \geq 1$. To separate them one must cut some edge of that path, and the cheapest choice is the lightest tree edge on it, which still carries $m-1$

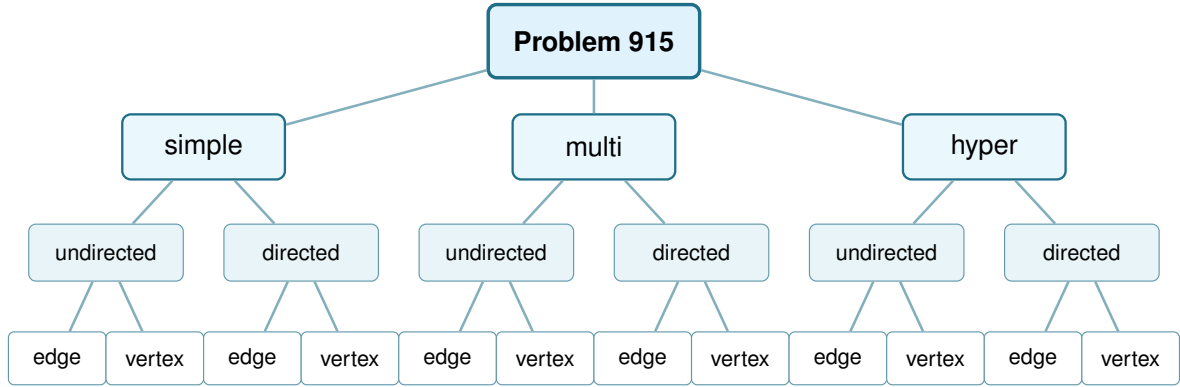


Figure 1.7. The whole space this thesis studies, drawn as one tree. Three model choices (simple graph, multigraph, r -uniform hypergraph), each splitting into undirected or directed, each in turn splitting into edge-disjoint or internally vertex-disjoint routes: twelve leaves, twelve variants. [Figure 4.7](#) redraws this exact tree once every leaf's status is known.

copies, so $\lambda(u, v) = m - 1$ for every pair and $\lambda^{\max} = m - 1$. The graph sits right at the ceiling.

For the upper bound, suppose $\lambda_G^{\max} \leq m - 1$ and build a Gomory–Hu tree T of G ([Theorem A.2](#)), the same $(n - 1)$ -edge summary of all connectivities used for Mader's theorem. Each tree edge e stands for a minimum cut of G , of capacity $c(e) = \lambda_G(u, v) \leq m - 1$ for the pair (u, v) it represents. Now charge each edge of G to a tree edge: an edge $\{x, y\}$ crosses the cut of the lightest tree edge on the tree path from x to y , so it is counted inside that tree edge's capacity. Summing the $n - 1$ capacities therefore counts every edge of G at least once, which already gives $|E(G)| \leq \sum_{e \in T} c(e) \leq (m - 1)(n - 1)$. Unlike Mader's simple-graph count there is no second charge to halve, because parallel edges are now allowed and nothing forces an edge to be counted twice. The tree-of-multiplicity- $(m - 1)$ construction above meets this bound exactly, which is why the answer is the clean product $(m - 1)(n - 1)$ and not a floor. $\square$

The vertex-disjoint question is just as quick at the first non-trivial value. Forbidding two internally vertex-disjoint paths is a strong constraint: it forbids cycles.

Proposition 1.4. *For simple graphs, $k_2(n) = n - 1$, attained exactly by trees.*

Proof. If G contains a cycle, a closed loop of distinct vertices, then any two vertices on that cycle are joined by the two paths of the cycle, which are internally vertex-disjoint, so $\kappa_G^{\max} \geq 2$. Hence $\kappa_G^{\max} \leq 1$ forces G to be acyclic, free of cycles, and an acyclic graph on n vertices has at most $n - 1$ edges, with equality for trees. A tree indeed has $\kappa^{\max} = 1$, since removing the single intermediate vertex on any path disconnects its endpoints. $\square$

Here $k_m(n)$ denotes the vertex-disjoint extremal function, the counterpart of $\ell_m(n)$ for $\kappa^{\max}$. We have just seen $k_2(n) = n - 1 = \ell_2(n)$. The agreement of the edge and vertex problems is no accident at small m , and the moment it breaks is the first place the proofs begin to strain.

1.5 The first divergence: vertex-disjoint paths

Replacing edge-disjoint by internally vertex-disjoint routes changes nothing at first. Leonard proved that the edge and vertex problems give the same answer while m stays small.

Theorem 1.5 (Leonard [6]). *For simple graphs, all $n \geq m$ and $m \leq 4$,*

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

The hypothesis $n \geq m$ is the same one Mader's theorem carries, and it is not decoration. Below it the complete graph K_n is itself feasible, since $\kappa_{K_n}^{\max} = \lambda_{K_n}^{\max} = n-1 \leq m-1$, so the answer is the trivial $\binom{n}{2}$ and the formula overshoots it. At $n=3$, $m=4$ the formula gives 4 where a graph on three vertices has only 3 edges. Every closed form in this thesis carries the same caveat, stated or inherited from its constructions, and the figures cap each curve at the trivial maximum for exactly this reason.

For $m \leq 4$ the extremal graphs for the two problems coincide: Whitney's inequality $\kappa \leq \lambda$ is tight where it matters and the harder vertex constraint costs nothing (Figure 1.9, the three overlapping blue curves). One is tempted to expect this forever. It fails at the very next value, and the failure is sharp.

Theorem 1.6 (Sørensen-Thomassen [7]). *For simple graphs and all $n \geq 13$,*

$$k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 3 > \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n).$$

The restriction $n \geq 13$ is the range over which Sørensen and Thomassen certify that $k_5(n)$ equals the formula, as the Erdős Problems database records it [3]. Below it the true $k_5(n)$ needs the case-by-case structural analysis [7] carries out rather than the closed form, which is why the stated domain begins there.

Remark 1.7 (A counting convention that shifts every quoted constant by one). The additive constant above deserves a warning, because two conventions are in circulation and they differ by exactly one. This thesis defines $\ell_m(n)$ and $k_m(n)$ as the *largest* number of edges a graph can carry while *no* pair is joined by m independent routes. The Erdős Problems database [3] states the same results the other way round, as the *smallest* number of edges that *forces* such a pair, which is one more.

The shift is visible in every entry that both sources state. The database gives $k_2(n) = n$ where a tree has $n-1$ edges, $k_3(2n) = 3n-1$ against $\lfloor 3(2n-1)/2 \rfloor = 3n-2$ here, $k_4(n) = 2n-1$ against $2n-2$, $\ell_5(2n) = 5n-2$ against $5n-3$, $\ell_6(n) = 3n-2$ against $3n-3$, and it writes the Bollobás-Erdős conjecture as $1 + \binom{m}{2}n$ where the convention here gives $\binom{m}{2}n$. Every one of those is exactly one apart, and the values in this thesis are the ones an exhaustive search returns: $k_3(4) = 4$, $k_3(5) = 6$ and $k_4(5) = 8$ were each confirmed by walking every graph of that

size.

Applied to [Theorem 1.6](#) the same shift would read $\lfloor 8n/3 \rfloor - 4$ rather than the $\lfloor 8n/3 \rfloor - 3$ printed above, which is the database's figure carried over unadjusted. Both candidates respect $k_5 \geq \ell_5$, which Whitney's inequality forces, so the thesis's own arithmetic cannot decide between them, and the original paper is the only thing that can. Nothing here depends on which is right: k_5 enters only as the cited first divergence between the edge and vertex problems, and that divergence holds under either constant. A reader who needs the exact value should take it from [\[7\]](#) directly and should note which convention that paper uses before comparing it with anything above.

At $m = 5$ the vertex problem genuinely diverges from the edge problem, visible in [Figure 1.9](#) as the shaded gap that opens between the dashed blue and solid orange curves. Leonard had already found the first crack, an explicit 57-vertex, 141-edge counterexample to the general conjecture at $m = 5$ [\[8\]](#), before Sørensen and Thomassen pinned the exact value down. The reason the crack opens at all is structural. Leonard's earlier method, the one that closes $m \leq 4$, leans on the extremal graphs being assembled from cliques, vertex sets joined in all pairs, glued along shared cliques. A k -tree is built up one vertex at a time: start from a complete graph on $k + 1$ vertices, then again and again add a single new vertex and join it to all k vertices of some clique of size k that is already there. [Figure 1.8](#) grows a 2-tree, hanging each new vertex on an edge already present. These clique scaffolds are exactly the building blocks the method uses. Through $m = 4$ those scaffolds are forced, and counting their edges gives the bound.

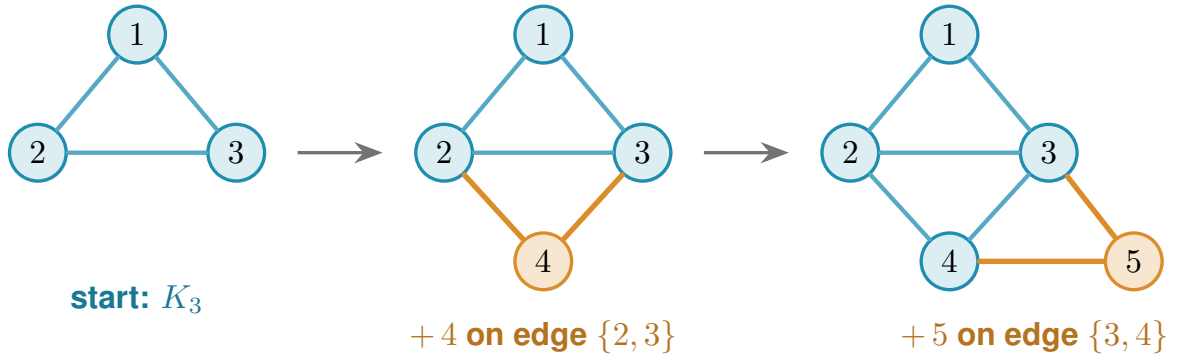


Figure 1.8. A 2-tree grown one vertex at a time, read left to right. The blue triangle on $\{1, 2, 3\}$ is the starting K_3 . Vertex 4 is then joined to the two ends of the edge $\{2, 3\}$, and vertex 5 to the two ends of the edge $\{3, 4\}$ (the new vertex and its two orange edges at each step), each new vertex hung on an edge already present. A general k -tree follows the same recipe with K_{k+1} in place of the triangle and a k -clique in place of the edge. These are the clique scaffolds Leonard's count rests on.

At $m = 5$ they are no longer the unique extremal shape. Additional configurations appear, and the clean count breaks. Sørensen and Thomassen recover the exact value by a delicate structural induction, which we cite rather than reproduce. For $m \geq 6$ even that much is unknown: the vertex-disjoint problem has stood open since 1974, and the extremal structure is genuinely not understood. It is the one direction this thesis does not attempt to force, and [Chapter 4](#) explains why.

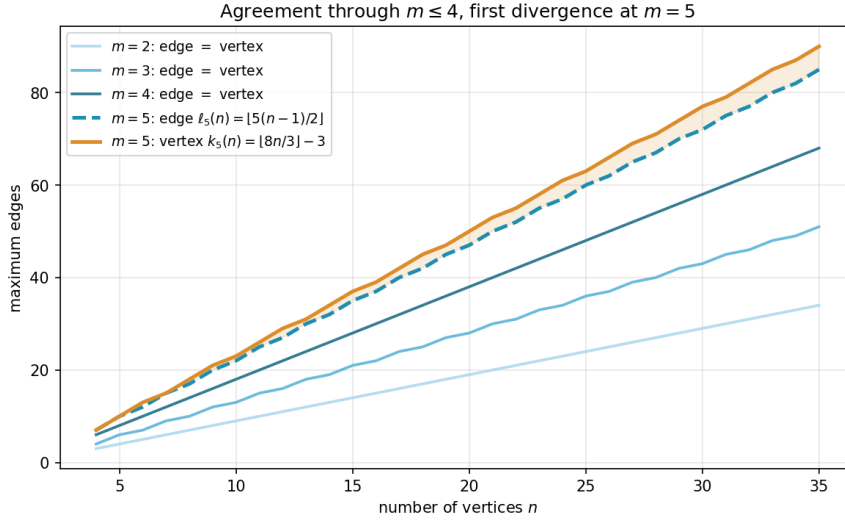


Figure 1.9. The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (dashed blue for edge, solid orange for vertex). Through $m = 4$ the two problems give identical answers, and at $m = 5$ the vertex formula $k_5(n) = \lfloor 8n/3 \rfloor - 3$ separates above the edge formula $\ell_5(n) = \lfloor 5(n-1)/2 \rfloor$, with the gap shaded.

1.6 The directed case and the quadratic phenomenon

Direction breaks a symmetry that undirected graphs never had. In an undirected graph the number of edges is tied to the connectivity through every vertex's degree. In a digraph a vertex can have many arcs in and none out, and a whole graph can lean entirely one way. Send every arc from one half of the vertices to the other and nothing comes back: between any two vertices there is at most one route, yet the number of arcs is quadratic.

Construction 1.8 (One-directional bipartite digraph). Split V into parts A and B with $|A| = \lfloor n/2 \rfloor$ and $|B| = \lceil n/2 \rceil$, and include every arc from A to B . All arcs run from A to B , and none within A , within B , or from B back to A . Since B has no outgoing arcs, a vertex in A cannot reach any other vertex of A , and two vertices of B cannot communicate at all. For any $a \in A$ and $b \in B$ the only directed path from a to b is the single arc $a \rightarrow b$, so $\lambda(a, b) = 1$ and $\lambda^{\max} = 1$. The arc count is $|A| \cdot |B| = \lfloor n^2/4 \rfloor$.

Figure 1.10 draws the construction. Every arrowhead lands in B and none leaves it, so the picture is a wall of one-way traffic: from any a on the left there is exactly one way to reach any b on the right, the single arc between them, and there is no way at all to travel between two vertices on the same side. The arc count $|A| \cdot |B|$ is quadratic, yet no pair carries more than one route. This is the lopsidedness direction buys, and it has no undirected analogue. It competes with a second construction built around a single centre.

That second construction is the directed hub. It keeps each vertex's in-degree and out-degree balanced, the directed measure already met in Figure 1.5, and spends its arcs on a central vertex rather than a one-way wall.

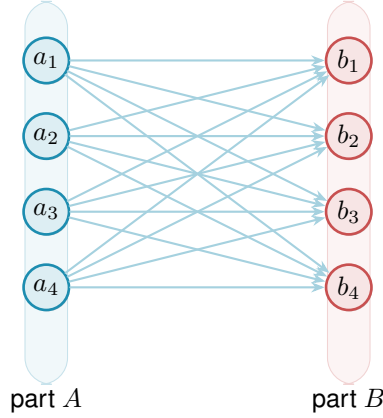


Figure 1.10. The one-directional complete bipartite digraph $B_{k,k}$ of Construction 1.8, drawn here at $n = 8$ with $|A| = |B| = 4$. Every one of the $|A| \cdot |B| = \lfloor n^2/4 \rfloor$ arcs runs left to right, from A to B . There are no arcs inside A , inside B , or back from B to A . Because B has no outgoing arcs and A no incoming ones, the only route from any a_i to any b_j is the single arc $a_i \rightarrow b_j$, so every pair has local edge-connectivity 1 while the arc count grows quadratically.

Construction 1.9 (Directed hub). For $n \geq m \geq 2$, take a hub vertex h and label the remaining $N = n - 1$ vertices $0, 1, \dots, N - 1$. Add the $2N$ hub arcs $h \rightarrow i$ and $i \rightarrow h$ for every i , and on the non-hub vertices add the circulant arcs $i \rightarrow i + j \pmod{N}$ for $j = 1, \dots, m - 2$, the target label wrapping around past N back to 0 (that ring pattern is what *circulant* means, and “mod N ” is the wrap-around). Counting these gives $2N$ hub arcs (each of the N spokes joined to h in both directions) plus $(m - 2)N$ circulant arcs ($m - 2$ layers, one arc per spoke), so the total is $2N + (m - 2)N = mN = m(n - 1)$ arcs. Every non-hub vertex has $d^+ = d^- = 1 + (m - 2) = m - 1$, and any ordered pair of vertices contains a non-hub endpoint, so the directed degree bound (Lemma A.3, which says $\lambda(u, v) \leq \min(d^+(u), d^-(v))$), a route out of u must use one of u ’s out-arcs and one of v ’s in-arcs) gives $\lambda^{\max} \leq m - 1$.

At $m = 2$ this construction is easiest to picture: the circulant layer is empty and what remains is the *bidirected star* of Figure 1.11, a hub joined to each spoke by a matched pair of opposite arcs. Every spoke can reach every other only by going in to the hub and back out, two hops, so no pair is ever joined by two independent routes, and the arc total is exactly $2(n - 1)$. For $m \geq 3$ not all $m(n - 1)$ arcs can touch the hub, since a simple digraph has at most $2(n - 1)$ arcs incident to one vertex: the surplus lives in the circulant layers among the spokes, tuned so that no degree exceeds the budget. The two constructions compete on different scales, linear against quadratic, and which one wins depends on n . At $m = 2$ the competition resolves exactly.

Theorem 1.10 (Directed arc value at $m = 2$). *For simple digraphs,*

$$\ell_2^{\text{dir}}(n) = \max\left(2(n - 1), \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

The hub construction leads for $n < 7$, the one-directional bipartite construction leads for $n > 7$, and the two tie at $n = 7$, where both reach 12 arcs.

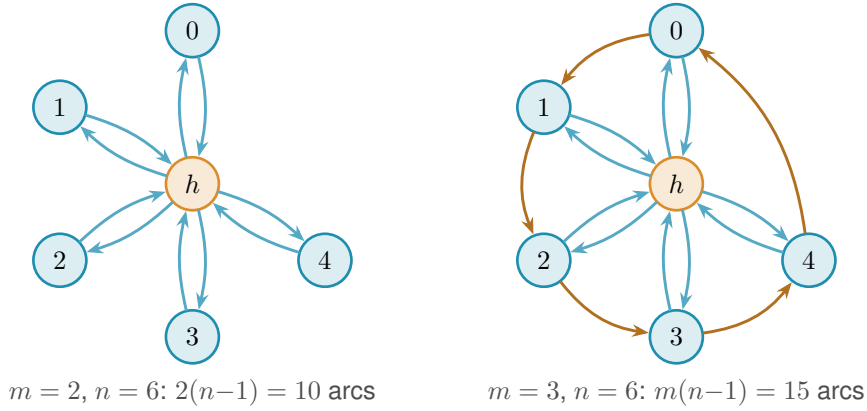


Figure 1.11. The directed hub at $n = 6$ for two values of m . *Left* ($m = 2$): the bidirected star. The hub h is joined to each spoke by one arc in each direction, giving $2(n - 1) = 10$ arcs. No circulant layer is added, so all spoke communication routes through h , so $\lambda^{\max} = 1$. *Right* ($m = 3$): a single circulant layer is added among the five spokes. The blue arcs are the same hub-spoke pairs as on the left. The orange arcs are that one circulant layer, the cycle $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$, so each spoke has $d^+ = d^- = m - 1 = 2$. The total is $m(n - 1) = 15$ arcs and $\lambda^{\max} = 2$. For $m \geq 3$ the spokes are no longer isolated from one another, and tracing all the routes becomes the argument that fills the rest of the chapter.

The lower bound is the better of the two constructions. The upper bound is where the trouble starts.

Idea of the upper bound at $m = 2$. The value $M(n) = \max(2(n - 1), \lfloor n^2/4 \rfloor)$ is a contest between a linear term and a quadratic one, and which leads is pure arithmetic: $2(n - 1) \geq \lfloor n^2/4 \rfloor$ precisely for $n \leq 7$, with equality at $n = 7$, where both give 12. So M follows the bidirected star up to the crossover and the bipartite digraph past it, and $n = 7$ is the single value where the two constructions tie. The proof that no digraph beats $M(n)$ is an induction on n (Appendix A). Delete a vertex of *minimum* total degree: since the degrees sum to twice the arc count, an averaging argument bounds that degree by $2a/n$, where a is the number of arcs, and the remaining digraph on $n - 1$ vertices still respects the ceiling, so the inductive bound applies to it. Feeding the two facts together gives $a \leq \frac{n}{n-2} \lfloor (n - 1)^2/4 \rfloor$, and a short parity check shows the right-hand side never exceeds $\lfloor n^2/4 \rfloor$ once $n \geq 8$: for even n it lands exactly on the bipartite value, and for odd n a sub-unit slack is swallowed by the fact that a is an integer. The averaging closes the step with no separate hub case, which is why the difficulty is not depth but bulk. Nothing in the argument is deep. It is a careful walk through two parity regimes, with the base cases $n \leq 7$ settled by exhaustive computer search rather than by hand. That experience of a correct but long and almost entirely mechanical case-check is exactly what motivates the chapters that follow. One genuinely wants to hand the case-checking to a machine.

The vertex-disjoint version of the same question has the same answer at $m = 2$, but it is worth being careful about why, because Whitney's inequality is easy to read backwards here. Since $\kappa \leq \lambda$ pairwise, a digraph that respects the arc ceiling automatically respects the vertex one, so the vertex-feasible family *contains* the arc-feasible one and is in general strictly larger. The free consequence therefore runs one way only: the two constructions above keep $\kappa^{\max} = 1$

as readily as $\lambda^{\max} = 1$, which makes them lower bounds for the vertex problem too. No upper bound comes with them. What settles the vertex value is that the same induction goes through unchanged, because deleting a vertex lowers $\kappa^{\max}$ for exactly the reason it lowers $\lambda^{\max}$, and because the base cases $n \leq 7$ were re-run under the stricter separation and returned the same six numbers. [Appendix A](#) gives both halves, and [Remark A.16](#) there spells out the direction trap in full.

Theorem 1.11 (Directed vertex value at $m = 2$). *For simple digraphs,*

$$k_2^{\text{dir}}(n) = \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right) = \ell_2^{\text{dir}}(n).$$

1.7 The failure of direct proof for $m \geq 3$

For $m \geq 3$ the induction no longer closes, and the natural guess is wrong. The natural initial conjecture was that the two constructions above continued to be the whole story, giving $\max(m(n-1), \lfloor n^2/4 \rfloor)$. A single graph on ten vertices refutes it.

Remark 1.12 (A thirty-arc counterexample). Take $A = \{a_1, \dots, a_5\}$ and $B = \{b_1, \dots, b_5\}$ with all 25 arcs from A to B , and add a directed five-cycle $b_1 \rightarrow b_2 \rightarrow b_3 \rightarrow b_4 \rightarrow b_5 \rightarrow b_1$ inside B . This digraph has 30 arcs. Every vertex of B has exactly one extra in-arc from inside B , so two arc-disjoint routes from a_i to b_j exist: the direct arc $a_i \rightarrow b_j$, and the detour $a_i \rightarrow b_{j-1} \rightarrow b_j$ using the 5-cycle predecessor of b_j (indices mod 5). For the matching upper bound, the only in-arcs of b_j reachable on a path from a_i are the direct arc $a_i \rightarrow b_j$ and the cycle arc $b_{j-1} \rightarrow b_j$. Removing both disconnects a_i from b_j , while neither alone does, so the minimum cut has size 2. Hence $\lambda(a_i, b_j) = 2$. Routes between two vertices of B never leave B and give $\lambda(b_i, b_j) = 1$. Hence $\lambda^{\max} = 2$, so the graph is feasible at $m = 3$, yet $30 > \max(3 \cdot 9, 25) = 27$.

[Figure 1.12](#) draws this thirty-arc digraph and traces the mechanism. The twenty-five faint arcs are the complete one-directional bipartite layer from A to B , the construction we already understand. On top of them the orange five-cycle gives each vertex of B a single extra in-arc from inside B . That one extra arc is exactly what doubles the connectivity. For the marked pair a_1, b_2 the two heavy routes show how: the blue arc is the direct hop $a_1 \rightarrow b_2$, and the green path $a_1 \rightarrow b_1 \rightarrow b_2$ borrows the cycle arc $b_1 \rightarrow b_2$ for its second step. The two share no arc, so $\lambda(a_1, b_2) = 2$, and the same pair of routes exists for every a_i, b_j . Five extra arcs lift the whole digraph from one independent route per pair to two, and the thirty arcs overshoot the old prediction of twenty-seven.

The counterexample generalises: the five-cycle gave every vertex of B exactly one extra in-arc from inside B , and the budget for such arcs is $m - 2$ per vertex, not one. [Figure 1.12](#) is the instance $m = 3$, where that budget is one and a single cycle suffices. For larger m each b_j collects $m - 2$ cyclic predecessors and the same picture acquires more concentric chords inside B .

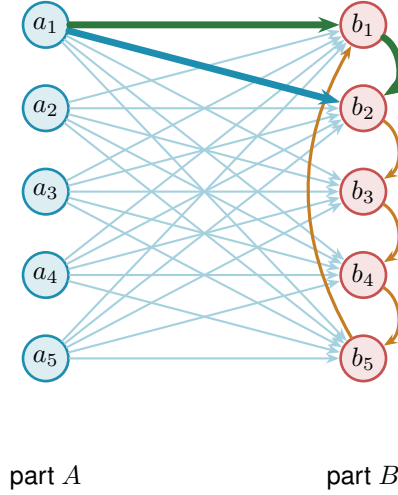


Figure 1.12. The thirty-arc counterexample of [Remark 1.12](#): the augmented bipartite digraph ([Construction 1.13](#)) at $m = 3$, $n = 10$, $|A| = |B| = 5$. The 25 faint arcs are the complete one-directional layer from A to B , and the 5 orange arcs form the directed cycle $b_1 \rightarrow b_2 \rightarrow b_3 \rightarrow b_4 \rightarrow b_5 \rightarrow b_1$ inside B , one extra in-arc per vertex of B . The heavy blue arc and the heavy green detour show the pair a_1, b_2 carrying 2 arc-disjoint routes: the direct arc $a_1 \rightarrow b_2$ (blue) and the two-step detour $a_1 \rightarrow b_1 \rightarrow b_2$ (green) through the cycle predecessor b_1 . Hence $\lambda^{\max} = 2$, feasible at $m = 3$, while $30 > \max(3 \cdot 9, 25) = 27$.

Construction 1.13 (Augmented bipartite digraph). For $m \geq 2$ and $n \geq m$, set $|B| = \lceil (n + m - 2)/2 \rceil$ and $|A| = n - |B| = \lfloor (n - m + 2)/2 \rfloor$, include every arc from A to B , and inside B give each vertex b_j the $m - 2$ in-arcs from its cyclic predecessors $b_{j-1}, \dots, b_{j-(m-2)}$ (indices mod $|B|$). The total is $\lfloor (n + m - 2)^2/4 \rfloor$ arcs.

For the connectivity, fix $a \in A$ and $b \in B$: the direct arc and the $m - 2$ two-step detours $a \rightarrow b' \rightarrow b$ through the in-neighbours b' of b inside B are $m - 1$ arc-disjoint routes, and they are all there are, because the only in-arcs of b reachable from a are exactly those $m - 1$ arcs, which therefore form a cut. Pairs inside B are limited by the in-degree $m - 2$, pairs inside A and pairs from B to A have no routes at all, so $\lambda^{\max} = m - 1$.

At $m = 2$ the inside of B is empty and the construction reduces to [Construction 1.8](#), with the balanced partition $(\lfloor n/2 \rfloor, \lceil n/2 \rceil)$. At $m = 3$ the formula gives $|B| = \lceil (n + 1)/2 \rceil$, which coincides with $\lceil n/2 \rceil$ at odd n but gives $|B| = n/2 + 1$ at even n . Both the balanced and the shifted partition achieve the same arc count $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$ in this case, so the two are interchangeable at $m = 3$. The shift to a strictly larger B is structurally significant only at $m \geq 4$, where the balanced partition is suboptimal. At $m = 3$, $n = 10$ it is the thirty-arc counterexample of [Remark 1.12](#), drawn in [Figure 1.12](#) (using the equivalent balanced partition $|A| = |B| = 5$).

The two rival shapes invite a political picture. The hub is a single king through whom all traffic must pass, cheap to build but limited by the one throne. The bipartite family is a flat cabinet of ministers, each handling its own dossier with no central seat, and it scales quadratically. Which arrangement packs more arcs under the ceiling depends on n , and for $m \geq 3$ the cabinet overtakes the king once n is large. This is the corrected shape, and it leads to the standing

conjecture for the directed arc problem.

Conjecture 1.14. *For simple digraphs and all $n \geq m \geq 2$,*

$$\ell_m^{\text{dir}}(n) = \max\left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right).$$

The hypothesis $n \geq m$ is the one both constructions already carry, and it is needed for the same reason as in [Theorem 1.5](#): below it the complete digraph is feasible and the answer is the trivial $n(n-1)$, which the formula exceeds. At $n = 3, m = 4$ the formula gives 8 where three vertices carry only 6 arcs. The exhaustive search confirms the value is 6 there.

For $m = 2$ this is [Theorem 1.10](#). For $m = 3$ the formula $\lfloor (n+1)^2/4 \rfloor$ equals $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$, so the conjecture reduces to the previous formula for $m = 2$ and $m = 3$. For $m = 3, n = 10$ the second branch gives $\lfloor 121/4 \rfloor = 30$, matching [Remark 1.12](#). For $m \geq 4$ the new formula and the old balanced-partition count $\lfloor n^2/4 \rfloor + (m-2)\lceil n/2 \rceil$ no longer agree: the quadratic branches already differ (for $m = 4$ the new one is larger by exactly one at every even n), but the hub term $m(n-1)$ dominates both at small n , so the full conjectured value first overtakes the old one at $n = 12$ for $m = 4$. The construction of [Construction 1.13](#) with the shifted partition witnesses the difference. The lower bound holds for all m , witnessed by [Construction 1.9](#) on the first branch and [Construction 1.13](#) on the second. The upper bound is open, and [Chapter 4](#) returns to it with what structural progress the machine and the hand can jointly supply.

1.8 The hypergraph variant

The third model deserves its own short treatment, not because its base case is hard but because it is the one variant whose objects are stored and manipulated differently from all the rest, and it is worth meeting that difference now rather than at the moment of computation. A hypergraph keeps no multiplicity matrix, only the list of its hyperedges, each a set of r vertices. We fix the edge size r , following the extremal literature, for a structural reason: under a connectivity ceiling a small edge is always at least as efficient per route as a large one, so a hypergraph free to mix sizes would maximise its edge count by using pairs only, and the question would collapse back to the graph case. Fixing r is what makes the hypergraph question a new question.

The route between two vertices is a *Berge path*, which alternates vertices and hyperedges so that each consecutive pair of vertices lies in the hyperedge listed between them, and two such routes are edge-disjoint when they share no hyperedge. Concretely, in a three-uniform hypergraph a route from u to w might be the single step $u, \{u, x, w\}, w$ that crosses one hyperedge, or a longer alternation $u, \{u, x, y\}, y, \{y, z, w\}, w$ that switches hyperedge at the intermediate vertex y .

With routes redefined this way the same Gomory–Hu machinery that proved the multigraph value carries over, and the base case has the same linear shape. The picture to keep in mind is a metro map: each hyperedge is a line, and a route from one vertex to another rides a line and

changes to another line at a station the two share.

A single hyperedge may serve only one route, so two Berge routes that share no hyperedge are edge-disjoint, the hypergraph counterpart of two edge-disjoint paths in a graph. Reading routes off a metro map is a matter of seeing which lines share track: two hyperedges that share several vertices run side by side along the stretch between them, as in [Figure 1.3](#), and two routes are hyperedge-disjoint exactly when the lines they ride share no track at all.

The separation axis needs its own definition here, and it is worth fixing now because the natural guess is not the one this thesis uses. Two Berge routes are called *internally disjoint* when they share neither an intermediate vertex nor a hyperedge. Both halves are required. Demanding only that the routes avoid each other's intermediate stations would let two routes ride the same line, and a line that can run only one train at a time cannot serve two routes at once, so the resulting count would not correspond to anything the metro map can carry. Requiring both is also what makes the vertex measure the exact analogue of the graph one under the translation of [Appendix A](#), where a hypergraph becomes its incidence graph, each hyperedge becomes a node, and internally disjoint routes become internally disjoint paths in the ordinary sense. It has the pleasant side effect that Whitney's inequality survives the move to hypergraphs by definition, since an internally disjoint family is in particular hyperedge-disjoint. Write $\kappa_{\mathcal{H}}(u, v)$ for the largest such family and $\kappa_{\mathcal{H}}^{\max}$ for its maximum over pairs.

Proposition 1.15 (Hypergraph edge bound). *An r -uniform hypergraph on n vertices with Berge edge-connectivity at most $m - 1$ has at most $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. A simple hypergraph attains the bound whenever $m - 1 \leq \binom{n-2}{r-2}$, and a multihypergraph attains it whenever $(r - 1) \mid n - 1$.*

The proof, given in [Appendix A](#), runs the multigraph argument through the hypergraph Gomory–Hu theorem of Bérczi, Chandrasekaran, Király and Kulkarni [9], charging each hyperedge to the at least $r - 1$ tree cuts it crosses. The star hypertree ([Figure 1.13](#)) attains the $m = 2$ case, a hub joined to disjoint blocks of $r - 1$ vertices, with Berge connectivity one everywhere and $\lfloor (n - 1)/(r - 1) \rfloor$ hyperedges, exactly as a tree attained the multigraph bound. One genuine surprise is worth recording here: for $r \geq 3$ the bound is attained by *simple* hypergraphs, with no repeated hyperedge ([Theorem A.42](#)). In the graph case simplicity is exactly what separates Mader's $\lfloor m(n - 1)/2 \rfloor$ from the multigraph value $(m - 1)(n - 1)$. One edge size higher, it costs nothing.

The proposition counts hyperedges, but it does not say how to *measure* the Berge connectivity of a hypergraph we are handed, and that measurement is the genuinely new part. The metro picture shows why. An ordinary edge is a one-stop link between a single pair, so policing edge-disjointness means watching each pair on its own. A hyperedge is a whole line: it stops at all r of its vertices and so touches $\binom{r}{2}$ pairs at once, yet like a line that can run only one train at a time it may serve only one route. When several routes want to board the same line at different stations, the checker must wave exactly one through and turn the rest away, and it must do this for every

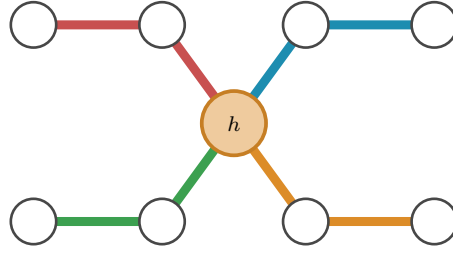


Figure 1.13. The star hypertree that attains the $m = 2$ hypergraph bound, here $r = 3$ and $n = 9$, drawn as a metro map with the hub h centred between two rows of stations. The single hub station lies on every line, and each of the $\lfloor (n-1)/(r-1) \rfloor = 4$ hyperedges is one metro line of $r = 3$ stations: it leaves the hub diagonally to an inner station and then runs horizontally to its outer station, hanging a fresh block of $r-1 = 2$ vertices off the hub. Because every line meets every other only at h , no pair has two hyperedge-disjoint routes, so the Berge connectivity is one everywhere, exactly as a spanning tree attained the multigraph bound.

line simultaneously. That bookkeeping is the one new piece of machinery the next chapter builds, and it is why the hypergraph rejoins the family only once the checker is in place.

1.9 Random graphs as intuition

The short proofs above hinted that the edge and vertex problems travel together for a while, agreeing through $m = 4$ before the split at $m = 5$. That separation is a statement about extremal graphs, the densest ones the ceiling allows, and it is worth asking whether the same split can be seen in ordinary graphs drawn at random, with no extremal tuning at all. The random model is the natural place to build intuition, because it shows the qualitative behaviour of the connectivity notions before any construction is pushed to its limit, and we will return to it whenever a picture of the typical graph clarifies a result. It is not a detour from the program to come, either: a sampled $G(n, p)$ is an ordinary graph, measured by the very same connectivity checker the next chapter builds for every other variant, so the random model is simply the one pipeline turned to *observe* the typical case rather than to prove or to discover the extremal one.

Each of the $\binom{n}{2}$ possible edges of $G(n, p)$ is included independently with probability p , and on the resulting graph we read off both binding connectivities, the edge version $\lambda_G^{\max}$ and the vertex version $\kappa_G^{\max}$, from the very same sample. Whitney's inequality $\kappa_G^{\max} \leq \lambda_G^{\max}$ guarantees the vertex value never exceeds the edge one, but it is silent on how often the two actually come apart.

Figure 1.14 looks directly at the distributions, fixing $n = 16$ and histogramming both connectivities over many samples at three densities. Two things are visible at once. As the density rises the whole distribution marches rightward, from a binding connectivity around five at $p = \frac{1}{4}$ to around thirteen at $p = \frac{3}{4}$, so by $n = 16$ the typical graph already sits well past the $m = 4$ at which the proofs agreed. And comparing the edge row to the vertex row, the vertex distribution sits at or just to the left of the edge one, never to its right, which is Whitney's inequality seen across a whole distribution rather than a single graph.

The two pull apart on a fraction of samples that shrinks as the density grows, from about a

fifth of the samples at $p = \frac{1}{4}$ to only a handful by $p = \frac{3}{4}$. The reason is structural. For $\kappa^{\max}$ to fall below $\lambda^{\max}$ some pair must carry more edge-disjoint routes than internally vertex-disjoint ones, which happens only when several of those routes are forced through one shared intermediate vertex. In a sparse graph routes are scarce and a single low-degree vertex can be exactly such a bottleneck, but as p grows every vertex gains degree and the graph offers many alternative intermediate vertices, so independent routes almost never need to reuse one. In the dense limit the graph approaches the complete graph, where $\kappa^{\max} = \lambda^{\max} = n - 1$ for every pair and the gap closes entirely. The divergence the thesis records at $m = 5$ is therefore not an artefact of the extremal construction. It is already there, in miniature, in the random model.

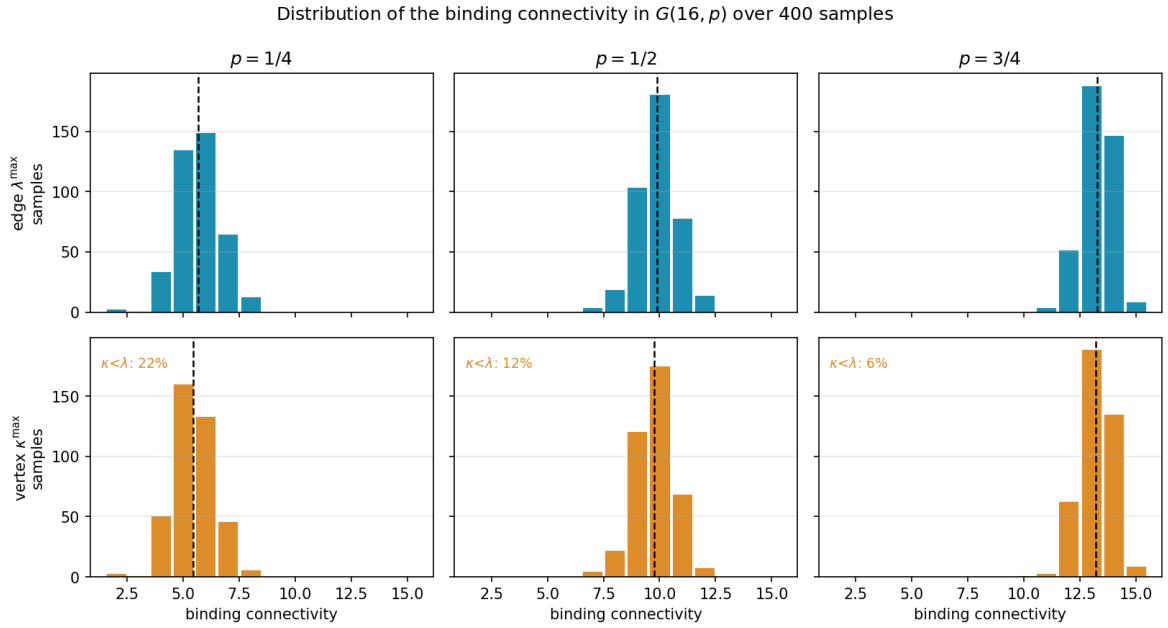


Figure 1.14. Distributions of the binding connectivity in $G(16, p)$ over 400 samples, at densities $p = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$. There are $2^{\binom{16}{2}} = 2^{120}$ labelled graphs on 16 vertices, far too many to enumerate, so these are Monte Carlo samples rather than the full population, and 400 is plenty to show the shape of each distribution. The top row histograms the edge-connectivity $\lambda^{\max}$, the bottom row the vertex-connectivity $\kappa^{\max}$, measured on the same samples, with the mean marked by a dashed line and the fraction of samples on which $\kappa^{\max} < \lambda^{\max}$ printed on each vertex panel. Reading a column top to bottom, the vertex distribution sits at or to the left of the edge one, which is Whitney's inequality, and the gap is largest at the sparsest density (22% of samples) and closes as p grows (6%). All six panels share one axis range.

The random model also supplies a landmark of a different flavour. Below a sharp density the forbidden configuration is almost surely absent, and above it almost surely present, and the location of that threshold is the same for every one of the connectivity notions above.

Theorem 1.16 (Appearance threshold). *Let $m = m(n) \geq 2$ grow with n so that $m/\ln n \rightarrow \infty$ and $m = o(n)$, and let $c > 0$ be a constant. In the binomial random graph $G(n, p)$ with $p = c \cdot m/n$,*

$$\lim_{n \rightarrow \infty} \Pr[\lambda^{\max} \geq m] = \begin{cases} 1 & \text{if } c > 1, \\ 0 & \text{if } c < 1, \end{cases}$$

so the threshold for the appearance of a pair with local edge-connectivity at least m is $p^* = m/n$. The same threshold governs local vertex-connectivity and the directed analogues ([Remark A.59](#)). It does not govern the hypergraph models, which live on a different density scale ([Remark A.60](#)).

The proof is a concentration argument with two halves ([Appendix A](#)). Above the threshold the edge count concentrates past Mader's bound, so [Theorem 1.2](#) itself forces a high pair into existence. Below it the maximum degree stays under m , and the degree bound caps every local connectivity. The two hypotheses earn their keep. The first, $m/\ln n \rightarrow \infty$, says m must outgrow the natural logarithm of n , and that is what lets a union bound over the n vertices close, the union bound being nothing more than adding up the n individual failure chances and needing even their sum to vanish. The second, $m = o(n)$, says m stays negligible next to n , which keeps the density $p = cm/n$ below one and in the sparse regime where the threshold lives. Two caveats on scope, and the second is the one that limits how far the result may be quoted. The upper half quoted here uses Mader's bound, which is the simple undirected edge case, so it is that variant the proof settles directly. The vertex-disjoint and directed analogues sit at the same threshold $p^* = m/n$ but reach it through their own extremal bounds rather than Mader's, possibly with a different leading constant, as [Remark A.59](#) records. The hypergraph models are a different matter and are *not* covered. A threshold of this kind sits where a vertex's expected degree reaches m , and in a binomial r -uniform hypergraph a vertex lies in an expected $p\binom{n-1}{r-1}$ hyperedges, so the balance point is $m/\binom{n-1}{r-1}$, smaller than m/n by a factor of order n^{r-2} once $r \geq 3$. [Remark A.60](#) works this out and [Figure B.5](#) plots each model against its own scale. So the collapse of complexity is real but bounded: within a fixed edge size the distinctions between edges and vertices and between the two directions wash out and one density decides the matter, while changing the edge size changes the density at which everything happens.

One limitation must be stated plainly, lest the threshold be read as evidence about the problem this thesis actually studies. The growth hypothesis $m/\ln n \rightarrow \infty$ forces m to climb with n , so the theorem says nothing whatever about a fixed small m , the regime of every exact value above, where m is 2, 3, or 4 and n runs to infinity around it. There the union bound is vacuous and the appearance of a dense pair is governed by rare low-degree configurations that need finer, Poisson-type tools. The random model therefore enters as contrast and not as evidence: it shows how cleanly the question collapses in one limit, which throws into relief how stubbornly it resists in the fixed- m limit the rest of the thesis must attack by hand and by machine.

1.10 Where this work sits

It is worth pausing, before the machine, to place the question and the method against the two traditions they descend from: the extremal graph theory that posed the problem and the computational mathematics that will help settle it.

The mathematical lineage is the older of the two. The problem belongs to the family of extremal connectivity questions opened by Bollobás and Erdős, who asked how dense a graph can

be while every pair of vertices stays below a fixed connectivity [1], logged today as Erdős Problem 915 [3]. Mader’s theorem [4] closed the simple undirected edge case with the clean floor $\lfloor m(n-1)/2 \rfloor$ recalled in [Theorem 1.2](#), and it is the floor every variant in this thesis is measured against. The vertex-disjoint branch has a longer and rougher history: Leonard [6] showed the edge and vertex problems coincide through $m \leq 4$, and Sørensen and Thomassen [7] found the first divergence at $m = 5$ with $k_5(n) = \lfloor 8n/3 \rfloor - 3$, after which the exact values $k_m(n)$ remain unknown. That gap between a settled edge case and a stalled vertex case is exactly the terrain [Chapter 1](#) has been mapping, and the directed and hypergraph axes extend it further. The thesis adds to this lineage not a single new closed form but a uniform way of pushing each axis as far as exact computation allows.

The computational tradition this thesis’s pipeline belongs to is the practice of settling combinatorial questions by machine in a way a human can audit, and it splits into a few schools worth contrasting with the certify-then-prove loop built next. The oldest is *exhaustive generation*: enumerate every object of a given size up to isomorphism and check each one. McKay and Piperno’s nauty/geng toolkit [10] is the standard engine for this, and its flagship results include exact small Ramsey numbers such as $R(4, 5) = 25$ [11]. The enumeration of [Chapter 2](#) is the same idea, run on the twelve variants of Problem 915. The second school is *SAT-assisted proof*, in which a question is encoded in propositional logic and a solver emits a machine-checkable certificate: Heule, Kullmann and Marek’s resolution of the Boolean Pythagorean triples problem [12], whose verified proof runs to some two hundred terabytes, and Heule’s determination of Schur number five [13] are the canonical examples of certified solver output at scale. The third is *flag algebras*, Razborov’s framework [14] that converts asymptotic extremal bounds into semidefinite-programming certificates, the dominant tool for density problems in the large- n limit. Closest of all to the method here is the fourth, *certification by linear and integer programming* (LP/ILP), where a feasibility or optimality argument over the rationals or integers is written as a linear program, an optimisation with linear constraints, and the proof is the certificate of optimality the solved program hands back, a short list of numbers anyone can verify by direct arithmetic. This is precisely the form the cut-counting certifier of [Chapter 2](#) takes.

Against that backdrop, what is distinctive here is consolidation rather than a new solver. A single checker measures connectivity exactly across all twelve problem variants on one data model, so search and certification share one codebase rather than living in separate tools. The certificates it emits are exact rational or integral objects, not floating-point estimates, so an upper bound it reports is a proof and not evidence. And because the same program both hunts for dense graphs and certifies the bounds they suggest, the discover step and the prove step are two modes of one loop, which is the arrangement the next chapter builds. What makes the consolidation honest is that it never blurs three kinds of knowledge: a value can be *proved* (by hand, or by exhausting every graph of a size), *certified* (by the same optimisation, reaching a vertex or two further), or only *conjectured* (witnessed by a dense graph the search exhibits, with no matching upper bound). This trichotomy, named in [Section 3.6](#) and obeyed by the summary of [Chapter 4](#), is the through-line of everything that follows.

Computer note

We are now stuck in three different ways: a fifty-year-old open problem at $m \geq 6$ for vertices, a long proof we would rather not check by hand at $m = 2$ for arcs, and a conjecture with no proof at all at $m \geq 3$. The base cases of the long induction were verified by exhaustive search, and the thirty-arc counterexample was found the same way. The next chapter builds a single program that models all of these cases at once, measures their connectivity exactly, and proves the small-case upper bounds outright.

Chapter 2

Certifying Bounds by Machine

The last chapter ended in three different kinds of stuckness, and all three share a shape. The objects are graphs of one flavour or another, the constraint is always a ceiling on connectivity, and the work is always either measuring a given graph or checking many small ones. That shared shape is what a single program can exploit. This chapter builds it, and then uses it for one purpose only: to *prove*. It gives every variant one representation, measures connectivity exactly through a max-flow checker, checks small cases by honest enumeration, and turns the finite checks the proofs of [Chapter 1](#) relied on into a single auditable certifier. The hunt for dense examples, which proves nothing on its own, waits until [Chapter 3](#).

A word on what the program is for. It is a microscope, not a source of final answers. It measures connectivity exactly, so what it reports about a given graph is a fact. It enumerates graphs of a fixed size, so for that size it can prove a theorem. It will later discover dense graphs, which are only lower bounds, but not here. Each of these has a different epistemic weight, and the program is built so that the three never get confused. [Figure 2.1](#) draws this spine: a single *model* holds every variant, one exact *measure* reads connectivity off it, and from that measurement a *prove* step bounds a fixed size from above, with the *discover* step of the next chapter and a final *observe* step for the random model added later. Every routine introduced from here on is shown as a small card carrying its name, its inputs and output, a one-line account of how it works, and a coloured badge naming its place in the spine. The function bodies are not reproduced, because the point is the interface, not the code.

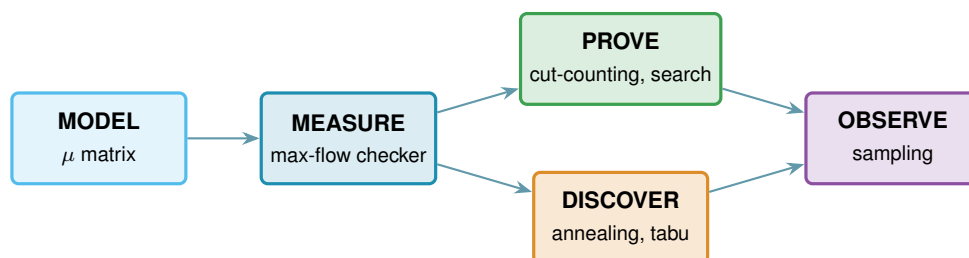


Figure 2.1. The architecture as one pipeline. A single data model, the multiplicity matrix μ , feeds an exact measurement, the max-flow checker. That measurement feeds the two bound-producing roles: a prover for upper bounds (the cut-counting optimisation and the exhaustive search), built in this chapter, and a discoverer for lower bounds (simulated annealing, or tabu search), built in [Chapter 3](#). A final sampling role studies the random model. The badge colours on the code cards below match the boxes here.

2.1 The solver

Everything in this chapter is a single function, `solve`, and it helps to hold its most basic version in mind from the start, because every later idea is a refinement of it and nothing is ever a separate program for a separate case. Given n and m , the basic strategy is as plain as it sounds. Walk through every graph on n vertices, measure the largest local connectivity $\lambda^{\max}$ of each with a max-flow computation, and keep the densest graph that stays at or below the ceiling $m - 1$. Because it has looked at every graph, the count it returns is the exact answer, not an estimate.

That one loop already settles the smallest cases, and the rest of the chapter is built on top of it without ever leaving `solve`. We first make the measurement step trustworthy and quick, via the connectivity checker and a Gomory–Hu shortcut for undirected graphs. We then make the enumeration step skip whole families of hopeless graphs, via a pruned search. Finally, for the directed multigraph case, we replace enumeration altogether with one small optimisation that proves every m at once, which is the certifier. Each of these is the same `solve` doing less work for the same guarantee, and the choice among them is made automatically from the booleans it receives.

```
solve(n, m, directed, simple, separation, exhaustive, max_seconds)
```

DRIVER

in the variant booleans `directed` and `simple` (or hypergraph with uniformity r), the `separation` (edge or vertex), n , m , the method, and a time budget `max_seconds`

out a `SolveResult` carrying a value with an honest label: `exact`, `lower bound`, or `upper bound`

how picks the method the case allows: brute-force enumeration, the pruned search, or the cut-counting certifier when proving, or the temperature search or tabu search when discovering (Chapter 3). It always respects the budget, and it is the one function the rest of the thesis ever calls.

2.2 A single representation for all variants

Two of the three axes from Chapter 1, the model and the direction, are properties of the object, and a single data structure holds almost all of them. Of the three structural models, simple graphs and multigraphs fit the matrix directly, while the hypergraph keeps the separate storage promised in Chapter 1 and rejoins at the end of this chapter. The hypergraph could in principle be pressed into the same shape as a higher-order tensor, an r -dimensional array marking which r -sets are present, but that array is almost entirely zero and costs n^r entries, so the program keeps the compact list of hyperedges instead and pays nothing for the empty cells. Sampled $G(n, p)$ graphs are ordinary simple graphs and therefore use this same representation without any change. A graph or digraph on n vertices is stored as an integer matrix μ , where $\mu(u, v)$ is the multiplicity of the edge or arc from u to v . A simple graph is the case $\mu \in \{0, 1\}$, an undirected graph satisfies $\mu = \mu^\top$, and a multigraph allows larger entries. The third axis, edge

against vertex separation, is a property of the question rather than the object, so it lives in how we measure μ , not in μ itself. Figure 2.2 illustrates the encoding.

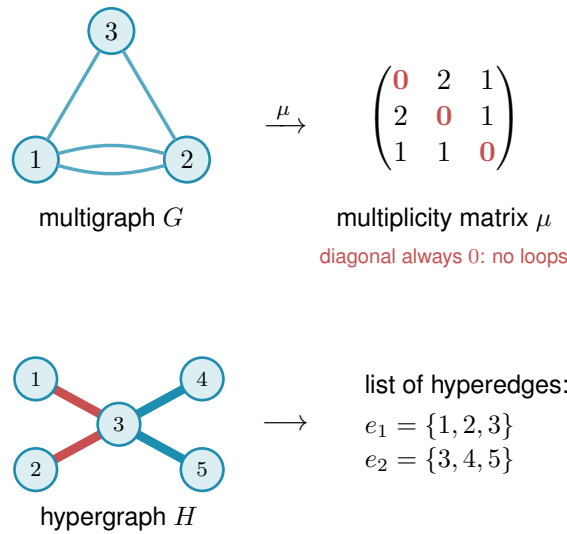


Figure 2.2. How the program stores a graph. Top: a graph or digraph on n vertices is one $n \times n$ integer matrix μ , with $\mu(u, v)$ the number of parallel edges or arcs from u to v . The double edge between 1 and 2 gives $\mu(1, 2) = 2$. The diagonal is always zero (red), since no vertex carries an edge to itself, an undirected graph keeps $\mu = \mu^\top$, a directed graph allows $\mu(u, v) \neq \mu(v, u)$, and a simple graph caps every entry at one. Bottom: the hypergraph is the one model that does not use the matrix at all, since a matrix has no room for an edge on three vertices, so it is kept as a plain list of its hyperedges, here two metro lines that share vertex 3.

```
Graph(n, Variant(directed, simple))
```

MODEL

in the vertex count n and a `Variant`, the small record holding the two booleans
out a graph whose whole state is the matrix μ , stored as the numpy array `self.mu`
how one $n \times n$ integer numpy array is the object, an undirected graph keeps $\mu = \mu^\top$, and a simple graph caps every entry at one.

Keeping a single representation is what lets one measurement routine, one certifier, and one search serve every case. Nothing downstream asks which of the twelve problems it is solving. It asks only the two booleans (directed or undirected, simple or multi) and the matrix. Everything that builds or edits a graph is a thin wrapper around μ that respects those booleans, and these operations are so routine that they need no individual exposition, and Table 2.1 lists them together.

The two booleans are all the mirroring above needs, and the whole trick fits in one short routine. Every write to μ passes through it, so this is the single place the undirected symmetry is kept, not an invariant a caller has to remember to maintain.

```
1 def _assign(self, u, v, value):
2     self.mu[u, v] = value
3     if not self.variant.directed:
```

Operation	Meaning
<code>addEdge(G, u, v, k)</code>	add k parallel edges or arcs $u \rightarrow v$ (a simple graph saturates at one)
<code>removeEdge(G, u, v, k)</code>	remove k copies, never below zero
<code>setMultiplicity(G, u, v, c)</code>	set $\mu(u, v) = c$ directly
<code>multiplicity(G, u, v)</code>	read $\mu(u, v)$
<code>hasEdge(G, u, v)</code>	whether $\mu(u, v) > 0$
<code>degree(G, v)</code>	total degree of v (in- plus out-degree if directed)
<code>edgeCount(G)</code>	number of edges or arcs, counted with multiplicity
<code>edges(G), vertices(G)</code>	iterate the present edges, and the vertex set
<code>copy(G)</code>	an independent duplicate

Table 2.1. The routine operations on the graph model. Each is a one-line wrapper that reads or writes the matrix μ while keeping an undirected graph symmetric and a simple graph binary. They are bookkeeping, and the chapter’s attention belongs to the routines that follow.

```
4 | self.mu[v, u] = value    # keep undirected graphs symmetric
```

Listing 2.1. How every write keeps an undirected graph symmetric. A directed graph simply skips the mirrored write.

2.3 The connectivity checker built on maximum flow

The whole question reduces to a flow computation. A *maximum flow* asks how much can be pushed from one point to another through a network whose links each carry a limited capacity. Menger’s theorem says the most edge-disjoint u – v paths a graph offers equals its smallest u – v cut, the fewest links whose removal severs every route between the two. The max-flow / min-cut theorem says that smallest cut is exactly the value of a maximum flow, once each multiplicity $\mu(u, v)$ is read as the capacity of the link $u \rightarrow v$. So the route count we are after is just a maximum-flow value: measure the flow and you have measured the connectivity. The routine that runs this flow and reports the route count is the connectivity *checker*. It is the single most important routine in the program: it is the only place graph theory meets computation, and every later claim about connectivity passes through it. Because the search of [Chapter 3](#) calls it by the million on tiny graphs, the program ships an optional compiled C helper that runs this same flow (named later in this chapter’s reproducibility section). It is a speed-up only: it returns the identical value the pure-Python checker does, verified against it, so no number in the thesis depends on whether the helper is built.

```
local_edge_connectivity(G, s, t) → ℕ
```

MEASURE

in a graph G and an ordered pair (s, t)

out $\lambda_G(s, t)$, the maximum number of edge-disjoint s – t paths

how a single `scipy.sparse.csgraph.maximum_flow` on the network whose arc capacities are the multiplicities μ , so by Menger the flow value is the path count.

`max_edge_connectivity(G) → ℕ`

MEASURE

in a graph G

out $\lambda^{\max}(G)$, the largest local edge-connectivity over all pairs

how one max-flow per pair (ordered if directed), take the largest. This is the quantity the edge problem caps at $m - 1$.

The vertex-disjoint version follows the same idea after one construction: split each vertex v into an in-copy v^{in} and an out-copy v^{out} joined by a single unit-capacity arc (Figure 2.3). Every arc $u \rightarrow v$ of the original graph becomes $u^{\text{out}} \rightarrow v^{\text{in}}$ in the split network. The construction is not special to digraphs: for an undirected graph each edge $\{u, v\}$ is first read as the two opposite arcs $u \rightarrow v$ and $v \rightarrow u$, and then the same split applies, so one routine serves the undirected and directed vertex problems alike. Routing through v now consumes the one unit of capacity on its internal arc, so two paths sharing v as an intermediate vertex compete for that unit and cannot coexist in any maximum flow. A max-flow from u^{out} to v^{in} in the split network therefore equals the number of internally vertex-disjoint u – v paths in the original graph.

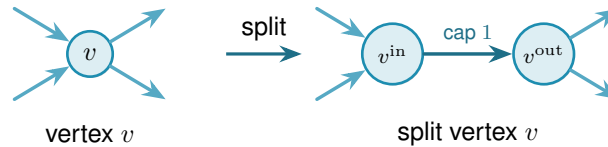


Figure 2.3. Vertex splitting for the vertex-disjoint checker. Incoming arcs attach to v^{in} and outgoing arcs leave from v^{out} , connected by a single unit-capacity arc. Any directed path that uses v as an intermediate vertex passes through this bottleneck, and two such paths would need two units of capacity and are therefore separated.

Figure 2.4 applies the split to a whole digraph. Three arc-disjoint routes are forced through one shared vertex w , so on the split network all three must cross the single unit-capacity arc inside w , which caps the number of internally vertex-disjoint routes below the edge-disjoint count.

`max_vertex_connectivity(G) → ℕ`

MEASURE

in a graph G

out $\kappa^{\max}(G)$, the largest local vertex-connectivity over all pairs

how the same sweep as the edge checker, but each flow runs on the split capacity matrix `_split_capacity_matrix(G)` of Figure 2.3, so the unit internal arcs count internally vertex-disjoint paths. Parallel edges are given capacity one here, so they never raise κ .

One convention in that codecard deserves to be said in prose, because it does not merely change how the checker computes, it changes what the multigraph vertex question asks. In vertex mode every adjacency carries capacity one, so a bundle of parallel edges counts as a single direct route. The rationale is that vertex-disjointness is about intermediate vertices, and parallel copies offer no new intermediate vertices to be disjoint over.

Remark 2.1 (What parallel edges mean in vertex mode). That choice has a consequence which

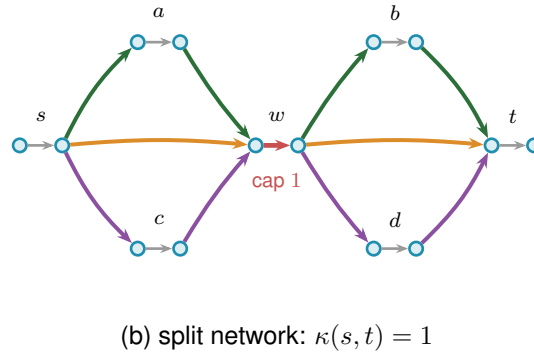
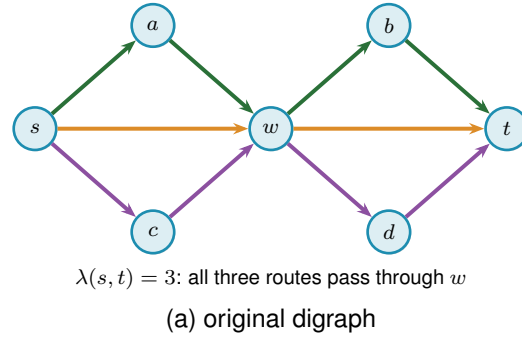


Figure 2.4. The vertex-split checker on a larger example. **(a)** A digraph in which three arc-disjoint s – t routes, $s \rightarrow w \rightarrow t$ (orange), $s \rightarrow a \rightarrow w \rightarrow b \rightarrow t$ (green), and $s \rightarrow c \rightarrow w \rightarrow d \rightarrow t$ (purple), all pass through the single vertex w , so $\lambda(s, t) = 3$. **(b)** The split network replaces every vertex V by an arc $V^{\text{in}} \rightarrow V^{\text{out}}$ of capacity one. All three routes must now traverse the single unit arc inside w (red), which carries only one unit, so at most one internally vertex-disjoint route gets through and $\kappa(s, t) = 1$. Splitting every vertex this way turns the vertex-disjoint count into an ordinary maximum flow.

has to be stated rather than left implicit. If parallel copies never raise κ , they never break feasibility either, so one could pile arbitrarily many copies onto any feasible multigraph and stay under the ceiling forever. Counted with multiplicity, the maximum number of edges would simply be infinite, and the extremal question would be empty. The two multigraph vertex variants are therefore posed here with the objective counting *adjacencies*, that is, pairs joined by at least one edge, rather than edges with multiplicity. Under that reading the question is exactly the simple question on the underlying graph, which is why those two rows of [Table 4.1](#) reduce to their simple counterparts and why every figure in this thesis reports the reduced problem for them.

The other convention is perfectly available and asks something genuinely different. A single edge is a path with no interior, so q parallel copies between the same pair are q internally vertex-disjoint paths, and feasibility caps every multiplicity at $m - 1$ by itself. Nothing runs away, and the multigraph vertex problem separates from the simple one immediately: at $n = 2$ and $m = 3$ two parallel edges are feasible where the simple graph allows one. That is a new extremal problem rather than a variant of a solved one, and [Chapter 4](#) lists it among the open problems. The appendix uses precisely this second reading whenever it argues about a general multigraph, most visibly in [Lemma A.47](#), whose hypothesis (iv) caps a multiplicity at two for exactly this

reason. The two readings never meet, because that lemma is applied only to incidence graphs, which carry no parallel edges at all, but a reader moving between the chapters and the appendix should know that the word “multigraph” is doing different work in the two places.

The two checkers are the entire interface between graph theory and computation in this thesis. As a sanity check, applied to the bidirected star of [Figure 1.11](#) on n vertices, the edge checker reports $\lambda^{\max} = 1$, and the arc count $2(n - 1)$ matches the proved values $\ell_2^{\text{dir}}(n) = 2, 4, 6, 8$ for $n = 2, 3, 4, 5$.

The same checker answers two different questions. Throughout this chapter it runs in its exact-value mode, returning $\lambda^{\max}$ or $\kappa^{\max}$ on the nose, and every proved or reported number rests on that exact reading. The search of [Chapter 3](#) needs only the cheaper yes-or-no question “is the binding connectivity above the bound?”, so there the same flow runs in a capped mode that stops as soon as it has seen one route too many. The capped mode never changes a reported value: it is a faster way to ask for a single bit, and [Chapter 3](#) shows that the search built on it reproduces the exact search step for step.

The Gomory–Hu tree from [Chapter 1](#) reappears here as an efficiency, and it is the honest kind. For undirected graphs all $\binom{n}{2}$ edge-connectivities are read off a single weighted tree, so $\lambda^{\max}$ is the heaviest tree edge rather than the maximum over $\binom{n}{2}$ separate flows. The speed-up is real, but it is bought with a theorem, not a heuristic.

`max_edge_connectivity_via_tree(G) → ℕ`

MEASURE

in an undirected graph G

out $\lambda^{\max}(G)$, identical to the pairwise sweep

how build one Gomory–Hu tree with `networkx.gomory_hu_tree`, then return its heaviest edge, which costs $n - 1$ flows in place of $\binom{n}{2}$ and cross-checks the checker.

The hypergraph is the one model that never used the multiplicity matrix, since it is stored as a plain list of hyperedges, each one a set of r vertices. The checker still has to read it, and the trick is to run the same flow computation as before on a slightly larger network that we build on the spot. The difficulty is that one hyperedge ties several vertices together at once. By the definition of a Berge route in [Chapter 1](#), only one route may use a hyperedge at a time, so we cannot just treat it as a bundle of ordinary links. Instead, for every hyperedge we add one extra helper point and connect each of that hyperedge’s member vertices to it ([Figure 2.5](#)). A route that wants to use the hyperedge now enters through one member vertex, passes through the helper point, and leaves through another.

The key is to make the helper point carry a single unit of capacity, built by the very same device as the vertex bottleneck of [Figure 2.3](#): the helper point is split into an in-copy and an out-copy joined by one unit-capacity arc, and every route boarding the hyperedge must squeeze through that one arc. A second route would need a second unit of capacity there and so is shut out. That one unit is what captures the rule that one hyperedge serves one route.

Counting routes between two vertices is then the very same maximum flow as before, run on this enlarged network, and when a hyperedge happens to hold just two vertices the helper point collapses back to an ordinary link, so the answer agrees with the graph case. That the flow on the enlarged network counts exactly the hyperedge-disjoint Berge routes is the hypergraph version of Menger’s theorem, proved in [Appendix A \(Theorem A.40\)](#), so the checker rests on a theorem rather than an analogy.

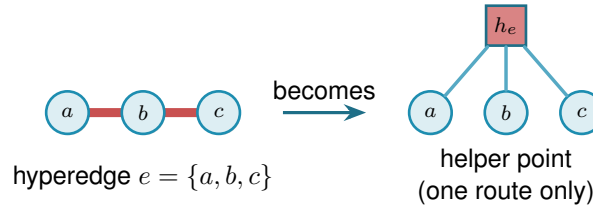


Figure 2.5. Measuring Berge connectivity by maximum flow. A hyperedge e , drawn as a metro line through its members (left), is replaced by a helper point h_e joined to each member (right). The helper point holds a single unit of capacity inside it, so at most one route may pass through h_e and the line runs one train at a time, which is what makes the hyperedge serve only one route. Counting independent routes between two vertices is then an ordinary maximum flow on this enlarged network, and a two-vertex hyperedge collapses to a single ordinary edge.

[Figure 2.5](#) shows one hyperedge in isolation. [Figure 2.6](#) carries the idea to a whole hypergraph in two panels. Panel (a) is the hypergraph drawn as a metro map: each of the eight hyperedges is one coloured line threading its three member stations, vertex to vertex, and a station on several hyperedges shows several coloured lines running through it. Panel (b) reads the same map as the flow network. For one pair of stations, u and v , it adds the helper point of every hyperedge a route boards, drawn as a small gate h_e in that line’s colour, and traces the edge-disjoint Berge routes between u and v as thin black lines passing through those gates, with every hyperedge rail faded to the background. The checker reads $\lambda(u, v) = 4$: two routes run straight through the two hyperedges that contain both u and v , and two longer ones change lines at an intermediate station, four in all, while a fifth would have to reuse a gate.

One way to picture the helper point is the metro map of [Chapter 1](#): each hyperedge is a metro line, its member vertices are the stations on it, and the helper is the line itself. Boarding at any station counts as taking that line. Because only one route may pass through the helper, only one train runs on the line at a time, so a second route must reach its destination without ever boarding it. The same flow computation that counts edge-disjoint paths through ordinary links now counts routes that share no line, which is exactly what Berge edge-disjointness means.

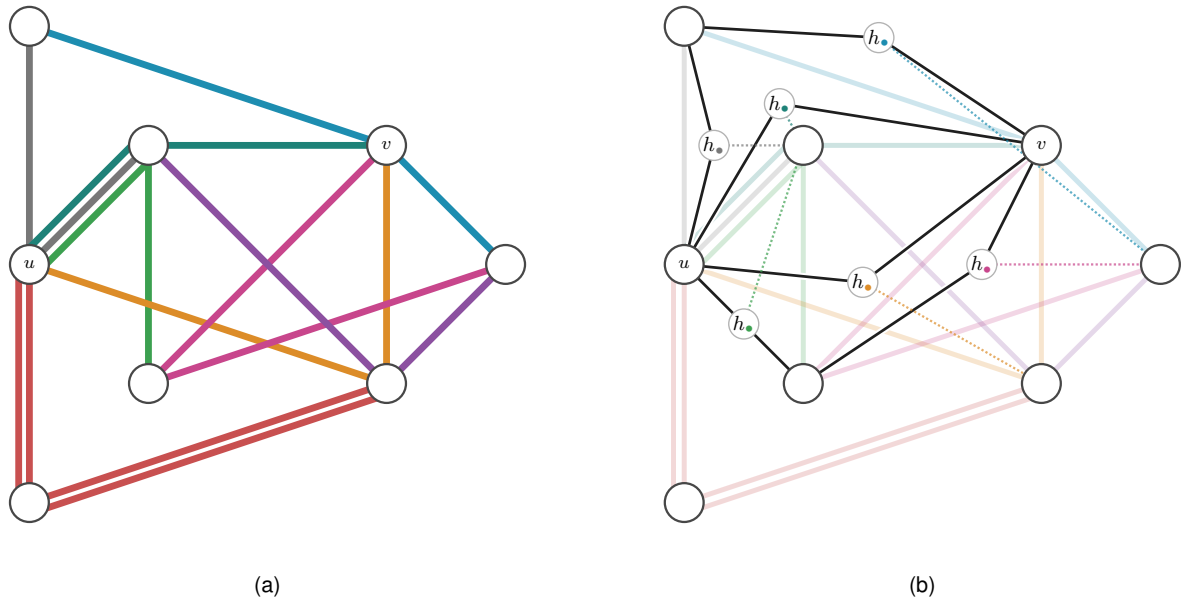


Figure 2.6. A 3-uniform hypergraph on eight stations, with the pair u, v singled out. Panel (a) draws it as a metro map: every hyperedge is one coloured line threading its three members, vertex to vertex, the metro reading of [Chapter 1](#). Panel (b) reads the same picture as the helper-point flow network. Each hyperedge a route boards becomes its one-route gate $h_\bullet$ of [Figure 2.5](#), shown in that line's colour and placed on the route, and the thin black lines are the edge-disjoint Berge routes from u to v threading those gates. A dotted leg ties each gate to the hyperedge's third member, the station that route does not use, and every hyperedge rail is faded to the background so the black routes read clearly. Two routes pass straight through the two hyperedges that hold both u and v , and two longer ones transfer at an intermediate station, so the checker reads $\lambda(u, v) = 4$. A fifth route would have to reuse a gate.

`max_hyperedge_connectivity(H) → ℕ`

MEASURE

in an r -uniform hypergraph H

out $\lambda^{\max}(H)$, the largest Berge edge-connectivity over all pairs

how build the enlarged capacity matrix `_hyper_capacity_matrix(H)` in which every hyper-edge becomes a helper point that only one route may pass through, then take the same maximum flow as the graph checker. On two-vertex hyperedges it matches ordinary edge-connectivity.

On the complete graph K_n the hypergraph checker returns $n - 1$, and on the complete three-uniform hypergraph $K_4^{(3)}$ it returns 3, larger than the two hyperedges through each pair because longer Berge routes also carry flow. The self-test confirms both of these values, so the construction is checked rather than merely asserted. The same helper-point network also answers the vertex-disjoint hypergraph question without alteration: splitting each ordinary vertex as in [Figure 2.3](#) counts internally vertex-disjoint Berge routes. The one remaining direction, the *directed* hypergraph, needs the helper point to learn which way is which, and that is the subject of the next subsection.

2.3.1 The directed hypergraph checker

A *directed* hyperedge carries an orientation: it is a single *tail* vertex together with $r - 1$ *head* vertices, drawn as the star of [Figure A.8](#), and a directed Berge route may only enter it at the tail and leave it at one of its heads. The undirected helper point treated a hyperedge as a line any route could board at any station. The directed one must instead be a one-way turnstile: a route may step onto the hyperedge only at its tail and step off only at a head. The gadget that enforces this is the helper point made directional. For a hyperedge $e = (a; b, c, \dots)$ we add a gate node g_e , send one capacity-one arc from the tail into it, $a \rightarrow g_e$, and send an arc out of it to each head, $g_e \rightarrow b$, $g_e \rightarrow c$, and so on ([Figure 2.7](#)). A route reaches g_e only through the tail a , and the single unit of capacity on $a \rightarrow g_e$ lets just one route use the hyperedge, exactly as the undirected turnstile did, but now it can leave only towards a head. The gate is built once per hyperedge, directed or not, by the same loop:

```
1 gate_in, gate_out = base + 2 * index, base + 2 * index + 1
2 cap[gate_in, gate_out] = 1 # one route through this hyperedge
3 if hypergraph.directed:
4     tails, heads = _dir_tails_heads(edge)
5     for tail in tails:
6         cap[leave(tail), gate_in] = _UNBOUNDED
7     for head in heads:
8         cap[gate_out, enter(head)] = _UNBOUNDED
```

Listing 2.2. The gate loop behind [Figure 2.5](#) and [Figure 2.7](#): one capacity-one gate per hyperedge, wired one-way for a directed hyperedge and both ways for an undirected one.

[Figure 2.8](#) reads the same hypergraph directed, again in two panels. Panel (a) gives the whole

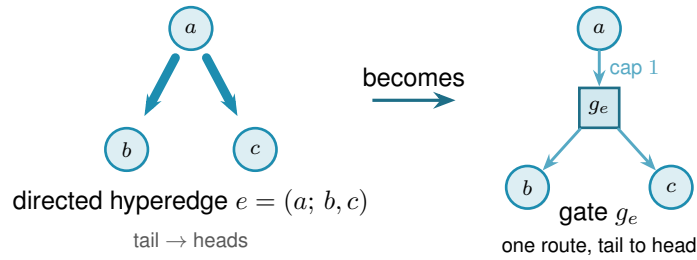


Figure 2.7. The directed helper point. A directed hyperedge $e = (a; b, c)$, the tail a fanning to its heads b, c (left), becomes a gate node g_e fed by one capacity-one arc from the tail and feeding an arc to each head (right). A route may reach g_e only through the tail and may leave only towards a head, so the gate is a one-way version of the turnstile of Figure 2.5, and the single unit of capacity still lets only one route use the hyperedge.

hypergraph oriented, every hyperedge a one-way line whose arrows point away from its tail, so the middle-tailed lines fan out from their centre and the end-tailed ones run one way along their length. Panel (b) traces the routes from v to u , the gate rule of Figure 2.7, fading every hyperedge rail to the background and threading a black arrow from the route's entry station through the one-route gate $h_\bullet$, set in a clear gap off the rails, to its exit, with a dotted leg to the hyperedge's third member. Orientation costs connectivity. Of the four undirected routes between u and v only two survive the arrows, $\lambda(v, u) = 2$, because v can set out through only the two hyperedges it heads, and the two survivors cross at the interchange l so they share no hyperedge. The same vertex-splitting trick of Figure 2.3 laid on top then counts internally vertex-disjoint directed routes, so this one construction measures all four hypergraph variants and `solve` reaches every one of the twelve problems through a single checker.

The gate is indifferent to how a hyperedge's vertices are split between tails and heads. It draws one capacity-one arc in from every tail and one out to every head, so the single forward tail it was built for is only the simplest case: give it a hyperedge with several tails and one head, or several of each, and it still counts the routes that enter at a tail and leave at a head. The same checker therefore measures every *orientation model* of a directed Berge edge, not just the one-tail one, and Section 4.3.1 takes up which of those models are worth distinguishing and what the program finds for them.

With the checker in place the proposed value of Proposition 1.15 can be tested on any small hypergraph the way every other variant is, which is the subject of the next section.

2.4 Checking every graph of a fixed size

The checker measures one graph. To bound every graph of a given size, the most direct method is the one the base cases of Chapter 1 already needed: enumerate them all. A simple graph on n vertices is a choice of $\mu(u, v) \in \{0, 1\}$ on each off-diagonal cell, a multigraph a choice in $\{0, \dots, m-1\}$, and a digraph varies every ordered pair rather than only the upper triangle. Walking that product, measuring each candidate with the checker, and keeping the densest feasible one settles the value exactly, and because it has looked at every graph it is a genuine upper

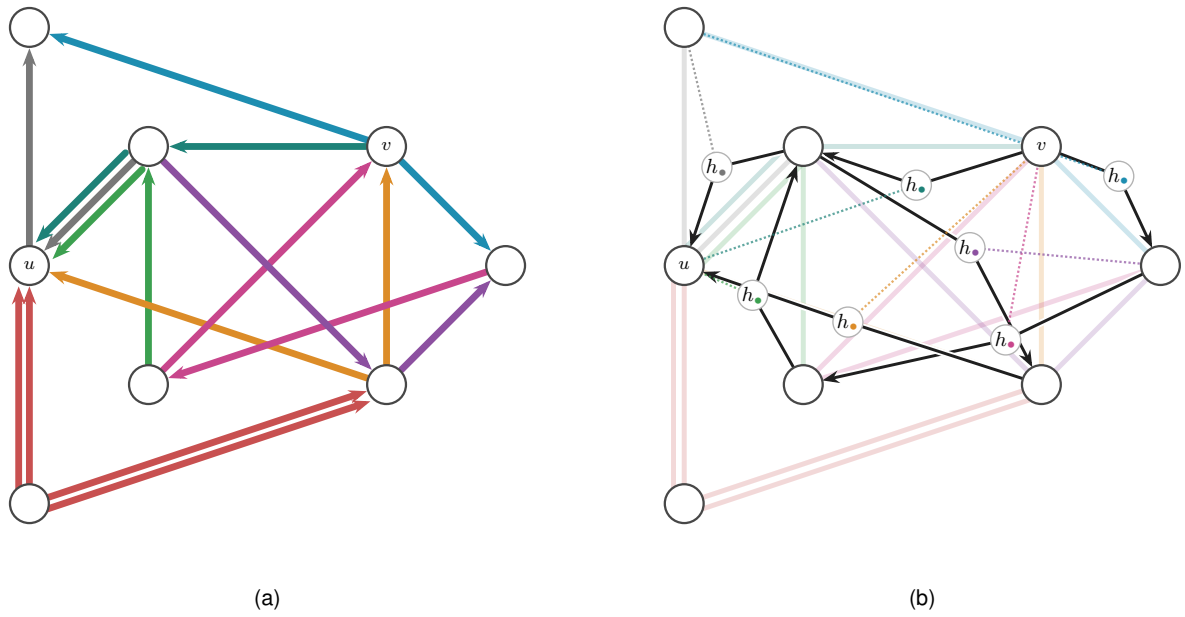


Figure 2.8. The hypergraph of Figure 2.6 read directed. Panel (a) gives the whole directed hypergraph, every hyperedge a one-way line with its arrows pointing away from the tail station, so a route may join a line at its tail and ride it to a head. Panel (b) traces the routes from v to u . Every hyperedge rail is faded to the background, and a black arrow runs from the route's entry station through the hyperedge's one-route gate $h_\bullet$ to its exit, the directed turnstile of Figure 2.7, with the gate set in a clear gap off the rails. A dotted leg in the line's colour ties each gate to the hyperedge's third member, the station the route does not visit. Two edge-disjoint directed Berge routes from v to u survive. The long route $v \rightarrow j \rightarrow c \rightarrow l \rightarrow u$ rides nmj , mcj , elc and nel in turn, and the short route $v \rightarrow l \rightarrow d \rightarrow u$ rides elm , ldj and edm . They cross only at the interchange l , so they share no hyperedge, and the checker reads $\lambda(v, u) = 2$, since v can set out through only the two hyperedges it heads. Against the undirected four of Figure 2.6, the arrows have halved the count.

bound, not evidence.

This is the first thing `solve` does when it is asked to prove a small case. With its exhaustive flag set it walks that product, measures each candidate with the checker, and keeps the densest feasible one. The value it returns is exact and exhausted, a true upper bound for that n , but only while the count of graphs stays small. No new routine is involved, only `solve` choosing to enumerate.

Run on the small cases this confirms the proved values directly, with no induction and no cleverness. [Table 2.2](#) collects a spread of variants where the enumeration finishes: in every row the machine returns exactly the value [Chapter 1](#) proved by hand. This is the first payoff of the uniform representation. One driver, `solve`, pointed at six different problems, settles all six.

Variant	m	n	<code>solve</code>	Theory
simple undirected, edge	2	5	4	$\lfloor m(n-1)/2 \rfloor = 4$
simple undirected, edge	3	5	6	$\lfloor m(n-1)/2 \rfloor = 6$
multigraph undirected, edge	3	5	8	$(m-1)(n-1) = 8$
simple undirected, vertex	2	6	5	$k_2(n) = n-1 = 5$
simple directed, arc	2	5	8	$\max(2(n-1), \lfloor n^2/4 \rfloor) = 8$
simple directed, arc	2	6	10	$\max(2(n-1), \lfloor n^2/4 \rfloor) = 10$

Table 2.2. Variants that converge under plain enumeration. For each small case `solve`, in its exhaustive mode, walks every graph of the variant, measures it with the checker, and returns the densest feasible one. Every value matches the hand proof of [Chapter 1](#) exactly.

The honesty of the method is also its limit. The number of graphs it must visit grows as $2^{\binom{n}{2}}$ for simple undirected graphs and $2^{n(n-1)}$ for digraphs, and capping multiplicities at $m-1$, so that each cell ranges over the m values $\{0, \dots, m-1\}$, raises the base from two to m , giving $m^{\binom{n}{2}}$ undirected multigraphs and $m^{n(n-1)}$ directed ones. The enumeration is exact wherever it finishes, and it finishes only for the smallest sizes. [Chapter 3](#) opens with exactly how fast this wall arrives. For now the point is that the wall arrives precisely where [Chapter 1](#) needed help, at the directed base cases $n = 6, 7$, so the rest of this chapter is about reaching a little further than blind enumeration can.

2.5 Reaching further: a pruned search for the directed base cases

The base cases $n \leq 7$ of the directed $m = 2$ theorem ([Theorem 1.10](#)) are the finite checks the induction of [Appendix A](#) depends on, and they sit just past where blind enumeration of digraphs is tractable. They can be reached by enumerating more cleverly. Feasibility is monotone: adding an arc can never lower $\lambda^{\max}$, so once a partial digraph is infeasible every completion of it is infeasible too and the whole branch is abandoned at once. A counting bound prunes further. At any partial digraph, count the arcs already placed and add the most the still-undecided cells could ever contribute. That sum is a ceiling on every completion of the branch, so if even the ceiling cannot beat the best feasible digraph found so far, the branch is dropped unexplored ([Figure 2.9](#)). What remains is a branch and bound over simple digraphs with the connectivity checker as the feasibility test, exact and complete at these sizes where the blind walk is not.

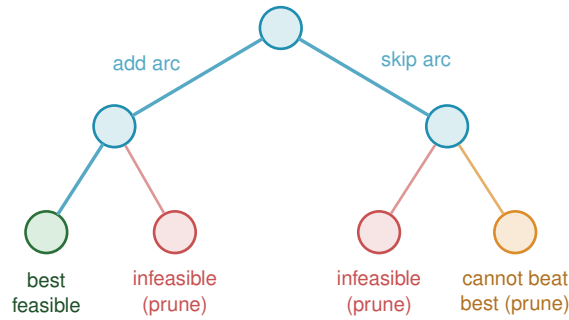


Figure 2.9. Branch and bound over digraphs. Each node fixes one more cell, arc present or absent, so the leaves are whole digraphs. Two rules cut entire subtrees before they are walked. A branch whose partial digraph already breaks the connectivity ceiling is infeasible, and so is every completion of it (red). A branch whose best possible completion still cannot beat the densest feasible digraph found so far is dropped on the counting bound (orange). Only the surviving branches are explored, which is what lets the search reach the base cases $n \leq 7$ the blind walk cannot.

Here `solve` changes its implementation without changing its interface. For a simple digraph it runs this branch and bound in place of the blind walk, still with the connectivity checker as the feasibility test, and so reaches the base cases $n \leq 7$ that blind enumeration cannot. The booleans alone decide which of the two it uses, and the caller never asks.

Run on the directed $m = 2$ problem `solve` returns the complete sequence 2, 4, 6, 8, 10, 12 for $n = 2, \dots, 7$, each proved complete at its size. The densest digraph it finds at each n is exactly the extremal shape [Chapter 1](#) named, the bidirected star (the hub) below the crossover at $n = 7$ and the one-directional bipartite digraph above it, so the search confirms not just the values but the winning constructions. These are exactly the base cases the induction requires, and with them the long proof of [Theorem 1.10](#) rests on a single auditable search rather than on anything done by hand. The exhaustive-search transcript is in [Appendix A](#).

2.6 Generating the directed cases faster

The $m = 3$ directed multigraph result reduces, through [Remark A.39](#), to a handful of finite statements about multigraphs on six and seven vertices, and proving them means listing every such multigraph up to a renaming of its vertices. The program does this in two independent ways, on purpose: when two different methods return the same list, that agreement is itself a check.

The first way is a search written from scratch. It is a depth-first walk over multiplicity matrices that abandons a branch the moment it turns infeasible or can no longer beat the best graph found, and it keeps just one representative of each isomorphism class by reducing every finished graph to a canonical relabelling, the smallest matrix in dictionary order over all relabellings. The canonical form keeps the memory small, but the time is spent walking labelled partial graphs, and time is what runs out: the full classification at $n = 5$ costs this in-house search about 456 seconds, and it climbs steeply from there.

The second way ([Table 4.2](#) names it) does not search at all, it generates, and it was suggested by Prof. Jan Goedgebeur. The idea is to hand the hard part, listing one graph per isomorphism

class, to a tool that already does it supremely well: the `geng` program from the `nauty` toolkit of McKay and Piperno [10], invoked through a subprocess and decoded from its `graph6` output (a compact one-line text encoding of a graph). `geng` lists the non-isomorphic *undirected* graphs on n vertices, and the program then decorates each one with the directed multiplicities the problem needs, so the isomorphism reduction is inherited for free from the support graph. The decoration itself is plain to state. For one support graph, walk its edges in a fixed order and assign each edge $\{u, v\}$ an ordered pair of multiplicities $\mu(u, v), \mu(v, u)$, each between 0 and $m-1$ but not both zero, since an edge of the support must carry at least one arc. Along the way the checker measures the partly decorated multigraph, and the moment some pair already exceeds the connectivity ceiling the whole branch of assignments is abandoned, since adding further arcs can never lower a connectivity, the same monotonicity prune as Figure 2.9. A decoration that survives to the end with the target arc count is kept, reduced to its canonical form so that each isomorphism class is reported once. Different supports share nothing, so they are processed independently across the processor cores. The companion orientation tools `directg` and `watercluster2` are not used, because each gives only one orientation per edge and so cannot express the bidirected arcs and repeated multiplicities the extremal graphs are built from. Layering the multiplicities in-house on top of `geng` is what fits the problem. To be precise about credit: Prof. Goedgebeur is not an author of `nauty`, which is the work of McKay and Piperno cited above. He suggested the generation route, and it is his suggestion that the speed-up rests on.

The two ways agree to the last graph. The generated pipeline reproduces the in-house classification exactly at every settled size, two non-isomorphic extremal families at $n = 4$ and three at $n = 5$, and it runs substantially faster, since the isomorphism work it never repeats is precisely what `geng` has already done. One difference decides which to trust further out: the in-house search leans on a conjectured counting bound once $n \geq 7$, while the generated enumeration prunes only with proved bounds, so it is a sound and complete search at every n . That soundness has now paid off. Pointed at the degree-bounded classification at $n = 7$, fact (b) of Remark A.39, the generation enumerator completed the sweep in about seven hours across ten processor cores and returned exactly the three predicted extremal families, settling fact (b) outright and leaving the value statement, fact (a), as the $m = 3$ argument's one remaining computation.

2.7 Proving every m at once

The pruned search proves one (n, m) at a time. For the directed multigraph arc problem `solve` has a sharper implementation, one that proves every m at a fixed n in a single pass, and it is worth the extra machinery because it closes the directed multigraph case outright for the sizes it reaches.

The key is that m can be scaled out of the problem completely, so that it never appears in the computation at all. Start with any directed multigraph that is feasible at level m , so $\lambda^{\max} \leq m-1$, and divide every multiplicity by $m-1$. Two things happen at once. The multiplicities become weights $w(u, v)$ between 0 and 1, and since dividing every capacity by $m-1$ divides every

flow by the same factor, the maximum flow between each pair drops to at most one. The arc count, meanwhile, turns into the total weight $\sum_{u \neq v} w(u, v) = |A|/(m-1)$. So a dense feasible multigraph becomes a heavy feasible weight matrix, and the other way round. Write

$$M^*(n) = \max \left\{ \sum_{u \neq v} w(u, v) : w(u, v) \in [0, 1], \text{ maxflow}(s, t; w) \leq 1 \text{ for all } s, t \right\}$$

for the largest total weight any such matrix can carry. It is one number for each n , with no m inside it, and scaling a feasible multigraph down gives

$$L_m^{\text{dir}}(n) \leq (m-1) M^*(n) \quad \text{for every } m \geq 2.$$

That inequality holds always, and it is the direction that does the proving. The reverse does not come for free, and it is worth being exact about why, because the two are easy to run together. Scaling *up* needs an optimal weight matrix whose entries are whole multiples of $1/(m-1)$, and a real optimum has no reason to oblige. What closes the gap is not an argument but an observation about the answer the solve returns: the heaviest matrix it finds is the *double star*, the bidirected star carrying weight one on both arcs $h \rightarrow v$ and $v \rightarrow h$ between the hub and every other vertex. Its weights are all 0 or 1, so multiplying them by $m-1$ gives an honest integer multigraph, not a fractional one, with $2(n-1)(m-1)$ arcs and $\lambda^{\max} = m-1$, for every m at once. Whenever the optimum is attained by such a $\{0, 1\}$ matrix, the two bounds meet and

$$L_m^{\text{dir}}(n) = (m-1) M^*(n) \quad \text{for every } m \geq 2,$$

so that one solve settles infinitely many values. This is exactly what happens at every $n \leq 6$, which is what [Theorem 2.2](#) rests on. It is a verified property of those optima rather than a theorem about all n , and where it is not verified the safe reading is the inequality: an integral question such as fact (a) of [Remark A.39](#) is then asked of the integral program `prove_integral_arc_bound` directly, not deduced from a fractional value.

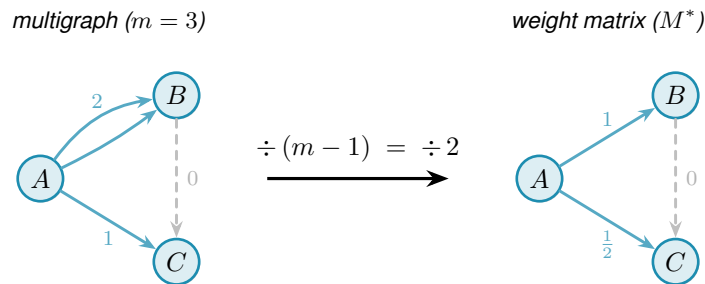


Figure 2.10. The scaling step for $m = 3$. Dividing each multiplicity by $m-1 = 2$ maps the integer multigraph (left) to a weight matrix with all values in $[0, 1]$ and max-flow at most one between every pair (right). The double arc $A \rightarrow B$ (multiplicity 2) becomes weight 1, the single arc $A \rightarrow C$ (multiplicity 1) becomes weight $\frac{1}{2}$, and the absent arc $B \rightarrow C$ (multiplicity 0, shown dashed) stays absent. Crucially, m drops out on the right, so a single optimisation over all weight matrices pins $L_m^{\text{dir}}(n) = (m-1) M^*(n)$ for every m simultaneously.

What remains is to compute $M^*(n)$, and the difficulty is that the constraint “maximum flow at

most one” is itself the answer to a flow problem rather than a plain inequality on w . The max-flow / min-cut theorem removes it. The maximum s – t flow equals the smallest capacity of any *cut* separating s from t . A cut just splits the vertices into a source side holding s and a sink side holding t , and its capacity is the total weight of the arcs that cross from the source side to the sink side. So $\text{maxflow}(s, t) \leq 1$ holds exactly when some such cut has capacity at most one. That lets us drop the flow entirely: instead, for every ordered pair (s, t) , the optimisation picks one cut and we require its capacity to stay at or below one.

A cut is described by a yes/no label x_u^{st} on each vertex, equal to one when u is on the source side, with s fixed on the source side and t on the sink side. An arc $u \rightarrow v$ crosses the cut exactly when u is on the source side and v is not, and a helper variable p_{uv}^{st} then picks up its weight $w(u, v)$. Forcing the crossing weights to sum to at most one for every pair forces every pair to admit a cut of capacity one, hence maximum flow at most one. This is the exact feasible region, not a relaxation of it, a relaxation being a loosened version that would also admit weight matrices no multigraph realises, and that exactness is what turns the result into a proof. The optimisation reports both the best total weight it has found and a value it has shown no matrix can exceed, and when those two meet the gap is zero and $M^*(n)$ is certified.

For the directed multigraph arc problem, then, `solve` runs this cut-counting optimisation in place of any enumeration: it chooses the minimum cut for every ordered pair, and a zero gap makes the reported $M^*(n)$ a proved upper bound, from which $L_m^{\text{dir}}(n) \leq (m - 1)M^*(n)$ delivers a ceiling for every m at once, met from below by the double star whenever the reported optimum is integral.

`prove_directed_multigraph(n) → ProofResult`

PROVE

in the vertex count n (and tightening flags such as `use_deletion_cuts`)

out $M^*(n)$ with a proof status, giving $L_m^{\text{dir}}(n) \leq (m - 1)M^*(n)$ for every m , with equality when the optimum found is integral

how builds the cut-counting optimisation, the mixed-integer linear program written out below, once with `_cut_counting_model` as a pulp model, then `_pick_solver` runs it on CBC, the free solver bundled with pulp, or on the commercial Gurobi solver by a one-line switch. A zero optimality gap is a proof. The integer-weight companion `prove_integral_arc_bound(n, m, target)` answers the single feasibility question behind fact (a) of [Remark A.39](#).

`solve` assembles this optimisation from n alone, and a reader content to take it on the strength of the cut argument just given can skip to [Theorem 2.2](#) without loss. Written out it reads more heavily than it behaves, so we first name every symbol in plain words, then show the one inequality that looks cryptic is a simple yes-or-no switch.

The optimisation is a *Mixed-Integer Linear Program*, or MILP for short. “Linear” means every constraint and the objective are linear functions of the variables: no products, no squares, no

exponentials. “Mixed-integer” means the variables split into two kinds: most are ordinary real numbers that may take any value in a continuous range, but a few are forced to be whole numbers, either 0 or 1. The reason for the split is conceptual. Arc weights w are naturally continuous: they are scaled multiplicities and can sit anywhere in $[0, 1]$. But the cut assignment for a vertex is a hard binary choice: a vertex is either on the source side of a given cut or on the sink side, with no meaningful notion of being “half on each side”. Allowing the cut labels to be fractional would describe a different, weaker condition and could report an optimistic value that no integer-weight multigraph can actually achieve. Keeping those labels in $\{0, 1\}$ is what keeps the optimal gap honest, and it is what makes the program’s result a proof rather than an estimate.

Three quantities appear, and only three. The first is already familiar.

- ★ $w(u, v) \in [0, 1]$ is the *scaled multiplicity* of the arc $u \rightarrow v$, the amount of arc weight the matrix places on $u \rightarrow v$ after the division by $m - 1$ of the previous section. It is what we are maximising the total of. This is a continuous variable: it may be any real number in $[0, 1]$.
- ★ $x_u^{st} \in \{0, 1\}$ is the *side label* for vertex u with respect to the ordered pair (s, t) . The optimisation chooses one cut for each pair, and x_u^{st} records whether u is on the source side (1) or the sink side (0). The two endpoints are pinned: $x_s^{st} = 1$ and $x_t^{st} = 0$. This is the integer variable: a vertex cannot straddle the cut.
- ★ $p_{uv}^{st} \geq 0$ is the *crossing weight*, the part of $w(u, v)$ that the pair (s, t) ’s chosen cut is forced to count. It equals $w(u, v)$ when the arc $u \rightarrow v$ crosses from source to sink, and is otherwise left at zero. This is a continuous variable.

With those three named, the program is short:

$$\begin{aligned}
& \text{maximise} && \sum_{u \neq v} w(u, v) \\
& \text{subject to} && p_{uv}^{st} \geq w(u, v) + x_u^{st} - x_v^{st} - 1 && \text{for all } (s, t), (u, v), \\
& && \sum_{u \neq v} p_{uv}^{st} \leq 1 && \text{for all } (s, t), \\
& && w(s, t) + \sum_x \min(w(s, x), w(x, t)) \leq 1 && \text{for all } (s, t), \\
& && d(v_0) \leq d(v_1) \leq \dots \leq d(v_{n-1}), && \text{with } d(v) = \sum_{u \neq v} (w(u, v) + w(v, u)), \\
& && w \in [0, 1], \quad x \in \{0, 1\}.
\end{aligned}$$

The first line is the only one that looks cryptic, and it is just an indicator written as an inequality. Its right-hand side $w(u, v) + x_u^{st} - x_v^{st} - 1$ takes one of four shapes according to which side the cut puts the two endpoints u and v on, and [Table 2.3](#) works all four out. Because $w(u, v) \leq 1$, the right-hand side is positive in one case only, the crossing case, where it equals $w(u, v)$ and so forces $p_{uv}^{st} \geq w(u, v)$. In the other three cases it is zero or negative, so the constraint reads $p_{uv}^{st} \geq (\text{something} \leq 0)$ and leaves p_{uv}^{st} free to stay at zero, which the maximisation is happy to

let it do.

x_u^{st}	x_v^{st}	meaning	$w + x_u^{st} - x_v^{st} - 1$	forces
1	0	u source side, v sink side: arc <i>crosses</i>	$w(u, v)$	$p_{uv}^{st} \geq w(u, v)$
1	1	both on the source side	$w(u, v) - 1 \leq 0$	$p_{uv}^{st} \geq 0$
0	0	both on the sink side	$w(u, v) - 1 \leq 0$	$p_{uv}^{st} \geq 0$
0	1	u sink side, v source side	$w(u, v) - 2 \leq 0$	$p_{uv}^{st} \geq 0$

Table 2.3. The first constraint, case by case. This is where the bound $w \leq 1$ earns its keep: it makes the three non-crossing rows land at or below zero, so the helper p_{uv}^{st} is pushed up to the arc weight in exactly one case, the crossing one. The constraint is therefore an exact indicator, “count $w(u, v)$ when the arc crosses this cut, and nothing otherwise”, with no slack and no fudge.

Figure 2.11 shows one chosen cut and the weights it counts. The second line of the program now reads plainly. It says the total crossing weight of this pair’s cut, $\sum_{u \neq v} p_{uv}^{st}$, is at most one. By max-flow / min-cut a cut of capacity at most one exists for (s, t) exactly when $\text{maxflow}(s, t) \leq 1$, so the two lines together say “every ordered pair admits a cut of capacity at most one”, which is precisely the feasibility we want. This is the true feasible region, not a relaxation of it, so an optimal solution reported with zero gap is a proof and not merely evidence.

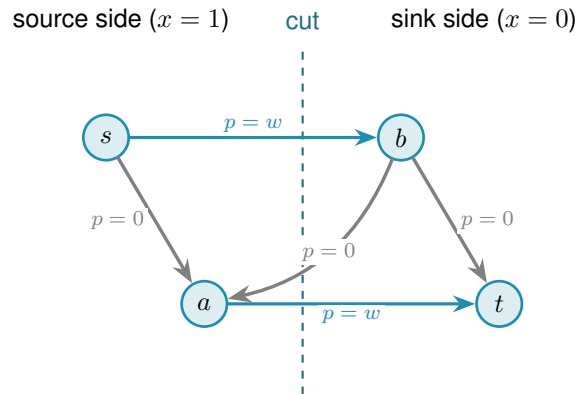


Figure 2.11. One ordered pair’s chosen cut. The dashed line splits the vertices into the source side, where $x_u^{st} = 1$, and the sink side, where $x_u^{st} = 0$. Only arcs running from source side to sink side cross the cut, and the first constraint forces their helper p_{uv}^{st} up to the arc weight w (the two thick arcs). Arcs that stay on one side, and arcs that run the other way across the line, contribute nothing. The second constraint then caps the total counted weight, the thick arcs, at one.

The last two lines change no answer. They only help the solver finish in time. The third is a redundant shortcut. The direct arc $s \rightarrow t$, together with one two-step detour $s \rightarrow x \rightarrow t$ through each intermediate vertex x , already forms that many routes between s and t . So $w(s, t) + \sum_x \min(w(s, x), w(x, t)) \leq 1$ holds of any feasible matrix anyway, and stating it outright just sharpens the continuous picture the solver reasons from.

The $\min$ in that line is not a linear function, so it cannot appear in the MILP as written. To keep the program linear, each $\min(w(s, x), w(x, t))$ is replaced by a fresh helper variable z_x^{st} together with a binary selector $b_x^{st} \in \{0, 1\}$. The selector records which of the two arguments is smaller: $b_x^{st} = 1$ means $w(s, x)$ is no larger than $w(x, t)$. Four linear inequalities then pin z_x^{st} exactly to the minimum. Two upper bounds, $z_x^{st} \leq w(s, x)$ and $z_x^{st} \leq w(x, t)$, prevent z from exceeding

either argument. Two lower bounds, one of the form $z_x^{st} \geq w(s, x) - (1 - b_x^{st})$ and the other $z_x^{st} \geq w(x, t) - b_x^{st}$, each become active only when the selector chooses that argument: at any integer value of b exactly one lower bound bites and forces z up to the chosen weight. Together with the upper bounds this pins $z_x^{st} = \min(w(s, x), w(x, t))$ exactly. This technique is called the McCormick linearisation of a minimum. It adds one continuous variable z and one binary variable b per intermediate vertex per ordered pair, all of them appearing only in the tightening constraint and nowhere else, so the proof structure of the program is unchanged.

The fourth line orders the vertices by weighted degree. Renaming vertices cannot change any count, so insisting on this order rules out no graph, and it spares the solver the many relabelled copies of a graph that differ only in vertex names.

Theorem 2.2 (Directed multigraph value, small n). *For $n \leq 6$, $M^*(n) = 2(n - 1)$, and the optimum is attained by a $\{0, 1\}$ weight matrix. Hence for all $m \geq 2$, $L_m^{\text{dir}}(n) = 2(n - 1)(m - 1)$, attained by the double star.*

It is worth being clear about what “for all m ” rests on, because the theorem proves infinitely many values from finitely many solves. It is not that each m was tested. In this mode `solve` never sees m at all. It proves the single m -free number $M^*(n)$, and because that optimum turns out to be attained by a $\{0, 1\}$ matrix, $L_m^{\text{dir}}(n) = (m - 1)M^*(n)$ then carries that one fact to every m at once. The only quantity that is checked case by case, and the only genuine limit on how far the result reaches, is n .

`solve` proves [Theorem 2.2](#) by returning $M^*(3) = 4$, $M^*(4) = 6$, $M^*(5) = 8$, and $M^*(6) = 10$, each optimal with zero gap, so that $M^*(n) = 2(n - 1)$ over this range. The first three finish quickly, $M^*(3)$ and $M^*(4)$ in well under a second and $M^*(5)$ in a few seconds, while $n = 6$ takes about twenty minutes. At $n = 7$ the same encoding does not close, and we do not pretend otherwise: it is a finite obstacle of scale, not a new idea, and it marks the current edge of certified knowledge for this variant. The solver transcript is in [Appendix A](#).

2.8 What the solver may claim

Now that the pieces inside `solve` are built, it is worth stating plainly, because the rest of the thesis depends on it, what each is allowed to claim. The checker *measures*: its connectivity values are exact. Its three proving implementations, the exhaustive enumeration, the pruned search, and the cut-counting, *prove*: an exhausted enumeration or an optimal, zero-gap solve is an upper bound for that fixed n . The search of [Chapter 3](#) will *discover*: it produces concrete graphs, hence lower bounds, and never an upper bound. These three claims are exactly what `solve` labels its answer with, so that an exact value, a lower bound, and an upper bound can never be quietly confused with one another.

With the microscope built and aimed at proof, the variants with small answers are settled and the directed base cases the hand proof needed are in hand. What proof by enumeration cannot do is reach the sizes where the answers are only conjectured, because the number of graphs ex-

plodes long before the interesting cases arrive. The next chapter measures that explosion exactly, and then turns the same program loose to *discover* the dense graphs that blind enumeration can never reach.

2.9 Reproducibility

Every claim in this thesis that rests on computation can be re-run from a single Python driver. All of the program’s logic lives in `program/erdos915_unified.py`, one self-contained file. Its core needs only `numpy` and `scipy`, because every connectivity value it reports is a single `scipy` integer maximum flow on a capacity matrix. The model, the checker, the search, the enumeration, and the self-test all run on those two libraries alone.

A few more libraries sit on top, each for one job. The certifier uses `pulp`, a Python library for writing an optimisation once and handing it to any solver, and runs it on CBC, the free solver `pulp` bundles, or on the commercial Gurobi solver by a one-line switch. Two others are optional and only for presentation: `matplotlib` draws the figures, and `networkx` supplies the Gomory–Hu tree behind one of them. An optional compiled helper, `_erdos_fast.so` (built from `_erdos_fast.c` by `build_fast.sh`), speeds up the hot-path `_tiny_maxflow` and `_canonical_form` calls, but the pure-Python path returns the same answers.

A handful of named entry points reach the whole pipeline. `solve` drives every variant. The measures are `max_edge_connectivity` and `max_vertex_connectivity`, with the hypergraph and Gomory–Hu views beside them. The provers are `prove_directed_multigraph` and `enumerate_extremal_directed_multigraphs`, and `search_for_dense_graph` is the discoverer. Running `python erdos915_unified.py` executes the built-in self-test, the suite of invariant checks that confirms the connectivity values quoted throughout this chapter, including the sanity checks on the bidirected star and on K_n and $K_4^{(3)}$.

The figures and proofs are just as reproducible. Every figure is regenerated by the companion driver `program/make_figures.py`, which imports the same file and fixes every random seed, so the pictures are reproduced exactly rather than merely resembled, and the table in `program/README.md` records which routine produces which figure. The solver transcripts behind this chapter’s certified values, the pruned-search log for [Theorem 1.10](#) and the cut-counting certificate for [Theorem 2.2](#), are reproduced verbatim in [Appendix A](#), so a reader may check the proofs without running anything at all.

2.9.1 What a machine-checked claim in this thesis means

Reproducibility is not the same as certification, and it is worth stating plainly what standard the computed results here meet, since a solver reporting OPTIMAL is not by itself a mathematical certificate. Four practices carry the weight.

First, the exhaustive searches are self-certifying in a way the optimisations are not. A finished include-or-exclude sweep has visited a superset of the objects it needed to visit, and the two

prunes are proved sound in [Chapter 3](#), so its verdict depends on no solver and no floating-point arithmetic, only on the checker. Where a value in this thesis is called *proved*, that is usually the reason.

Second, every reported connectivity is exact and integral. The checker is a single integer maximum flow, so no value in the thesis is a rounded floating-point quantity. The one place real numbers enter is the fractional optimisation of [Theorem 2.2](#), and there the reported optimum is a $\{0, 1\}$ matrix, which is checked exactly.

Third, the load-bearing computations are run twice, by implementations that share no code. The base cases of [Theorem 1.10](#) and [Theorem 1.11](#) were each confirmed by the program and by separately written checkers, agreeing on every size. The values of [Section A.8](#) were computed by the program's routine and by an independent script, agreeing on every cell. Fact (b) of [Remark A.39](#) was produced by the generation enumerator and re-verified graph by graph with the exact checker. Two implementations agreeing is not a proof, but it is the practical guard against the failure mode a certificate would catch, namely a correct method with a wrong program.

Fourth, the self-test at the foot of the file is a standing regression check on every invariant the text quotes, and it is run after every change to the program.

What this does not amount to is a formally checkable certificate of the kind a SAT solver can emit, such as a resolution proof a small independent verifier could replay. For the one result here that rests on a solver's infeasibility verdict rather than on an exhaustion, fact (a) of [Remark A.39](#), that would be the right standard to aim at, and it is recorded among the open problems rather than claimed.

Chapter 3

Discovering Bounds by Search

The previous chapter proved what could be proved by looking at every graph of a fixed size. That method is honest and exact, and it stops almost at once. This chapter begins by measuring exactly how fast it stops, because the speed of that collapse is the whole reason a second kind of tool is needed. Then it builds that tool. Where the certifier walks all graphs, the search walks a few of them well, cooling like a physical system toward the densest feasible shape. What it returns is never a proof. It is a concrete graph, hence a lower bound and a conjecture, a stepping stone toward an upper bound that a later proof or certificate must supply. The chapter ends by asking the question any such tool must face: how good is it, and how far from the truth might its answers be.

3.1 The wall: how enumeration explodes

The brute force of [Chapter 2](#) visits every graph of a given size. The trouble is how many that is. A simple undirected graph on n vertices is a yes-or-no choice on each of the $\binom{n}{2}$ possible edges, so there are $2^{\binom{n}{2}}$ of them. A digraph chooses on each of the $n(n-1)$ ordered pairs, giving $2^{n(n-1)}$, and allowing multiplicities up to a cap c raises the base from two to $c+1$. These are not numbers that merely grow. They detonate. [Figure 3.1](#) draws the growth on a logarithmic scale, where each step up the axis multiplies the count tenfold, and the message is the same in every panel: the count passes any reachable budget while n is still in the single digits.

Three readings of the figure matter for what follows. Adding vertices is the sharpest cost: each new vertex multiplies the count by an exponential in n , so a single extra vertex can move a problem from seconds to centuries. Raising the connectivity budget is the next cost, lifting the base of the exponent rather than its height, which is why the multigraph curves sit above the simple one in every panel. Direction is the third: a digraph has twice as many cells to fill as the undirected graph on the same vertices (and a directed 3-uniform hyperedge, a tail with two heads, has three times as many), so the directed panel climbs fastest of all, and those are precisely the variants whose answers [Chapter 1](#) could not prove. The separation, by contrast, costs nothing here, because edge and vertex disjointness search the same graphs and differ only in what the checker measures, which is why a single pair of panels suffices for all of them. The certifier of [Chapter 2](#) reaches a few vertices further than blind enumeration, because its cut-counting optimisation is polynomial in the size of the encoding rather than in the number of graphs, but solving that encoding is itself hard and the gain is measured in single vertices, not orders of magnitude. Past that, no exhaustive method reaches the interesting sizes. Something

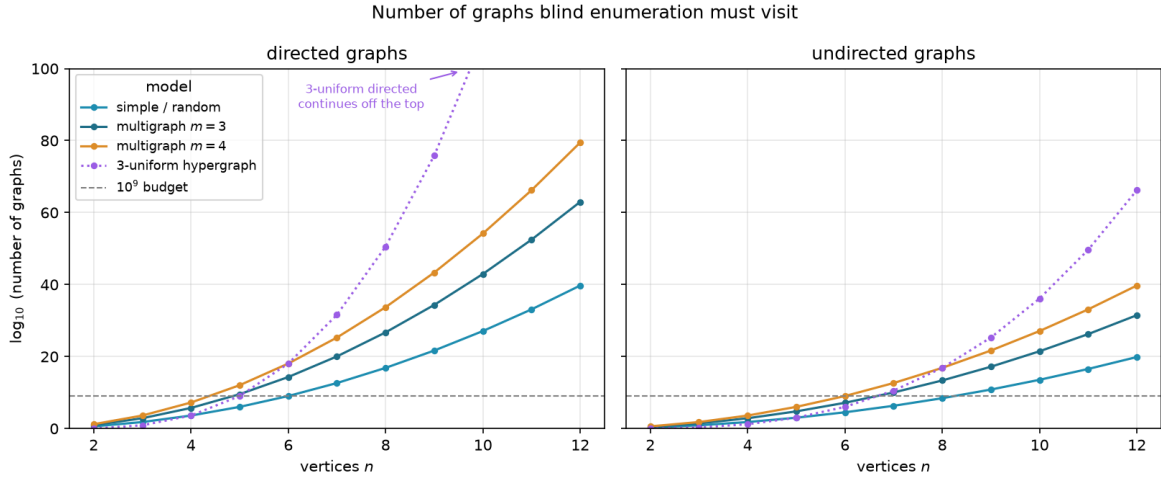


Figure 3.1. The number of graphs blind enumeration must visit, on a logarithmic vertical axis, for directed graphs (left) and undirected graphs (right). Each panel shows the simple model, two multigraph connectivity caps, and the three-uniform hypergraph (dotted), with a dashed line at a generous budget of 10^9 candidates, and every curve crosses that budget while n is still in the single digits. Writing E for the number of off-diagonal cells a model may fill, a simple graph offers 2^E candidates, a multigraph with multiplicity cap $m - 1$ (the connectivity ceiling) offers m^E (one choice per cell from $\{0, \dots, m - 1\}$), and the 3-uniform hypergraph offers $2^{\binom{n}{3}}$ undirected or $2^{n \binom{n-1}{2}}$ directed, since a directed hyperedge is a tail together with two heads. The count depends only on the model and the direction, never on whether one later asks for edge- or vertex-disjoint routes, so a single pair of panels covers all of those variants at once. Direction multiplies the number of cells to fill, by two for graphs (ordered versus unordered pairs) but by three for the 3-uniform hypergraph, so the directed panel sits far above the undirected one, and the directed cases, exactly the ones [Chapter 1](#) left open, are the first to go.

that does not look at every graph is required.

3.2 The temperature search

We want, for fixed n and m , a graph with as many edges as possible subject to $\lambda^{\max} \leq m - 1$ (or $\kappa^{\max} \leq m - 1$ for the vertex variants). The space of graphs on n vertices is enormous and its feasible region has no helpful shape: adding an edge can be harmless until, together with edges added long before, it suddenly creates a forbidden pair. A greedy walk that only ever accepts an improving move stalls at the first such collision, in a graph that is locally maximal but globally far from extremal.

The classical method for this kind of landscape is *simulated annealing* (SA), named after the metallurgical process of cooling a heated material slowly so that its atoms settle into a low-energy crystal rather than a disordered glass. Applied to graphs, annealing escapes local optima by sometimes accepting a worsening move, and by accepting fewer of them as the run proceeds. Each graph G is a state with energy

$$E(G) = -|E(G)| + \Lambda \cdot \max(0, \lambda^{\max}(G) - (m - 1)),$$

Algorithm 1: Temperature search for a dense feasible graph

Input: vertices n , value m , start T_0 , cooling α , penalty Λ , steps S
Output: the densest feasible graph encountered

```
1  $G \leftarrow$  empty graph on  $n$  vertices;  
2  $T \leftarrow T_0$ ;  
3 for  $S$  steps do  
4   propose  $G'$  by toggling one adjacency, biasing removals toward low-sensitivity edges;  
5    $\Delta \leftarrow E(G') - E(G)$ ;  
6   if  $\Delta \leq 0$  or  $\text{rand}() < e^{-\Delta/T}$  then  
7      $G \leftarrow G'$ ;  
8   if  $G$  is feasible and denser than the best so far then  
9     record  $G$  as the best;  
10   $T \leftarrow \alpha T$ ;  
11 return the best recorded graph;
```

which rewards edges and penalises any excess connectivity. The penalty is a soft constraint rather than a hard one, so the walk may pass briefly through infeasible graphs rather than being walled inside the feasible region, and we return below to why that freedom is worth having. A move toggles one adjacency. The Metropolis rule accepts an improving move always and a worsening move only with probability $e^{-\Delta E/T}$, a chance that shrinks both as the move gets worse and as the temperature drops, and the temperature T falls geometrically, multiplied by a fixed factor just below one at every step. Algorithm 1 is the whole loop, the one place in this chapter where a body is worth showing, because the loop *is* the method.

This loop is what `solve` runs in its discovery mode. With the exhaustive flag off it anneals instead of enumerating, following the Metropolis walk of Algorithm 1 under geometric cooling, removals biased toward low-sensitivity edges, and returns the densest feasible graph it encountered together with the full temperature trace. What it returns is a witness, hence a lower bound, never a proof.

`search_for_dense_graph(n, m, variant, separation) → SearchResult`

DISCOVER

in the size n , the ceiling m , the `Variant`, the separation, and a step budget

out a `SearchResult`: the densest feasible graph found, with the full step-by-step trace

how one Metropolis cooling run of Algorithm 1. The driver `best_of_searches` dispatches several independent runs across processor cores and keeps the densest, so a single local optimum never decides the answer.

It is fair to ask why the walk is allowed above the ceiling at all. Reaching every feasible graph does not require it. Removing an edge can only lower $\lambda^{\max}$ and $\kappa^{\max}$, never raise them, so every subgraph of a feasible graph is again feasible, and the empty graph is feasible trivially. From any feasible target we can therefore delete edges one at a time down to the empty graph without ever crossing the ceiling, and reading that path backward builds the target up the same way. The

feasible region is connected through the empty graph by single-edge moves that stay inside it, so a walk that never went infeasible could in principle visit every feasible graph there is.

We let it go infeasible anyway, because reachability and efficiency are different things. A feasible graph can be locally maximal, meaning every edge we might add creates a forbidden pair, and yet still sit far below the densest feasible graph. To improve such a graph the walk has to add an edge and then remove a different one, a swap whose first half is infeasible. A walk confined to feasible graphs could only find that better graph by retreating most of the way back to the empty graph and climbing a different slope, a detour so long that the cooling schedule usually ends first. Allowing a brief excursion above the ceiling lets the walk tunnel straight across instead, trading one load-bearing edge for a better one in two moves rather than dozens. The soft penalty also turns the search into a single smooth landscape with no hard walls for the Metropolis rule to bounce off, and the same machinery then carries over unchanged to variants whose feasible region is not so conveniently connected. Going above the bound is never the goal, then. It is the shortcut the annealer takes on its way back down to it.

3.3 Temperature and edge sensitivity

The phrase “biasing removals toward low-sensitivity edges” in [Algorithm 1](#) is where the temperature acquires its meaning, and it is the heart of the method. For an edge e define its *sensitivity*

$$\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e),$$

the amount by which deleting e relaxes the binding connectivity. A load-bearing edge sits on a tightest cut and has $\sigma(e) > 0$. Removing it would drop $\lambda^{\max}$ and so loosen the very constraint that limits how many edges the graph may hold. A free edge has $\sigma(e) = 0$: it can be removed without easing the binding pair, which usually means it can be put to better use elsewhere.

`edge_sensitivity(G, u, v) → ℕ`

MEASURE

in a graph G and one of its edges $e = (u, v)$

out $\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e)$

how delete e and let the checker remeasure, where a positive value marks a load-bearing edge. The whole-graph version `sensitivity_map(G)` reuses one shared baseline measurement. This single number is the dial the cooling turns.

When the search proposes a removal it draws the edge with probability proportional to $e^{-\sigma(e)/T}$. The dial is now explicit. At high temperature these weights flatten and the walk toggles edges regardless of how load-bearing they are, exploring widely and willing to dismantle essential structure. As the temperature falls the weights concentrate on $\sigma = 0$, so the walk almost never disturbs a load-bearing edge and instead rearranges only the free ones, until the graph crystallises onto an extremal shape. This is exactly the behaviour one wants, and it is the behaviour the formulation asks for: the temperature tracks how strongly removing an edge

changes the bound. Figure 3.2 shows one such run settling onto a certified optimum. The other variants behave identically, and their traces are collected in Section B.2.

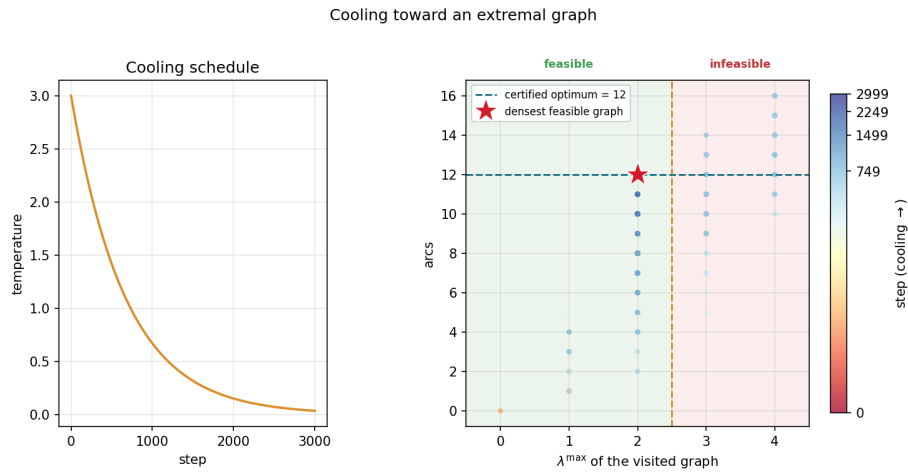


Figure 3.2. A single cooling run on the directed multigraph problem at $n = 4$, $m = 3$. *Left:* the temperature schedule, cooling geometrically from its starting value. *Right:* every visited graph plotted by its binding connectivity $\lambda^{\max}$ (horizontal) against its arc count (vertical), coloured by the step at which it was visited. Annealing uses a soft penalty rather than a hard wall, so while hot (dark points) the walk strays into the infeasible region $\lambda^{\max} > m - 1 = 2$, picking up extra arcs it is not allowed to keep. As the temperature falls (bright points) the penalty drives the walk back across the feasibility boundary and onto the densest graph that stays feasible, the certified optimum of 12 arcs at $\lambda^{\max} = 2$ (star).

The same sensitivities also explain a finished graph. Colouring a graph's arcs by σ shows at a glance which arcs are structural and which are slack. Figure 3.3 illustrates all four sensitivity levels on one directed multigraph: the darkest arcs carry the most flow and lower $\lambda^{\max}$ the most when removed, the cool arc contributes nothing. The picture is not decoration: it is the certificate of tightness made visible, and it tells the search exactly which arcs to leave alone as the temperature falls.

A single cooling schedule can still settle into a local optimum, so in discovery mode `solve` runs several independent schedules from different starting seeds and keeps the densest result across all of them. Because the restarts share no state and none depends on the outcome of another, they run in parallel. The program dispatches them across the available processor cores, so the wall-clock cost of k restarts is a small multiple of a single run rather than k times it, bounded below by the slowest restart and by the cost of starting the worker processes. A handful of parallel restarts is enough to reach the known optima at the sizes we examine. This is a reliability measure, not a change of method, and the parallelism is invisible to every number the thesis reports: the same best graph would be found by running the schedules sequentially, just more slowly.

Parallel arcs drawn explicitly, coloured by sensitivity σ :
cool ($\sigma = 0$, free) $\rightarrow$ warm (load-bearing)

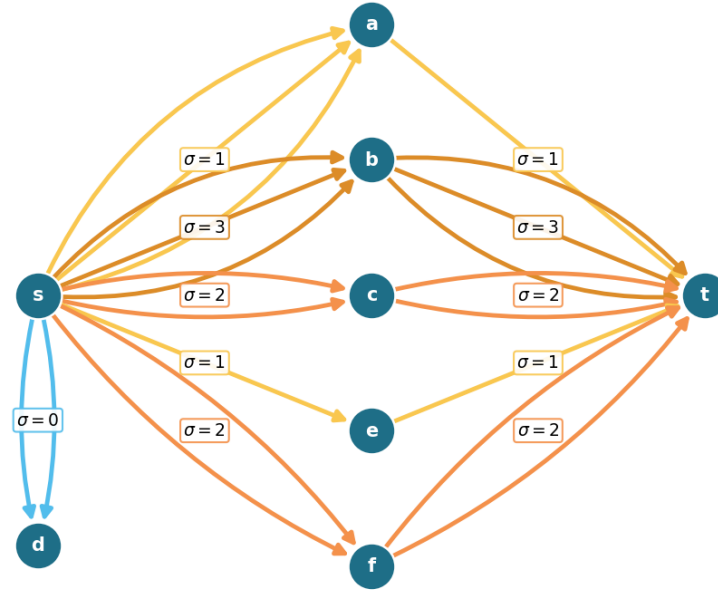


Figure 3.3. A directed multigraph on eight vertices with $\lambda^{\max} = 9$, every parallel arc drawn explicitly and coloured by its sensitivity σ , so multiplicity is read by counting arcs rather than from a label. The point is the contrast between $s \rightarrow a$ and $s \rightarrow b$: both carry three parallel arcs, yet $s \rightarrow a$ is cool ($\sigma = 1$) because the single arc $a \rightarrow t$ downstream throttles that route to one unit, while $s \rightarrow b$ is hot ($\sigma = 3$) because all three of its copies carry flow. The arcs $s \rightarrow d$ are coolest of all ($\sigma = 0$): d is a dead end, so deleting them changes nothing. Removing a hot adjacency drops $\lambda^{\max}$ by its σ while removing a cool one does not, which is exactly what tells the search which arcs to leave alone as the temperature falls.

3.4 Keeping the checker cheap inside the search

The energy of [Algorithm 1](#) reads the binding connectivity $\lambda^{\max}(G)$, and a literal reading of the loop would call the checker afresh on the whole graph at every step. That checker is the most expensive routine in the program, one maximum flow per ordered pair, and the walk runs for thousands of steps with several restarts, so such a search would spend nearly all of its time remeasuring graphs that barely changed from the step before. Two elementary facts remove almost all of that work without altering a single number the search reports.

Proposition 3.1 (Monotonicity of the binding connectivity). *Let G be a graph or directed multigraph, and let G' be obtained from G by changing one adjacency by a single unit of multiplicity. Then*

- (1) (monotonicity) *removing the unit cannot raise $\lambda^{\max}$, and adding it cannot lower $\lambda^{\max}$, and*
- (2) (unit step) *adding the unit raises $\lambda^{\max}$ by at most one.*

Both statements hold verbatim for the vertex measure $\kappa^{\max}$.

Proof. Fix an ordered pair (s, t) . By the max-flow / min-cut theorem $\lambda_G(s, t)$ is the smallest capacity of any cut separating s from t , where the capacity of a cut is the total multiplicity of the arcs that cross it from the source side to the sink side.

For part (1), reducing one adjacency by a unit lowers the capacity of every cut that the changed arc crosses and leaves all other cut capacities untouched, so no cut capacity rises. The smallest cut capacity therefore cannot rise either, which gives $\lambda_{G'}(s, t) \leq \lambda_G(s, t)$. Taking the maximum over all pairs yields $\lambda^{\max}(G') \leq \lambda^{\max}(G)$. Adding a unit is the same statement read backwards.

For part (2), write G'' for G with one unit added to a single adjacency. That extra unit of capacity crosses any fixed cut at most once, so every cut has capacity at most one greater in G'' than in G . The smallest cut capacity thus grows by at most one, giving $\lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$, and together with part (1) we have $\lambda_G(s, t) \leq \lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$. The maximum over pairs gives $\lambda^{\max}(G'') \leq \lambda^{\max}(G) + 1$.

For $\kappa^{\max}$ the argument is identical on the vertex-split network of [Figure 2.3](#), where changing an adjacency changes the capacity of its single arc $u^{\text{out}} \rightarrow v^{\text{in}}$ in the same way. Parallel copies leave that capacity at one and so change $\kappa^{\max}$ not at all, which only strengthens both parts. $\square$

These two facts settle almost every step without a flow. The energy depends on $\lambda^{\max}$ only through the excess $\max(0, \lambda^{\max} - (m - 1))$, which is zero exactly when the graph is feasible. So a removal applied to a feasible graph leaves it feasible by part (1), with excess still zero, and needs no measurement at all. An addition made while the graph sits strictly below the bound, at $\lambda^{\max} \leq m - 2$, stays feasible by part (2), again with excess zero and no measurement. Only an addition made right at the boundary $\lambda^{\max} = m - 1$ can tip the graph over, and only there does the search run a single flow, capped so that it stops the instant it finds one route too many. The exact connectivity value is recomputed only on the rare step whose proposal is genuinely infeasible, where the penalty term needs it. The three cases of [Proposition 3.1](#) read directly off the code:

```

1 if (is_add and lam_ub <= m - 2) or (not is_add and feasible_current):
2     excess = 0                                # part (2) or part (1): provably feasible
3 elif not exceeds_bound(proposal, m - 1, separation=separation):
4     excess = 0                                # at the boundary, but the flow clears it
5 else:
6     excess = measure_full(proposal) - (m - 1) # infeasible: exact penalty

```

Listing 3.1. The three cases behind [Proposition 3.1](#), in the order the proof gives them: removal from a feasible graph (part 1), addition below the bound (part 2), and only otherwise a single capped flow.

To know which case applies, the search carries a running upper bound on $\lambda^{\max}$, raised by one whenever it accepts an addition (part (2)) and left untouched when it accepts a removal (part (1)), and resynchronised to the exact value at the cadence where it already pays for flows to refresh

Construction	n	m	Search	Theory
spanning tree (simple undirected, $m=2$)	6	2	5	5
multiplicity- $(m-1)$ star (multi undirected, $m=3$)	6	3	10	10
bidirected star (simple directed, $m=2$)	4	2	6	6
bidirected star (simple directed, $m=2$)	6	2	10	10
hub-bipartite tie (simple directed, $m=2$)	7	2	12	12
double star (multi directed, $m=3$)	4	3	12	12
double star (multi directed, $m=3$)	5	3	16	16

Table 3.1. Validation by rediscovery. For each variant the temperature search, run from scratch with a handful of restarts, recovers exactly the extremal value proved or certified in the earlier chapters. The “Search” column is an honest lower bound found by annealing, and the “Theory” column is the proved or certified value.

the sensitivity map. The bound can only ever sit at or above the truth, so any step it clears is certainly feasible.

The key point is that the energy returned on every step is, by [Proposition 3.1](#), exactly the value the naive implementation would have computed. Every accept-or-reject decision, every random draw, and the graph the walk finally returns are therefore unchanged, and the speed-up is invisible to every number the thesis reports. This is verified rather than asserted. A regression test runs both implementations, the fast one and a reference that measures exactly at every step, across all four graph variants and both separations, and checks that the two agree step for step in energy, in every accept-or-reject decision, and in the final graph, and that the returned graph is feasible under the exact checker. Nothing about the optimisation hides inside the bound: the witness the search reports is always certified by an exact measurement, exactly as in the slower search.

It is worth noting where this method earns its keep. For undirected graphs one could resynchronise the exact value cheaply from the Gomory–Hu tree of [Chapter 2](#), which encodes every pairwise edge-connectivity in $n - 1$ flows. The directed and vertex variants admit no such tree, so there the monotone bound and the capped predicate are the only inexpensive handle on feasibility, and they are what let one annealer serve every variant at speed.

3.5 Rediscovering the known extremisers

A search method is only worth trusting if it recovers what is already known before it is pointed at what is not. Run from the empty graph on the cases whose answers [Chapter 1](#) and [Chapter 2](#) settled, and told only to pack in edges without breaching the connectivity ceiling, the cooled search arrives unprompted at the named constructions. [Table 3.1](#) reports the densest graph it found against the value the theory predicts, and in every case the two agree.

What the table hides is that the graphs themselves are the named constructions, not merely graphs of the right size. The simple undirected run at $m = 2$ settles on a spanning tree, the only shape with $n - 1$ edges and no cycle. The undirected multigraph run settles on a star with every

edge at multiplicity $m - 1$, an instance of the multi-tree extremiser behind [Theorem 1.3](#). The directed multigraph runs settle on the double star of [Chapter 2](#), both directions of every spoke at full multiplicity. The simple directed runs find the bidirected star at $n = 4$ and, at $n = 7$, land on twelve arcs, the value at which the hub and the one-directional wall tie, exactly the crossover [Theorem 1.10](#) predicts. The cooled search does not know these names. It rediscovers the structures because, under the connectivity ceiling, there is nowhere denser to go.

Some extremisers are past the size at which a quick search lands reliably, but the exact checker of [Chapter 2](#) settles them immediately by measuring their connectivity. The one-directional bipartite digraph on eight vertices has 16 arcs at $\lambda^{\max} = 1$, and 16 already exceeds the linear hub value $2(n - 1) = 14$: the smallest size at which the quadratic phenomenon strictly wins, made concrete. It is also a measured illustration of the search's limit: repeated cooling runs at $n = 8$, $m = 2$ stall at the hub value of 14, because dismantling a finished hub and rebuilding a one-directional wall is far more than the single-arc moves of the walk can tunnel through. The known answer there comes from the construction and the checker, not from the search, which is exactly the division of labour this chapter argues for. The augmented bipartite digraph on ten vertices at $m = 3$ has 30 arcs at $\lambda^{\max} = 2$, the thirty-arc counterexample of [Remark 1.12](#), confirmed pair by pair against the naive bound of 27 it refutes. The directed double star on seven vertices at $m = 4$ has 36 arcs at $\lambda^{\max} = 3$, matching $2(n - 1)(m - 1)$. None of these is a proof of optimality. Each is an exactly measured feasible graph, a lower bound the checker stamps as genuine.

3.5.1 Simulated annealing versus tabu search

The annealer is not the only way to walk this landscape. A natural alternative, suggested by Jan Goedgebeur, is *tabu search* ([Table 4.2](#) names the implementation), which keeps the energy and the neighbourhood of [Algorithm 1](#) but replaces the cooling schedule with a deterministic best-improving step. At each step it evaluates every single-edge move from the current graph and takes the one that lowers the energy most, then forbids the pair it just changed for a fixed number of steps, the *tabu tenure*, so the walk cannot immediately undo its last move and stall in a two-step cycle. When no improving move is left and the search stalls, it perturbs the incumbent with a few random moves and carries on. There is no temperature and no accept-worse coin: apart from the perturbation kicks the walk is deterministic. Where annealing explores by occasionally stepping uphill and trusts the cooling to settle it, tabu climbs greedily and trusts the tabu list to keep it from retracing its path.

The two engines rediscover the same values, but at very different speeds, and on the harder directed cases the difference is decisive. [Table 3.2](#) runs them head to head at an equal budget of six seconds each, both restarted within that budget. On the simple undirected problem both reach the optimum at once. On the two directed problems past the certified range tabu reaches the optimum within the budget while the annealer plateaus below it, assembling the eighteen-arc simple-directed extremiser and the full twenty-four-arc double star at $n = 7$ where the annealer stalls at sixteen and twenty. The cause is the one already met at the $n = 8$, $m = 2$ hub stall above. The annealer cools onto whatever dense shape it first reaches and then cannot dismantle

it within the budget, whereas tabu's best-improving step follows the steepest density gradient and the tabu list stops it sliding back, so it assembles the structured extremiser the cooled walk struggles to find.

Variant	n	optimum	best in 6 s		time to optimum	
			SA	tabu	SA	tabu
simple undirected	7	9	9	9	1.3 s	0.1 s
simple directed	7	18	16	18	—	2.6 s
directed multigraph	7	24	20	24	—	2.5 s

Table 3.2. Simulated annealing (SA) against tabu search at an equal six-second budget per method, the annealer restarted from fresh seeds and tabu restarted on stalls. Only the simple undirected value is a proved optimum (Mader). The two directed values are the best known rather than proved, found by search and construction past the certified range that stops below $n = 7$ for those variants (Section 3.6), so here the “optimum” column is the target each engine is trying to reach. Both reach the simple undirected optimum. On the two directed problems tabu reaches the best known value while the annealer plateaus below it. The last two columns give the wall-clock time each method first reached that value on the reference machine, with a dash where it did not reach it in the budget.

Figure 3.4 shows the divergence directly, plotting the densest feasible graph each engine has found against wall-clock time on the two directed multigraph cases. Tabu climbs in a few decisive steps and holds at the optimum, while a single annealing schedule rises quickly and then flattens a few arcs below it. The annealer's independent restarts recover the optimum on the smaller case, as Table 3.1 records, so a single plateau is not the whole story there. But at $n = 7$ the plateau persists within the budget, and one tabu schedule already reaches what the restarted annealer does not.

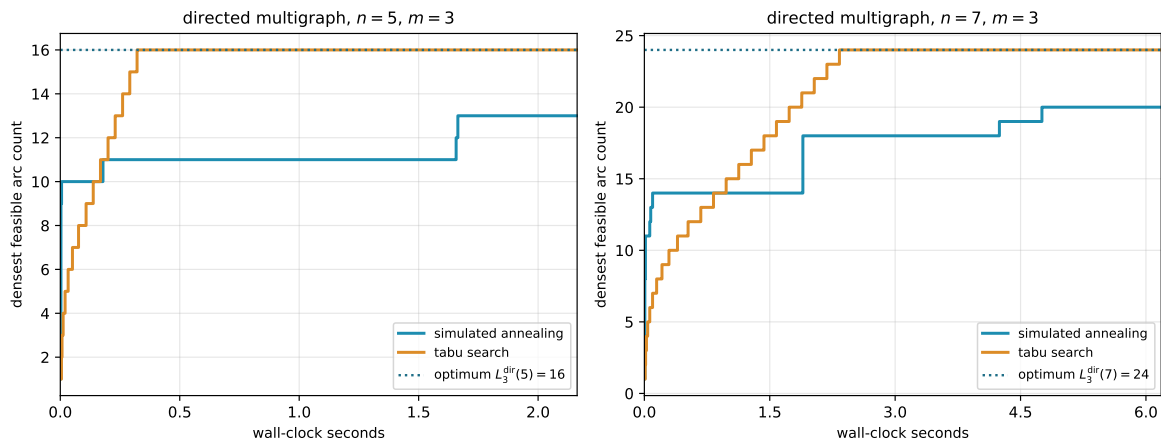


Figure 3.4. Densest feasible graph found against wall-clock time, for one simulated-annealing schedule and one tabu schedule on the directed multigraph at $n = 5$ and $n = 7$, $m = 3$. Tabu climbs in a few best-improving steps to the optimum (dotted) and holds, while a single annealing schedule rises quickly and then flattens below it. The time axis is wall-clock on the reference machine, so unlike the seed-reproducible figures elsewhere this one is a representative timed run.

This advantage comes at a cost the equal-time comparison already absorbs but is worth naming: each tabu step scans the whole neighbourhood, so it is far more expensive than the an-

nealer's single random proposal, and tabu takes many fewer but better-aimed steps in the same wall-clock. Tabu search is therefore the program's default for solving the problems, reaching every value this thesis reports, the rediscovery of [Table 3.1](#) and the conjectural lower bounds alike, and reaching the harder directed optima past the certified range where the annealer plateaus. Simulated annealing is kept for the self-check and verification, where the small values fall at once, the lighter per-step cost is all that is needed, and its independent restarts parallelise trivially across cores.

3.6 The trichotomy: proved, certified, conjectured

It is worth stopping to lay out, for every variant, the three kinds of knowledge the computation can hold about it, because the whole value of the pipeline is that they are kept apart and never quietly merged. About any given value the program can do one of three things. It can *prove* the value, closing a fixed size from above by exhausting every graph of that size, an upper bound with the standing of a hand proof. It can *certify* it, the same kind of upper bound but obtained by the cut-counting optimisation rather than blind exhaustion, reaching a vertex or two further. Or, where proof and certificate both run out, it can only *conjecture* the value, exhibiting a concrete dense graph that pins down a lower bound and leaving the matching upper bound to a later argument.

Proof and certificate bound the answer from above. The search bounds it only from below. Where a closed *formula* exists it can tie the two together, threading through the certified small cases and the conjectured large ones as a single curve. This is the discipline the whole thesis runs on, and it is worth stating as one: the search proposes a lower bound, a certificate or a hand proof disposes the upper bound, and only when both meet is a value called settled.

We begin with the single variant where all three kinds of knowledge are non-trivial and visibly distinct, the directed simple arc problem at a fixed $m \geq 3$, and then sweep the whole family. For that variant the knowledge splits cleanly by the size of the graph.

For small n the answer is not conjectured at all. The pruned exhaustion inside `solve` walks every simple digraph on n vertices and returns the exact maximum, complete and certain, an upper bound with the standing of a hand proof. At $m = 3$ it certifies 6, 9, 12 for $n = 3, 4, 5$, and there the exhaustion stops within a reasonable budget, the pruned walk no longer closing $n = 6$ in tractable time. That is the certified region, finite and closed.

Beyond it the ceiling is gone and only lower bounds remain, each witnessed by a concrete feasible graph. The search finds the extremal 15 arcs at $n = 6$ on its own. Past that size it falls behind: across six restarts it reaches 17, 17, 20, 23 for $n = 7, \dots, 10$, while the named constructions of [Chapter 1](#), measured exactly by the checker, supply 18, 21, 25, 30, the last being the thirty-arc graph of [Remark 1.12](#). The reported lower bound at each size is the better of the two, exactly as the rediscovery section combined search and checker by hand. These are conjectures with an honest floor and no matching ceiling. That is the searched region, unbounded and forever one-sided.

The two regions are tied together by a single closed form. The conjectured value

$$\ell_m^{\text{dir}}(n) = \max\left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right) \quad (n \geq m)$$

of [Conjecture 1.14](#) is not a fresh guess bolted on past the certified range. It is one formula that already passes through every certified value at small n and then continues, without a seam, through every searched value at large n . The directed-arc panels of [Figure 3.5](#) show this directly, the certified squares at the small sizes and the open lower-bound circles beyond both sitting on the one red conjecture curve, proof below the formula and witnessed graphs above it. The formula is the through-line that makes the small certified cases and the large conjectured ones a single statement rather than two.

Two features of those directed-arc panels are worth naming. The crossover from the linear hub term to the quadratic bipartite term happens at $n = 9$, where the second argument of the maximum first wins, and that is past the last certified size. The shape of the answer therefore changes in exactly the region where only the search can see it, which is precisely why a discovery tool was needed and a certifier alone would not do. And at $n = 10$ the formula reads 30, the count of the very counterexample that overturned the natural initial conjecture, so the closed form is not a smooth fit to easy data but one that survives the case that broke the previous guess.

Asking the same three questions of every variant, the answers sort into three regimes, and [Figure 3.5](#) shows all twelve at once, one panel per variant, in exactly this proved, conjectured, and guessed language.

In the first regime the closed form is already a theorem for every n , proved by hand in [Chapter 1](#) or cited, and the computation only confirms it. The undirected families live here, edge and vertex alike, together with the directed problems at $m = 2$. For these the certified small cases and the rediscovered constructions of [Table 3.1](#) are validation, not new knowledge: the formula was never in doubt, and the program's role is to check that the pipeline reproduces it. That the two multigraph vertex rows measure exactly as their underlying simple graphs is a matter of the counting convention of [Remark 2.1](#) rather than a theorem, so those rows need no computation of their own at all. The directed vertex rows do need their own, since Whitney's inequality gives a lower bound there but no upper one, and they get it: every exact point on those panels comes from the search and the exhaustion run under the vertex test.

In the second regime the three columns genuinely differ and the computation is load-bearing. This is the regime of the worked example above. The directed simple arc and vertex problems at $m \geq 3$ are certified only at the small n the exhaustion reaches and conjectured by the search beyond, with the formula of [Conjecture 1.14](#) the through-line.

The directed multigraph arc problem sits here too, with one extra economy: in its cut-counting mode `solve` proves the m -free number $M^*(n) = 2(n-1)$ outright for $n \leq 6$, and the scaling identity $L_m^{\text{dir}}(n) = (m-1)M^*(n)$ of [Chapter 2](#) carries that one certified number to every m at once, so the closed form $2(n-1)(m-1)$ is proved across the whole m axis at those sizes. Past

the crossover at $n = 7$ the conjectured value switches to the bipartite branch $(m - 1) \lfloor n^2/4 \rfloor$, where the even sizes are proved conditional on the odd ones and only a single mixed-regime minimum-degree statement keeps the induction from closing ([Appendix A](#)).

In every row of this regime the formula agrees with proof where proof reaches and with the search where it does not, exactly the linkage the worked example displayed.

In the third regime there is no formula at all, and the honesty is in drawing the guess as a guess. The undirected vertex problem at $m \geq 6$ has been open since 1974 with its extremal structure genuinely unknown, and the hypergraph variants beyond the proved edge case are unexplored, the directed ones needing a one-tail, many-head reading of a Berge edge that this thesis defines itself only so that `solve` can measure it at all. Here the program still computes exact values at the few sizes that finish, and its search still exhibits feasible graphs at a handful more, but no closed form is in sight. The panels of [Figure 3.5](#) carry, instead of a red conjecture curve, a yellow line that simply interpolates the machine points in straight segments, trusted no further than the eye. We draw it *through* the points rather than fitting a smooth curve above them: the points are honest lower bounds, the truth lies on or above them, and a line riding above the points would claim more than the search ever found.

Two cautions govern how to read the dotted curves. At small n the complete graph or hypergraph is itself feasible, its binding connectivity being only $n - 1$, so the earliest points merely trace the trivial maximum and reveal nothing of the open behaviour: the undirected-vertex points run up $\binom{n}{2}$ until n passes m , and only then bend toward the linear growth the proved edge value predicts.

Past that bend the dotted line is as strong as the best lower bound behind it, which at each size is the better of the quick search and a named construction the checker verifies. The search alone stalls, exactly as it stalled on the quadratic wall in the rediscovery above: where it cannot build a denser construction its own reports climb only linearly. The clearest case is the directed hypergraph, whose bipartite family is quadratic, reaching $\sim (m - 1)n^2 / (4(r - 1))$ hyperarcs by [Proposition A.50](#): there the walk slips below that construction at larger n , so it is the construction that carries the circles, with the search free to beat it wherever it can. We therefore read the dotted trends as honest lower bounds and not as fitted laws, and the one closed-form hypothesis we record, the quadratic for the directed hypergraph, comes from the construction and not from the shape of the dots.

So to the question of whether one formula links what is proved to what is conjectured, the answer is three-valued and depends on the variant, and the grid of [Figure 3.5](#) shows all three at a glance. Where the problem is already solved the curve is blue, a theorem the computation merely agrees with. Where the problem is open but tractable the curve is red, a conjecture the certified squares confirm below and the searched circles confirm above, and that is the case the program was built for. Where the problem is genuinely hard the curve is yellow, a guess interpolated through the search points and an open question of what, if anything, the dots truly trace.

One feature stands out in the bottom row: the directed hypergraph panels, both arc and vertex, climb steeply at $m = 3$, the yellow guess curving upward rather than running straight. This is genuine and not search instability: the directed Berge edge packs a tail and $r - 1$ heads into a single object, and the bipartite construction of [Proposition A.50](#) already reaches roughly $(m - 1)n^2/(4(r - 1))$ hyperarcs, so the value grows with n^2 . The same quadratic shape sits one column over, in the directed arc and vertex panels, whose conjectured red curves carry the $\left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor$ term and rise just as fast. Super-linear growth is the signature of *direction* across the grid, not of the hypergraph alone: every directed column bends upward while the two undirected columns stay linear. [Figure 3.6](#) redraws the same grid at $m = 6$: the proved curves climb linearly with m exactly as their formulas predict, while the open variants' search points and their yellow interpolation climb higher without a proved curve to meet, so raising the threshold does not change which regime a variant lives in, only how far the red and yellow curves have travelled.

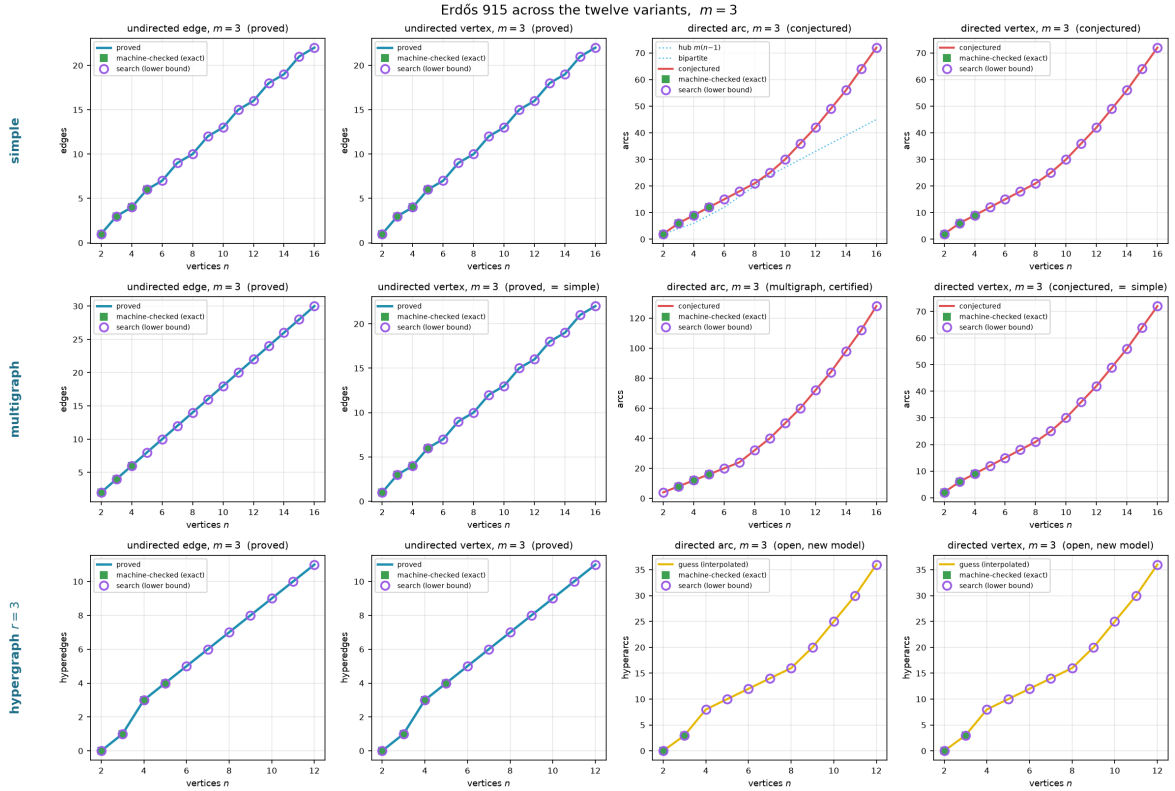


Figure 3.5. All twelve variants at $m = 3$, edges against n , laid out by model (rows: simple, multigraph, 3-uniform hypergraph) and by separation and direction (columns: undirected edge, undirected vertex, directed arc, directed vertex). A blue curve is a value proved for all n , a red curve a conjectured formula, and a yellow curve a bare interpolation of the machine points where no formula is known, the three told apart by colour rather than by dash pattern. Filled squares are the sizes `solve` proved or certified exactly. Open circles are lower bounds at larger n , the better at each size of the search's densest find and a named construction measured by the checker, reported as a running maximum so a feasible graph at one size is never lost at the next. On the open variants the yellow curve is the piecewise-linear interpolation *through* those circles, never above them, and the axes are scaled to the data rather than to a loose certain interval. The directed hypergraph panels in the bottom row rise steeply because their value is quadratic in n , not linear.

The same grid answers the harder question the trichotomy raises, which is what the search is

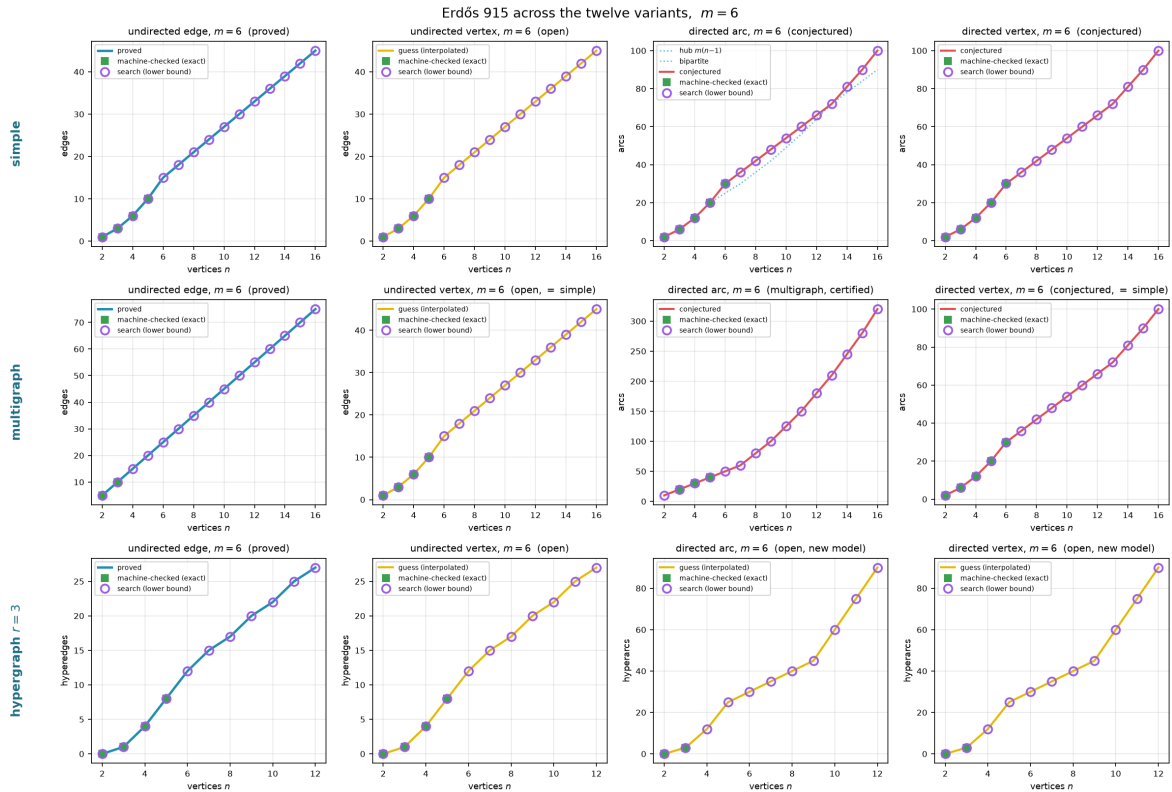


Figure 3.6. The same twelve variants at $m = 6$. The undirected edge and multigraph edge cases remain proved (Mader’s theorem holds for all m), while the undirected vertex separation is the famous open problem at $m \geq 5$ and the directed cases carry the same conjectured formula scaled to $m = 6$. On the open vertex panels the circles never sit below the proved edge value at the same size, because every graph feasible for the edge problem is feasible for the vertex problem (Whitney), so the edge extremiser is an honest lower bound there, and the search is free to climb past that floor where the vertex problem is genuinely richer. Comparing this figure with the $m = 3$ version above shows that the proved cases grow linearly with m as the formulas predict, while on the open cases the circles and their yellow interpolation climb steadily without a proved curve to meet.

worth where no upper bound exists to check it against. It says two things at once. First, wherever an upper bound exists, proved or certified, the search lands exactly on it and never above it at every plotted size, which is strong evidence that the search is not leaving edges on the table, with the one instructive exception recorded above: the directed $m = 2$ run at $n = 8$, where it stalls on the hub and the bipartite optimum must be supplied by the construction instead.

Second, on the directed arc problem at $m \geq 3$, where no upper bound is known past the certified range, the plotted lower bounds land exactly on the conjectured value $\max(m(n - 1), \lfloor (n + m - 2)^2/4 \rfloor)$ at every reachable n , and the certified points agree with it as far as they go, but past the first open size it is the constructions, not the cooled walk, that carry the bounds there. Run from scratch the search confirms the conjecture at $n = 6$ and then plateaus a few arcs short (17, 17, 20, 23 against 18, 21, 25, 30 at $n = 7, \dots, 10$, across six restarts), for the same reason as the $m = 2$ stall recorded earlier: the extremal shapes past the crossover are bipartite walls, and a walk that has crystallised onto a hub cannot rebuild itself into a wall by single-arc moves within the cooling budget.

So to the question “do we believe the reported lower bounds are best possible,” the grid answers: wherever a proof or a certificate can check them, yes, and on the open sizes they sit on the only value anyone has conjectured, with the search confirming the first open size and the constructions carrying the rest. The gap between those lower bounds and what is proved is exactly the gap between the conjecture and the missing upper bound, which [Chapter 4](#) names precisely. The search has not closed that gap, and it cannot. What the pipeline supplies is reproducible evidence for which side of the gap the truth lies on.

Honesty

A number found by the search is reported as a lower bound, and is marked as certified or proved only when a separate cut-counting optimisation or a hand proof supplies the matching upper bound. No extremal value is ever asserted on the strength of the search alone. The figures in this chapter use fixed random seeds, so every search point is reproducible.

Chapter 4

Synthesis, Results, and Open Problems

The journey set out from one clean theorem and followed it across twelve variants, proving what proved cleanly, building a machine where the proofs ran out, certifying the small cases the machine could close, and using the search to discover the dense graphs the proofs had singled out and to point at the ones still open. This closing chapter draws the threads together. It maps the directed frontier, where proof and certificate both still fall short of the full answer, names the two genuine surprises the landscape contains, gathers the twelve answers into one picture, and says plainly what the program changed and where it stops.

4.1 The directed frontier and the single open lemma

Two of the open directions are directed, and they share a structure worth stating exactly rather than hiding. The directed arc problem is a race between two constructions on different scales. The directed hub ([Construction 1.9](#)) reaches $m(n-1)$ arcs and grows linearly. The augmented bipartite construction ([Construction 1.13](#)) packs $\lfloor (n+m-2)^2/4 \rfloor$ arcs and grows quadratically. Which leads depends on n , and the two cross at an order linear in m , near $n = 9$ at the $m = 3$ of [Figure 4.1](#). [Conjecture 1.14](#) asserts that together they are the whole story.

[Conjecture 1.14](#) is proved as a lower bound for all m , and it is a theorem at $m = 2$ by [Theorem 1.10](#). Past $m = 2$ it sits squarely in the second regime of the trichotomy of [Section 3.6](#): certified by exhaustion at the small sizes `solve` can close for $m = 3$, conjectured by the temperature search beyond, and never beaten anywhere the search reaches. What is missing is a general upper bound for $m \geq 3$, and the obstacle is structural rather than computational.

The route to the upper bound is to show that an extremal digraph, once it is past the hub regime, must look like the augmented bipartite construction: its vertices split into a part A and a part B with every arc running $A \rightarrow B$, and B thickened only mildly. Three structural facts, each proved in [Appendix A](#), push in this direction. At $m = 2$ out-neighbourhoods are mutually unreachable and second out-neighbourhoods are disjoint, and for every m a complete $A \rightarrow B$ partition forces each vertex of B to absorb at most $m-2$ arcs from inside B . With the partition free to vary, the total arc count $|B|(|A| + m - 2) = |B|(n - |B| + m - 2)$ is maximised at $|B| = \lceil (n + m - 2)/2 \rceil$, which gives $\lfloor (n + m - 2)^2/4 \rfloor$, precisely the conjectured upper bound.

The reduction is clean except at a single point, and it is worth naming the point exactly rather

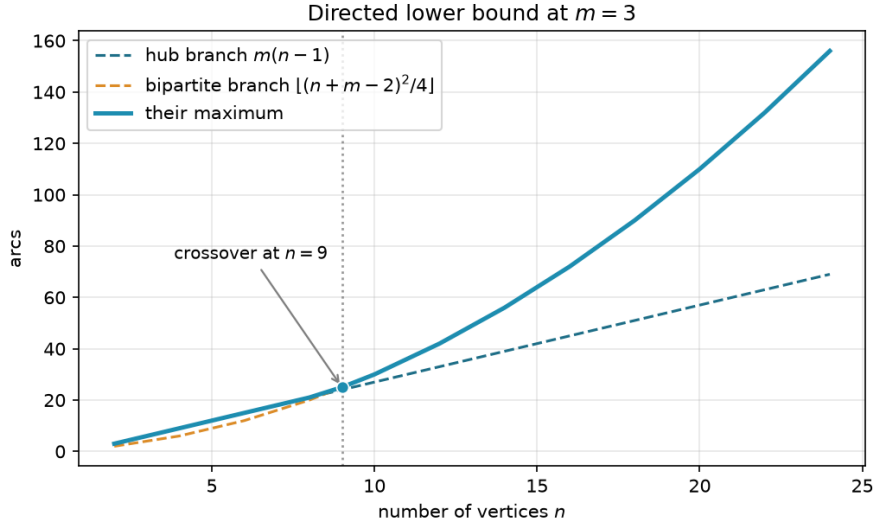


Figure 4.1. The two directed lower bounds at $m = 3$. The linear hub branch leads for small n . The quadratic augmented-bipartite branch overtakes it near $n = 9$ and leads thereafter. [Conjecture 1.14](#) asserts their maximum is the exact value.

than hiding it. The count goes through provided the extremal digraph decomposes the way [Figure 4.2\(a\)](#) draws it, and that shape is four conditions rather than one: a partition $V = A \cup B$ exists, every arc from A to B is present, no arc lies inside A , and no arc runs from B back to A . The last condition is the one with a name and a picture. Under it a route between two vertices of B can never escape B , so the connectivity inside B is governed entirely by B 's own arcs and the two-hop budget applies, and [Figure 4.2\(b\)](#) shows how a single backward arc destroys exactly that. It is also the condition where the natural delete-and-compensate exchange fails to be monotone, which is why it is the one usually named as the obstacle.

Naming it alone would nonetheless overstate the position, so the appendix states the whole hypothesis as [Conjecture A.21](#): an extremal non-hub digraph admits such a partition, complete forwards, empty inside A , and with nothing coming back. Nothing proved here forces an arbitrary extremal digraph into that shape, and the small-case evidence for it is exactly the kind of thing the search of [Chapter 3](#) can probe. So the honest summary is that [Conjecture 1.14](#) for $m \geq 3$ awaits one structural decomposition theorem, of which the backward-arc clause is the hardest quarter.

What does not wait is the leading constant. [Proposition A.22](#) proves, with no hypothesis at all beyond feasibility, that a digraph with $\lambda^{\max} \leq m - 1$ has at most $\lfloor n^2/4 \rfloor + \sqrt{m} n^{3/2}$ arcs, and that deleting at most $\sqrt{m} n^{3/2}$ of them leaves a one-directional bipartite digraph. Its engine is a single observation the rest of the chapter has been circling: for any ordered pair, the direct arc and the two-step detours through distinct midpoints are automatically arc-disjoint, so feasibility caps their number, and summing that cap over every pair caps $\sum_x d^+(x)d^-(x)$. A vertex with both a large in-degree and a large out-degree is therefore expensive, every vertex is nearly a pure source or a pure sink, and throwing away each vertex's smaller side costs only $O_m(n^{3/2})$ arcs and leaves the bipartite wall standing. So $\ell_m^{\text{dir}}(n) = n^2/4 + O_m(n^{3/2})$ for every fixed m , and what [Conjecture 1.14](#) still claims beyond that is the second-order term $(m-2)n/2$, not the shape

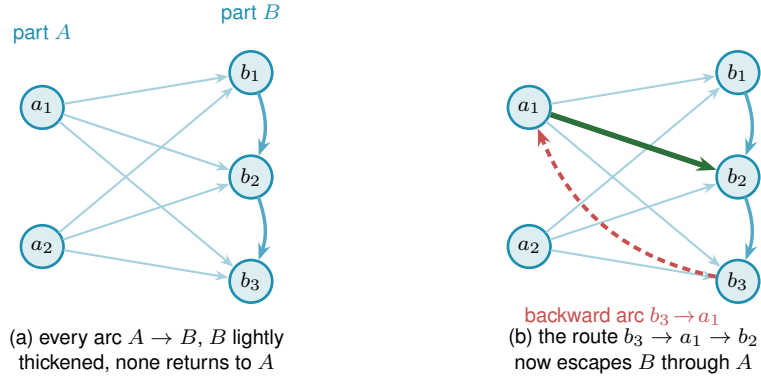


Figure 4.2. The directed frontier reduction and the single obstruction. **(a)** The structure the upper bound wants. Every arc runs from A to B (faint), B carries only a light internal thickening (the $m - 2$ budget, dark blue), and no arc returns to A , so a route between two vertices of B never leaves B and the count $|A||B| + (m - 2)|B|$ is complete. **(b)** A single backward arc $b_3 \rightarrow a_1$ (red) breaks this. The route $b_3 \rightarrow a_1 \rightarrow b_2$ (red leg then green leg) now escapes B through A , an extra route the budget never counted. Ruling out backward arcs in an extremal digraph is the one missing lemma behind the $m \geq 3$ upper bound, and the checker refuted a proposed counterexample by finding exactly such a hidden route.

of the answer.

The machine has been useful here in the negative direction too: a proposed counterexample to the conjecture, a backward arc bolted onto the thirty-arc graph, was refuted by the checker, which found the third route the construction's author had missed.

The directed vertex-disjoint problem shares the constructions but not the proofs, and the distinction matters more than it looks. Whitney's inequality $\kappa \leq \lambda$ says a digraph obeying the arc ceiling obeys the vertex one, so the arc constructions are vertex-feasible and every lower bound carries across for free. It says nothing in the other direction, and the vertex-feasible family is the larger of the two, so no arc upper bound restricts it. At $m = 2$ the vertex value is nonetheless the same, and [Theorem 1.11](#) proves it by running the induction a second time under the stricter separation, with its own exhaustive base cases through $n = 7$. For $m \geq 3$ the vertex value is conjectured equal to the arc value, on the strength of the same constructions and of exhaustive small-case checks run under the vertex test rather than inherited from the arc one. What it will not do is follow from the arc case, even once [Conjecture A.21](#) is supplied, because an upper bound proved over the smaller family says nothing over the larger one. The vertex problem needs its own upper bound at $m \geq 3$, and [Table 4.1](#) lists it as a separate conjecture rather than a corollary. The directed multigraph problem has its own two-branch story, parallel to the simple one. The conjectured value is

$$L_m^{\text{dir}}(n) = (m - 1) \max(2(n - 1), \left\lfloor \frac{n^2}{4} \right\rfloor),$$

where the double star realises the linear branch and the one-directional bipartite multigraph, every arc at multiplicity $m - 1$, realises the quadratic one.

The quadratic branch here uses the balanced partition $|A| = \lfloor n/2 \rfloor$, $|B| = \lceil n/2 \rceil$, giving $(m - 1) \lfloor n^2/4 \rfloor$ arcs. The multigraph and the simple digraph disagree about the best partition,

and the reason is worth spelling out. In a multigraph the $m - 1$ routes per pair are supplied directly by $m - 1$ parallel copies of every $A \rightarrow B$ arc, so nothing needs to be added inside B , and the arc count $(m - 1)|A||B|$ is largest at the balanced split. In a simple digraph each ordered pair carries at most one arc, so the extra routes must instead come from arcs inside B , and feeding B those in-arcs is what pushes its optimal size up to $|B| = \lceil (n + m - 2)/2 \rceil$ once $m \geq 4$ (at $m \leq 3$ the balanced and shifted splits happen to give the same count, as [Construction 1.13](#) records). One consequence is that the multigraph's two branches cross at $n = 7$ for every m , because the factor $m - 1$ multiplies both branches and cancels, whereas the simple crossing point grows with m and already sits near $n = 9$ at $m = 3$.

On the linear side the certifier closes the problem, proving $L_m^{\text{dir}}(n) = 2(n - 1)(m - 1)$ for $n \leq 6$ and every m at once ([Theorem 2.2](#)), and at $m = 2$ the whole problem collapses onto the simple digraph theorem for every n , because two parallel arcs are already two routes ([Theorem A.31](#)). On the quadratic side, a scaling-and-deletion induction proves the bipartite bound for even $n \geq 8$, conditional on the bound one vertex below. The odd case once reduced to a single minimum-degree statement ([Conjecture A.29](#)), which the program confirms in the purely saturated and purely unsaturated regimes but leaves open in the mixed one, where multiplicities strictly between 0 and $m - 1$ defeat the integrality the two pure regimes lean on ([Appendix A](#)).

A sharper route now bypasses that conjecture at $m = 3$, and the way it does so is the cleanest instance in the thesis of the proof handing its remaining work to the machine. An *attachment lemma* ([Lemma A.34](#)) bounds the degree of a vertex glued onto the bipartite extremiser, and a *conditional odd step* built on it ([Theorem A.36](#), for $m \in \{3, 4\}$) carries the value and extremal characterisation from each even level to the next odd one. At $m = 3$ that characterisation propagates through both parities ([Remark A.39](#)), so the entire problem for all $n \geq 8$ collapses onto two finite, integral statements at $n = 7$:

- (a) $L_3^{\text{dir}}(7) = 24$, and
- (b) the only 24-arc feasible multigraphs on 7 vertices with maximum degree at most 8 are $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, and the doubled bidirected path P_7 .

[Conjecture A.29](#) is thereby off the critical path for $m = 3$, replaced by two machine-checkable targets: a MILP infeasibility check for fact (a), and an exhaustive enumeration of degree-bounded multigraphs for fact (b). The enumeration has since closed fact (b): the sound generation enumerator of [Chapter 2](#) completed the sweep and returns exactly the three predicted extremisers ([Remark A.39](#)). The value statement (a), $L_3^{\text{dir}}(7) = 24$, is now the one computation still standing between the $m = 3$ problem and a complete theorem, and there the MILP remains runtime-limited on the bundled solver.

The linear branch turns out to be richer than the double star first suggested. Every doubled bidirected spanning tree is extremal, confirmed exhaustively at $n = 4$ and 5, and the saturated attachment lemma ([Lemma A.37](#)) explains why: the doubled trees are exactly the closure of a single doubled edge under maximal attachment. Several distinct extremisers tie at 24 arcs for

$n = 7$, and sifting and classifying them is precisely what the enumeration must do.

For $m \geq 4$ the conditional step still yields the value but uniqueness no longer propagates through odd levels, and for $m \geq 5$ even the value step stalls. These residual gaps are stated in full among the open problems below. They are finite, identified targets rather than mysteries, and natural goals for the next round of certified computation.

4.2 Two principal phenomena

Across all the variation, two phenomena stand out as the ones a first guess would not have predicted.

The first is the divergence at $m = 5$. The edge and vertex problems agree for $m \leq 4$, and it would be natural to expect them to agree forever. Instead they part company at the very next value, with $k_5(n) = \lfloor 8n/3 \rfloor - 3$ pulling strictly above $\ell_5(n)$, and beyond it the vertex problem becomes genuinely hard and has stayed open since 1974.

The second is the quadratic bipartite phenomenon. In an undirected graph every edge contributes to the degree of both its endpoints, and Mader's theorem turns that symmetry into a hard ceiling: $\lfloor m(n-1)/2 \rfloor$ edges, linear in n . A digraph breaks the symmetry. Place all n vertices into two equal parts A and B and draw every arc from A to B , one direction only. The result has $\lfloor n^2/4 \rfloor$ arcs, quadratic in n , yet $\lambda^{\max} = 1$: every pair has at most one arc-disjoint route, because every route must cross the A -to- B wall exactly once and no arc crosses the other way to offer a second path. A graph whose arc count grows like n^2 can sit at connectivity one, something no undirected graph can do past a handful of vertices.

The initial conjecture for the directed case was linear, matching the undirected intuition. The one-directional bipartite construction refutes it outright, already at $m = 2$: for $n = 8$ the wall holds 16 arcs while the linear hub holds only 14. At $m = 3$ the refutation is the thirty-arc augmented bipartite graph of [Remark 1.12](#), which packs 30 arcs at $\lambda^{\max} = 2$ against the naive prediction of 27. The construction generalises to every m via [Construction 1.13](#), thickening the larger part B slightly so that each vertex of B absorbs $m - 2$ extra in-arcs from inside B : those extra arcs add exactly $m - 2$ short detours from A to each $b \in B$, bringing $\lambda^{\max}$ from 1 to $m - 1$ while the arc count climbs to $\lfloor (n + m - 2)^2/4 \rfloor$.

The quadratic term is what makes the directed cases the deepest in the thesis. The upper bound argument must explain why no configuration beats the bipartite wall, which requires producing the partition in the first place, ruling out backward arcs, and controlling the internal structure of the large part B . That structural task is exactly what [Conjecture A.21](#) has not yet resolved for $m \geq 3$. What [Proposition A.22](#) shows is that the wall is at least the right shape: every feasible digraph is within $O_m(n^{3/2})$ arcs of being one, so the difficulty is the second-order term rather than the phenomenon. The undirected problem has no analogous obstacle, because Mader's degree-counting argument closes it in a few lines. The quadratic phenomenon is therefore not just a curiosity: it is the geometric reason the directed and undirected problems live on

different levels of difficulty.

That same scarcity of extremal graphs is the thread through the remaining figures. [Figure 4.3](#) shows where the frontier runs: it plots every labelled graph at small n as a single point, its binding connectivity $\lambda^{\max}$ on the horizontal axis and its edge count on the vertical, so the whole population of graphs is visible at once. The points crowd into the low-connectivity, low-edge corner and thin out upward, where the blue extremal envelope, an integer staircase carrying one dot per connectivity level, traces the densest graph available at each level. The empty region above that staircase is precisely the part the main theorems forbid a graph to occupy. The extremal problem is the question of where that staircase runs, and the figure shows it is a thin frontier with almost nothing pressed against it from below.

[Figure 4.4](#) steps inside each graph to the level of individual pairs. For every graph in the full enumeration it records the connectivity of every vertex pair and pools all the observations, so the histogram counts not graphs but (graph, pair) observations. Each bar is split and stacked by the connectivity of the graph the pair came from: blue for pairs drawn from the lower-connectivity graphs, red for pairs from the higher-connectivity ones. The boundary between the two is set per variant at the midpoint of the connectivity range that variant's enumeration can reach, since the twelve variants span very different ranges and a single fixed boundary would leave some panels entirely one colour. A pair whose own connectivity exceeds the boundary can only come from a graph above it, so the bars to the right of the dashed line are wholly red, while to the left the two colours mix, since a high-connectivity graph still carries many low-connectivity pairs. The blue mass sits almost entirely at the lowest levels: high-connectivity pairs are rare even inside the graphs that have one.

[Figure 4.5](#) then shows how the same graphs distribute by sheer edge count, with the same per-variant stacked colouring. At each edge count the blue part counts the lower-connectivity graphs and the red part the higher-connectivity ones, so the bar height is the total number of graphs there and the split shows how the two populations separate. The red mass sits to the right, since denser graphs tend to carry higher connectivity, and the dotted line marks the densest graph still on the low-connectivity side, landing far out in the sparse tail. The densest low-connectivity graphs, the ones the extremal problem is about, are exceptional in every variant.

Finally, [Figure 4.6](#) lifts the whole study onto the (n, m) grid at once, drawing the best feasible edge count as a bar surface across every variant. The three proved variants rise as smooth linear or quadratic sheets that the formulas describe exactly, while the open variants show a visible step between the machine-proved blue bars and the search's purple lower bounds, so the surface makes the proved-versus-open divide of the trichotomy a single readable shape. Reading along the m axis recovers the linear scaling in the threshold, and reading along n recovers each variant's growth rate, which is the whole of [Table 4.1](#) compressed into one object.

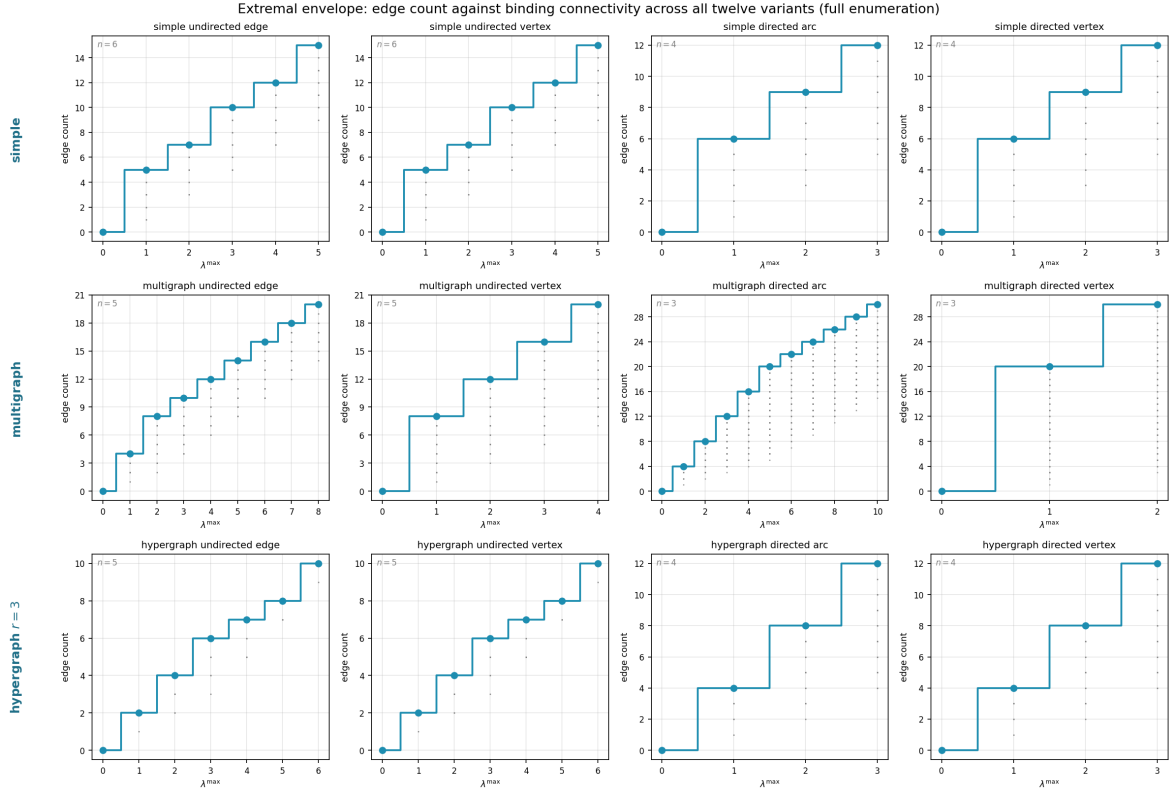


Figure 4.3. The complete landscape of all labelled graphs at small n (see panel annotations for each variant's n): every point is one graph, plotted at its binding connectivity $\lambda^{\max}$ horizontally and its edge count vertically. The blue staircase marking the densest graph at each connectivity level is the extremal envelope, the frontier the extremal problem asks about, with one dot per level so its exact value is legible. Graphs crowd the low-connectivity, low-edge region. The extremal graphs live at the top of the staircase. The emptiness above the envelope is exactly the region the main theorems forbid. The horizontal reach differs by panel because the connectivity that fits in a variant's enumeration n differs: arc-disjoint multigraph routes grow with edge multiplicity (the multigraph directed arc panel reaches $\lambda^{\max} = 10$ at $n = 3$), whereas vertex-disjoint routes are capped by the available internal vertices, so the multigraph directed *vertex* panel reaches only $\kappa^{\max} = 2$ at the same $n = 3$ (with a single internal vertex, a pair has at most the direct route plus one detour). This is a property of the separation, not of the search.

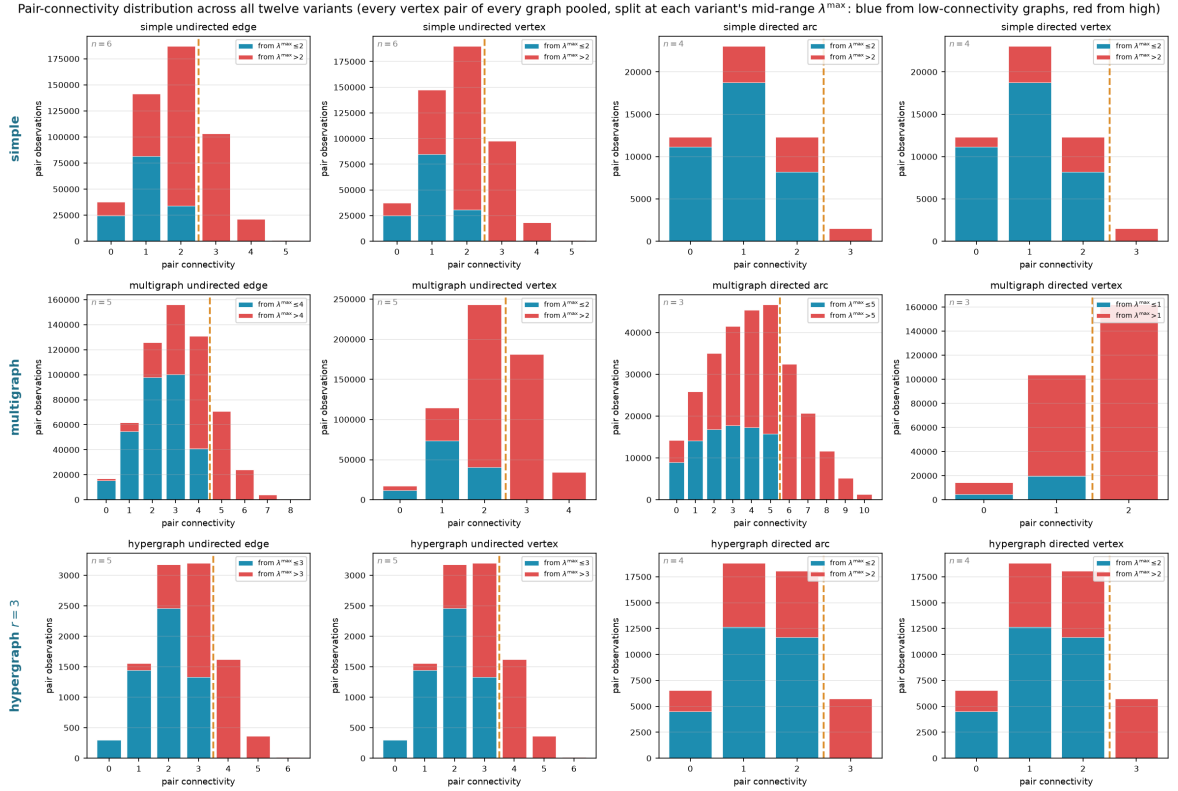


Figure 4.4. Pair-connectivity distribution across all twelve variants, pooling every vertex pair of every graph in the full enumeration. The horizontal axis is the connectivity of a single pair, and a bar's height is the number of (graph, pair) observations at that level. Each bar is split and stacked by the connectivity of the graph the pair was drawn from, blue from graphs at or below a per-panel boundary T and red from graphs above it, with the dashed line at T . The boundary is set per variant to the midpoint of the connectivity range that variant's enumeration reaches (each panel's legend gives its T), so the split stays informative in every panel instead of collapsing to one colour. Bars to the right of the dashed line are wholly red, since a pair of connectivity above T forces its graph above T . To the left the colours mix.

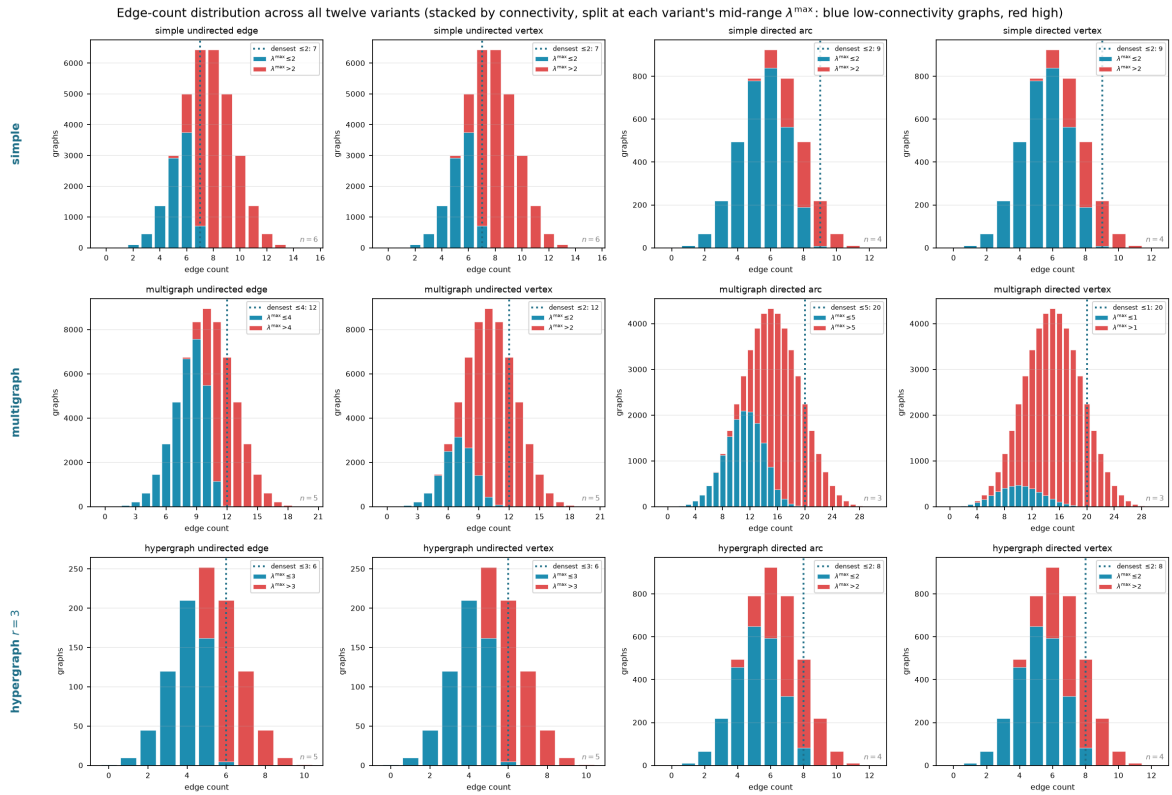


Figure 4.5. Edge-count distribution across all twelve variants, the same enumeration viewed by edge count rather than connectivity. Each bar is split and stacked by graph connectivity, blue for graphs at or below the per-panel boundary T and red above it (each panel's legend gives its T), so the full height is the total number of graphs at that edge count. The dotted vertical line marks the densest graph on the low-connectivity side. The reader can see directly how the two populations separate by edge count and how far low-connectivity graphs sit from the bulk.

Erdős 915: optimal bound over (n, m) , blue where proved, purple where only a search lower bound is known

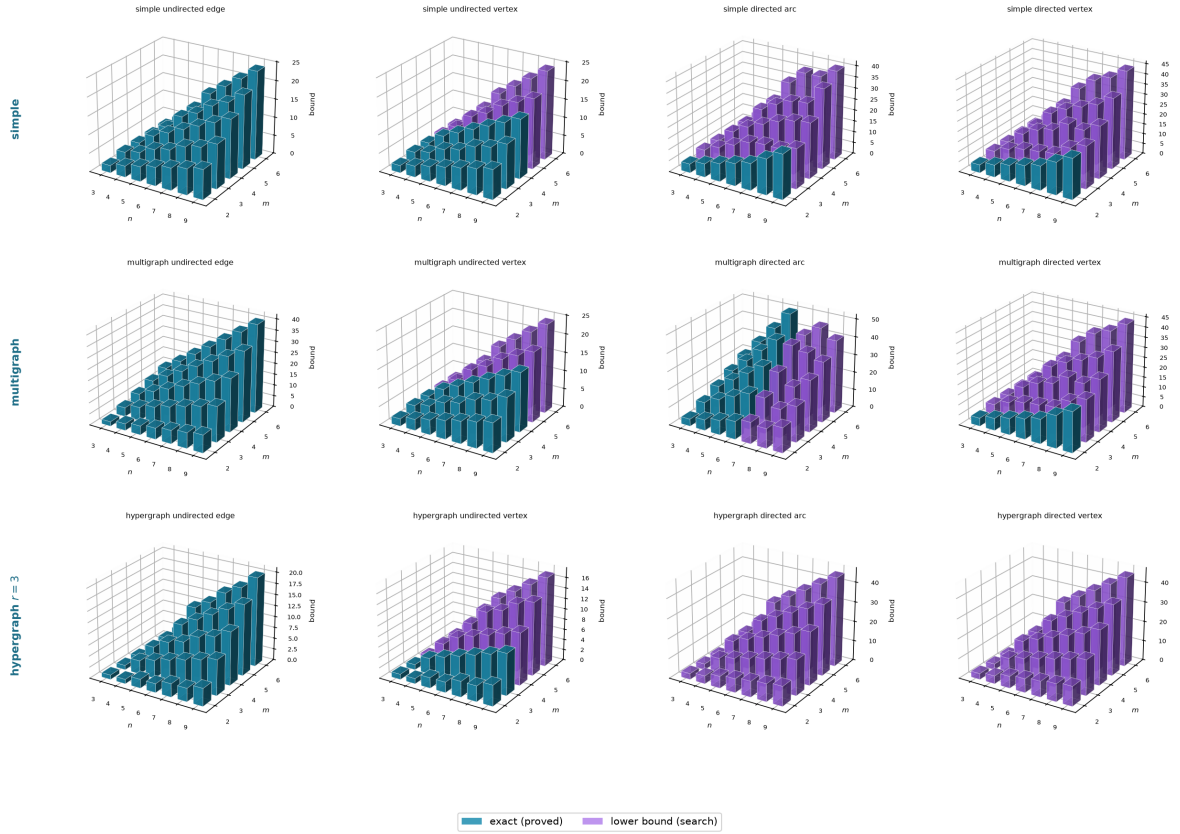


Figure 4.6. The optimal bound as a surface over the (n, m) grid for all twelve variants, every panel drawn from the same camera so the shapes compare directly. Each bar's height is the largest feasible edge count found by `solve` at that vertex count and forbidden threshold m . The single thing to read off is colour: a bar is *blue* exactly where the value is machine-proved and *purple* exactly where only a search lower bound is known, so each panel is a little map of what is settled. The three fully proved variants (Mader's undirected edge cases and the multigraph analogue) are solid blue sheets, rising linearly or quadratically as their formulas predict, while the open variants turn purple precisely where proof runs out. Reading along the m axis shows how the bound scales with the forbidden threshold, and reading along the n axis shows the growth rate for a fixed variant.

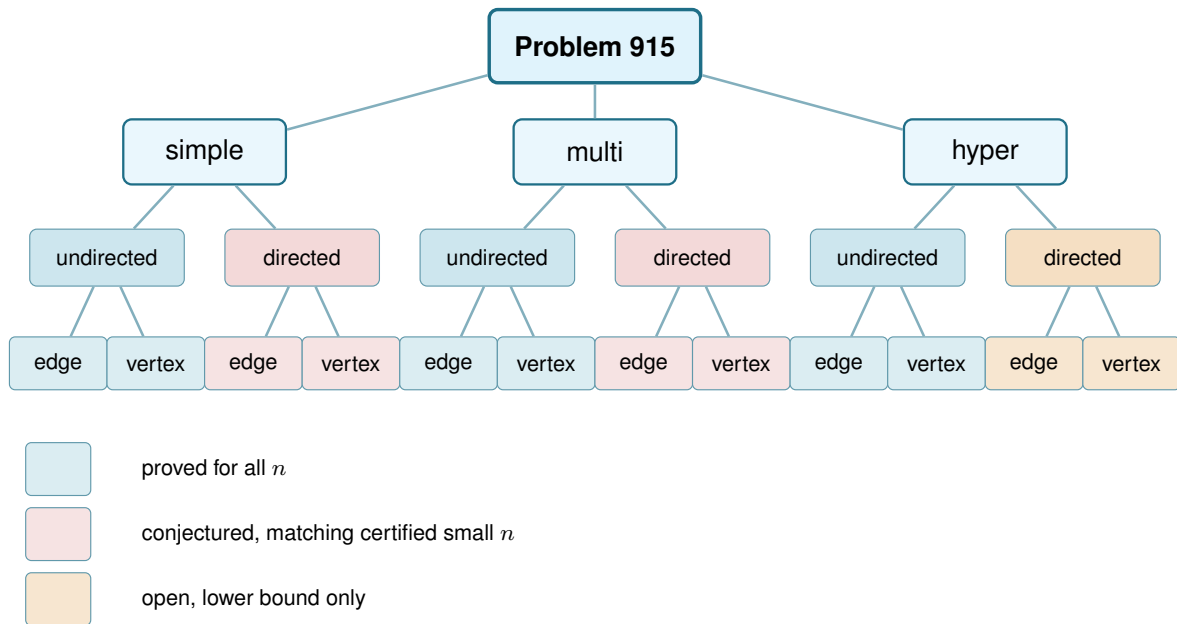


Figure 4.7. The same branching as Figure 1.7, now with every leaf resolved by the trichotomy of Section 3.6: blue is proved for all n , red is a conjectured formula backed by certified small cases and a matching search, and amber is open with only a lower bound known. Colour marks the status at general m , the harder regime this thesis mostly studies, and hides m -dependent exceptions and one convention that Table 4.1 states precisely. Every red leaf is a proved theorem at $m = 2$ and becomes conjectured only at $m \geq 3$, the directed edge leaves by Theorem 1.10 and the directed vertex leaves by the separate Theorem 1.11, which does not follow from it. Three blue leaves are proved only up to a finite m : the undirected vertex problem for simple graphs and for multigraphs ($m \leq 4$) and for hypergraphs ($m \leq 3$), open beyond. The two multigraph vertex leaves inherit whatever their simple counterparts have, because the counting convention of Remark 2.1 makes those two questions the same question.

4.3 Summary of the twelve variants

Table 4.1 records the status of all twelve structural variants. Each entry is marked by how it is known: proved by hand here or in Appendix A, cited to the literature, certified by the cut-counting optimisation for the stated sizes, or conjectured with a proved lower bound. The distinction is the honesty contract of Chapter 2 carried through to the summary. Figure 4.7 shows the same distinction as a picture before the table spells out every case.

Table 4.2 answers a different question about the same twelve variants: not what is known, but which routine of the program knows it. Most variants share machinery rather than each needing its own, so the table is grouped by family, and every name in it has its own codecard box where it is built in Chapters 2 and 3.

4.3.1 Orientation models for the directed hypergraph

The directed rows of Table 4.1 read a directed Berge edge in one way, the *forward* one of a single tail fanning to $r - 1$ heads. It is not the only reading. In general a directed r -uniform hyperedge is a split of its vertices into a non-empty tail set T and head set H , a route entering at a tail and

Separation	Model	Value	Status
<i>Undirected</i>			
edge	simple	$\lfloor m(n-1)/2 \rfloor$	proved (Mader)
edge	multi	$(m-1)(n-1)$	proved
edge	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$	proved (cited, Gomory-Hu)
vertex	simple	$\lfloor m(n-1)/2 \rfloor$ ($m \leq 4$), $\lfloor 8n/3 \rfloor - 3$ ($m = 5$, $n \geq 13$, see Remark 1.7)	cited, open $m \geq 6$
vertex	multi	adjacency count, so the simple value $\lfloor m(n-1)/2 \rfloor$ ($m \leq 4$)	convention (Remark 2.1)
vertex	hyper	$\lfloor \frac{(m-1)(n-1)}{r-1} \rfloor$ ($m \leq 3$), $\geq$ that for all m	proved $m \leq 3$ (Theorems A.45 and A.48), open $m \geq 4$
<i>Directed</i>			
arc	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m = 2$)	proved, conj. $m \geq 3$
arc	multi	$(m-1) \max(2(n-1), \lfloor n^2/4 \rfloor)$	proved $n \leq 6$ and $m=2$, cond. $m=3$
arc	hyper	quadratic in n , exact value open	lower bd. (Proposition A.50)
vertex	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m = 2$)	proved separately (Theorem 1.11), conj. $m \geq 3$
vertex	multi	adjacency count, so the simple digraph value	convention (Remark 2.1)
vertex	hyper	quadratic in n , exact value open	lower bd. (Proposition A.50)

Table 4.1. The twelve structural variants of Problem 915 and their status. Every simple-graph closed form here is stated for $n \geq m$, the hypothesis [Theorem 1.2](#), [Theorem 1.5](#) and [Conjecture 1.14](#) carry: below it the complete graph or digraph is itself feasible and the answer is the trivial maximum, which the formulas exceed. “Certified” means proved by the cut-counting optimisation for the stated sizes. “conj.” means a conjecture with a proved matching lower bound. “cond.” means proved conditional on a finite, identified statement (for $m = 3$ directed multigraphs, the one remaining $n = 7$ computation, fact (a) of [Remark A.39](#)). “lower bd.” means only a proved lower bound is known, with the exact value open. “Convention” marks the two rows whose value is fixed by the counting convention of [Remark 2.1](#) rather than by a theorem. The appearance threshold $p^* = m/n$ of [Theorem 1.16](#) is deliberately not a column here: it is proved for $G(n, p)$, extends to the vertex and directed analogues by [Remark A.59](#), and does *not* transfer to the hypergraph models, whose natural density scale is different ([Remark A.60](#)).

leaving at a head, and the gate checker of [Chapter 2](#) measures any such split without change. Three models are worth naming, *forward* ($|T| = 1$), *backward* ($|H| = 1$), and *general* (any split, with genuinely mixed ones appearing once $r \geq 4$). The natural question, and one a reader of the directed figures tends to ask, is whether they multiply the table.

They do not, and two results say why. First, backward is not a new problem. Reversing every hyperarc turns a backward hypergraph into a forward one and reverses every route along the way, so the two models share every extremal number, edge and vertex, for all r , m and n ([Lemma A.51](#)). The backward column is the forward column read upside down. Second, the general model, though it admits more hyperedges on a single vertex set, obeys the same first-order bound. A hyperedge (T, H) occupies $|T||H| \geq r-1$ ordered tail-head pairs, each usable by at most $m-1$ disjoint one-step routes, so $(r-1)|E| \leq (m-1)n(n-1)$, exactly the forward cap of [Proposition A.50](#) ([Proposition A.52](#)). The forward construction is itself a set of general hyperedges, so general is squeezed between the same lower and upper bounds as forward, and its value too is quadratic in n . Those two bounds differ by a factor of four, so what this fixes is the order and not the leading constant, and it does not by itself force the two models to agree asymptotically.

Where the models part company is only at small sizes, and the program shows exactly where. [Table 4.3](#) lists the extremal hyperedge counts the branch-and-bound checker ([Table 4.2](#)) proves for the forward and general models. The method is the include-or-exclude search of [Chapter 2](#) ([Figure 2.9](#)) walked over the candidate hyperedges instead of the arc cells: a hyperedge is included only while the checker confirms the growing hypergraph stays feasible, a branch is

Variant family	Measure	Prove	Discover
Undirected, edge (all three models)	max_edge_connectivity max_hyperedge_connectivity	exhaustive enumeration (solve), Gomory–Hu cross-check	search_for_dense_graph
Undirected, vertex (all three models)	max_vertex_connectivity max_hyper_vertex_connectivity	exhaustive enumeration (solve)	search_for_dense_graph
Directed arc, simple	max_edge_connectivity	_exhaustive_directed: a pruned branch-and-bound	tabu_search_for_dense_graph
Directed arc, multigraph	max_edge_connectivity	the cut-counting optimisation (prove_directed_multigraph, prove_integral_arc_bound); the $n = 7$ classification (enumerate_extremal_directed_multigraphs_via_generation)	tabu_search_for_dense_graph
Directed vertex (simple and multi)	max_vertex_connectivity	reduces to the arc problem above	tabu_search_for_dense_graph
Directed hyper (arc and vertex)	hyper_connectivity	max_feasible_hyperedges: a pruned branch-and-bound	search_for_dense_graph

Table 4.2. The machinery of [Chapters 2](#) and [3](#) indexed by variant family rather than by function name. This is the cross-reference from a variant to the routine that handles it; the routine’s own signature, inputs, and outputs are in its codecard box.

dropped once even including everything that remains cannot beat the best count already found, and a sweep that finishes has considered every candidate set, so the value it reports is proved exact rather than searched. The general one is strictly denser precisely when vertices are scarce, $n \approx r$, where the richer set of orientations lets a single vertex-set carry more hyperedges. At $n = r = 4$ it reaches eight hyperarcs against forward’s four, and the surplus grows with the connectivity budget. Once n exceeds r the computed gap closes, proved equal already at $r = 3$, $n = 4$, and larger searches find no separation thereafter. The reading that suggests itself is that once the hypergraph spreads past a single edge-set, the one-tail model already saturates the Menger budget and extra orientations buy nothing. That reading is computational evidence and a conjecture, not a theorem: what is proved is that both models obey the same upper bound and admit the same quadratic construction ([Proposition A.52](#)), and those two bounds differ by a factor of four, which is too coarse to force the values to agree. On the strength of the evidence the orientation axis is treated as collapsing, which is why Problem 915 stays a twelve-variant question, and the general model enters as a refinement at the margin rather than a thirteenth column. Proving that it collapses, or exhibiting the first $n > r$ where it does not, is left open.

4.4 Contributions and limitations of the program

The program turned a per-case craft into a uniform pipeline. Before it, each variant came with its own one-off computation and its own hand argument. After it, one representation holds every variant, one exact checker measures connectivity, one certifier proves small-case upper bounds, and one cooled search discovers extremisers and the lower bounds they witness. The rediscovery of [Chapter 3](#) is the evidence that the pipeline is faithful: pointed at solved cases, it returns the solved answers.

r	m	n	forward	general
3	3	3	3	4
3	3	4	8	8
4	3	4	4	6
4	4	4	4	8

Table 4.3. Extremal hyperedge counts the branch-and-bound checker proves exact for the forward and general directed r -uniform models, at the small sizes where they can differ. The general model is strictly denser only when $n \approx r$, doubling the forward count at $n = r = 4$, $m = 4$, and the gap closes once n exceeds r (proved equal at $r = 3$, $n = 4$). Edge- and vertex-disjoint counts agree throughout, and backward equals forward by [Lemma A.51](#).

Its limits are as clear as its reach, and stating them is part of the honesty the thesis insists on. The search proves no upper bounds. It only ever exhibits graphs. The certifier proves upper bounds, but only for a fixed size, and the directed multigraph encoding closed unconditionally only up to $n = 6$. Of the two $m = 3$ computations at $n = 7$, the generation enumerator has completed the classification, fact (b), while the value statement, fact (a), remains runtime-limited on the bundled solver and is the one certificate still outstanding. And no amount of either replaces the genuinely structural gaps the thesis leaves open: the decomposition of [Conjecture A.21](#) that would turn [Conjecture 1.14](#) into a theorem, of which the backward-arc clause is the hardest part but not the whole, the directed vertex problem at $m \geq 3$, which Whitney’s inequality pointedly does not hand over with the arc case, and the undirected vertex-disjoint problem at $m \geq 6$, which has resisted for half a century and which the program can probe but not crack.

4.5 Open problems

- ★ The decomposition of [Conjecture A.21](#): prove that an extremal non-hub digraph admits a partition $A \cup B$ with every arc from A to B present, no arc inside A , and no arc from B back to A . That last clause is the backward-arc condition, and it is the one where the natural delete-and-compensate exchange fails to be monotone, but the other three are hypotheses too, and no argument here forces an arbitrary extremiser into that shape. The decomposition completes the upper bound in [Conjecture 1.14](#) for $m \geq 3$. Unconditionally, [Proposition A.22](#) already gives $\ell_m^{\text{dir}}(n) = n^2/4 + O_m(n^{3/2})$, so what the decomposition is needed for is the second-order term, and sharpening the error there from $n^{3/2}$ to $O_m(n)$ by a stability argument is a smaller and possibly easier target than the decomposition itself.
- ★ The directed *vertex* problem at $m \geq 3$, which is a separate question and not a corollary of the arc one. Whitney’s inequality makes the vertex-feasible family the larger of the two, so an arc upper bound does not restrict it, and [Conjecture A.21](#) would not settle it either. At $m = 2$ the two values agree ([Theorem 1.11](#)), proved by re-running the induction and its base cases under the stricter separation rather than by transfer. Whether they agree for $m \geq 3$ is open. The exhaustive search run under both tests at $m = 3$ returns the same value at every n it reaches, tabulated in [Remark A.17](#), so there is no small counterexample, but that is evidence

and not an argument. The useful next step is structural rather than numerical: find the first digraph in which some pair carries strictly more arc-disjoint routes than internally disjoint ones *and* that slack buys extra arcs, or show that it never can.

- ★ The multigraph vertex problem under the other counting convention, worked out in [Section A.8](#). Reading parallel copies as distinct routes makes this a genuinely new question, and the section settles $m = 2$ and exhibits several competing constructions. The natural guess from the $m \leq 4$ data, that a thickened tree becomes optimal once $n \geq m + 2$, is false for every $m \geq 5$: [Theorem A.56](#) shows that a bouquet of thickened cliques sharing one vertex beats the tree by an amount that grows linearly in n , and the true growth rate in m is $\Theta(m^2)$ rather than the $\Theta(m)$ the tree would give. That construction is a lower bound of the right order and not the answer: at $m = 5$, $n = 7$ it gives 27 where an unfinished exhaustive search has already found 28, so the exact value of $K_m^{\text{multi}}(n)$ is open.
- ★ The last finite $n = 7$ computation that closes the directed multigraph problem at $m = 3$. By the attachment lemma and the conditional odd step ([Lemma A.34](#), [Corollary A.35](#), [Theorem A.36](#)), the entire $m = 3$ problem (value *and* extremal uniqueness for all $n \geq 8$) reduces along a single induction to two integral statements about multigraphs with multiplicities in $\{0, 1, 2\}$ ([Remark A.39](#)): (a) $L_3^{\text{dir}}(7) = 24$, and (b) every 24-arc feasible multigraph on 7 vertices with maximum degree at most 8 is $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, or the doubled bidirected path P_7 . Fact (b) is settled: the sound generation enumeration completed and returns exactly those three multigraphs, each re-verified by the exact checker ([Remark A.39](#)). What remains is fact (a) alone, where the MILP infeasibility check still reaches its time limit on the bundled solver, so the gap is a single finite computation rather than a structural one. [Conjecture A.29](#) is thereby bypassed for $m = 3$, though it remains open and of independent interest as the natural minimum-degree statement for the directed multigraph problem.
- ★ Extremal uniqueness at odd levels for directed multigraphs with $m \geq 4$. The conditional odd step ([Theorem A.36](#)) and the propagation of [Remark A.39](#) are stated for $m \in \{3, 4\}$. For $m = 4$ the *value* step goes through, but at an odd extremal level the counting no longer forces a vertex of degree $(m - 1)k$ (all degrees $\geq (m - 1)k + 1$ are arithmetically consistent), so the uniqueness statement does not yet propagate. For $m \geq 5$ the same escape climbs one level and even the *value* step stalls: a surplus extremiser could be degree-regular (every vertex at one and the same degree) strictly above the deletion bound, so no vertex deletion is forced to land on an extremal multigraph. Closing either hole asks for an arithmetic or structural input the present averaging argument does not supply.
- ★ The undirected vertex-disjoint problem for $m \geq 6$, open since 1974, where even the extremal structure is unknown.
- ★ The hypergraph vertex problem at $m \geq 4$. The problem is now settled at $m = 2$ and $m = 3$ for every uniformity and all n , with $k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor = \ell_m^{(r)}(n)$ ([Theorems A.45](#) and [A.48](#)). The incidence reduction stands for every m , but the rank bound that drives it ([Lemma A.47](#))

leans on the triconnected (Tutte/SPQR) decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$ it would need a 4-connectivity analogue, where no comparably clean decomposition is available. Exhaustive computation finds no counterexample at $m = 4$ at any of eleven sizes, confirming both that $\lfloor 3(n-1)/(r-1) \rfloor$ is attained and that one more hyperedge is infeasible, so the agreement $k_m^{(r)} = \ell_m^{(r)}$ survives well past the theorem. That it must break eventually is not a guess: at $r = 2$ this problem is exactly the multigraph vertex problem of [Section A.8](#), where [Theorem A.56](#) breaks the formula for every $m \geq 5$. So the question is not whether the pattern fails but whether $r \geq 3$ postpones the failure past $m = 4$, which leaves the $m = 4$ row sitting on the boundary, and where it first breaks for $r \geq 3$ is unknown.

- ★ A machine-checkable certificate for fact (a) of [Remark A.39](#). It is the one computed claim in this thesis that would rest on a solver's infeasibility verdict rather than on a finished exhaustion ([Section 2.9.1](#)), so it is the one place where a replayable proof object, of the kind a pseudo-Boolean or SAT solver emits, would raise the standard rather than merely repeat the work. Encoding the question as a pseudo-Boolean constraint problem over the 42 multiplicities in $\{0, 1, 2\}$, with the cut conditions written out per ordered pair, is the natural route.
- ★ The directed hypergraph variants. The tail-and-heads model has first bounds and a quadratic lower-bound family ([Proposition A.50](#)): counting how many hyperedges may serve the same ordered pair of vertices caps the total at $(m-1)n(n-1)/(r-1)$, and a bipartite construction reaches $\approx (m-1)n^2/(4(r-1))$, so the value is quadratic in n . The two bounds differ by a factor of four, so the order is settled and the leading constant $\lim_{n \rightarrow \infty} \text{ex}(n)/n^2$ is not, for any of the orientation models. That constant is the right first target, ahead of exact small values, and [Conjecture A.53](#) says which way it should go: the factor of four is exactly the gap between the n^2 ordered pairs the upper bound allows and the $n^2/4$ the bipartite construction uses, the same gap that appears in the simple directed problem, where [Theorem A.24](#) now proves the bipartite pattern right to first order. So the upper bound is the side that should improve. The appendix records where the proof of that theorem fails to transfer, which is that a hyperedge carries $r-1$ heads at once, so two-step routes through different midpoints need not be edge-disjoint. The orientation axis does not visibly widen the gap: the backward model coincides with forward by reversal ([Lemma A.51](#)) and the general model obeys the same cap ([Proposition A.52](#)), beating forward only in the scarce-vertex regime $n \approx r$ ([Section 4.3.1](#)). That the three models share a leading constant is a conjecture supported by small-case computation, not a consequence of sharing two bounds four apart.

The map is filled in where honest mathematics could fill it, and where it could not, the blanks are marked precisely, with a machine standing ready at each of them. That, finally, is the use of building the microscope: not that it answers every question, but that it shows exactly which questions are left, and hands the next reader a clean place to start.

Appendix A

Full Proofs and Computational Transcripts

This appendix holds the arguments deferred from the body: the ones that are either standard machinery or long case-checking, and whose length, not whose difficulty, kept them out of the narrative. The body carries the ideas. Here they are carried out in full. Because graphs are visual objects, the longer arguments here are accompanied by figures: each is drawn rather than merely described, and the prose points the reader to the picture at the moment it is needed.

How to read this appendix. The sections divide into standard machinery and the thesis's own proofs, and a reader chasing one thread of the body need not read all of them. The first two sections, on [Theorem A.1](#), [Theorem A.2](#), and Mader's theorem, are classical results included only so the body can cite them in the exact form it uses, and a reader who already trusts Menger, Gomory–Hu, and Mader may skip them. Three sections carry the load-bearing original arguments. The directed arc value at $m = 2$ supplies the finite base cases the body's induction rests on. The directed multigraph problem at large n is the longest argument here and the one behind the main directed contribution. The hypergraph vertex problem proves the incidence-rank lemma that settles the $m = 3$ hypergraph case. The remaining sections, on the hypergraph edge bounds and the random threshold, each support a single claim of the body and can be read on their own when that claim is reached. Finally, everything the program asserts is reproduced verbatim in the computational transcripts and the self-test section, so the machine proofs can be audited without running anything.

A.1 Menger, Gomory–Hu, and the degree bound

The two classical theorems below are the foundation of everything the thesis measures, and we state them at the level of generality at which the program actually applies them. A *capacitated graph*, or *weighted graph*, is a graph, directed or not, carrying a nonnegative capacity $c(e)$ on each edge or arc. A plain graph is read as the *unit-capacity* case $c \equiv 1$, and a multigraph as the integer-capacity case, one unit per parallel copy. A *flow* from u to v sends units along edges without exceeding any capacity, and the *capacity of a cut* is the total capacity of the edges crossing it. Both theorems below are theorems about capacitated graphs, and the thesis uses them mostly at unit and integer capacities, where the maximum flow is exactly the connectivity λ .

Theorem A.1 (Menger, as max-flow min-cut [2]). *In any capacitated digraph, and for distinct*

vertices u, v , the maximum flow from u to v equals the minimum capacity of a cut separating u from v . At unit capacities this is the combinatorial statement we cite throughout: the maximum number of pairwise edge-disjoint u – v paths equals the minimum number of edges whose removal separates them,

$$\lambda_G(u, v) = \min\{|C| : C \subseteq E(G), G - C \text{ has no } u\text{--}v \text{ path}\}.$$

A directed capacitated graph is the most general object here, and every graph model the thesis uses is a special case of it. An undirected graph is the symmetric case, each edge a pair of opposite unit-capacity arcs. A multigraph is the integer-capacity case, with $\mu(u, v)$ units on the pair uv . The hypergraph version ([Theorem A.40](#)) is this very theorem applied to the helper-point network of [Figure 2.5](#), a capacitated digraph. The internally vertex-disjoint version is obtained not by changing the theorem but by changing the graph it runs on, a device called vertex splitting: replace each internal vertex w by two copies joined by a single unit-capacity arc $w^- \rightarrow w^+$, send every arc that entered w into w^- and every arc that left w out of w^+ . A flow may now pass through w at most once, so edge-disjoint paths in the split graph are exactly internally vertex-disjoint paths in the original, and a minimum edge cut there is a minimum vertex cut here. This split graph is literally what the checker of [Chapter 2](#) builds, which is why a single maximum-flow routine measures all four separations.

Theorem A.2 (Gomory–Hu [[15](#)]). Every undirected capacitated graph G has a weighted tree T on the same vertex set, its Gomory–Hu tree, such that for all u, v the value $\lambda_G(u, v)$ equals the minimum edge weight on the u – v path in T , and each tree edge e corresponds to a minimum cut of G whose capacity is the weight of e . All $\binom{n}{2}$ pairwise values are thereby stored in the $n - 1$ weights of a single tree.

Two features of the statement fix how far it reaches, and both are used below. Being a theorem about capacities, it applies verbatim to a simple graph (unit weights), to a multigraph (integer weights, the multiplicity μ as capacity), and, in the form of [Theorem A.41](#), to the hyperedge-connectivity of a hypergraph. But it needs the cut function to be *symmetric*, which forces G to be undirected: a digraph has $\lambda(u, v) \neq \lambda(v, u)$ in general and admits no such tree, and the vertex-connectivity function fails symmetry for the same reason. So the Gomory–Hu tree is precisely the tool for the undirected edge problems, where it drives Mader’s theorem and its multigraph and hypergraph cousins ([Theorems 1.2](#) and [1.3](#) and [proposition 1.15](#)) through one argument, and precisely the tool that vanishes once arcs turn one-way.

Primer: why a Gomory–Hu tree exists

The theorem is far from obvious, so here is why such a tree can always be built, in enough detail that nothing is taken on faith. Everything rests on one property of cuts. For a set of vertices S write $d(S)$ for the total capacity of the edges with exactly one endpoint in S , which is the capacity of the cut $(S, V \setminus S)$. The cut function is *submodular*: for any two vertex sets

A and B ,

$$d(A) + d(B) \geq d(A \cap B) + d(A \cup B).$$

One checks this one edge at a time. An edge contributes to each side of the inequality according to where its two endpoints fall relative to A and B , and a short case check over the possible placements shows its contribution on the right never exceeds its contribution on the left. The only edges that could contribute more on the right have one endpoint in $A \setminus B$ and the other in $B \setminus A$, and those contribute 0 on the right and 2 on the left, so the inequality only gains.

Submodularity buys the single lemma the construction needs, the *uncrossing* lemma: if $(S, V \setminus S)$ is any cut and u, v lie on the same side of it, then some minimum $u-v$ cut keeps all of S (or all of $V \setminus S$) on one side, that is, it does not cross S . The picture is that a cheapest $u-v$ cut which sliced through S could be pushed clear of S without ever gaining capacity, and submodularity is exactly the inequality guaranteeing the push is free.

With that lemma the tree is built in $n - 1$ rounds, one minimum-cut computation each. Begin with the trivial tree, a single node holding all n vertices. In each round pick a node of the current tree that still holds two or more vertices, choose two of its vertices u and v , and compute a minimum $u-v$ cut in the graph in which every neighbouring tree node, itself already a bag of several vertices, is shrunk to one vertex placed on whichever side the cut assigns it. Split the chosen node along the two sides of the cut and join the two halves by a new tree edge weighted by the cut's capacity. The uncrossing lemma is what makes the shrinking legitimate, because it says the minimum $u-v$ cut can be chosen to respect every cut found in earlier rounds, so no earlier tree edge is spoiled. After $n - 1$ rounds every node holds a single vertex and T is a weighted tree on V with $n - 1$ edges.

Two facts finish the argument, and they are exactly what the body uses. Each tree edge is by construction the capacity of an actual cut of G , so its weight is a genuine minimum $u-v$ cut value $\lambda_G(u, v)$ for the pair it separates. And for any u, v the value $\lambda_G(u, v)$ equals the lightest edge on the $u-v$ path of T . One inequality holds because every tree edge on that path is itself a cut separating u from v , so none can be lighter than the global minimum $u-v$ cut. The other holds because the lightest tree edge on the path is a genuine $u-v$ cut of exactly that weight. The reader who wants the uncrossing induction spelled out in full will find it in Gomory and Hu [15]. The point here is only that the tree is an honest theorem, computed by $n - 1$ maximum-flow calls, and not a convenient fiction.

The most basic constraint on local connectivity is local: a pair cannot have more independent routes than either endpoint has edges. The constructions below use it constantly.

Lemma A.3 (Degree bound). *For every pair of vertices of a graph G , $\lambda_G(u, v) \leq \min(\deg u, \deg v)$. In a digraph D , $\lambda_D(u, v) \leq \min(d^+(u), d^-(v))$.*

Proof. Every edge-disjoint $u-v$ path uses a distinct edge at u , so there are at most $\deg u$ of them, and likewise at most $\deg v$. In the directed case each arc-disjoint path leaves u by a distinct

out-arc and enters v by a distinct in-arc. □

A.2 Mader's theorem in full

The upper bound of [Theorem 1.2](#) rests on three short lemmas about a Gomory–Hu tree T of G . For an edge $e = \{u, v\}$ of G , write $\text{dist}_T(e)$ for the number of tree edges on the unique $u-v$ path in T . Since T is a tree on the very same vertex set, $\text{dist}_T(e) \geq 1$ for every edge. [Figure A.1](#) shows a small graph beside one of its Gomory–Hu trees and marks which edges have $\text{dist}_T(e) = 1$ and which have $\text{dist}_T(e) \geq 2$. It is worth keeping in view, since the three lemmas below are exactly statements about that picture. The first lemma is the one place the proof uses that G is *simple*, and it is exactly where the simple-graph bound improves on the multigraph bound.

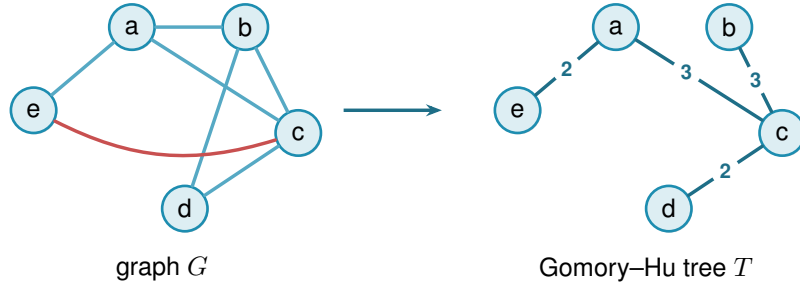


Figure A.1. A simple graph G (left) and a Gomory–Hu tree T of it (right). Both live on the same five vertices. The tree T has edges ea (2), ac (3), cb (3), cd (2), where each weight equals the local edge-connectivity of its endpoints in G . These four edges are also edges of G , so each has $\text{dist}_T(e) = 1$: these are the E_1 edges of [Lemma A.4](#), and a *simple* graph admits at most $n - 1 = 4$ of them, one per tree edge. The remaining edges ab , bd , and the red chord ec all join tree-distance-2 pairs: the path in T from e to c is $e-a-c$, so $\text{dist}_T(ec) = 2$. Reading $\text{dist}_T(e)$ as the number of tree cuts an edge crosses makes $\sum_e \text{dist}_T(e) = \sum_t c(t)$ ([Lemma A.6](#)). Every edge crosses at least one cut, but only the $n - 1$ tree edges cross exactly one, which is the surplus that drives [Lemma A.5](#) and ultimately the factor of two in Mader's bound.

Lemma A.4 (Distance-one edges). *At most $n - 1$ edges of a simple graph G satisfy $\text{dist}_T(e) = 1$.*

Proof. An edge $e \in E(G)$ has $\text{dist}_T(e) = 1$ if and only if its endpoints are adjacent in T . The tree T has exactly $n - 1$ edges, and since G is simple each pair of adjacent tree vertices supports at most one edge of G . □

Lemma A.5 (Sum of tree distances).
$$\sum_{e \in E(G)} \text{dist}_T(e) \geq 2|E(G)| - (n - 1).$$

Proof. Partition $E(G)$ into $E_1 = \{e : \text{dist}_T(e) = 1\}$ and $E_{\geq 2} = \{e : \text{dist}_T(e) \geq 2\}$. Then

$$\sum_{e \in E(G)} \text{dist}_T(e) \geq |E_1| + 2|E_{\geq 2}| = 2|E(G)| - |E_1| \geq 2|E(G)| - (n - 1),$$

using [Lemma A.4](#) in the last step. □

Lemma A.6 (Double-counting identity). *Let G carry unit capacities, so that each weight $c(t)$ of a Gomory–Hu tree T is the plain number of edges of G crossing the cut that t induces. Then*

$$\sum_{t \in E(T)} c(t) = \sum_{e \in E(G)} \text{dist}_T(e).$$

This is the form used for Mader’s theorem, where G is a simple graph and so unit-capacitated by default.

Proof. By [Theorem A.2](#) each tree edge t induces a cut of G of capacity $c(t)$, so $c(t) = \sum_{e \in E(G)} \mathbf{1}_{e \text{ crosses } t}$. Exchanging the order of summation,

$$\sum_{t \in E(T)} c(t) = \sum_{e \in E(G)} \sum_{t \in E(T)} \mathbf{1}_{e \text{ crosses } t} = \sum_{e \in E(G)} \text{dist}_T(e),$$

where the last equality holds because an edge $\{u, v\}$ crosses exactly those tree cuts induced by the tree edges on the unique u – v path in T . $\square$

Proof of [Theorem 1.2](#), upper bound. Let G be a simple graph on n vertices with $\lambda_G^{\max} \leq m - 1$ and let T be a Gomory–Hu tree of G . Each tree edge $t = \{x, y\}$ has weight $c(t) = \lambda_G(x, y) \leq m - 1$, so summing over the $n - 1$ tree edges, $\sum_t c(t) \leq (m - 1)(n - 1)$. Combining with [Lemmas A.5](#) and [A.6](#),

$$2|E(G)| - (n - 1) \leq \sum_{e \in E(G)} \text{dist}_T(e) = \sum_{t \in E(T)} c(t) \leq (m - 1)(n - 1).$$

Rearranging gives $2|E(G)| \leq m(n - 1)$, and since $|E(G)|$ is an integer, $|E(G)| \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. $\square$

Note where the factor of two came from: every edge crosses at least one tree cut, but a simple graph can afford at most $n - 1$ edges that cross only one, so on average each edge is charged closer to twice. For multigraphs that refinement is unavailable, since parallel edges between tree-adjacent pairs each have $\text{dist}_T = 1$, and the same chain of inequalities stops at $|E| \leq \sum_e \text{dist}_T(e) \leq (m - 1)(n - 1)$, which is exactly the multigraph bound of [Theorem 1.3](#).

The lower bound is a pair of constructions, drawn side by side in [Figure A.2](#). The first is the shape the bound was guessed from, and works when $m - 1$ divides $n - 1$. The second generalises it to every $n \geq m \geq 2$.

Construction A.7 (Shared-clique graph). For $m \geq 3$ with $(m - 1) \mid (n - 1)$, take $k = (n - 1)/(m - 1)$ copies of the complete graph K_m and identify one vertex of each copy into a single shared hub vertex h . The result has $k(m - 1) + 1 = n$ vertices and $k \binom{m}{2} = \frac{m(n-1)}{2}$ edges. Every non-hub vertex has degree $m - 1$, and any pair of vertices contains a non-hub vertex, so $\lambda^{\max} \leq m - 1$ by [Lemma A.3](#).

Construction A.8 (Hub-and-spokes graph). For $n \geq m \geq 2$, let $H_{m,n}$ have vertex set $\{h, v_1, \dots, v_{n-1}\}$ with (i) all $n - 1$ *hub edges* $\{h, v_i\}$, and (ii) a *spoke graph* on $\{v_1, \dots, v_{n-1}\}$ with $\left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor$ edges in which every v_i has spoke degree at most $m - 2$. Such a spoke graph exists by Lemma A.9.

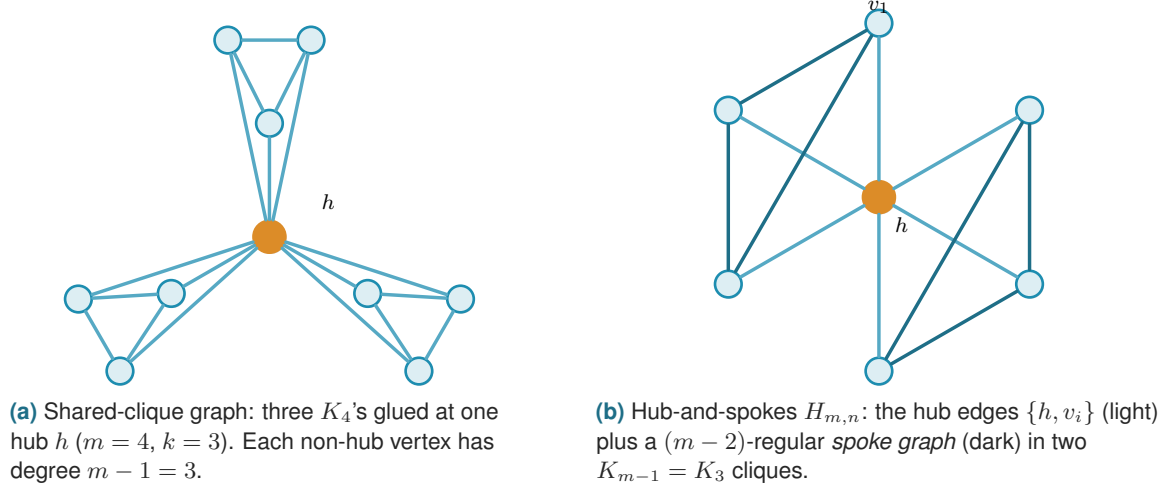


Figure A.2. The two extremal families. (a) The shared-clique graph of Construction A.7 works when $(m - 1) \mid (n - 1)$: every pair contains a non-hub vertex of degree exactly $m - 1$, so $\lambda^{\max} \leq m - 1$ by the degree bound, while the edge count hits $\frac{m(n-1)}{2}$. (b) The hub-and-spokes graph $H_{m,n}$ of Construction A.8 reaches the same bound for every $n \geq m$: each spoke vertex v_i has total degree $1 + (m - 2) = m - 1$, and choosing the spoke graph as disjoint cliques K_{m-1} recovers (a). The hub is drawn in orange in both.

Lemma A.9 (Near-regular graphs exist). For $N \geq 1$ and $0 \leq r \leq N - 1$ there is a graph on N vertices with $\left\lfloor \frac{rN}{2} \right\rfloor$ edges and maximum degree at most r .

Proof. If rN is even we build an r -regular circulant graph on vertex set $\{0, \dots, N - 1\}$: for even r connect each i to $i \pm j \pmod{N}$ for $j = 1, \dots, r/2$, and for odd r (then N is even) use $j = 1, \dots, (r-1)/2$ together with the diameter distance $N/2$, which contributes degree one. Both are r -regular with $rN/2$ edges. If rN is odd, then r and N are both odd: take the $(r - 1)$ -regular circulant with distances $1, \dots, (r - 1)/2$ and add the near-perfect matching that pairs vertex i with $i + (N + 1)/2 \pmod{N}$ for $i = 0, \dots, (N - 3)/2$, which covers all vertices but one. The total is $\frac{(r-1)N}{2} + \frac{N-1}{2} = \frac{rN-1}{2} = \left\lfloor \frac{rN}{2} \right\rfloor$ edges, with one vertex of degree $r - 1$ and all others of degree r . $\square$

Theorem A.10 (Extremal graphs exist). For all $n \geq m \geq 2$ the graph $H_{m,n}$ has $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ edges and $\lambda^{\max} \leq m - 1$.

Proof. The edge count is $(n - 1) + \left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, since $(m - 2)(n - 1)$ and $m(n - 1)$ have the same parity. For the connectivity, any two distinct vertices include at least one spoke vertex v_i , whose total degree is $1 + (m - 2) = m - 1$. Lemma A.3 then gives $\lambda(u, v) \leq m - 1$. When $(m - 1) \mid (n - 1)$, choosing the spoke graph as k disjoint cliques K_{m-1} recovers Construction A.7, so the hub-and-spokes family genuinely generalises the shared-clique one. $\square$

Together the upper bound and [Theorem A.10](#) prove [Theorem 1.2](#). The proof also reveals which graphs are extremal: equality forces both inequalities in the chain to be tight.

Theorem A.11 (Characterisation of equality). *Let G be simple with $\lambda_G^{\max} \leq m - 1$ and $|E(G)| = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and let T be a Gomory–Hu tree of G . If $m(n - 1)$ is even, then every tree edge has weight exactly $m - 1$, every tree edge is also an edge of G , and every edge of G joins vertices at tree distance at most 2. If $m(n - 1)$ is odd, there is exactly one unit of slack, located in exactly one of three places: one tree edge has weight $m - 2$, or one tree edge is not an edge of G , or one edge of G joins vertices at tree distance 3. Everything else is tight.*

Proof. Write the chain of the upper-bound proof as three nonnegative slacks. The first, $S_1 = (m - 1)(n - 1) - \sum_t c(t)$, counts tree weights below $m - 1$. The second, $S_2 = \sum_{e: \text{dist}_T(e) \geq 2} (\text{dist}_T(e) - 2)$, counts edges beyond tree distance two. The third, $S_3 = (n - 1) - |E_1|$, counts tree edges that support no edge of G . Retracing the proof, $\sum_e \text{dist}_T(e) = 2|E| - |E_1| + S_2$ and $\sum_e \text{dist}_T(e) = \sum_t c(t) = (m - 1)(n - 1) - S_1$, so altogether $2|E| = m(n - 1) - (S_1 + S_2 + S_3)$. If $m(n - 1)$ is even, equality $|E| = m(n - 1)/2$ forces $S_1 = S_2 = S_3 = 0$. If $m(n - 1)$ is odd the floor absorbs exactly one unit, so $S_1 + S_2 + S_3 = 1$ and exactly one of the three slacks equals one. $\square$

For $m = 2$ the characterisation says the extremal graphs are exactly the spanning trees: every tree edge must carry weight one and lie in G , so G contains its own Gomory–Hu tree and has only $n - 1$ edges to spend. For $m = 4$ and $n \equiv 1 \pmod{3}$ it is consistent with the trees of K_4 blocks glued at shared cut vertices, the case of [Construction A.7](#) that the search of [Chapter 3](#) rediscovers.

A.3 The directed arc value at $m = 2$

Three short lemmas feed the induction. The first shows a hub forces a linear bound for every m , and is the proof behind the hub branch of [Conjecture 1.14](#). The three lemmas below and the proof of [Theorem 1.10](#) that they support are the author’s own, and its finite base cases $2 \leq n \leq 7$ were verified by the author’s program, through the exhaustive search of [Chapter 2](#) whose transcript is reproduced at the end of this appendix.

Lemma A.12 (Two-hop lemma). *Let D be a simple digraph with $\lambda_D^{\max} \leq m - 1$ containing a vertex r with $d^+(r) = n - 1$. Then every $v \neq r$ has at most $m - 2$ in-arcs from $V \setminus \{r, v\}$.*

Proof. If $u_1, \dots, u_{m-1}$ were distinct in-neighbours of v other than r , that is, distinct vertices each sending an arc to v , the paths $r \rightarrow v$ and $r \rightarrow u_i \rightarrow v$ for $i = 1, \dots, m - 1$ would be m arc-disjoint routes, since the u_i are distinct and $d^+(r) = n - 1$ supplies every starting arc. That contradicts $\lambda_D(r, v) \leq m - 1$. $\square$

Theorem A.13 (Hub upper bound). *If a simple digraph D with $\lambda_D^{\max} \leq m - 1$ has a vertex r with $d^+(r) = n - 1$, then $|A(D)| \leq m(n - 1)$.*

Proof. Split the arcs into those leaving r (exactly $n - 1$), those entering r (at most $n - 1$), and the internal arcs avoiding r . By [Lemma A.12](#) each of the $n - 1$ other vertices receives at most $m - 2$ internal arcs, so there are at most $(m - 2)(n - 1)$ internal arcs, and the total is at most $m(n - 1)$. $\square$

Lemma A.14 (No transitive triangle). *A simple digraph with $\lambda^{\max} \leq 1$ contains no three vertices x, y, z with all of $(x, y), (y, z), (x, z)$ present, since $x \rightarrow z$ and $x \rightarrow y \rightarrow z$ would be two arc-disjoint routes.*

Lemma A.15 (Deletion monotonicity). *Write $D - v_0$ for the digraph obtained from D by deleting the vertex v_0 together with all arcs incident to it. For any digraph D and vertex v_0 ,*

$$\lambda^{\max}(D - v_0) \leq \lambda^{\max}(D) \quad \text{and} \quad \kappa^{\max}(D - v_0) \leq \kappa^{\max}(D),$$

since arc-disjoint paths in $D - v_0$ are still arc-disjoint paths in D , and internally vertex-disjoint paths in $D - v_0$ are still internally vertex-disjoint paths in D . The two statements are proved by the same one-line observation because deleting a vertex only removes routes, never creates one, and it cannot make two surviving routes share anything they did not share before.

Proof of Theorem 1.10. Write $M(n) = \max(2(n - 1), \lfloor n^2/4 \rfloor)$. A direct computation shows $2(n - 1) \geq \lfloor n^2/4 \rfloor$ exactly for $n \leq 7$, with equality at $n = 7$, so $M(n) = 2(n - 1)$ for $n \leq 7$ and $M(n) = \lfloor n^2/4 \rfloor$ for $n \geq 7$. The lower bound $\ell_2^{\text{dir}}(n) \geq M(n)$ is witnessed by the directed hub construction ([Construction 1.9](#) at $m = 2$, the bidirected star) and the one-directional bipartite construction ([Construction 1.8](#)).

For the upper bound we show by strong induction on n that $\lambda_D^{\max} \leq 1$ forces $|A(D)| \leq M(n)$.

Base cases $2 \leq n \leq 7$. All six are settled uniformly by the pruned exhaustive search of [Chapter 2](#), which walks every simple digraph with $\lambda^{\max} \leq 1$ on up to seven vertices and reports the maxima 2, 4, 6, 8, 10, 12, matching $M(n)$ at each size. The transcript is reproduced below, and the cut-counting certifier independently confirms the values within its reach through the $m = 2$ reduction of [Theorem A.31](#). The two smallest are also immediate by hand, as a check on the machine rather than a separate argument: at $n = 2$ there are at most 2 arcs, and at $n = 3$ a fifth arc would leave at most one arc of the complete bidirected triangle missing, so three of the rest form a transitive triangle, which [Lemma A.14](#) forbids. The point of the theorem is precisely that $n = 4, 5, 6, 7$ are *not* immediate by hand, which is what makes handing the case-check to the program the right move.

Inductive step $n \geq 8$. Let D be a simple digraph on n vertices with $\lambda_D^{\max} \leq 1$ and put $a := |A(D)|$. Choose v_0 minimising the total degree $d(v) = d^+(v) + d^-(v)$. Since $\sum_v d(v) = 2a$, averaging gives $d(v_0) \leq 2a/n$. By [Lemma A.15](#) the digraph $D - v_0$ on $n - 1 \geq 7$ vertices satisfies the induction hypothesis, so $a - d(v_0) \leq M(n - 1) = \lfloor \frac{(n-1)^2}{4} \rfloor$. Combining,

$$a \leq \frac{2a}{n} + \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \quad \implies \quad a \leq \frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor.$$

It remains to check that this right-hand side never exceeds $\lfloor n^2/4 \rfloor$, and the two parities behave differently enough to treat them in turn.

For even $n = 2k$ with $k \geq 4$, first simplify $\lfloor (n-1)^2/4 \rfloor$. Expanding $(2k-1)^2 = 4k^2 - 4k + 1$ and dividing by 4 gives $k^2 - k + \frac{1}{4}$, whose floor is $k(k-1)$. The bound then reads

$$a \leq \frac{2k}{2k-2} k(k-1) = \frac{k}{k-1} k(k-1) = k^2,$$

and since $\lfloor n^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$ as well, the bound lands exactly on the target with no slack left over.

For odd $n = 2k+1$ with $k \geq 4$, the same simplification gives $\lfloor (n-1)^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$, so the bound is $a \leq \frac{2k+1}{2k-1} k^2$. To compare this fraction with the target, split off its integer part by dividing $(2k+1)k^2$ by $2k-1$. The remainder is exactly k , because $(2k-1)(k^2+k) = 2k^3+k^2-k$ and $(2k+1)k^2 = 2k^3+k^2$ differ by k , so

$$\frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor = \frac{(2k+1)k^2}{2k-1} = k(k+1) + \frac{k}{2k-1}.$$

The leftover fraction $\frac{k}{2k-1}$ lies strictly between 0 and 1 for every $k \geq 1$, so the bound puts a at most an integer $k(k+1)$ plus a positive amount below one. Since a is itself an integer it cannot reach the next integer up, so $a \leq k(k+1)$. The target $\lfloor n^2/4 \rfloor = \lfloor (2k+1)^2/4 \rfloor = k^2 + k = k(k+1)$ agrees, so for odd n it is integrality, not the raw arithmetic, that closes the last fraction of a unit. In both parities $a \leq \lfloor n^2/4 \rfloor \leq M(n)$, completing the induction. $\square$

Two remarks on the shape of this proof. First, the minimum-degree deletion closes the step on its own, and no separate hub case is needed, because the averaging bound holds whether or not D contains a hub, although [Theorem A.13](#) shows directly that a hub digraph never exceeds the linear branch. Second, for even n the bound is exactly tight against the one-directional bipartite construction, which is $\frac{n}{2}$ -regular in total degree, so every step of the chain is achieved by the conjectured extremiser, and for odd n the slack $\frac{k}{2k-1} < 1$ is absorbed by integrality. The proof is long not because it is deep but because the two regimes of $M(n)$ must be carried through the arithmetic separately, and that bookkeeping is precisely what [Chapter 2](#) hands to the machine at the base cases.

Remark A.16 (Which way Whitney runs). It is worth pausing on the direction of Whitney's inequality before using it, because the tempting reading is the wrong one. From $\kappa_D(u, v) \leq \lambda_D(u, v)$ for every pair it follows that $\kappa_D^{\max} \leq \lambda_D^{\max}$, hence that

$$\{D : \lambda_D^{\max} \leq m-1\} \subseteq \{D : \kappa_D^{\max} \leq m-1\}.$$

Every digraph feasible for the *arc* problem is feasible for the *vertex* problem, and not the other way round. The vertex-feasible family is the larger of the two, so the inequality between the extremal

numbers it hands over for free is

$$k_m^{\text{dir}}(n) \geq \ell_m^{\text{dir}}(n),$$

and an upper bound for the arc problem says nothing at all about the vertex problem. The gap is not vacuous: two routes that share an interior vertex but no arc are counted by λ and not by κ , so there really are digraphs in the difference of the two families. What Whitney gives is therefore exactly one thing, that an arc construction is an honest vertex *lower* bound, and the vertex upper bound has to be earned separately. The proof below earns it, and [Chapter 4](#) keeps the same discipline for $m \geq 3$, where the vertex problem stays open even though the arc conjecture is stated.

Proof of Theorem 1.11. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ as before.

Lower bound. The directed hub and the one-directional bipartite construction both have $\lambda^{\max} = 1$, so by [Remark A.16](#) both have $\kappa^{\max} \leq 1$ and are feasible for the vertex problem. Hence $k_2^{\text{dir}}(n) \geq M(n)$.

Upper bound. We show by strong induction on n that $\kappa_D^{\max} \leq 1$ forces $|A(D)| \leq M(n)$. The induction is the one that proved [Theorem 1.10](#), and every step of it survives the change of separation, because the only two properties of the ceiling that argument uses are that it is inherited by vertex deletion and that it holds at the base sizes.

The base cases $2 \leq n \leq 7$ are settled by the same pruned exhaustive search of [Chapter 2](#), run with the vertex feasibility test in place of the arc one. It walks every simple digraph with $\kappa^{\max} \leq 1$ on up to seven vertices and reports the maxima 2, 4, 6, 8, 10, 12, matching $M(n)$ at each size. The transcript is reproduced in [Section A.10](#) beside the arc one.

For the inductive step $n \geq 8$, let D be a simple digraph on n vertices with $\kappa_D^{\max} \leq 1$ and put $a := |A(D)|$. Choose v_0 of minimum total degree, so $d(v_0) \leq 2a/n$ by averaging. By [Lemma A.15](#), which holds for κ exactly as it does for λ , the digraph $D - v_0$ on $n-1 \geq 7$ vertices again satisfies $\kappa^{\max} \leq 1$, so the induction hypothesis gives $a - d(v_0) \leq M(n-1) = \lfloor (n-1)^2/4 \rfloor$. From here the two parity computations are verbatim those of [Theorem 1.10](#): they use only the two displayed inequalities and the integrality of a , no property of the separation. They yield $a \leq \lfloor n^2/4 \rfloor \leq M(n)$, completing the induction.

Hence $k_2^{\text{dir}}(n) = M(n) = \ell_2^{\text{dir}}(n)$. The two problems have the same answer at $m = 2$, but this is a computed coincidence of the two searches meeting at every base size, not a formal consequence of Whitney's inequality. $\square$

Remark A.17 (The directed vertex problem at $m \geq 3$). The proof above transfers the $m = 2$ induction but nothing more, and for $m \geq 3$ the vertex problem is genuinely separate: the arc conjecture, and the decomposition of [Conjecture A.21](#) that would prove it, both concern the smaller family and leave $k_m^{\text{dir}}(n)$ untouched. What can be said is computational. Running the

exhaustive search of [Chapter 2](#) under both feasibility tests at $m = 3$ gives

n	2	3	4	5	6
$\ell_3^{\text{dir}}(n)$	2	6	9	12	15
$k_3^{\text{dir}}(n)$	2	6	9	12	15

with every entry proved complete, so the two separations agree at every size the exhaustion reaches, and from $n = 3$ onwards both match the conjectured $\max(m(n-1), \lfloor (n+m-2)^2/4 \rfloor)$. At $n = 2$ the conjecture's formula gives 3 where only 2 arcs exist at all, the usual degeneracy of the formula below $n = m$. Agreement over five sizes is evidence and not a proof, and the natural next question is structural rather than numerical: to separate the two problems one needs a digraph where some pair carries strictly more arc-disjoint routes than internally disjoint ones *and* where that slack pays for extra arcs. The two demands pull against each other, which is perhaps why no small example does both.

A.4 Structural propositions on the directed frontier

These are the proved ingredients of the reduction discussed in [Chapter 4](#). The section ends with what the reduction still assumes, stated as [Conjecture A.21](#), and with the one quadratic bound that assumes nothing, [Proposition A.22](#).

Proposition A.18 (Mutually unreachable out-neighbourhoods). *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$. For any vertex u and distinct out-neighbours v, w of u , there is no directed path from v to w in $D - u$.*

Proof. If P were a directed v – w path avoiding u , then $u \rightarrow w$ and $u \rightarrow v \xrightarrow{P} w$ would be two arc-disjoint u -to- w routes: the first uses only the arc (u, w) , the second starts with (u, v) and continues along P , none of whose arcs touch u . This contradicts $\lambda_D(u, w) \leq 1$. The restriction to $D - u$ is not cosmetic, and [Figure A.3](#) draws both the argument and the reason for the restriction: in the digraph with arcs (u, v) , (v, u) , (u, w) every local connectivity is 1, yet $v \rightarrow u \rightarrow w$ reaches w from v through u . $\square$

Proposition A.19 (Disjoint second out-neighbourhoods). *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$, let $u \in V$, and let $v_1 \neq v_2$ be out-neighbours of u . Then $(N^+(v_1) \setminus \{u\}) \cap (N^+(v_2) \setminus \{u\}) = \emptyset$.*

Proof. A common out-neighbour $w \neq u$ would give the two arc-disjoint routes $u \rightarrow v_1 \rightarrow w$ and $u \rightarrow v_2 \rightarrow w$, contradicting $\lambda_D(u, w) \leq 1$. $\square$

Proposition A.20 (Two-hop budget on bipartite partitions). *Let D be a simple digraph with $\lambda_D^{\max} \leq m - 1$. If V admits a partition $A \cup B$ with every arc from A to B present, then every $b \in B$ has in-degree at most $m - 2$ from inside B , and consequently the number of arcs inside B is at most $(m - 2)|B|$.*



(a) If a directed $v \rightsquigarrow w$ path (red) existed in $D - u$, then $u \rightarrow w$ and $u \rightarrow v \rightsquigarrow w$ would be two arc-disjoint $u-w$ routes, against $\lambda(u, w) \leq 1$.

(b) The counterexample $\{(u, v), (v, u), (u, w)\}$: $\lambda^{\max} = 1$, yet w is reachable from v , but only *through* u .

Figure A.3. Why [Proposition A.18](#) must exclude paths through u . (a) For out-neighbours v, w of u in a digraph with $\lambda^{\max} \leq 1$, no $v \rightsquigarrow w$ path can avoid u : such a path would combine with the lone arc $u \rightarrow w$ into two arc-disjoint routes. (b) The restriction to $D - u$ is essential. In the three-vertex digraph $\{(u, v), (v, u), (u, w)\}$ we have $\lambda^{\max} = 1$, yet v reaches w via $v \rightarrow u \rightarrow w$, a route that reuses the vertex u , which the proposition allows.

Proof. Fix $a \in A$ and $b \in B$, and let $b'_1, \dots, b'_d$ be the in-neighbours of b inside B , where $d = \deg_B^-(b)$. The direct arc $a \rightarrow b$ together with the two-step detours $a \rightarrow b'_i \rightarrow b$ are $1 + d$ arc-disjoint a -to- b routes, all present because $A \rightarrow B$ is complete. Feasibility forces $1 + d \leq m - 1$, so $d \leq m - 2$, and summing over B bounds the internal arcs by $(m - 2)|B|$. $\square$

We now combine [Proposition A.20](#) with the best choice of partition. Suppose every arc of D either runs from A to B or stays inside B : the partition $A \rightarrow B$ is complete, no arc lies inside A , and crucially *no arc runs from B back to A* . Write $b = |B|$, so that $|A| = n - b$. The arcs then come in exactly two kinds. The arcs crossing the partition number $|A||B| = (n - b)b$, one for each ordered A - B pair. The arcs inside B number at most $(m - 2)b$: since no route between two vertices of B ever needs to leave B (there is no arc back to A to leave by), [Proposition A.20](#) applies to B on its own and caps the in-degree of each $b \in B$ from inside B at $m - 2$. There are no other arcs, so, adding the two kinds,

$$|A(D)| \leq (n - b)b + (m - 2)b = b(n + m - 2 - b).$$

The right-hand side is a downward parabola in b . Writing $s = n + m - 2$, it is $b(s - b) = -b^2 + sb$, maximised at $b = s/2$, so over integers the best partition takes $b = \lceil s/2 \rceil$ (equivalently $\lfloor s/2 \rfloor$), and the maximum value is

$$\max_b b(s - b) = \left\lfloor \frac{s^2}{4} \right\rfloor = \left\lfloor \frac{(n + m - 2)^2}{4} \right\rfloor.$$

This is the quadratic branch of [Conjecture 1.14](#), reached uniformly in m . It is worth being exact about how much was assumed, because the count is often summarised as resting on the absence of backward arcs alone, and that summary is too generous. Four things were granted at once: that a partition $V = A \cup B$ exists at all, that every arc from A to B is present, that no arc lies inside A , and that no arc runs from B back to A . Only the last of the four is the backward-arc condition. The first three describe the shape of the extremiser, and nothing proved so far forces

an arbitrary extremal digraph to have that shape. The honest statement of the gap is therefore a structural one:

Conjecture A.21 (The missing decomposition). *Let n be large enough that the quadratic branch of [Conjecture 1.14](#) strictly exceeds the linear one, that is, $\lfloor (n + m - 2)^2/4 \rfloor > m(n - 1)$. Then every extremal digraph D on n vertices with $\lambda_D^{\max} \leq m - 1$ admits a partition $V(D) = A \cup B$ such that every arc from A to B is present, no arc lies inside A , and no arc runs from B to A .*

The hypothesis on n has to be there, and it is worth seeing why, because the obvious phrasing is too narrow. On the linear branch the extremisers are *not* just the hub. At $m = 2$, every bidirected spanning tree is feasible and has exactly $2(n - 1)$ arcs, since between two vertices the only route follows the unique tree path, so the whole extremal set on that branch is the set of bidirected trees. An exhaustive classification confirms it: at $n = 5$ and $n = 6$ there are exactly 3 and 6 extremal digraphs up to isomorphism, which are precisely the numbers of trees on 5 and 6 vertices, and none of them admits the decomposition. Excluding “the hub” would leave all the other trees behind, so the clean hypothesis is on the size rather than on the shape.

Given [Conjecture A.21](#) the count above closes [Conjecture 1.14](#) for $m \geq 3$. The backward-arc clause is the part where the natural delete-and-compensate exchange fails to be monotone, which is why it is the clause usually named, but the partition itself is not free either, and [Chapter 4](#) states the position that way. Testing the conjecture by exhaustion is harder than it looks: the crossover sits at $n = 8$ for $m = 2$ and at $n = 9$ for $m = 3$, so every size an exhaustive sweep can reach at $m = 3$ still lies on the linear branch, where the statement says nothing. That is the practical reason this conjecture has resisted a computational probe rather than a proof.

What can be proved without any of the four is the leading constant, and the argument is short enough to give in full. It also says something the conjecture does not: that every feasible digraph is within a lower-order number of arcs of being one-directional bipartite.

Proposition A.22 (Unconditional quadratic bound and stability). *Let D be a simple digraph on $n \geq 2$ vertices with $\lambda_D^{\max} \leq m - 1$. Then*

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + \sqrt{m} n^{3/2}.$$

Moreover D can be made one-directional bipartite, meaning every remaining arc runs from one part to the other and none returns, by deleting at most $\sqrt{m} n^{3/2}$ arcs. Consequently, for each fixed m ,

$$\ell_m^{\text{dir}}(n) = \frac{n^2}{4} + O_m(n^{3/2}),$$

so the leading constant $\frac{1}{4}$ of [Conjecture 1.14](#) holds unconditionally, with no hypothesis on the structure of the extremiser.

Proof. Step 1, a budget on two-step routes. Fix an ordered pair (u, v) with $u \neq v$, and let $p_2(u, v)$ count the vertices x with both $u \rightarrow x$ and $x \rightarrow v$ present. Such an x is automatically distinct from

u and from v , since D has no loops, so each one gives a genuine two-step route $u \rightarrow x \rightarrow v$. Two of these routes with different midpoints $x \neq x'$ share no arc, since the first uses (u, x) and (x, v) and the second uses (u, x') and (x', v) , and neither uses the arc (u, v) , which has no midpoint at all. So the direct arc, when present, together with all the two-step routes, is a family of pairwise arc-disjoint u -to- v routes, and feasibility caps its size:

$$\mathbf{1}_{(u,v) \in A(D)} + p_2(u, v) \leq \lambda_D(u, v) \leq m - 1.$$

Summing over all $n(n - 1)$ ordered pairs gives $|A(D)| + \sum_{u \neq v} p_2(u, v) \leq (m - 1) n(n - 1)$.

Step 2, from pairs to degrees. Counting directed walks of length two by their midpoint, $\sum_{u,v} p_2(u, v) = \sum_x d^-(x) d^+(x)$, where this sum runs over *all* ordered pairs, the diagonal $u = v$ included. The diagonal terms count the closed walks $u \rightarrow x \rightarrow u$, that is, twice the number of digons of D . Distinct digons use disjoint pairs of arcs, so twice the number of digons is at most $|A(D)|$, and that is exactly the slack Step 1 left over. Combining the two,

$$\sum_{x \in V} d^+(x) d^-(x) \leq [(m - 1) n(n - 1) - |A(D)|] + |A(D)| = (m - 1) n(n - 1).$$

This is the one inequality the whole section runs on, and it is worth reading in words: a vertex with a large in-degree *and* a large out-degree is expensive, because it manufactures two-step routes for many pairs at once, and the budget for those routes is only quadratic in n .

Step 3, the smaller half-degree is small on average. Since $\min(a, b) \leq \sqrt{ab}$ for non-negative a, b , and by the Cauchy-Schwarz inequality in the form $\sum_x \sqrt{a_x} \leq \sqrt{n \sum_x a_x}$,

$$\begin{aligned} \sum_{x \in V} \min(d^+(x), d^-(x)) &\leq \sum_{x \in V} \sqrt{d^+(x) d^-(x)} \leq \sqrt{n \sum_{x \in V} d^+(x) d^-(x)} \\ &\leq \sqrt{n \cdot (m - 1) n(n - 1)} = n \sqrt{(m - 1)(n - 1)} \leq \sqrt{m} n^{3/2}. \end{aligned}$$

Step 4, deleting the smaller half of every vertex. Call x *source-like* if $d^+(x) \geq d^-(x)$ and *sink-like* otherwise, and write S and T for the two classes, so $V = S \cup T$. Delete every in-arc of a source-like vertex and every out-arc of a sink-like vertex. A source-like x loses $d^-(x) = \min(d^+(x), d^-(x))$ arcs and a sink-like x loses $d^+(x) = \min(d^+(x), d^-(x))$ arcs, so the number of deleted arcs is at most $\sum_x \min(d^+(x), d^-(x))$, which Step 3 bounds by $\sqrt{m} n^{3/2}$. An arc (a, b) survives only if a is not sink-like and b is not source-like, that is, only if $a \in S$ and $b \in T$. So what remains runs entirely from S to T , which is the one-directional bipartite shape, and it has at most $|S| |T| \leq \lfloor n^2/4 \rfloor$ arcs. Adding the two counts gives the bound.

The asymptotic statement follows because the augmented bipartite construction ([Construction 1.13](#)) already gives $\ell_m^{\text{dir}}(n) \geq \lfloor (n + m - 2)^2/4 \rfloor = n^2/4 + O_m(n)$, and $n^{3/2}$ dominates n . $\square$

The error term $n^{3/2}$ is larger than the $(m - 2)n/2$ that [Conjecture 1.14](#) predicts, so the proposition as it stands does not separate the conjectured formula from its near neighbours. It can be

sharpened to the right order, and the extra idea is small enough to be worth giving, because it explains where the $\sqrt{n}$ was being wasted.

Step 3 is lossy exactly when every vertex carries a middling degree. The Cauchy-Schwarz step is tight only if $d^+(x)d^-(x)$ is about $(m-1)n$ at every vertex, which forces every degree down to about $\sqrt{mn}$. But a digraph with $\Theta(n^2)$ arcs has average degree $\Theta(n)$, and at a vertex of large degree the constraint $d^+(x)d^-(x) \leq (m-1)n$ bites much harder: with $d^+ + d^-$ of order n , the smaller of the two must be of order m rather than $\sqrt{mn}$. The dense case and the tight case of Step 3 are therefore incompatible, and the way to exploit that is a dichotomy on the minimum degree, with vertex deletion to handle the sparse side.

Lemma A.23 (Small side of a high-degree vertex). *For any vertex x of a digraph with total degree $d(x) = d^+(x) + d^-(x) > 0$,*

$$\min(d^+(x), d^-(x)) \leq \frac{2 d^+(x) d^-(x)}{d(x)}.$$

Proof. Write $\mu = \min(d^+(x), d^-(x))$ and $M = \max(d^+(x), d^-(x))$, so $\mu + M = d(x)$ and $\mu M = d^+(x)d^-(x)$. Since $\mu \leq M$ we have $M \geq d(x)/2$, hence $d^+(x)d^-(x) = \mu M \geq \mu d(x)/2$, which rearranges to the claim. $\square$

Theorem A.24 (The quadratic bound with a linear error). *For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \leq m-1$ satisfies*

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1),$$

so for each fixed m ,

$$\ell_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n),$$

and both the leading constant $\frac{1}{4}$ and the order of the second-order term of [Conjecture 1.14](#) hold unconditionally.

Proof. Induction on n . For $n = 2$ a simple digraph has at most 2 arcs and the right-hand side is $1 + 4(m-1) \geq 5$, so the base holds. Let $n \geq 3$ and split on the minimum total degree.

Case 1: some vertex v has $d(v) < n/2$. Since $d(v)$ is an integer, $d(v) \leq \lfloor n/2 \rfloor$. By [Lemma A.15](#) the digraph $D - v$ on $n-1$ vertices is still feasible, so the induction hypothesis applies to it and

$$|A(D)| \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 4(m-1)(n-2) + \left\lfloor \frac{n}{2} \right\rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-2),$$

using [Lemma A.27](#) for the last step, and this is below the claimed bound.

Case 2: every vertex has $d(x) \geq n/2$. By [Lemma A.23](#) and Step 2 of [Proposition A.22](#),

$$\sum_{x \in V} \min(d^+(x), d^-(x)) \leq \sum_{x \in V} \frac{2d^+(x)d^-(x)}{d(x)} \leq \frac{4}{n} \sum_{x \in V} d^+(x)d^-(x) \leq \frac{4}{n} (m-1)n(n-1),$$

which is $4(m-1)(n-1)$. Step 4 of [Proposition A.22](#) deletes exactly that many arcs and leaves at most $\lfloor n^2/4 \rfloor$, so the bound holds in this case too.

For the two-sided asymptotic statement, [Construction 1.13](#) gives

$$\ell_m^{\text{dir}}(n) \geq \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor = \frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1),$$

so the second-order term is linear in n from both sides. □

What is left open is therefore a constant factor in one coefficient, not a shape. [Conjecture 1.14](#) says the second-order coefficient is exactly $(m-2)/2$, and the theorem pins it between that and $4(m-1)$, a factor of roughly eight. Closing that gap is where the structure of [Conjecture A.21](#) lives, and [Remark A.25](#) rules out the most obvious way to try.

Remark A.25 (Case 2 cannot be sharpened without a new inequality). One might hope to cut Case 2's constant using the observation that in the conjectured extremiser, one whole side has $\min(d^+, d^-) = 0$, so summing [Lemma A.23](#)'s worst case at every vertex looks wasteful. It is not: Case 2's bound is already the exact maximum obtainable from the two facts it uses, the budget $\sum_x d^+(x)d^-(x) \leq (m-1)n(n-1)$ of Step 2 in [Proposition A.22](#) and the floor $d(x) \geq n/2$, and no cleverer use of those two facts alone can do better.

At a vertex with $d(x) = s \geq n/2$, achieving $\min(d^+(x), d^-(x)) = t$ costs $d^+(x)d^-(x) = t(s-t)$ at least, cheapest at $s = n/2$, giving the cost function $c(t) = t(n/2 - t)$ on $t \in [0, n/4]$. Since c is concave with $c(0) = 0$, for any two vertices with values $0 < a \leq b$ strictly below the cap, shifting mass from a to b preserves $a+b$ while lowering $c(a) + c(b)$, because c' is decreasing. Repeating this exchange shows that the true maximum of $\sum_x t_x$ under a shared budget Q is attained by concentrating the budget on as few vertices as the cap allows, each at $t_x = n/4$, and the rest at $t_x = 0$, exactly the qualitative picture the conjectured extremiser suggests, and it already gives $4Q/n = 4(m-1)(n-1)$ once the number of vertices at the cap is below n . The reason the crude per-vertex bound loses nothing is that it is tight at exactly the same point: [Lemma A.23](#) is an equality only when $d^+(x) = d^-(x)$, and the floor substitution $d(x) \rightarrow n/2$ is an equality only when $d(x) = n/2$, so both hold simultaneously exactly at $d^+(x) = d^-(x) = n/4$, the vertex profile the true optimum concentrates on. Cutting the constant therefore needs a second, independent inequality, one that constrains interference among the handful of expensive near-balanced vertices themselves rather than a sharper reading of the one already in hand.

A.5 The directed multigraph problem at large n

The body proves $L_m^{\text{dir}}(n) = 2(n-1)(m-1)$ for $n \leq 6$ (Theorem 2.2). This section records what is known beyond, where the one-directional bipartite construction takes over. Throughout, the scaling reduction of Chapter 2 is used in the following form.

Lemma A.26 (Scaling reduction). *Let D be a directed multigraph with $\mu(u, v) \leq m-1$ on every ordered pair and $\lambda^{\max}(D) \leq m-1$. Then $w := \mu/(m-1)$ takes values in $[0, 1]$, satisfies $\text{maxflow}(s, t; w) \leq 1$ for all $s \neq t$, and $|A(D)| = (m-1) \sum_{u \neq v} w(u, v)$.*

Proof. Set $w = \mu/(m-1)$. Every multiplicity obeys $0 \leq \mu(u, v) \leq m-1$, so each $w(u, v)$ lies in $[0, 1]$. Dividing every capacity by the constant $m-1$ divides every flow value by $m-1$, and by Theorem A.1 the maximum flow under the capacities μ is $\lambda_D(s, t) \leq m-1$, so under w it is at most 1. For the arc count, recall that the number of arcs is the total multiplicity, $|A(D)| = \sum_{u \neq v} \mu(u, v)$, since each parallel copy is one arc. Substituting $\mu = (m-1)w$,

$$|A(D)| = \sum_{u \neq v} \mu(u, v) = \sum_{u \neq v} (m-1) w(u, v) = (m-1) \sum_{u \neq v} w(u, v),$$

which is the claimed identity. $\square$

Lemma A.27 (Arithmetic identity). *For all $n \geq 2$, $\left\lfloor \frac{(n-1)^2}{4} \right\rfloor + \lfloor n/2 \rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor$.*

Proof. For $n = 2k$ the left side is $k(k-1) + k = k^2$, and for $n = 2k+1$ it is $k^2 + k = k(k+1)$. Both match the right side. $\square$

Proposition A.28 (Bipartite upper bound for even $n \geq 8$, conditional). *Let $n = 2k \geq 8$ be even. Suppose every fractional weighting w' on $n-1$ vertices with values in $[0, 1]$ and all pairwise maximum flows at most one has total weight at most $\left\lfloor \frac{(n-1)^2}{4} \right\rfloor$. Then every directed multigraph D on n vertices with $\lambda^{\max}(D) \leq m-1$ satisfies $|A(D)| \leq (m-1) \left\lfloor \frac{n^2}{4} \right\rfloor$.*

Proof. By Lemma A.26 the arc count is $|A(D)| = (m-1)M$ with $M := \sum_{u \neq v} w(u, v)$, so it suffices to bound $M \leq \lfloor n^2/4 \rfloor$ for the scaled weighting w , all of whose pairwise maximum flows are at most 1.

Averaging. Give each vertex its weighted total degree $d(v) = \sum_u (w(u, v) + w(v, u))$. Summing over all vertices counts the weight of each ordered pair exactly twice, once at its tail and once at its head, so $\sum_v d(v) = 2M$, and some vertex v_0 has $d(v_0) \leq 2M/n$.

Deletion. The number $d(v_0)$ is exactly the total weight carried by pairs touching v_0 , so deleting v_0 leaves a weighting on $n-1$ vertices of total $M - d(v_0)$. Removing a vertex cannot raise any maximum flow, so that restricted weighting still has all pairwise flows at most 1, and the hypothesis applies to it:

$$M - d(v_0) \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor = \left\lfloor \frac{(2k-1)^2}{4} \right\rfloor = k(k-1),$$

the last step because $(2k - 1)^2/4 = k^2 - k + \frac{1}{4}$ floors to $k(k - 1)$.

Combining. Since $n = 2k$ we have $d(v_0) \leq 2M/n = M/k$, hence

$$M - \frac{M}{k} \leq M - d(v_0) \leq k(k - 1), \quad \text{that is} \quad M \cdot \frac{k - 1}{k} \leq k(k - 1).$$

Dividing by $(k - 1)/k > 0$ gives $M \leq k^2 = \lfloor n^2/4 \rfloor$, and multiplying back by $m - 1$ gives $|A(D)| \leq (m - 1) \lfloor n^2/4 \rfloor$. $\square$

The small cases $n \in \{4, 6\}$ genuinely do not satisfy the conclusion, and must not: there $2(n - 1) > \lfloor n^2/4 \rfloor$ and the double star beats the bipartite weighting, which is why the certified values of [Theorem 2.2](#) sit on the linear branch. For odd n the same deletion gives $M \leq k(k + 1) + \frac{k}{2k-1}$, an overshoot of less than one that integrality no longer absorbs, because fractional weights need not have integer totals. The induction would close if some vertex always had weighted degree at most $(m - 1)k$ before scaling, which is the missing minimum-degree statement.

Conjecture A.29 (Minimum-degree lemma for odd n). *In any directed multigraph D on $n = 2k + 1$ vertices ($k \geq 3$) with $\lambda^{\max}(D) \leq m - 1$ and $m \geq 3$, some vertex has total degree $d(v) \leq (m - 1)k$, counting arcs with multiplicity.*

At $m = 2$ the statement is a proposition rather than a conjecture, through a reduction that also settles the whole multigraph problem at $m = 2$.

Proposition A.30 (Parallel-arc elimination at $m = 2$). *A directed multigraph with $\lambda^{\max} \leq 1$ has no parallel arcs, since two parallel arcs are already two arc-disjoint routes. Hence it is a simple digraph.*

Theorem A.31 (Directed multigraphs at $m = 2$). *For all $n \geq 2$, $L_2^{\text{dir}}(n) = \ell_2^{\text{dir}}(n) = \max(2(n - 1), \lfloor \frac{n^2}{4} \rfloor)$.*

Proof. Immediate from [Proposition A.30](#) and [Theorem 1.10](#). $\square$

Proposition A.32 (Minimum degree at $m = 2$, odd n). *In any directed multigraph on $n = 2k + 1 \geq 7$ vertices with $\lambda^{\max} \leq 1$, some vertex has $d(v) \leq k$.*

Proof. By [Proposition A.30](#) and [theorem 1.10](#), $|A(D)| \leq \max(2(n - 1), \lfloor n^2/4 \rfloor) = \lfloor n^2/4 \rfloor = k(k + 1)$, the maximum being the quadratic branch because $n \geq 7$. The average total degree is $\frac{2k(k+1)}{2k+1} < k + 1$, and some integer degree is at most k . The hypothesis $n \geq 7$ is needed: on the linear branch the statement can fail, as the bidirected star on three vertices has minimum total degree $2 > 1$. $\square$

For $m \geq 3$ the two extreme regimes of [Conjecture A.29](#) are settled. In the *purely saturated* regime, where every multiplicity is 0 or $m - 1$, the scaled weighting is the indicator of a simple digraph with $\lambda^{\max} \leq 1$, its degrees are integers, and the argument of [Proposition A.32](#) applies

verbatim after multiplying back by $m - 1$. In the *all-unsaturated* simple regime at $n = 7$, $m = 3$, the program certifies by a cut-counting infeasibility that no simple digraph satisfies all two-hop inequalities with every total degree at least 7. The open case is the *mixed* regime, where some multiplicity lies strictly between 0 and $m - 1$: there the scaled degrees are genuinely fractional, the averaging bound $\frac{2k(k+1)}{2k+1}$ lands strictly between k and $k + 1$, and no integrality rescues it. This is the single identified gap that separates the certified values of [Theorem 2.2](#) from the full conjecture

$$L_m^{\text{dir}}(n) = (m - 1) \max(2(n - 1), \left\lfloor \frac{n^2}{4} \right\rfloor),$$

whose lower bound is unconditional: the double star realises $2(n - 1)(m - 1)$ and the one-directional bipartite multigraph, every arc at multiplicity $m - 1$, realises $(m - 1) \lfloor n^2/4 \rfloor$ with $\lambda^{\max} = m - 1$.

Remark A.33 (Why the mixed regime resists, and its kinship with the back-arc gap). Two facts explain why the mixed regime is genuinely hard rather than merely unfinished, and both are worth recording so the conjecture is not mistaken for a routine averaging exercise.

First, averaging alone cannot close it, even granted the strongest input available. Work in the scaled weighting $w = \mu/(m - 1)$ of [Lemma A.26](#), with total weight $M = \sum_{u \neq v} w(u, v)$ and weighted total degree $d(v)$, so the conjecture asks for a vertex with $d(v) \leq k$. Granting the fractional even bound that drives [Proposition A.28](#), the odd deletion already gives $M \leq k(k + 1) + \frac{k}{2k-1} = \frac{(2k+1)k^2}{2k-1}$. Since the minimum degree never exceeds the average $\frac{2M}{2k+1}$,

$$\min_v d(v) \leq \frac{2M}{2k+1} \leq \frac{2k^2}{2k-1} = k + \frac{k}{2k-1} < k + 1.$$

In the saturated and $m = 2$ regimes the degrees are integers, so this rounds down to k and the conjecture follows, exactly as in [Proposition A.32](#). In the mixed regime $d(v)$ is genuinely fractional and the leftover $\frac{k}{2k-1}$ cannot be rounded away. The obstruction is not the particular bound chosen. To force $\min_v d(v) \leq k$ by averaging one would need $M \leq k^2 + \frac{k}{2}$, yet the one-directional bipartite weighting with $|A| = k$ and $|B| = k + 1$ already carries $M = k(k + 1) = k^2 + k$, so no averaging argument can finish for any m . Closing the gap needs a structural step that exhibits a specific low-degree vertex, which is precisely what [Lemma A.34](#) supplies at $m = 3$ in order to *bypass* the conjecture rather than prove it.

Second, the conjecture is tight, and the natural way to break it fails for a reason that ties this gap to the back-arc obstruction of [Conjecture 1.14](#). At the bipartite extremiser the minimum degree is exactly k , attained on the B side. The obvious attempt to raise it bolts a little backward weight onto the saturated bipartite point, sending $w(b, a_0) = \epsilon$ from every $b \in B$ to one fixed source $a_0 \in A$, which would lift each B -degree to $k + \epsilon$. It is infeasible for every $\epsilon > 0$. With the forward layer at full weight the backward arcs open the recirculating routes $a \rightarrow b \rightarrow a_0 \rightarrow b'$, so the flow from a source $a \neq a_0$ to a sink b' rises to $1 + k\epsilon > 1$. A single backward arc laid on a saturated forward layer manufactures an extra route, the same mechanism that keeps the upper bound of [Conjecture 1.14](#) open and the reason the bipartite extremisers are locally rigid under

the searches of [Chapter 3](#). The minimum-degree gap and the back-arc gap are two faces of one obstruction. A systematic fractional hill-climb (`fractional_search`), a search that repeatedly nudges one weight at a time and keeps a change only when the objective improves while every pairwise flow stays at most one, run with three initialisation types and eight seeds each at $n = 9, 11, \text{ and } 13$, finds no feasible weighting with minimum weighted degree above k , confirming the bipartite point's rigidity for $k = 4, 5, \text{ and } 6$. Feasibility of a fractional weighting is checked by `fractional_flows_feasible`, which calls `_float_maxflow_value`, a textbook maximum-flow routine (Edmonds–Karp) run directly on the fractional capacities, since `scipy's csgraph` max-flow accepts only integer capacities.

Lemma A.34 (Attachment lemma). *Let $m \geq 2$ and let D_0 be the one-directional complete bipartite multigraph with parts A and B , $|A| = p \geq 2$, $|B| = q \geq 2$, and $\mu(a, b) = m - 1$ for every $a \in A, b \in B$, with no other arcs. Let D be obtained from D_0 by adding one new vertex v together with any multiset of arcs incident to v , a multiset because parallel copies are allowed (multiplicities at most $m - 1$). If $\lambda_D^{\max} \leq m - 1$, then $d(v) \leq (m - 1) \max(p, q)$.*

The proof is a bookkeeping argument on the arcs at v , and [Figure A.4](#) is its scaffold. Sort those arcs into the four classes i_A, o_A, i_B, o_B of the figure, by direction and by which part they touch. Three combinations are impossible, because each completes an m -th arc-disjoint route between two old vertices on top of the $m - 1$ parallel arcs they already share. An i_A arc with an o_B arc makes $a \rightarrow v \rightarrow b$. An i_A arc with an o_A arc to a *different* source makes $a \rightarrow v \rightarrow a' \rightarrow b$. An i_B arc with an o_B arc to a *different* sink makes $a \rightarrow b \rightarrow v \rightarrow b'$. Forbidding these forces v into one of a few rigid shapes, a pure sink touching a single source, a pure source touching a single sink, or a small mixed remnant, and in each shape a count of two-step routes through v caps its degree. The three cases below are exactly those shapes, and the bound $(m - 1) \max(p, q)$ is whichever of them is largest.

Proof. The strategy is to sort the arcs at the new vertex v into four groups by their direction and which side they touch, rule out the combinations that would manufacture an m -th route between two old vertices, and then bound whatever survives. Write $i_A = \sum_a \mu(a, v)$, $i_B = \sum_b \mu(b, v)$, $o_A = \sum_a \mu(v, a)$, $o_B = \sum_b \mu(v, b)$ for the total multiplicities into v from A , into v from B , out of v to A , and out of v to B , so that $d(v) = i_A + i_B + o_A + o_B$. Three route patterns are forbidden outright, because each adds an m -th arc-disjoint route on top of the $m - 1$ parallel arcs of a bipartite pair:

- (1) $\mu(a, v) \geq 1$ and $\mu(v, b) \geq 1$: the path $a \rightarrow v \rightarrow b$ together with the $m - 1$ arcs $a \rightarrow b$ gives $\lambda(a, b) \geq m$.
- (2) $\mu(a, v) \geq 1$ and $\mu(v, a') \geq 1$ with $a' \neq a$: the path $a \rightarrow v \rightarrow a' \rightarrow b$ (any $b \in B$) gives $\lambda(a, b) \geq m$.
- (3) $\mu(b, v) \geq 1$ and $\mu(v, b') \geq 1$ with $b' \neq b$: the path $a \rightarrow b \rightarrow v \rightarrow b'$ (any $a \in A$) gives $\lambda(a, b') \geq m$.

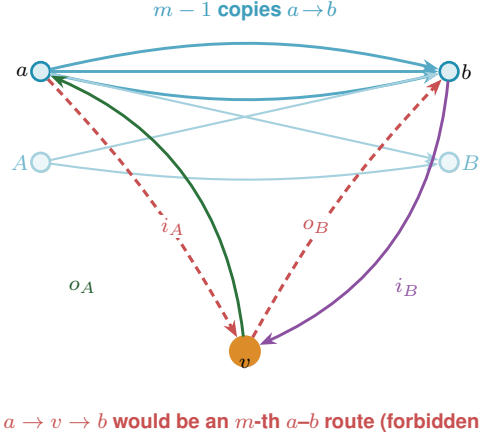


Figure A.4. The attachment lemma (Lemma A.34): a new vertex v is glued onto $(m - 1) \cdot B_{p,q}$, the one-directional complete bipartite multigraph in which every $a \in A$ sends $m - 1$ parallel arcs to every $b \in B$ (the bundle along the top, faint for the pairs other than a, b). The arcs at v fall into four classes by direction and side: i_A into v from A , o_A out of v to A , i_B into v from B , and o_B out of v to B . The two red legs i_A and o_B are drawn together because they are the worst case: the route $a \rightarrow v \rightarrow b$ laid on top of the $m - 1$ existing $a \rightarrow b$ arcs is an m -th arc-disjoint a - b route, so $\lambda(a, b) \geq m$, which feasibility forbids. Ruling this out, with two siblings like it, caps $d(v) \leq (m - 1) \max(p, q)$.

Each bound below rests on one principle. The local connectivity $\lambda(s, t)$ is at least the number of pairwise arc-disjoint s -to- t routes we can exhibit, because any such family of routes is an integral flow from s to t . Feasibility says $\lambda \leq m - 1$, so the moment we can name m arc-disjoint routes between two vertices we have a contradiction.

Case I: $i_A \geq 1$. Some arc enters v from a vertex $a \in A$. Pattern (1) then forces $o_B = 0$, because any arc $v \rightarrow b$ would complete the route $a \rightarrow v \rightarrow b$, an m -th route on the pair (a, b) laid on top of its $m - 1$ parallel arcs.

Suppose first that $o_A \geq 1$, so v also sends an arc to some vertex of A . Pattern (2) makes this rigid: an arc into v from a and an arc out of v to a different $a' \neq a$ would give the route $a \rightarrow v \rightarrow a' \rightarrow b$ for any b , again an m -th route. So every arc between v and A , in either direction, must touch one single vertex a^* . Counting arc-disjoint routes from a^* to v , there are the $\mu(a^*, v)$ direct arcs, and for each $b \in B$ the two-step route $a^* \rightarrow b \rightarrow v$, which can be run $\min(\mu(a^*, b), \mu(b, v))$ times on disjoint arcs. Different b use different arcs, and none reuse a direct arc, so

$$\lambda(a^*, v) \geq \mu(a^*, v) + \sum_b \min(\mu(a^*, b), \mu(b, v)) = \mu(a^*, v) + i_B,$$

the last equality because $\mu(a^*, b) = m - 1$ is at least $\mu(b, v)$, so each minimum is $\mu(b, v)$ and the sum is i_B . Feasibility caps the left side at $m - 1$, so $\mu(a^*, v) + i_B \leq m - 1$. Every arc at v now touches either a^* or B , so $d(v) = \mu(a^*, v) + i_B + \mu(v, a^*)$, and with the first two summing to at most $m - 1$ and the last at most $m - 1$ we get $d(v) \leq 2(m - 1)$.

Suppose instead $o_A = 0$. With $o_B = 0$ already, v is now a pure sink. The same two-step count applies to every $a \in A$ at once, giving $\lambda(a, v) \geq \mu(a, v) + i_B$, hence $\mu(a, v) \leq (m - 1) - i_B$ for each of the p vertices of A . Summing over A bounds the in-flow from A by $i_A \leq p(m - 1) - p i_B$,

so

$$d(v) = i_A + i_B \leq p(m-1) - (p-1)i_B \leq p(m-1),$$

the final step because $(p-1)i_B \geq 0$.

Case II: $i_A = 0$ and $o_B \geq 1$. This is Case I read backwards. Reversing every arc turns $(m-1) \cdot B_{p,q}$ into $(m-1) \cdot B_{q,p}$, swaps in-arcs with out-arcs, and sends pattern (2) to pattern (3), so the same argument applies with the roles of A and B exchanged. It yields either $d(v) \leq 2(m-1)$, or v is a pure source for which $\lambda(v, b) \geq \mu(v, b) + o_A \leq m-1$ holds for every b , and summing over the q vertices of B gives $d(v) \leq q(m-1) - (q-1)o_A \leq q(m-1)$.

Case III: $i_A = o_B = 0$. Now v receives nothing from A and sends nothing to B , so its only arcs are the i_B in-arcs from B and the o_A out-arcs to A . For any $a \in A$ the two-step routes $a \rightarrow b \rightarrow v$ over all b give $\lambda(a, v) \geq i_B$, so $i_B \leq m-1$, and for any $b \in B$ the routes $v \rightarrow a \rightarrow b$ over all a give $\lambda(v, b) \geq o_A$, so $o_A \leq m-1$. Hence $d(v) = i_B + o_A \leq 2(m-1)$.

Every case bounds $d(v)$ by either $2(m-1)$ or $\max(p, q)(m-1)$. Since $p, q \geq 2$ we have $2(m-1) \leq (m-1)\max(p, q)$, so in all cases $d(v) \leq (m-1)\max(p, q)$. $\square$

Corollary A.35 (Attachment equality). *In the setting of Lemma A.34 with $\max(p, q) \geq 3$, suppose $d(v) = (m-1)\max(p, q)$. If $q > p$, then v is a pure source with $\mu(v, b) = m-1$ for every $b \in B$, so $D = (m-1) \cdot B_{p+1, q}$ with v joining the source side. If $p > q$, then v is symmetrically a pure sink at full multiplicity, and if $p = q$, it is one of the two.*

Proof. Rerun the cases of Lemma A.34. The mixed cases cap $d(v)$ at $2(m-1) < (m-1)\max(p, q)$. The pure-sink case gives $d(v) \leq p(m-1) - (p-1)i_B$, which reaches $(m-1)\max(p, q)$ only when $p \geq q$, $i_B = 0$ and every $\mu(a, v) = m-1$. The pure-source case symmetrically forces $o_A = 0$ and every $\mu(v, b) = m-1$ when $q \geq p$. Feasibility of the resulting digraph is the bipartite construction itself. $\square$

Theorem A.36 (Conditional odd step, $m \in \{3, 4\}$). *Fix $k \geq 4$ and $m \in \{3, 4\}$, and assume (i) $L_m^{\text{dir}}(2k) = (m-1)k^2$, and (ii) every directed multigraph attaining it is the one-directional complete bipartite multigraph $(m-1) \cdot B_{k, k}$. Then $L_m^{\text{dir}}(2k+1) = (m-1)k(k+1)$.*

Proof. The lower bound is $(m-1) \cdot B_{k, k+1}$. The upper bound is a counting squeeze. If the arc count were even one over the target, then deleting any single vertex would still leave a feasible multigraph on $2k$ vertices, which by (i) cannot exceed $(m-1)k^2$, so every vertex would have to carry a large degree. Yet the average degree is only barely above that floor, so some vertex must sit exactly on it, and deleting that one lands precisely on the even extremiser $(m-1) \cdot B_{k, k}$ of (ii). The attachment lemma then caps how much that vertex could have carried, below what the surplus requires, and the contradiction forces the surplus to vanish. We now make this precise.

For the upper bound let D on $n = 2k+1$ vertices have $\lambda^{\max} \leq m-1$ and $|A(D)| = (m-1)k(k+1) + j$ with $j \geq 1$. Deleting any vertex v leaves a feasible multigraph on $2k$ vertices,

so by (i) $d(v) \geq |A(D)| - (m-1)k^2 = (m-1)k + j$ for every v . The average degree is

$$\frac{2|A(D)|}{2k+1} = (m-1)k + j + \frac{(m-1)k - (2k-1)j}{2k+1},$$

by direct computation. If $(m-1)k - (2k-1)j < 0$ the average lies below the minimum, a contradiction. Otherwise the correction term is nonnegative and, because $m \leq 4$ and $j \geq 1$, strictly below one: $(m-1)k - (2k-1)j \leq (m-3)k + 1 \leq k + 1 < 2k + 1$. The average therefore lies in $[(m-1)k + j, (m-1)k + j + 1)$ while every degree is at least $(m-1)k + j$, so some vertex v has $d(v) = (m-1)k + j$ exactly, and $D - v$ has exactly $(m-1)k^2$ arcs. By (ii), $D - v = (m-1) \cdot B_{k,k}$, and Lemma A.34 caps $d(v) \leq (m-1)k < (m-1)k + j$, the final contradiction. Hence $j \leq 0$. $\square$

The attachment lemma bounds the degree of a vertex glued onto the *bipartite* extremiser. The linear branch has an exact analogue, worth stating once for the whole family it governs. Call a directed multigraph *everywhere-saturated* at level m when every ordered pair of distinct vertices has local connectivity exactly $m-1$. The doubled bidirected spanning trees at multiplicity $m-1$, the extremisers of the linear branch, are exactly such graphs: the $m-1$ routes between two vertices run along the doubled tree path, and cutting one doubled edge of that path is a cut of $m-1$ arcs.

Lemma A.37 (Saturated attachment). *Let $m \geq 2$ and let D_0 be a directed multigraph on at least two vertices with $\lambda_{D_0}(x, y) = m-1$ for every ordered pair of distinct vertices. Let D be obtained from D_0 by adding one new vertex v together with any multiset of arcs incident to v . If $\lambda_D^{\max} \leq m-1$, then $d(v) \leq 2(m-1)$. If $d(v) = 2(m-1)$, then every arc at v joins v to one single vertex u , with $\mu(v, u) = \mu(u, v) = m-1$, and D is again everywhere-saturated.*

Proof. No mixed pair. Suppose v has an in-arc from u and an out-arc to w with $u \neq w$. By Menger and saturation D_0 carries $m-1$ pairwise arc-disjoint u -to- w routes, and the route $u \rightarrow v \rightarrow w$ uses only arcs incident to v , so it is arc-disjoint from all of them and $\lambda_D(u, w) \geq m$, a contradiction. Hence v is a pure source, a pure sink, or every arc at v touches one single vertex u .

A pure source has $d(v) \leq m-1$. Let $t = \sum_x \mu(v, x)$ and fix u with $\mu(v, u) \geq 1$. Any cut $(S, \bar{S})$ with $v \in S$ and $u \in \bar{S}$ has capacity $\sum_{x \in \bar{S}} \mu(v, x) + e_{D_0}(S \setminus \{v\}, \bar{S})$. If S contains a vertex x of D_0 , the second term is at least $\lambda_{D_0}(x, u) = m-1$ and the first is at least $\mu(v, u)$. If $S = \{v\}$, the capacity is t . Every cut therefore has capacity at least $\mu(v, u) + \min(t - \mu(v, u), m-1)$, and by max-flow min-cut so does $\lambda_D(v, u)$, which feasibility caps at $m-1$. Were $t - \mu(v, u) \geq m-1$, this would force $\mu(v, u) = 0$, a contradiction, so $t - \mu(v, u) < m-1$ and the bound reads $t \leq m-1$.

A pure sink has $d(v) \leq m-1$. Reverse every arc: the reverse of an everywhere-saturated multigraph is everywhere-saturated, a pure sink becomes a pure source, and the previous step applies.

A single partner. If every arc at v touches one vertex u , then parallel arcs are already that many arc-disjoint one-step routes, so $\mu(v, u) \leq m-1$ and $\mu(u, v) \leq m-1$, giving $d(v) \leq$

$2(m - 1)$.

Equality. Since $2(m - 1) > m - 1$, a degree of $2(m - 1)$ rules out the pure cases, so v has a single partner at full multiplicity both ways. The result is feasible and again everywhere-saturated: a route through v must enter and leave through u , so it adds nothing to any flow between old vertices, and $\lambda_D(x, v) = \min(\lambda_{D_0}(x, u), \mu(u, v)) = m - 1$ for every old x , symmetrically for $\lambda_D(v, x)$. $\square$

Remark A.38 (The linear branch grows one leaf at a time). The equality case of [Lemma A.37](#) attaches a new leaf to an arbitrary vertex at full multiplicity, and that is the only attachment reaching $2(m - 1)$. Iterating from a single doubled edge, itself everywhere-saturated on two vertices, generates exactly the doubled bidirected spanning trees, so the extremal family of the linear branch is closed under maximal attachment and grows one leaf at a time. The program confirms the lemma exhaustively at $m = 3$: over every doubled tree on five and six vertices and every one of the 3^{2n} possible attachment patterns, the densest feasible attachment has degree exactly $4 = 2(m - 1)$, and the degree-4 patterns are precisely the single-partner attachments at full multiplicity, one per choice of partner.

Remark A.39 (The $m = 3$ chain, sharpened). [Theorem A.36](#) removes [Conjecture A.29](#) from the critical path: the odd case no longer needs a minimum-degree statement, only the extremal characterisation one level down, and for $m = 3$ that characterisation propagates along the same induction in both parities.

Odd levels ($n = 2k + 1 \geq 9$): an extremiser has $|A| = 2k(k + 1)$ arcs and every degree at least $2k$ (deletion bound). All degrees $\geq 2k + 1$ is impossible since $(2k + 1)^2 > 4k(k + 1) = \sum_v d(v)$, so some $d(v) = 2k$ exactly, $D - v$ is the even extremiser $2 \cdot B_{k,k}$, and [Corollary A.35](#) (the $p = q$ case) forces v to rejoin as a full-multiplicity pure source or sink: $D = 2 \cdot B_{k+1,k}$ or $2 \cdot B_{k,k+1}$.

Even levels ($n = 2k \geq 10$): equality in the averaging step forces $2k$ -regularity, every vertex at degree exactly $2k$, hence $D - v$ extremal at $2k - 1$ for every v . Odd uniqueness one level down leaves $2 \cdot B_{k,k-1}$ or $2 \cdot B_{k-1,k}$, and [Corollary A.35](#) with $d(v) = 2k = 2 \max(k - 1, k)$ (the $q > p$ case for $2 \cdot B_{k-1,k}$, its $p > q$ mirror for $2 \cdot B_{k,k-1}$) rebuilds exactly $2 \cdot B_{k,k}$ either way.

The seam at $n = 8$: the same regularity argument runs from $n = 7$, where the branches tie at 24 arcs and several extremisers coexist. On the linear branch the extremisers are richer than the double star alone. Every *bidirected spanning tree at full multiplicity $m - 1$* is feasible with $2(n - 1)(m - 1)$ arcs: deleting a tree edge splits the vertices in two, the only arcs from one side to the other are that edge's own $m - 1$ copies, and the $m - 1$ routes along the tree path meet that cut, so every local connectivity is exactly $m - 1$ and the doubled trees are everywhere-saturated in the sense of [Lemma A.37](#). Exhaustive enumeration confirms that at $m = 3$ these doubled trees are the *only* extremisers for $n = 4, 5, 6$ alike, with 2, 3, and 6 classes matching the tree shapes exactly: the in-house enumerator (`enumerate_extremal_directed_multigraphs`) settles $n \leq 5$, and the sound generation enumerator has now completed the uncapped $n = 6$ classification as well. A deleted vertex of the 8-regular D leaves a 24-arc extremiser with maximum degree at most 8: among doubled trees only the doubled path qualifies (tree degree ≤ 2), the double star

(hub degree 24) being excluded. The doubled path P_7 is everywhere-saturated, so [Lemma A.37](#) caps any feasible attachment to it at $d(v) \leq 2(m-1) = 4$, while 8-regularity demands $d(v) = 8$. [Figure A.5](#) draws what a degree-8 attachment would have to be, weight 4 on each path end, closing the path into the doubled bidirected cycle C_8 with $\lambda = 4 > 2$: the collapse the lemma rules out wholesale. So every deletion is bipartite-type, the attachment equality applies, and $D = 2 \cdot B_{4,4}$ again.

The whole $m = 3$ problem therefore rests on two finite integral statements about multigraphs with multiplicities in $\{0, 1, 2\}$:

- (a) $L_3^{\text{dir}}(7) = 24$, which $M^*(7) = 12$ would imply but which is not implied by it in reverse, so it is posed to the integral program directly, and
- (b) every 24-arc feasible multigraph on 7 vertices with maximum degree at most 8 is $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, or the doubled bidirected path P_7 .

Both targets sit at sizes the certifier and the enumerator can realistically reach. Fact (a) is the single feasibility query `prove_integral_arc_bound(7, 3, 25)`: the cut-counting model of [Chapter 2](#) with the weights read as integer multiplicities and the maximisation replaced by the fixed arc total 25, so that an INFEASIBLE verdict is a proof that no 25-arc feasible multigraph on seven vertices exists. The model is run with two extra families of rows (`use_deletion_cuts`), and each is a valid inequality, satisfied by every feasible weighting, so adding them speeds the solver without ever excluding a witness. The deletion bounds say that restricting a feasible weighting to any five or six of the seven vertices leaves a feasible weighting, whose total is therefore capped by the certified $M^*(5)$ or $M^*(6)$. The degree-pair inequality $d^+(s) + d^-(t) - w(s, t) \leq n - 1$ holds for every ordered pair because the direct arc and the two-step detours are arc-disjoint routes, so $\text{maxflow}(s, t) \geq w(s, t) + \sum_x \min(w(s, x), w(x, t)) \geq d^+(s) + d^-(t) - w(s, t) - (n - 2)$, the last step using $\min(a, b) \geq a + b - 1$ for weights in $[0, 1]$, and feasibility caps the left side at one. Fact (b) is the run `enumerate_extremal_directed_multigraphs_via_generation(7, 3, 24, max_degree=8)` on the generation enumerator of [Chapter 2](#), which prunes with proved bounds only and is therefore a sound complete search at $n = 7$, where the in-house depth-first twin is not. That run has now completed, in about seven hours across ten processor cores, and it returns exactly three isomorphism classes: $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, and the doubled bidirected path P_7 , each re-verified by the exact checker. *Fact (b) is therefore settled by machine*, and the whole $m = 3$ directed multigraph problem now rests on fact (a) alone. The $n = 6$ companion run (target 20, maximum degree 8) completed the same way and returns exactly one class, the doubled bidirected path P_6 , the pattern the seam argument uses one level up. The uncapped $n = 6$ classification also pays directly into fact (a): a hypothetical 25-arc feasible multigraph on 7 vertices has every degree at least 5 by the deletion bound against $L_3^{\text{dir}}(6) = 20$, and a vertex of degree exactly 5 would delete to a 20-arc extremiser on six vertices, a doubled tree, to which [Lemma A.37](#) forbids any attachment of more than 4 arcs. Any witness against fact (a) therefore has minimum total degree at least 6.

For $m \geq 4$ one hole remains: at an odd extremal level the counting no longer forces a vertex of degree $(m-1)k$ (all degrees $\geq (m-1)k+1$ is arithmetically consistent once $k(m-3) \geq 1$), so the uniqueness statement does not yet propagate through odd levels, although the *value* at odd levels follows from even uniqueness whenever that is available. For $m \geq 5$ the same escape climbs one level: a hypothetical extremiser carrying $j \geq 1$ surplus arcs could have *every* degree strictly above the deletion bound (all degrees $\geq (m-1)k+j+1$ is consistent exactly when $(2k-1)j \leq (m-1)k-2k-1$, which admits $j=1$ once $m \geq 5$), so no vertex deletion is forced to land on an extremal multigraph and even the *value* step stalls. This is why [Theorem A.36](#) is stated for $m \in \{3, 4\}$.

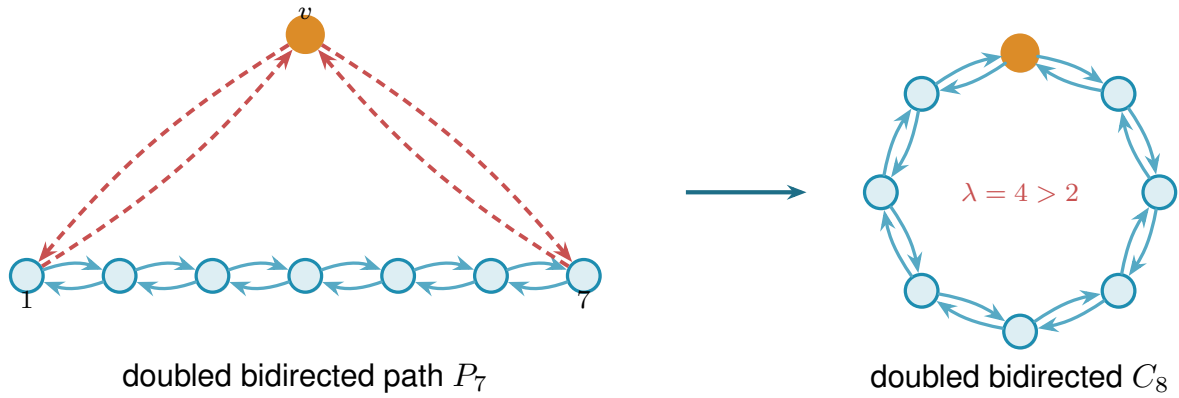


Figure A.5. The seam argument of [Remark A.39](#). A degree count forces a deleted vertex of the 8-regular extremiser D to leave the doubled bidirected path P_7 (left, each path edge a pair of opposing arcs). To reach degree 8 the new vertex v must attach with full multiplicity to the two ends 1 and 7 (red), bidirected. That closes the path into the doubled bidirected cycle C_8 (right), which is *infeasible*: the two directions around the cycle give two arc-disjoint routes each way, so $\lambda = 4 > 2$ between any pair. The contradiction rules out the path branch, forcing every deletion to be bipartite-type and hence $D = 2 \cdot B_{4,4}$.

A.6 The hypergraph bounds

The body's hypergraph claims rest on two theorems: a Menger theorem for Berge routes, which also certifies the helper-point checker of [Chapter 2](#), and a hypergraph Gomory–Hu theorem from the recent literature.

Theorem A.40 (Menger for hypergraphs). *For a hypergraph $\mathcal{H}$ and vertices u, v , the maximum number of hyperedge-disjoint Berge u – v paths equals the minimum size of a set C of hyperedges whose removal separates u from v .*

Proof. Build the helper-point network of [Figure 2.5](#): for each hyperedge e a helper arc $e^- \rightarrow e^+$ of capacity one, and arcs $x \rightarrow e^-$ and $e^+ \rightarrow x$ of unbounded capacity for each member $x \in e$. We match the two sides by translating in each direction.

Berge paths become flow. A Berge path $u, e_1, v_1, \dots, e_k, v$ becomes the directed walk $u \rightarrow e_1^- \rightarrow e_1^+ \rightarrow v_1 \rightarrow \dots \rightarrow v$, crossing one helper arc per hyperedge it uses. The e_i along a single

Berge path are distinct, and two hyperedge-disjoint Berge paths share no hyperedge at all, so they use disjoint sets of helper arcs and translate to *arc-disjoint* flow paths, paths that share no arc, each carrying one unit of flow.

Flow becomes Berge paths. Conversely, because every capacity is an integer, a maximum flow may be taken integral, and an integral flow of value k splits into k arc-disjoint unit paths. Reading a helper crossing $\cdots x \rightarrow e^- \rightarrow e^+ \rightarrow y \cdots$ back as the single hyperedge step x, e, y , that is, *suppressing* the two helper nodes, turns each unit path into a Berge path, and no two of them can share a hyperedge, since that would send two units across some capacity-one helper arc.

Cuts. Finally, the member arcs are uncapped, so any cut of finite capacity uses helper arcs only, and a set of helper arcs separates u from v in the network exactly when the corresponding hyperedges separate them in $\mathcal{H}$. The max-flow min-cut theorem equates the largest number of arc-disjoint flow paths with the smallest helper-arc cut, which is precisely the claim. $\square$

Theorem A.41 (Hypergraph Gomory–Hu, Bérczi et al. [9]). *Every r -uniform hypergraph $\mathcal{H}$ on vertex set V has a weighted tree T^* on V , its hypergraph Gomory–Hu tree, such that $\lambda_{\mathcal{H}}(u, v)$ equals the minimum weight on the u – v tree path, and each tree edge t induces a vertex partition $(S_t, V \setminus S_t)$ with $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$, where $\delta_{\mathcal{H}}(S)$ is the set of hyperedges incident to both sides, that is, having at least one vertex in S and at least one outside it.*

This is the exact hypergraph echo of [Theorem A.2](#), the same tree-of-cuts summary of all pairwise connectivities, with the single change that the capacity of a cut now counts the hyperedges straddling it, $|\delta_{\mathcal{H}}(S)|$, in place of the edges. Bérczi, Chandrasekaran, Király and Kulkarni [9] obtain the tree by the same uncrossing strategy sketched for the graph case above, applied now to the submodular hyperedge-cut function $|\delta_{\mathcal{H}}(\cdot)|$. We use only the two properties in the statement, the path-minimum reading of $\lambda_{\mathcal{H}}$ and the identity $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$.

Proof of Proposition 1.15. Upper bound. Let $\mathcal{H}$ be r -uniform with $\lambda_{\mathcal{H}}^{\max} \leq m - 1$, and let T^* be its hypergraph Gomory–Hu tree ([Theorem A.41](#)), a weighted tree on the same n vertices with weights c^* . The plan is to charge each hyperedge to tree edges and count the charges two ways, exactly as in the Mader proof, with the one change that a hyperedge spans r vertices rather than two.

Step 1: each hyperedge crosses at least $r - 1$ tree cuts. Fix a hyperedge e , a set of r vertices, and let $\text{ST}(e)$ be its *smallest connected subtree*, the minimal subtree of T^* that contains all r of them (its Steiner tree). A tree spanning at least r vertices has at least $r - 1$ edges, so $\text{ST}(e)$ has at least $r - 1$ of them. Each such edge t is a genuine cut for e : deleting t from T^* splits the vertices into the two sides $(S_t, V \setminus S_t)$, and because t lies on $\text{ST}(e)$ the hyperedge e has at least one vertex on each side. So e touches both sides, that is, $e \in \delta_{\mathcal{H}}(S_t)$. Therefore

$$|\{t \in E(T^*) : e \in \delta_{\mathcal{H}}(S_t)\}| \geq r - 1 \quad \text{for every hyperedge } e.$$

Step 2: count the incidences two ways. Sum that inequality over all hyperedges, then exchange the order of summation. This is legitimate because both of the middle sums below simply count the same pairs (e, t) with $e \in \delta_{\mathcal{H}}(S_t)$, once grouped by the hyperedge and once by the tree edge:

$$(r-1)|E(\mathcal{H})| \leq \sum_e |\{t : e \in \delta_{\mathcal{H}}(S_t)\}| = \sum_t |\{e : e \in \delta_{\mathcal{H}}(S_t)\}| = \sum_t |\delta_{\mathcal{H}}(S_t)|.$$

Step 3: read off the bound. By the Gomory–Hu property each tree edge has $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$, and each weight $c^*(t)$ is a local connectivity $\lambda_{\mathcal{H}}$ of its endpoints, hence at most $m-1$ by feasibility. Summing over the $n-1$ tree edges, $\sum_t |\delta_{\mathcal{H}}(S_t)| = \sum_t c^*(t) \leq (m-1)(n-1)$. Chaining Steps 2 and 3,

$$(r-1)|E(\mathcal{H})| \leq (m-1)(n-1),$$

and dividing by $r-1$, then rounding down because $|E(\mathcal{H})|$ is an integer, gives $|E(\mathcal{H})| \leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. *Lower bound.* When $(r-1) \mid (n-1)$, take a hub h and split the other $n-1$ vertices into groups $G_1, G_2, \dots$ of size $r-1$ each. Form the star hypertree whose hyperedges are $\{h\} \cup G_i$, one per group, and give each hyperedge multiplicity $m-1$. Every non-hub vertex lies in exactly $m-1$ hyperedges, which form a cut isolating it, so by [Theorem A.40](#) every pair has Berge connectivity at most $m-1$, while the count is $\frac{(m-1)(n-1)}{r-1}$. For general n under the hypothesis $m-1 \leq \binom{n-2}{r-2}$, [Theorem A.42](#) below attains the floor exactly, without even needing repeated hyperedges. $\square$

The multiplicities in the lower bound can in fact be avoided: for $r \geq 3$, *simple* r -uniform hypergraphs already attain the same bound, in sharp contrast to the graph case $r = 2$, where simplicity costs the factor that separates Mader’s $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$.

Theorem A.42 (Simplicity is free for $r \geq 3$). *Let $r \geq 3$, $n \geq r$, and $2 \leq m$ with $m-1 \leq \binom{n-2}{r-2}$. Then the maximum number of hyperedges in a simple r -uniform hypergraph on n vertices with Berge edge-connectivity at most $m-1$ is exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$.*

Proof. The upper bound is [Proposition 1.15](#), since a simple hypergraph is a multihypergraph. For the lower bound, fix a hub h and let $V' = V \setminus \{h\}$. By [Lemma A.44](#) applied with rank $r-1$ and degree ceiling $d = m-1$ on the $n-1$ vertices of V' , there is an $(r-1)$ -uniform simple hypergraph $\mathcal{G}^*$ on V' with maximum degree at most $m-1$ and exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. Adding h to each hyperedge produces distinct r -sets, so the result $\mathcal{H}$ is simple, and every non-hub vertex w lies in $\deg_{\mathcal{G}^*}(w) \leq m-1$ hyperedges, which form a cut isolating w . By [Theorem A.40](#), $\lambda_{\mathcal{H}}^{\max} \leq m-1$. $\square$

The sparse hypergraph the construction needs is a corollary of a classical partition theorem, which we state before using it.

Theorem A.43 (Baranyai [16]). *If r divides n , the $\binom{n}{r}$ distinct r -subsets of an n -element set can be partitioned into $\binom{n-1}{r-1}$ classes, each a perfect matching, meaning a family of n/r disjoint r -sets that together cover every vertex exactly once. More generally, for any prescribed class sizes $a_1 + \dots + a_t = \binom{n}{r}$, and with no divisibility assumed, the r -subsets can be partitioned into classes of exactly those sizes so that in the class of size a_i every vertex lies in either $\lfloor ra_i/n \rfloor$ or $\lceil ra_i/n \rceil$ of the sets, as evenly balanced as the size permits.*

The first statement is the memorable one. When r divides n the complete r -uniform hypergraph, the one holding every r -set, comes apart into $\binom{n-1}{r-1}$ perfect matchings, the exact hypergraph analogue of splitting a regular bipartite graph into perfect matchings by König's theorem. The proof is an induction that keeps the partial partition as balanced as possible at every step, an integer-flow argument in the spirit of Hall's marriage theorem rather than an explicit construction, and its full form is in Baranyai [16]. We need only the general equitable form, and only its conclusion about degrees.

Lemma A.44 (Sparse uniform hypergraphs exist). *Let $r \geq 2$, $n \geq r$, and $0 \leq d \leq \binom{n-1}{r-1}$. There exists an r -uniform simple hypergraph on n vertices with maximum degree at most d and exactly $\lfloor \frac{dn}{r} \rfloor$ hyperedges.*

Proof. Put $e := \lfloor \frac{dn}{r} \rfloor \leq \frac{n}{r} \binom{n-1}{r-1} = \binom{n}{r}$. Apply the equitable form of Theorem A.43 with two classes, of sizes e and $\binom{n}{r} - e$. Within each class every vertex is covered either $\lfloor re/n \rfloor$ or $\lceil re/n \rceil$ times, so the class of size e is a simple r -uniform hypergraph with $e = \lfloor \frac{dn}{r} \rfloor$ hyperedges and maximum degree at most $\lceil re/n \rceil \leq \lceil d \rceil = d$, the last step because $re/n \leq r(dn/r)/n = d$ and d is an integer. $\square$

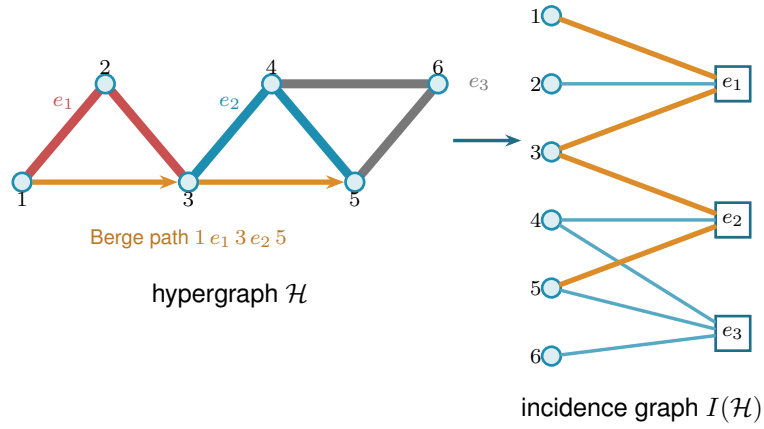
A.7 The hypergraph vertex problem

The whole vertex problem translates into a graph problem through one observation, the one the rest of this section rests on and which Figure A.6 draws in full. Write $I(\mathcal{H})$ for the bipartite incidence graph of a hypergraph $\mathcal{H}$: one node per vertex, one node per hyperedge, an edge when the vertex lies in the hyperedge. A Berge path with distinct hyperedges is exactly a path of $I(\mathcal{H})$ between two vertex-class nodes. The interior of such a path consists of both kinds of node, the hyperedges it rides and the vertices where it changes hyperedge, so two paths of $I(\mathcal{H})$ are internally disjoint in the ordinary graph sense exactly when the two Berge routes share neither a hyperedge nor an intermediate vertex. That is precisely the separation fixed in Chapter 1, and it is the reason both halves were required there. Hence

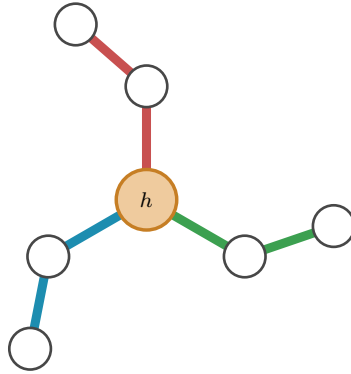
$$\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v),$$

the maximum number of internally disjoint u - v paths in $I(\mathcal{H})$, and the hypergraph vertex problem asks for the maximum number of degree- r nodes on one side of a bipartite graph whose other side holds n nodes, subject to no two of those n nodes being joined by m internally disjoint paths.

Figure A.6a traces a single Berge path through both pictures, and Figure A.6b shows the star hypertree of Theorem A.45, whose incidence graph is a tree.



(a) A 3-uniform hypergraph $\mathcal{H}$ (left) and its bipartite incidence graph $I(\mathcal{H})$ (right): circles for vertices, hollow squares for hyperedges, an edge when the vertex lies in the hyperedge. The Berge path $1-e_1-3-e_2-5$ is traced in orange in both pictures. In $\mathcal{H}$ it rides the red line e_1 and the blue line e_2 , changing at the shared vertex 3, while the grey line e_3 lies off it. In $I(\mathcal{H})$ it is the ordinary path $1-e_1-3-e_2-5$ alternating sides.



(b) A star hypertree: every hyperedge $\{h\} \cup G_i$ shares only the hub h (orange), and the other $r-1=2$ vertices of each block sit in a single hyperedge. Its incidence graph is a tree, so $\kappa^{\max} \leq 1$.

Figure A.6. The translation the vertex problem rests on. Internally vertex-disjoint Berge paths in $\mathcal{H}$ correspond exactly to internally disjoint paths in $I(\mathcal{H})$, so $\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v)$. **(top)** A hypergraph and its incidence graph with one Berge path drawn in both. **(bottom)** The star hypertree of Theorem A.45, whose incidence graph is a tree.

Theorem A.45 (Hypergraph vertex value at $m = 2$). *For $r \geq 2$ and $n \geq 1$, the maximum number of hyperedges of an r -uniform multihypergraph on n vertices with $\kappa^{\max} \leq 1$ is*

$$k_2^{(r)}(n) = \left\lfloor \frac{n-1}{r-1} \right\rfloor,$$

attained by the star hypertree. In particular $k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m = 2$ for every uniformity, and repeated hyperedges never help.

Proof. We claim $\kappa_{\mathcal{H}}^{\max} \leq 1$ holds if and only if $I(\mathcal{H})$ is a forest. If $I(\mathcal{H})$ contains a cycle $v_1, e_1, v_2, e_2, \dots, v_\ell, e_\ell, v_1$ (it alternates classes, so $\ell \geq 2$, all v_i and e_i distinct), then v_1, e_1, v_2 and $v_1, e_\ell, v_\ell, e_{\ell-1}, \dots, e_2, v_2$ are two Berge v_1-v_2 paths with disjoint hyperedge sets and disjoint

interior vertex sets, so $\kappa(v_1, v_2) \geq 2$. (At $\ell = 2$ this is the repeated pair: two hyperedges sharing two vertices.) Conversely, two such paths between u and v concatenate to a cycle of $I(\mathcal{H})$. This proves the claim. A forest on $n + q$ nodes has at most $n + q - 1$ edges, while $I(\mathcal{H})$ has exactly rq edges, so $rq \leq n + q - 1$, i.e. $q \leq (n - 1)/(r - 1)$, and q is an integer. The star hypertree has $\lfloor (n - 1)/(r - 1) \rfloor$ hyperedges and a tree as incidence graph (each block vertex lies in one hyperedge, every cycle would need two hyperedges sharing two vertices), so the floor is attained. $\square$

Proposition A.46 (General lower bound). *For $r \geq 3$, $m \geq 2$ and $m - 1 \leq \binom{n-2}{r-2}$,*

$$k_m^{(r)}(n) \geq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

attained by a simple hypergraph.

Proof. The simple construction of [Theorem A.42](#) has Berge edge-connectivity at most $m - 1$, and Whitney's inequality $\kappa \leq \lambda$ holds pairwise (an internally vertex-disjoint family is in particular hyperedge-disjoint under our definition), so the same hypergraph is feasible for the vertex problem. $\square$

The $m = 3$ case also closes completely. The engine is a rank bound for graphs in which one side of a partition must stay below local 3-connectivity. The hypergraph theorem then falls out of the incidence translation. Throughout, $\text{rank}(G) = |E(G)| - |V(G)| + 1$ denotes the cycle rank of a connected multigraph, that is, the number of independent cycles it carries. A tree has cycle rank 0, and each edge added beyond a spanning tree closes exactly one new cycle and raises the rank by one, so bounding the rank is the same as bounding how many independent cycles the graph can hold.

Primer: global connectivity and the pieces of a graph

The lemma below speaks of a graph being 2-connected or 3-connected and of taking it apart at small separators. These are the global cousins of the local $\kappa(u, v)$ from [Chapter 1](#), so we fix the words once. A graph on more than k vertices is *k-connected* if it stays connected after the removal of any $k - 1$ vertices. By Menger this is the same as asking that every pair of vertices be joined by at least k internally vertex-disjoint paths, so *k-connected* means $\kappa(u, v) \geq k$ for all pairs at once, the uniform version of the local measure. Thus 2-connected means no single vertex disconnects the graph, and 3-connected means no two vertices do. Three named obstructions go with this. A *cut vertex* is a vertex whose removal disconnects the graph, so a connected graph on at least three vertices is 2-connected exactly when it has no cut vertex. A *separating pair* is a set of two vertices whose removal disconnects the graph, the obstruction a 2-connected graph must lack in order to be 3-connected. Finally a *block* is a maximal piece with no cut vertex of its own, hence either a single bridge edge or a maximal 2-connected subgraph. Two blocks overlap in at most one vertex, always a cut

vertex, and the blocks and cut vertices hang together in a tree, the *block-cut tree*, with one node per block and one per cut vertex. Step 1 of the proof is precisely an induction over that tree.

Lemma A.47 (Incidence rank lemma). *Let G be a connected multigraph with vertex set $X \cup Z$ such that (i) Z is independent, (ii) no edge incident to Z is parallel to another edge, (iii) every $z \in Z$ has degree at least 2, and (iv) $\kappa_G(x, x') \leq 2$ for all distinct $x, x' \in X$. Then $\text{rank}(G) \leq |X| - 1$.*

Proof. Induction on $|V| + |E|$. The proof peels G down to a core that cannot exist. Step 1 handles a cut vertex by arguing block by block. Steps 2 and 3 remove a degree-two Z -vertex or X -vertex without changing the bound. What survives is 2-connected with minimum degree at least three, and Step 4 shows that such a graph already violates the connectivity ceiling (iv), so it never arises. *Base* $|V| \leq 2$. A single vertex lies in X by (iii), and $0 \leq |X| - 1$. On two vertices, (i)–(iii) force both into X , parallel edges are internally disjoint paths, so (iv) caps the multiplicity at 2 and $\text{rank} \leq 1 = |X| - 1$.

Step 1: a cut vertex. Suppose G has a cut vertex, so it is not yet 2-connected. Break it into its *blocks* $B_1, \dots, B_c$ with $c \geq 2$, a block being a maximal piece with no cut vertex of its own, that is, either a single bridge edge or a maximal 2-connected subgraph. Two blocks meet in at most one vertex, and any shared vertex is a cut vertex of G . No cycle can pass through two different blocks, so the cycle rank adds up over blocks, $\text{rank}(G) = \sum_i \text{rank}(B_i)$, and it is enough to bound each $\text{rank}(B_i)$ by $|X \cap B_i| - 1$ and sum.

Each block inherits the hypotheses. A bridge block is one edge, so its rank is 0, and by (i) at least one endpoint lies in X , giving $0 \leq |X \cap B| - 1$. Every other block is 2-connected, hence has minimum degree at least 2, so (i) to (iv) hold inside it and the induction hypothesis gives $\text{rank}(B_i) \leq |X \cap B_i| - 1$.

Adding these requires care, because a cut vertex is shared by several blocks and would be counted several times. Write $b(v)$ for the number of blocks containing v , so $b(v) = 1$ except at cut vertices. Counting each X -vertex once for every block it lies in,

$$\sum_i |X \cap B_i| = |X| + \sum_{x \in X \text{ cut}} (b(x) - 1).$$

The number of blocks satisfies the block–cut tree identity $c = 1 + \sum_{v \text{ cut}} (b(v) - 1)$, which holds because the blocks and cut vertices form a tree in which each cut vertex v is joined to the $b(v)$ blocks that contain it. Subtracting this count of blocks from the previous display and separating the cut vertices into their X - and Z -parts leaves

$$\text{rank}(G) = \sum_i \text{rank}(B_i) \leq \sum_i (|X \cap B_i| - 1) = |X| - 1 - \sum_{z \in Z \text{ cut}} (b(z) - 1) \leq |X| - 1,$$

where the last inequality holds because each $b(z) - 1 \geq 0$, so the Z -cut sum only pushes the bound further below $|X| - 1$.

Step 2: a Z -vertex of degree two. From now on G is 2-connected, every cut vertex having been handled. Suppose some $z_0 \in Z$ has degree exactly 2. Both its neighbours lie in X by (i), and they are distinct, since two edges to the same vertex would be parallel at a Z -vertex, which (ii) forbids. *Suppress* z_0 by deleting it and joining its two neighbours x, x' with one new edge. This loses two edges and one vertex and gains one edge, so $|E|$ and $|V|$ each drop by one and $\text{rank} = |E| - |V| + 1$ is unchanged. The new edge runs X to X , so $|X|$ is unchanged and (i), (ii), (iii) still hold for the remaining Z -vertices. Replacing the path x, z_0, x' by a direct edge alters no route among the surviving vertices, so every $\kappa(x_1, x_2)$ is preserved and (iv) holds. The graph is strictly smaller, so the induction hypothesis applies and returns the bound for G .

Step 3: an X -vertex of degree two. Now G is 2-connected and, Step 2 being used up, every Z -vertex has degree at least 3. Suppose some $x_0 \in X$ has degree 2, and delete it. Deleting one vertex from a 2-connected graph leaves it connected. Stripping a degree-2 vertex removes two edges and one vertex, so rank drops by exactly one. Each neighbour of x_0 loses one edge, and a neighbour in Z only falls from degree at least 3 to degree at least 2, so (iii) survives, while deleting a vertex cannot raise a connectivity, so (iv) survives. Here $|X|$ drops by one, and $|X| \geq 2$ to begin with, for if $|X| \leq 1$ then by (i) and (ii) every z would send all its edges to one X -vertex without repeats, hence have degree at most 1, against (iii). The smaller graph satisfies the hypotheses, so induction gives $\text{rank}(G) - 1 \leq (|X| - 1) - 1$, that is $\text{rank}(G) \leq |X| - 1$.

Step 4: the remaining case is impossible. Now G is 2-connected with $|V| \geq 3$ and every degree at least 3. First, G is simple: a parallel class lies inside X by (ii), and if $\mu(x, x') \geq 2$ then a third internally disjoint route exists: walk from x to any third vertex w inside $G - x'$, then from w to x' inside $G - x$. The concatenation visits x only first and x' only last, so it contains an x - x' path with at least one interior vertex, giving $\kappa(x, x') \geq 3$, against (iv). If G is 3-connected, Menger puts every pair at $\kappa \geq 3$, so $|X| \leq 1$ and every z has degree at most 1 by (i), absurd.

Otherwise G is 2-connected but not 3-connected, so it can be taken apart at its 2-vertex separators.

Primer: the triconnected (SPQR) decomposition

A 2-connected graph that is not 3-connected must have a separating pair, and cutting along every such pair sorts any 2-connected graph into a few standard shapes. This is a classical theorem of Tutte [17], made into a linear-time algorithm by Hopcroft and Tarjan [18].

Theorem (triconnected decomposition). Every 2-connected graph G on at least three vertices decomposes into a *tree of pieces*, unique up to relabelling, in which each piece is one of three kinds: a *cycle* of length at least three (an *S*-piece, for *series*), a *bond*, meaning two vertices joined by three or more parallel edges (a *P*-piece, for *parallel*), or a 3-connected simple graph on at least four vertices (an *R*-piece, for *rigid*). Two adjacent pieces share exactly one *virtual edge* $\{a, b\}$, a placeholder that records a separating pair of G at which the two pieces were split apart. The three initials *S*, *P*, *R*, together with *Q* for a trivial single-edge piece, give the structure its usual name, the *SPQR tree*.

The tree is built by the recursion that also defines it. If G is already a cycle, a bond, or 3-connected, it is a single piece and the tree is one node. Otherwise choose a separating pair $\{a, b\}$, split G into the parts that this pair separates, add the virtual edge $\{a, b\}$ to each part so the split is remembered, and recurse on the parts. Every split makes the parts strictly smaller, so the recursion halts, and Tutte's theorem is that the resulting collection of pieces does not depend on the order in which the separating pairs are taken. The one fact the proof draws from all this is that a virtual edge $\{a, b\}$ always stands for a genuine a – b path through the rest of G , so an argument that must cross from a to b outside a given piece can always do so on the far side of that piece's virtual edge.

The decomposition just stated is sketched in [Figure A.7](#), which also marks the leaf at which the argument bites. Two properties are all the proof needs from it, and both are visible in the figure. First, a vertex of a piece that touches none of that piece's virtual edges belongs to that piece alone, so it keeps in G exactly the degree it has in the piece. Second, for any virtual edge $\{a, b\}$ the part of G on the far side of the cut contains a genuine a – b path whose interior avoids the piece, so the virtual edge can always be replaced by a real route.

Now G itself is none of the three piece types, since it is not a single cycle (its degrees are at least three), not a bond (it is simple), and not 3-connected (the case just handled), so the tree has at least two leaves. Let L be one, with unique virtual edge $\{a, b\}$. If L is a cycle, a vertex of L off $\{a, b\}$ has degree 2 in G , absurd. If L is a bond, it holds at least two real parallel edges, absurd in a simple graph. If L is 3-connected, then any two of its vertices are joined by three internally disjoint paths inside L , at most one through the virtual edge, which the far-side path replaces: so $\kappa_G(u, v) \geq 3$ for all $u, v \in V(L)$, forcing $|X \cap V(L)| \leq 1$ by (iv). But $|V(L)| \geq 4$ leaves a Z -vertex $z^* \notin \{a, b\}$. Its neighbours lie in $X \cap V(L)$ by (i), at most one vertex, reached by at most one edge since G is simple, giving degree at most 1, which is absurd. $\square$

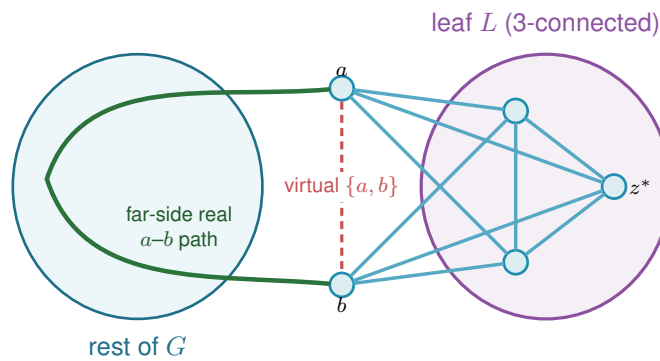


Figure A.7. The triconnected (Tutte/SPQR) decomposition in Step 4 of [Lemma A.47](#), schematic: G splits into a tree of pieces (blobs) glued along virtual edges, and the analysis works at a *leaf* L with its single virtual edge $\{a, b\}$ (dashed red). The far side of the tree supplies a *real* a – b path (green) that internally avoids L . If L is 3-connected, any two of its vertices are joined by three internally disjoint paths inside L , at most one using the virtual edge, and the far-side path replaces it, so $\kappa_G \geq 3$ for every pair in L . With $|V(L)| \geq 4$ that forces a hyperedge-node $z^* \notin \{a, b\}$ of degree at most one, which is the contradiction that closes the lemma.

Theorem A.48 (Hypergraph vertex value at $m = 3$). *For $r \geq 2$, every r -uniform multihypergraph on n vertices with $\kappa^{\max} \leq 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $r \geq 3$ with $2 \leq \binom{n-2}{r-2}$ the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) = \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor = \ell_3^{(r)}(n).$$

Proof. Apply [Lemma A.47](#) to each connected component of the incidence graph $I(\mathcal{H})$, with X the vertex class and Z the hyperedge class: Z is independent, incidences are simple even when hyperedges repeat, hyperedge-nodes have degree $r \geq 2$, components without an X -node are impossible, and $\kappa_I(u, v) = \kappa_{\mathcal{H}}(u, v) \leq 2$ for vertex pairs. Sum $\text{rank} \leq |X_c| - 1$ over the C components. The incidence graph $I(\mathcal{H})$ has rq edges (each of the q hyperedges contributes r incidences) and $n + q$ nodes, so $\sum_c \text{rank} = rq - (n + q) + C$, while $\sum_c (|X_c| - 1) = n - C$. The inequality reads $rq - (n + q) + C \leq n - C$, hence $q(r - 1) \leq 2(n - 1) + 2(1 - C) \leq 2(n - 1)$, the last step using $C \geq 1$. The lower bound is [Proposition A.46](#) at $m = 3$. $\square$

Remark A.49 (Edges as gates at $r = 2$, and the boundary of the method). At $r = 2$ the theorem speaks of multigraphs under the hyperedge-as-gate convention, in which parallel edges are distinct one-step routes, giving the multigraph value $2(n - 1)$ of [Theorem 1.3](#), complementing the body's convention for graphs, where parallel adjacencies count once. The proof of [Lemma A.47](#) leans on the triconnected decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$, the analogous question asks for a rank bound relative to $\kappa \leq 3$, where no comparably clean decomposition exists. That the pattern $k_m^{(r)} = \ell_m^{(r)}$ must fail eventually is not merely an analogy with the graph case ($m = 5$, Sørensen-Thomassen), which is stated for simple graphs under the body's convention. It already fails inside this convention, at $r = 2$. The identification in the first sentence of this remark runs both ways: the $r = 2$ case of the present problem is the multigraph vertex problem of [Section A.8](#), since a route there is a family of edge-disjoint and internally disjoint paths, which is exactly [Lemma A.54](#). So [Theorem A.56](#) applies to it, and the value exceeds $\lfloor (m - 1)(n - 1)/(r - 1) \rfloor = (m - 1)(n - 1)$ for every $m \geq 5$ and every $n \geq 3$, by a margin that grows with n . The formula is therefore not a candidate for all m and r together, and the live question is only whether $r \geq 3$ postpones the failure past $m = 4$.

For $m = 4$ it does not fail anywhere within reach. An exhaustive search over r -uniform multihypergraphs confirms both halves of the value, that $\lfloor 3(n - 1)/(r - 1) \rfloor$ is attained and that one more hyperedge is infeasible, at $(n, r) = (4, 3), (5, 3), (5, 4), (6, 4), (6, 5), (7, 5), (7, 6), (8, 6), (8, 7), (9, 7)$ and $(9, 8)$. That is a wider net than the single cell $k_4^{(3)}(5) = 6$, and it is evidence rather than proof: the missing ingredient remains the 4-connectivity rank bound, and the $r = 2$ failure at $m = 5$ shows the surrounding claim has a genuine boundary rather than extending indefinitely.

Proposition A.50 (Directed hypergraphs: first bounds). *Model a directed r -uniform hyperedge as a tail vertex together with $r - 1$ head vertices, a Berge route entering at the tail and leaving at a head (the program's convention, drawn in [Figure A.8](#)). If every pair of vertices has at most*

$m - 1$ arc-disjoint directed Berge routes, then

$$|E(\mathcal{H})| \leq \frac{(m - 1) n(n - 1)}{r - 1},$$

and for $m - 1 \leq \binom{n - \lceil n/2 \rceil - 1}{r - 2}$ there are feasible constructions with $\max_{1 \leq \alpha \leq n - 1} \alpha \left\lfloor \frac{(m - 1)(n - \alpha)}{r - 1} \right\rfloor \approx (m - 1)n^2 / (4(r - 1))$ hyperedges, so the value is quadratic in n .

Proof. Upper bound: a hyperedge with tail u serves the $r - 1$ ordered pairs (u, h) , h a head. If some ordered pair (u, v) were served by m distinct hyperedges, those would be m arc-disjoint one-step routes, so each ordered pair is served at most $m - 1$ times and $(r - 1)|E| \leq (m - 1)n(n - 1)$. Lower bound: split V into tails A , $|A| = \alpha$, and heads B . By Lemma A.44 there is a simple $(r - 1)$ -uniform hypergraph $\mathcal{G}^*$ on B with maximum degree $m - 1$ and $\lfloor (m - 1)|B| / (r - 1) \rfloor$ edges. Give every tail $a \in A$ the hyperedges (a, H) , $H \in \mathcal{G}^*$. No vertex of B is a tail, so every route is a single step and $\lambda(a, b) = \deg_{\mathcal{G}^*}(b) \leq m - 1$, while pairs inside A or starting in B have no routes. $\square$



Figure A.8. How the thesis draws a single hyperedge on its r vertices, in the metro colours of Figure 1.3. *Left:* an undirected hyperedge is one metro line threading all its members, with no arrows. *Right:* a directed hyperedge, a tail with $r - 1$ heads, is a star, the tail sending one thick arc to each head, so a Berge route enters at the tail and leaves at any head. The hypergraph panels of Figure 1.4 use this same star. The shape is a drawing convention, not a path. Each coloured line or star is a *single* edge on several vertices, not a sequence of ordinary edges, so the line on the left is one hyperedge and not three. One colour means one hyperedge, and where two of them meet they cross at a shared station rather than joining into a longer line. What might be mistaken for a hyperedge is a *route*, which does run edge by edge from one vertex to another, but a route is drawn as a thin thread laid over the lines, so the thick coloured hyperedge and the thin route never look alike.

The forward model is one of three ways to orient a Berge edge. In general a directed r -uniform hyperedge is a split of its r vertices into a non-empty *tail set* T and a non-empty *head set* H , a route entering at a tail and leaving at a head. The *forward* model is $|T| = 1$, the *backward* model is $|H| = 1$, and the *general* model allows any split (genuinely mixed splits, both parts at least two, exist from $r \geq 4$). The next two results place all three. The first is that backward needs no separate study, and the second is that the general model, despite admitting more hyperedges on a single vertex set, obeys the very same first-order bound as forward.

Lemma A.51 (Backward is the reverse of forward). *Reversing every hyperarc, $(T, H) \mapsto (H, T)$, is an edge-count-preserving bijection between the forward directed r -uniform hypergraphs and the backward ones on the same vertices, and the reverse $\mathcal{H}^R$ satisfies $\lambda_{\mathcal{H}}(u, v) = \lambda_{\mathcal{H}^R}(v, u)$*

and $\kappa_{\mathcal{H}}(u, v) = \kappa_{\mathcal{H}^R}(v, u)$. Hence $\lambda^{\max}$ and $\kappa^{\max}$ are invariant under reversal, and the backward edge and vertex extremal numbers equal the forward ones, for every r , m and n .

Proof. Reversal sends a backward hyperarc ($|T| = r - 1$, $|H| = 1$) to a forward one ($|T| = 1$) and is its own inverse, so it is an edge-count-preserving bijection of the two families. Let $u = x_0, e_1, x_1, \dots, e_k, x_k = v$ be a directed Berge route in $\mathcal{H}$, where x_{i-1} is a tail and x_i a head of e_i . The reversed edge $e_i^R = (H_i, T_i)$ has x_i as a tail and x_{i-1} as a head, so $v = x_k, e_k^R, x_{k-1}, \dots, e_1^R, x_0 = u$ is a directed Berge route from v to u in $\mathcal{H}^R$ on the same edges and the same internal vertices. This is a bijection between u - v routes in $\mathcal{H}$ and v - u routes in $\mathcal{H}^R$ that preserves which edges and which internal vertices any two routes share, so it carries an edge-disjoint (respectively internally vertex-disjoint) family to one of equal size. Therefore $\lambda_{\mathcal{H}}(u, v) = \lambda_{\mathcal{H}^R}(v, u)$ and $\kappa_{\mathcal{H}}(u, v) = \kappa_{\mathcal{H}^R}(v, u)$, and maximising over ordered pairs gives $\lambda^{\max}(\mathcal{H}) = \lambda^{\max}(\mathcal{H}^R)$ and the same for κ . Since reversal is a bijection preserving feasibility and edge count, the two models share every extremal number. $\square$

Proposition A.52 (The general model obeys the forward bound). *Let a directed r -uniform hyperedge be any split into a non-empty tail set T and head set H . If every ordered pair has at most $m - 1$ arc-disjoint, or at most $m - 1$ internally vertex-disjoint, directed Berge routes, then*

$$|E(\mathcal{H})| \leq \frac{(m - 1) n(n - 1)}{r - 1},$$

the same bound as the forward model of Proposition A.50. The forward construction of that proposition is itself a set of general hyperedges, so the general value obeys the same upper bound and inherits the same quadratic lower bound, and in particular is $\Theta(n^2)$.

The two models are therefore trapped between the same pair of bounds, which differ by a factor of 4. That is enough to fix the *order* of both values and not enough to fix either leading constant, so it does not follow that the forward and general models agree asymptotically, only that any disagreement between them lives inside that factor. Determining $\lim_{n \rightarrow \infty} \text{ex}(n)/n^2$ for even one of the two, and then asking whether the two limits coincide, is the open question [Chapter 4](#) records.

It is worth saying which end of that gap is the likely culprit, because the two bounds are not equally crude and the analogy with the simple digraph case is informative. The upper bound charges each hyperedge to the ordered tail-head pairs it occupies and then allows all $n(n - 1)$ ordered pairs to be used to capacity. The construction uses only the $\alpha(n - \alpha) \leq n^2/4$ pairs that run from the tail class to the head class, and leaves the pairs inside each class, and those running backwards, entirely idle. The factor of four is exactly that difference, n^2 against $n^2/4$, and it is the same difference that appears in the simple directed problem, where [Construction 1.13](#) also uses only a one-directional bipartite pattern. There, [Theorem A.24](#) now proves that the bipartite pattern is right to first order, that no arrangement recovers the idle pairs. If that phenomenon is a feature of directedness rather than of graphs specifically, then it is the upper bound here that

should come down by a factor of four, not the construction that should climb, which suggests the following.

Conjecture A.53 (The directed hypergraph constant). *For fixed $r \geq 3$ and $m \geq 2$, the maximum number of hyperedges of a directed r -uniform hypergraph on n vertices with $\lambda^{\max} \leq m - 1$ is $(1 + o(1)) (m - 1)n^2 / (4(r - 1))$, so the bipartite construction of [Proposition A.50](#) is asymptotically optimal.*

The proof of [Theorem A.24](#) does not transfer directly, and the place it breaks is worth recording for anyone who takes this up. There, two-step routes through distinct midpoints were automatically arc-disjoint. Here a single hyperedge carries $r - 1$ heads at once, so two two-step routes with different midpoints can enter through the very same hyperedge, and the family of two-step routes is no longer automatically edge-disjoint. Recovering a packing of them, or replacing the count by one that tolerates the sharing, is the technical step this conjecture waits on.

Proof. Fix an ordered pair (u, v) . A hyperedge (T, H) with $u \in T$ and $v \in H$ carries the one-step route $u \rightarrow v$ that enters at the tail u and leaves at the head v . Distinct such hyperedges give routes that share no edge and no internal vertex, so if k hyperedges had u in their tail set and v in their head set then $\lambda(u, v) \geq k$ and $\kappa(u, v) \geq k$, so feasibility forces $k \leq m - 1$. Each hyperedge (T, H) is counted by exactly the $|T| |H|$ ordered pairs (u, v) with $u \in T$ and $v \in H$, so summing the per-pair bound over all pairs gives $\sum_{(T, H)} |T| |H| \leq (m - 1)n(n - 1)$. As $|T| + |H| = r$ with both parts at least one, $|T| |H| \geq 1 \cdot (r - 1) = r - 1$, whence $(r - 1)|E| \leq (m - 1)n(n - 1)$. The forward family of [Proposition A.50](#) consists of general hyperedges, so it witnesses the same $\approx (m - 1)n^2 / (4(r - 1))$ lower bound. $\square$

A.8 The multigraph vertex problem under the other convention

[Remark 2.1](#) collapses parallel copies in vertex mode, which makes the two multigraph vertex variants restatements of their simple counterparts and is why they carry no theorem of their own. This section takes the other reading seriously, because under it the question is genuinely new, and records what can be proved and what the machine finds. Throughout, q parallel copies of the edge uv count as q internally vertex-disjoint u - v routes, since a single edge is a path with no interior. Write $K_m^{\text{multi}}(n)$ for the largest total multiplicity of a multigraph on n vertices with $\kappa^{\max} \leq m - 1$ under this reading.

The first thing to notice is that the two kinds of route never interfere, which splits the measure cleanly in two.

Lemma A.54 (Splitting the vertex measure). *Let G be a multigraph with multiplicity function μ and underlying simple graph G_0 . For any two vertices $u \neq v$, write $\pi(u, v)$ for the largest number of pairwise internally disjoint u - v paths of length at least two in G_0 . Then, under the convention above,*

$$\kappa_G(u, v) = \mu(u, v) + \pi(u, v).$$

Proof. A parallel copy of uv is a path with empty interior, so any two copies are internally disjoint from each other and from every longer route. A maximum family may therefore be assumed to contain all $\mu(u, v)$ copies, and what it can add to them is a family of pairwise internally disjoint routes of length at least two, whose interiors are disjoint sets of vertices. Since parallel copies are irrelevant to a route of length at least two, which is determined by its vertex sequence, the largest such family has size $\pi(u, v)$, computed in G_0 . $\square$

So feasibility says exactly that $\mu(u, v) \leq m - 1 - \pi(u, v)$ on every edge, and that $\pi(u, v) \leq m - 1$ on every pair. At $m = 2$ this is already decisive.

Theorem A.55 (The other convention at $m = 2$). $K_2^{\text{multi}}(n) = n - 1$, attained exactly by the spanning trees.

Proof. By Lemma A.54, $\kappa^{\max} \leq 1$ forces $\mu(u, v) + \pi(u, v) \leq 1$ on every edge. If G_0 contained a cycle, an edge uv of that cycle would have $\mu(u, v) \geq 1$ and $\pi(u, v) \geq 1$ from the rest of the cycle, a contradiction, so G_0 is a forest and every multiplicity is one. A forest has at most $n - 1$ edges, and a spanning tree attains it. $\square$

For $m \geq 3$ at least three constructions compete, and which one wins depends on how n compares with m , exactly as in every other variant of this thesis.

- ★ *The thickened tree.* Take a spanning tree and give every edge multiplicity $m - 1$. Tree edges have $\pi = 0$ and non-adjacent pairs have $\pi = 1$, so this is feasible for every $m \geq 2$, and it carries $(m - 1)(n - 1)$ edges, the multigraph edge value of Theorem 1.3.
- ★ *The thickened complete graph.* Give every edge of K_n the same multiplicity q . Here $\pi(u, v) = n - 2$, so feasibility asks $q \leq m + 1 - n$, and the count is $\binom{n}{2}(m + 1 - n)$, positive only when $n \leq m + 1$.
- ★ *The thickened theta.* Choose two poles and join each of the $n - 2$ remaining vertices to both of them at multiplicity $m - 2$, then join the poles to each other at multiplicity $m + 1 - n$ when that is positive. A pole and a middle vertex have $\pi = 1$, two middle vertices have $\pi = 2$, and the two poles have $\pi = n - 2$, so this is feasible exactly when $n \leq m + 1$, and it carries $2(n - 2)(m - 2) + \max(0, m + 1 - n)$ edges.

The theta family is the one that makes this problem genuinely different from the edge problem, and it was found by the search rather than by hand. Whenever it is feasible, which is to say for $n \leq m + 1$, it beats the thickened tree for every $m \geq 4$, and it is the extremiser in every such case the exhaustion has settled. Comparing the three counts, the tree leads once $n \geq m + 2$, where the other two die for lack of route budget.

Exhaustive search over all multigraphs with multiplicities in $\{0, \dots, m - 1\}$, run under this convention, gives the values in Table A.1. The routine is `max_multigraph_vertex_standard(n,`

m), which is the same include-or-exclude branch and bound as everywhere else in [Chapter 2](#), with the checker's vertex mode switched to the second convention by one flag, `parallel_routes`, on `_split_capacity_matrix`. Every value below was computed twice, once by that routine and once by a separately written script sharing no code with it, and on the sixteen cells both have finished they agree exactly.

		$n = 2$	3	4	5	6	7
$m = 2$	value	1	2	3	4	5	6
	tree	1	2	3	4	5	6
$m = 3$	value	2	4	6	8	10	12
	tree	2	4	6	8	10	12
$m = 4$	value		6	9	12	15	
	tree		6	9	12	15	
	theta		6	9	12		
$m = 5$	value			14	19		
	tree			12	16		
	theta			14	19		
$m = 6$	value			19			
	tree			15			
	theta			19			

Table A.1. The multigraph vertex problem under the convention that parallel copies are distinct routes, computed exhaustively by `max_multigraph_vertex_standard`. Entries in bold are the cases where the value exceeds the multigraph edge value $(m - 1)(n - 1)$, and there the thickened theta is the extremiser. A blank cell is a size not computed, or one where the construction in that row does not apply. Cells with $n \leq m + 1$ are the regime in which the theta and complete constructions are feasible at all.

Two things stand out. First, the problem is *not* the edge problem in disguise: at $m = 5, n = 5$ the value is 19 against the edge value 16, so the separation axis genuinely buys something for multigraphs under this reading. Second, the small- n regime $n \leq m + 1$, where the complete graph is feasible for the simple problems too, is tempting to read as the *only* place a cycle pays for itself, with the thickened tree optimal once $n \geq m + 2$. It is not: a single triangle grafted anywhere onto the thickened tree already beats it, for every $m \geq 5$ and every such n , which the next theorem makes precise.

By [Lemma A.54](#), a feasible multigraph on top of a fixed connected simple graph G_0 maximizes its total multiplicity by setting $\mu(u, v) = m - 1 - \pi(u, v)$ on every edge, giving total $(m - 1)|E(G_0)| - \sum_{uv \in E(G_0)} \pi(u, v)$. Writing $|E(G_0)| = n - 1 + \text{rank}(G_0)$, the thickened tree is beaten by any G_0 with

$$\sum_{uv \in E(G_0)} \pi(u, v) < (m - 1) \text{rank}(G_0).$$

The obvious guess is that cycles must always pay for themselves against this bound, each independent cycle costing $m - 1$ units of route capacity. The next theorem shows a single triangle

already fails to pay, for $m \geq 5$, and packing many of them compounds the failure into a gap that grows with n .

Theorem A.56 (Thickened cliques beat the thickened tree, for every $m \geq 5$). *Fix $3 \leq r \leq m$ and set $q = m + 1 - r \geq 1$, the multiplicity of a thickened K_r that saturates the vertex ceiling on r vertices (the “thickened complete graph” construction above, restricted to a block of size r rather than all of n). Let $1 \leq k \leq \lfloor (n - 1)/(r - 1) \rfloor$. Take k copies of K_r that all share one common vertex and are otherwise disjoint, thicken every edge of every copy to multiplicity q , and attach any leftover vertices as a pendant path at multiplicity $m - 1$. This multigraph is feasible, $\kappa^{\max} \leq m - 1$, and its total multiplicity is exactly*

$$(m - 1)(n - 1) + k \cdot \text{gain}(r, m), \quad \text{gain}(r, m) := \frac{(r - 1)(r - 2)(m - 1 - r)}{2}.$$

Proof. Write G_0 for the underlying simple graph.

Feasibility. The shared vertex is a cut vertex of G_0 , and so is every vertex along the pendant path, since the blocks meet only there. So a path of length ≥ 2 between two vertices of the same block cannot leave that block and come back: to do so it would have to pass through the shared vertex twice. Hence for an edge uv inside a block, every route counted by $\pi(u, v)$ stays inside its K_r , where deleting uv leaves u, v joined by exactly $r - 2$ pairwise internally-disjoint two-step routes, one through each of the other block vertices, and by Menger no more, since those same $r - 2$ vertices are a cut of that size. So $\pi(u, v) = r - 2$ and, by [Lemma A.54](#), $\kappa_G(u, v) = q + (r - 2) = m - 1$ exactly, never over. For u, v in different blocks, or with either endpoint on the pendant path, a single vertex separates them, the shared vertex in the first case and a path vertex in the second, so $\kappa_G(u, v) \leq 1$.

Count. Since $q \geq 1$, every block edge is genuinely present, so G_0 is connected and $|E(G_0)| = n - 1 + \text{rank}(G_0)$. This is where the hypothesis $r \leq m$ earns its keep: at $r = m + 1$ the multiplicity q would drop to zero, the blocks would carry no edges at all, and G_0 would fall apart into isolated vertices. Only the k blocks contribute to $\text{rank}(G_0)$ or to $\sum \pi$, since the pendant edges have $\pi = 0$ and lie in no cycle. Each K_r has rank $(r - 1)(r - 2)/2$ and contributes $\binom{r}{2}(r - 2)$ to the sum, since each of its $\binom{r}{2}$ edges has $\pi = r - 2$. Substituting into $(m - 1)(n - 1 + \text{rank}(G_0)) - \sum \pi$,

$$(m - 1) \left[n - 1 + k \frac{(r - 1)(r - 2)}{2} \right] - k \frac{r(r - 1)(r - 2)}{2} = (m - 1)(n - 1) + k \text{gain}(r, m),$$

$$\text{using } (m - 1) \frac{(r - 1)(r - 2)}{2} - \frac{r(r - 1)(r - 2)}{2} = \frac{(r - 1)(r - 2)[(m - 1) - r]}{2} = \text{gain}(r, m). \quad \square$$

At $r = 3$, $\text{gain}(3, m) = m - 4$: zero at $m = 4$, matching the exhaustive finding that $m \leq 4$ has no counterexample, and strictly positive for every $m \geq 5$. So the guess that the thickened tree is optimal once $n \geq m + 2$ is false for every $m \geq 5$: it held only within the reach of the exhaustive table, $m \leq 4$. The smallest witness is small enough to check by hand. Take $m = 5$ and $n = 7$, the first size the guess covers. Put a triangle on $\{u_1, u_2, u_3\}$ at multiplicity 3 and hang a path of four further vertices off u_1 at multiplicity 4. Two triangle vertices have three parallel copies plus one

route through the third vertex, so $\kappa = 4$, and every other pair is separated by a single cut vertex or joined only by its own parallel copies, so nothing exceeds $4 = m - 1$. The total multiplicity is $3 \cdot 3 + 4 \cdot 4 = 25$, one more than the $(m - 1)(n - 1) = 24$ the guess allows. Packing k blocks instead of one multiplies the excess by k , and k may be taken as large as $\lfloor (n - 1)/(r - 1) \rfloor$, so the gap over the thickened tree grows linearly in n . Sharing a vertex is what buys that count. Disjoint blocks joined by bridges would fit only $\lfloor n/r \rfloor$ of them, since each bridge spends a vertex on nothing but the join, and the difference is real rather than cosmetic: at $m = 10, r = 6$ and $n = 11$ the shared-vertex form carries 150 against the bridged form's 120. Optimising r widens the gap further. Treated as continuous, the gain per vertex $\text{gain}(r, m)/(r - 1)$ peaks near $r \sim (m + 1)/2$, where it is $\Theta(m^2)$, so $K_m^{\text{multi}}(n)/n \rightarrow (m - 1) + \Theta(m^2)$ as $n \rightarrow \infty$ at fixed large m , a different order in m from the guess above.

Even this is not the truth. At $m = 5, n = 7$ the construction above gives 27, and an exhaustive branch and bound, stopped by its time limit before it could finish, had already found a feasible multigraph with 28. So the value of $K_m^{\text{multi}}(n)$ is genuinely open rather than reduced to a single inequality about cycle rank, and what these theorems establish is a lower bound of the right order in both n and m , not the extremal number.

A.9 The random threshold

The probabilistic ingredients are two Chernoff bounds, quoted in the form we use them.

Primer: what a Chernoff bound says

Both proofs below need to say that a sum of many independent yes-or-no trials almost never lands far from its average. That is the content of a *Chernoff bound*, and the intuition is worth stating even though we only quote the result. Let $X = X_1 + \dots + X_N$ count the successes among independent trials, with mean $\mu = \mathbb{E}[X]$. A Chernoff bound says the chance that X overshoots μ by a factor $1 + \delta$, or undershoots it by a factor $1 - \delta$, decays *exponentially* in μ , not merely polynomially as Chebyshev's inequality would give. The mechanism is a single trick. Rather than bound $\Pr[X \geq a]$ directly, bound $\Pr[e^{tX} \geq e^{ta}]$ by Markov's inequality for a parameter $t > 0$, use independence to factor $\mathbb{E}[e^{tX}] = \prod_i \mathbb{E}[e^{tX_i}]$ into a product over the trials, and then choose t to make the resulting bound as small as possible. The exponential in the answer is exactly the e^{tX} that was fed through Markov. We quote the two tails in the shape the threshold proof consumes.

Theorem A.57 (Chernoff bounds, see [19, Cor. 4.4, Thm. 4.4]). *Let X be a sum of independent $\{0, 1\}$ -valued random variables with mean μ . For $\delta > 0$, $\Pr[X \geq (1 + \delta)\mu] \leq \exp(-\frac{\min(\delta, \delta^2)\mu}{4})$, and for $\delta \in (0, 1)$, $\Pr[X \leq (1 - \delta)\mu] \leq \exp(-\frac{\delta^2\mu}{2})$.*

Corollary A.58. *Let $X \sim \text{Binomial}(n - 1, p)$ with $p = cm/n$ for a constant $c \in (0, 1)$. Then for all sufficiently large m , $\Pr[X \geq m] \leq e^{-\alpha m}$ with $\alpha = \alpha(c) > 0$.*

Proof. The mean is $\mu = (n - 1)p \leq cm$, so $m \geq (1 + \delta)\mu$ with $\delta \geq \frac{1-c}{c} > 0$, and [Theorem A.57](#)

applies with $\mu \geq cm/2$ for $n \geq 2$, giving $\alpha = \min\left(\frac{1-c}{c}, \left(\frac{1-c}{c}\right)^2\right) \frac{c}{8}$. $\square$

Proof of Theorem 1.16. Recall the hypotheses: $m = m(n) \geq 2$ with $m/\ln n \rightarrow \infty$ and $m = o(n)$, and $p = cm/n$ for a constant $c > 0$.

Case $c > 1$. The edge count $|E|$ of $G(n, p)$ is binomial with mean $\mu = \binom{n}{2}p = \frac{cm(n-1)}{2}$. With $\delta = 1 - \frac{1}{c}$ the lower-tail bound of Theorem A.57 gives

$$\Pr\left[|E| \leq \frac{m(n-1)}{2}\right] \leq \exp\left(-\frac{(c-1)^2 m(n-1)}{4c}\right) \rightarrow 0,$$

so with high probability $|E| > \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and the contrapositive of Mader's theorem (Theorem 1.2) forces a pair with $\lambda \geq m$. Hence $\Pr[\lambda^{\max} \geq m] \rightarrow 1$.

Case $c < 1$. Each degree is $\text{Binomial}(n-1, p)$, so by Corollary A.58 and a union bound over the n vertices,

$$\Pr[\Delta(G) \geq m] \leq n e^{-\alpha m} = e^{\ln n - \alpha m} \rightarrow 0,$$

since $m/\ln n \rightarrow \infty$. By the degree bound (Lemma A.3) $\lambda^{\max} \leq \Delta(G)$, so $\Pr[\lambda^{\max} \geq m] \rightarrow 0$. $\square$

Remark A.59 (The vertex and directed analogues). Below the threshold, $\kappa^{\max} \leq \lambda^{\max}$ gives the vertex statement for free, and in the random digraph $D(n, p)$, where each ordered pair's arc is tossed independently with probability p , the out-degree version of the same union bound applies. Above the threshold, the minimum degree exceeds m with high probability by the lower-tail bound. Mader's theorem alone does not carry this to the vertex problem, since $\kappa^{\max} \leq \lambda^{\max}$ runs the wrong way to deduce $\kappa^{\max} \geq m$ from the edge bound. What supplies it instead are the theorems of Bollobás [20, Ch. 7] for $G(n, p)$, in hitting-time form due to Bollobás and Thomason [21], and its directed analogue [22], which state that in this regime the global connectivities equal the minimum degree (respectively the minimum semi-degree, the smaller of a vertex's in- and out-degree) with high probability, so $\kappa^{\max}$ and the directed $\lambda^{\max}$ also cross m at $p^* = m/n$. The growth hypothesis $m/\ln n \rightarrow \infty$ is what lets the union bound beat the factor n . For fixed m , the bound is vacuous and finer tools (Poisson approximation of low-degree vertex counts) would be needed.

Remark A.60 (How far the threshold reaches, and where it stops). Theorem 1.16 is a statement about $G(n, p)$, and Remark A.59 carries it to the vertex separation and to the directed model $D(n, p)$. It does not reach the hypergraph models, and the reason is a change of scale rather than a gap in the argument. The threshold sits where a vertex's expected degree reaches m , since below that the degree bound caps every local connectivity and above it the extremal bound forces a high pair. In $G(n, p)$ a vertex has expected degree $p(n-1)$, so the balance point is $p^* = m/n$ as stated. In the binomial r -uniform hypergraph, where each of the $\binom{n}{r}$ possible hyperedges is included independently with probability p , a vertex lies in an expected $p \binom{n-1}{r-1}$

hyperedges, so the balance point is

$$p^* = \frac{m}{\binom{n-1}{r-1}} = \Theta\left(\frac{m}{n^{r-1}}\right),$$

which for $r \geq 3$ is smaller than m/n by a factor of order n^{r-2} . Reading m/n as the hypergraph threshold would therefore place the transition far above where it happens: at $r = 3$ and $n = 8$, for instance, m/n is $3/8$ while $m/\binom{7}{2}$ is $3/21$, and by the former density the sampled hypergraph is long past the transition. [Figure B.5](#) accordingly sweeps each model in units of its own p^* rather than a shared p . A second limit of scope is worth repeating here: the theorem assumes $m/\ln n \rightarrow \infty$, so any picture at a fixed small m illustrates the shape of the transition without being an instance of the theorem.

A.10 Computational transcripts

The cut-counting optimisation formulation that certifies [Theorem 2.2](#) is given in [Chapter 2](#). Here are the solver transcripts that back it, together with the exhaustive-search output that settles the directed base cases. The complete commented source accompanies the document as a single self-contained file, `erdos915_unified.py` in the `program/` directory. Its core runs on `numpy` and `scipy` alone, the certifier adds `pulp`, and `matplotlib` and `networkx` are optional and used only for the figures. An optional compiled helper accelerates selected small hot paths but is not needed for correctness.

```

1 Cut-MILP certification of the directed multigraph problem
2 M*(n) is the scaled optimum; L_m^dir(n) = (m-1) M*(n).
3
4 n=3: status=OPTIMAL, M*=4.000, target=2(n-1)=4, proof=True, time=0.0s, L_3^
    ↳ dir(n)=8
5 n=4: status=OPTIMAL, M*=6.000, target=2(n-1)=6, proof=True, time=0.3s, L_3^
    ↳ dir(n)=12
6 n=5: status=OPTIMAL, M*=8.000, target=2(n-1)=8, proof=True, time=5.9s, L_3^
    ↳ dir(n)=16
7 n=6: status=OPTIMAL, M*=10.000, target=2(n-1)=10, proof=True, time=1259.9s,
    ↳ L_3^dir(n)=20

```

Listing A.1. Solver output: $M^*(n) = 2(n-1)$, optimal with zero gap, for $n = 3, 4, 5, 6$.

The cut-counting optimisation closes the gap up to $n = 5$ inside a short budget and reaches $n = 6$ in about twenty minutes. Past that it stalls, because a solver handles the yes-or-no cut labels by case-splitting, trying a label both ways and solving each half, and the number of cases it must open grows too fast at $n = 7$. The base cases of the directed induction ([Theorem 1.10](#)) run to $n = 7$, and those are settled instead by the pruned exhaustive search over all simple digraphs with $\lambda^{\max} \leq 1$. That search is exact and complete at these sizes, and it agrees with $\ell_2^{\text{dir}}(n) = M(n)$ throughout.

```

1 Exhaustive search over all simple digraphs with lambda^max <= 1 (directed,
    ↳ m=2).

```

```

2 Branch and bound; reported max arc count = ell_2^dir(n) = max(2(n-1), floor
  ↳ (n^2/4)).
3
4 n=2: max= 2, formula= 2, proved (complete), nodes=5, 0.0s
5 n=3: max= 4, formula= 4, proved (complete), nodes=22, 0.0s
6 n=4: max= 6, formula= 6, proved (complete), nodes=537, 0.0s
7 n=5: max= 8, formula= 8, proved (complete), nodes=17823, 0.1s
8 n=6: max=10, formula=10, proved (complete), nodes=788101, 9.1s
9 n=7: max=12, formula=12, proved (complete), nodes=45122129, 718.1s
10
11 The cut-MILP certifier of the multigraph problem closes n <= 6 (n = 6 in
  ↳ about
12 21 minutes); the directed m = 2 base cases through n = 7 that the induction
13 requires are settled by this exhaustive search instead.

```

Listing A.2. Exhaustive-search output for the base cases $2 \leq n \leq 7$: the maximum arc count equals $M(n) = 2, 4, 6, 8, 10, 12$, proved complete at each n .

The vertex problem needs its own base cases, since [Remark A.16](#) shows they cannot be inherited from the arc ones, and the same driver supplies them with only the inner feasibility test swapped. The two separations agree at every size, which is what closes [Theorem 1.11](#).

```

1 Exhaustive search over all simple digraphs with kappa^max <= 1 (directed, m
  ↳ =2).
2 Branch and bound; reported max arc count = k_2^dir(n) = max(2(n-1), floor(n
  ↳ ^2/4)).
3
4 n=2: max= 2, formula= 2, proved (complete), nodes=5, 0.0s
5 n=3: max= 4, formula= 4, proved (complete), nodes=22, 0.0s
6 n=4: max= 6, formula= 6, proved (complete), nodes=537, 0.0s
7 n=5: max= 8, formula= 8, proved (complete), nodes=17823, 1.1s
8 n=6: max=10, formula=10, proved (complete), nodes=788101, 77.0s
9 n=7: max=12, formula=12, proved (complete), nodes=45122129, 3124.8s
10
11 Same driver, same prunes, same sizes as the arc run above. Only the
12 inner test changes, from arc-disjoint to internally vertex-disjoint.
13 The two separations agree at every base case, which is what the
14 vertex induction needs and what Whitney's inequality cannot supply.

```

Listing A.3. The same search run with the internally vertex-disjoint feasibility test, settling the base cases of [Theorem 1.11](#): the maximum arc count is again $2, 4, 6, 8, 10, 12$, proved complete at each n .

A.11 How the program checks itself

The invariants are checked by the self-test at the foot of `erdos915_unified.py`, run simply with `python erdos915_unified.py`. It confirms the checker against the proven $m = 2$ values $2, 4, 6, 8$ and against the first three vertex base cases of [Theorem 1.11](#), the fast vertex-flow test against the exact checker it stands in for, the counting step and both bounds of [Proposi-](#)

tion A.22 and Theorem A.24 on sampled feasible digraphs, including the high-degree case of the latter’s proof, two values of the alternative-convention multigraph vertex problem of Section A.8 including the one that separates it from the edge problem, the named constructions against their predicted sizes and connectivities, the Gomory–Hu shortcut against brute-force max-flow, the Berge connectivities of the hypergraph checker, the Monte Carlo appearance threshold and Whitney’s inequality on every sample, the cut-counting optima for small n , the search reaching the certified optimum, and the driver returning the proven value or a feasible witness in each mode. Every check passes. A single companion script, `make_figures.py`, regenerates every figure in the thesis from the same file, with the mapping listed in `program/README.md`.

Appendix B

Supplementary Figures

This appendix collects the supplementary figures that the program produces but that the chapters reference rather than reproduce in full. Each one is named where it belongs in the body, and the file name printed in its caption matches the output of `make_figures.py` and the figure table in `program/README.md`. Figures already shown in the running text are not repeated here.

B.1 A gallery of extremal graphs

The extremal graphs are not abstractions. [Figure B.1](#) draws nine of the named families the body argues about, each at a deliberately large representative size rather than at the smallest case, so the structure is visible. These are the special, structured extremisers, not the trivial small trees: the directed double star and bidirected star (hub at the centre, with arcs in both directions to every leaf), the doubled bipartite wall $2 \cdot B_{3,4}$ that ties the double star at the crossover, the dense bidirected complete graph, the multigraph star at full multiplicity, and the star hypertrees, the last drawn as metro maps in which each hyperedge is one coloured line through its members. The program builds every one of these directly and, at the small sizes the enumeration reaches, also proves them optimal rather than merely feasible, so the gallery is the analysis made concrete at a size a reader can take in.

B.2 Search traces across the variants

[Figure 3.2](#) in [Chapter 3](#) follows one cooling run in detail. The two runs below repeat that experiment for two further, deliberately dissimilar variants, one undirected and simple, one directed and multigraph, to show the behaviour is the same across the family rather than special to one case. Each plots every visited graph by its binding connectivity (horizontal) against its size (vertical), colouring each point by the step at which it was visited, from the hot early exploration in warm tones to the converged tail in blue. In both the walk strays past the feasibility boundary while hot and is driven back onto the densest feasible graph as the temperature falls, exactly as in the body figure. The seeds are fixed, so each picture reproduces verbatim.

B.3 The variant landscape at $m = 6$ and in three dimensions

This appendix shows the per-graph connectivity distribution at the fixed forbidden threshold $m = 6$, alongside a three-dimensional view of the appearance threshold ([Figure B.5](#)). They con-

Extremal graphs of the named families, drawn large

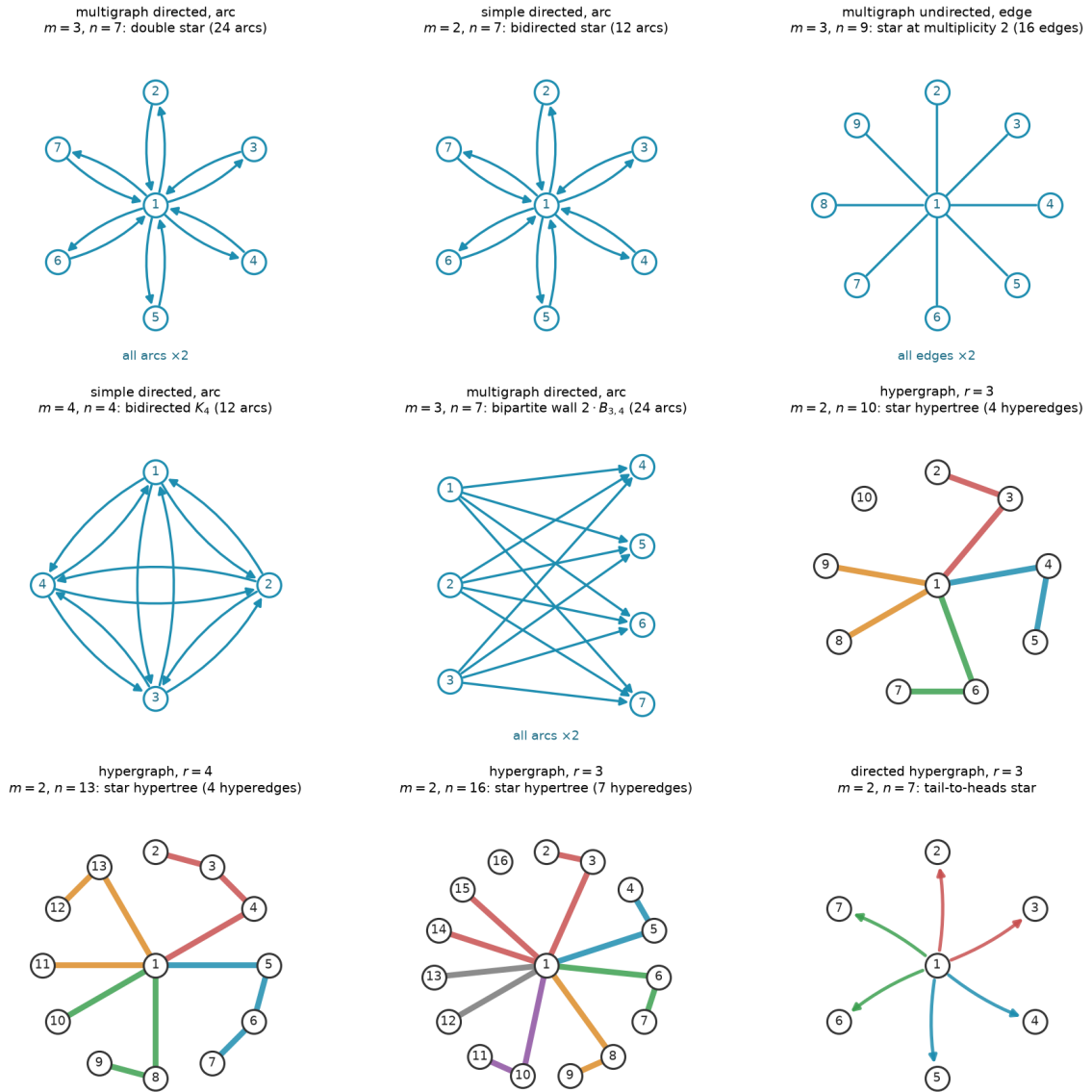


Figure B.1. Extremal graphs of the named families, drawn at a large representative size. *Top and middle rows:* matrix variants, with a central hub where the family has one. Every family here carries one uniform multiplicity, stated once per panel as “all arcs $\times k$ ” rather than repeated on each arc or drawn as parallel curves, and directed arcs carry arrowheads. *Hypergraph panels:* each hyperedge is a metro line through its member vertices, in its own colour, with the hub at the centre, and the directed hypergraph fans coloured arrows from each tail to its heads. Every graph shown is feasible at the stated m . The matrix and undirected-hypergraph families are extremal for their variant, the two directed stars drawn at the crossover $n = 7$, the largest size at which the linear star still ties the one-directional wall before the wall overtakes it. The bipartite wall $2 \cdot B_{3,4}$ is drawn at that same size to show the other side of the tie, matching the double star at 24 arcs as one of the three families the $n = 7$ classification of [Remark A.39](#) identifies. The final panel is the directed-hypergraph analogue, a tail-to-heads star, shown for the one variant whose exact value is still open ([Proposition A.50](#)). File `extremal_graphs_gallery.png`.

Search trace: simple undirected, $n = 7, m = 3$

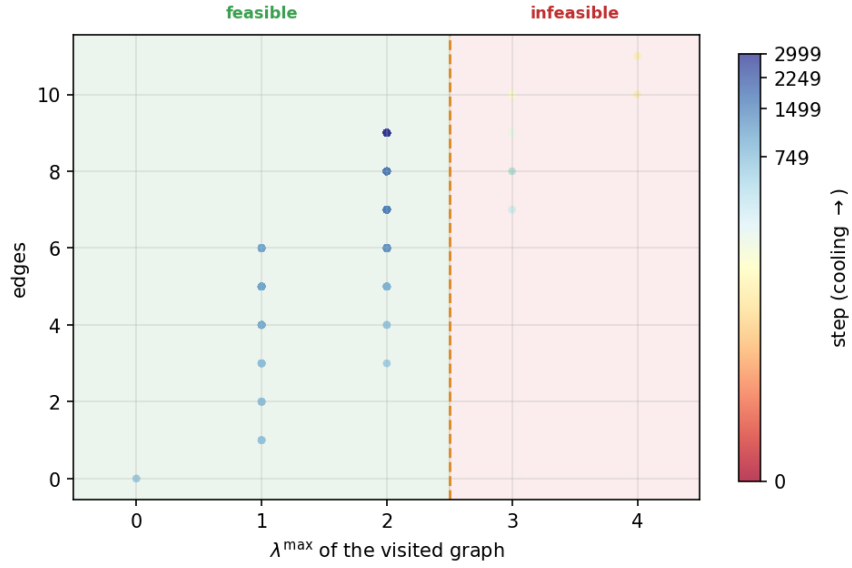


Figure B.2. Search trace for the simple undirected edge problem at $n = 7, m = 3$. File `trace_simple_undirected_n7_m3.png`.

Search trace: multi directed, $n = 5, m = 3$

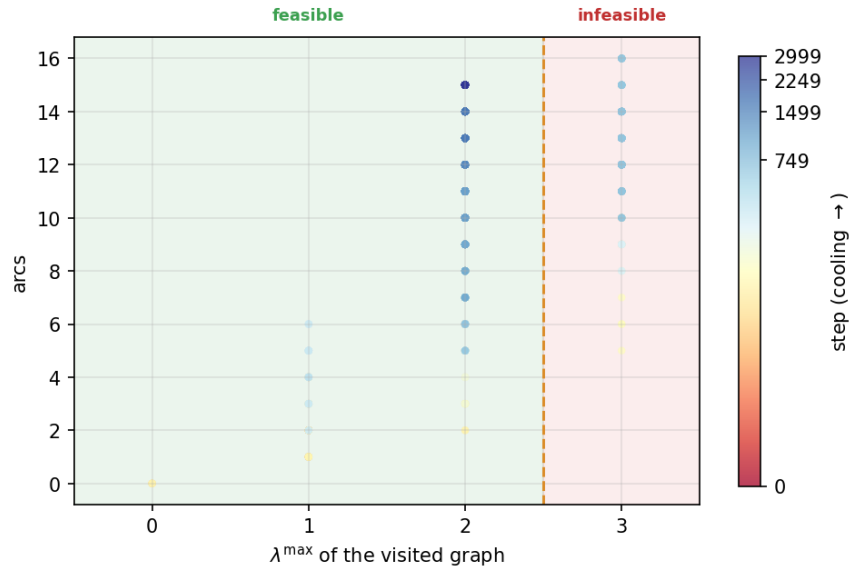


Figure B.3. Search trace for the multigraph directed arc problem at $n = 5, m = 3$. File `trace_multi_directed_n5_m3.png`.

firm that raising the forbidden threshold relocates the feasibility boundary without changing the qualitative picture. The pooled pair-connectivity and edge-count grids of [Chapter 4](#) ([Figures 4.4](#) and [4.5](#)) already split each variant at its own mid-range connectivity, so they carry no separate $m = 6$ companion.

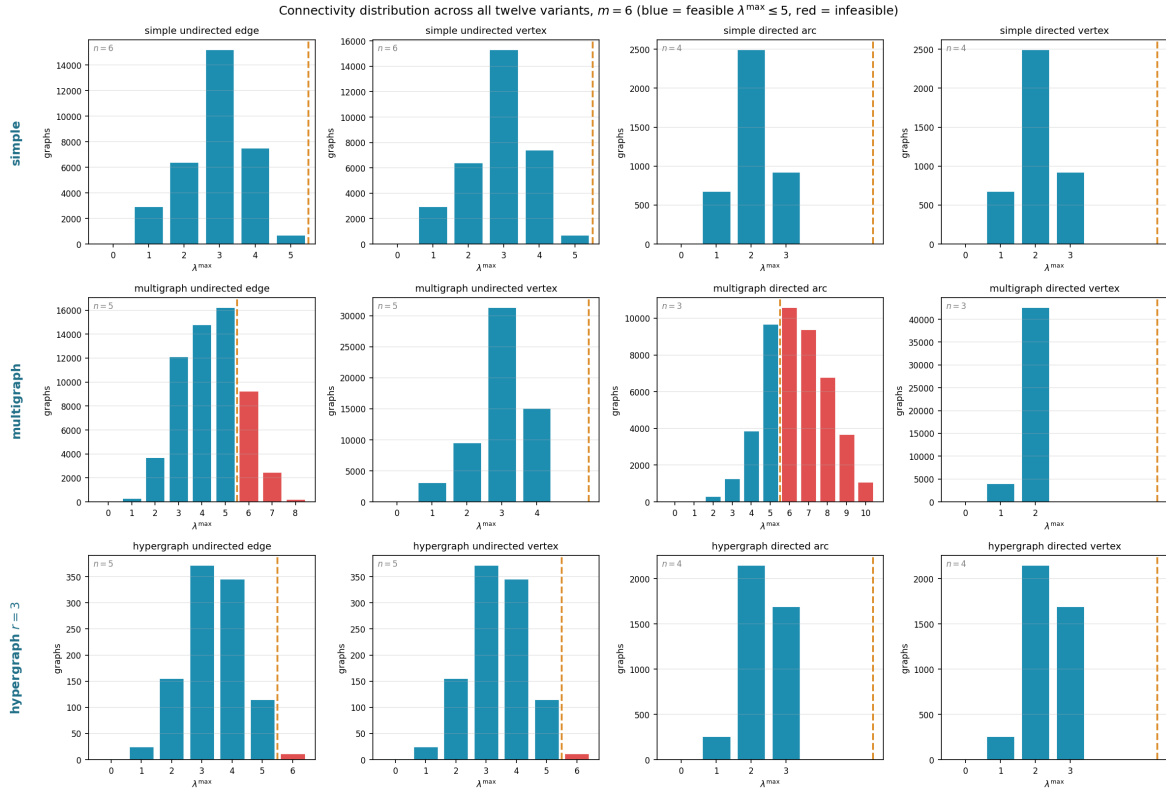


Figure B.4. How the full small- n enumeration distributes by binding connectivity $\lambda^{\max}$ across all twelve variants at the fixed forbidden threshold $m = 6$. Blue bars are feasible graphs ($\lambda^{\max} \leq 5$), red bars infeasible ones, and the dashed line marks the feasibility boundary. At this higher threshold most of the small graphs the enumeration reaches are feasible, so the blue mass dominates: raising m moves the boundary out past the bulk of the distribution. File `conn_dist_m6.png`.

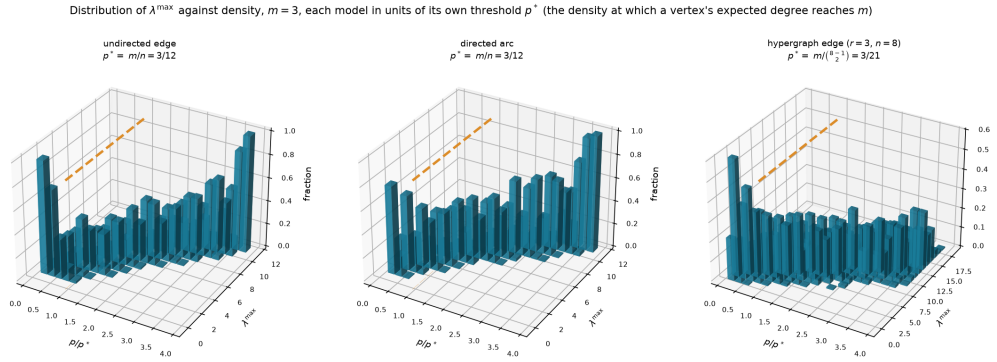


Figure B.5. The appearance threshold as a three-dimensional bar chart, for three representative variants (simple undirected edge and simple directed arc at $n = 12$, three-uniform hypergraph edge at $n = 8$), all at $m = 3$. Each panel plots the fraction of samples (height) reaching each binding connectivity $\lambda^{\max}$ (depth) against density (width), and the mass migrates from low to high connectivity as the density crosses the dashed line. The three models do *not* share a density scale, so each panel is swept in units of its *own* threshold p^* , the density at which a vertex's expected degree reaches m : that is m/n for the two graph models and $m/\binom{n-1}{r-1}$ for the hypergraph, smaller by a factor of order n^{r-2} (Remark A.60). Each panel title gives its own p^* . Only the two graph panels fall under Theorem 1.16, and even they sit outside its asymptotic regime $m/\ln n \rightarrow \infty$ at this fixed $m = 3$, so the figure illustrates the shape of the transition rather than instantiating the theorem. File threshold_3d.png.

Appendix C

The Complete Source Code

Every number in this thesis that was not proved by hand came out of the program printed here, and this appendix prints all of it, with nothing summarised away. The point is that a reader who doubts a computed value can find the exact lines that produced it, rather than taking a description of them on trust.

There is no need to read this front to back. It is laid out so that you can jump to whichever part you want to check. The listing is broken into the same four movements as the thesis itself, because the program is organised that way on purpose: it defines the objects, then measures them, then proves things about them, then searches for examples and draws the results. Each part below opens with a short account of what it does and which chapter it belongs to, and every listing carries line numbers so that a reference like “line 1412” has one unambiguous meaning. Where a line of code was too long for the page it is wrapped, and the continuation is marked with a $\hookrightarrow$ arrow rather than being silently cut.

Four files make up the whole of it. The main program, `erdos915_unified.py`, is the one the chapters discuss throughout and the only one that computes a reported result. `make_figures.py` redraws every picture in the thesis by calling into that program, so a figure can never drift away from the numbers behind it. `_erdos_fast.c` is a small optional accelerator: it speeds up two inner loops and changes no answer, and the program runs correctly without it. The `tests/` directory holds the suite that checks all of the above.

C.1 The main program

C.1.1 How the file opens

The program begins by saying what it is for, then imports what it needs and fixes a few constants. Two decisions made here shape everything after. The first is that the heavy scientific libraries are optional: only `numpy` and `scipy` are required, and `matplotlib`, `networkx` and `pulp` are each wrapped so that the file still loads without them, with the features that need them disabled rather than the whole program failing. The second is that every import is hoisted to the top, so there is one place to look to see what the program depends on.

```
1 """
2 erdos915_unified: the complete unified program for Erdos Problem 915, in one file.
3
4 This single module is the whole program, kept in one namespace so it can be read
```

```

5 top to bottom as a continuous story. It is meant to be read, not just run: every
6 non-trivial step carries an inline comment explaining *why* it is there, not
7 merely what it does. Running it ('python erdos915_unified.py') executes the
8 invariant self-check at the foot of the file; 'make_figures.py' next to it
9 regenerates the thesis figures.
10
11 -----
12 The problem, in one breath
13 -----
14 Erdos Problem 915 asks, in its many guises, for the densest graph on 'n'
15 vertices in which no pair of vertices has 'm' independent routes between them.
16 "Independent" can mean edge-disjoint or internally vertex-disjoint; the object
17 can be a simple graph, a multigraph, a digraph, or a hypergraph; and forbidding
18 'm' routes is exactly the constraint that the largest *local* connectivity
19 stays at or below 'm - 1'. Twelve concrete variants fall out of these
20 choices, and this program treats all of them with one representation and one
21 search.
22
23 -----
24 The epistemic spine: measure / prove / discover
25 -----
26 The whole architecture is organised around three different *kinds of claim* a
27 program can make about extremal values, and it keeps them scrupulously apart so
28 they can never silently masquerade as one another:
29
30 * MEASURE. The checker computes exact local connectivities by max-flow
31 (Menger's theorem). Whatever it reports is a fact about a concrete graph,
32 not an estimate. This is the trusted core; everything else is checked
33 against it.
34
35 * PROVE. The cut-counting method proves *upper* bounds for a fixed number of
36 vertices: it shows that no graph on 'n' vertices can be denser, with a
37 zero optimality gap, so an optimal solution is a genuine proof.
38
39 * DISCOVER. The random search finds concrete dense graphs, hence
40 *lower* bounds. A construction it returns is a witness that the extremal
41 value is at least so large; it never proves that no denser graph exists.
42
43 The closed-form 'bounds' library sits to the side as the theory's predictions
44 (proved, cited, or merely conjectured, each labelled honestly), so that "what
45 the program found" can always be held against "what the theory predicts" without
46 the two drifting apart. The Monte Carlo module sits in the DISCOVER/observe
47 corner: a sampled estimate is an estimate, used to make a proved threshold
48 *visible*, never to prove it.
49
50 -----
51 Chapter map (sections are grouped below by the thesis chapter they support)
52 -----
53 CHAPTER 1 The problem and its base cases
54     VARIANT, BOUNDS, HYPERGRAPH, CONSTRUCTIONS
55 CHAPTER 2 Proving bounds by machine
56     CHECKER, GOMORY-HU, PROVE (the brute-force and exhaustive provers
57     live inside the SOLVE driver, in Chapter 3)
58 CHAPTER 3 Discovering bounds by search
59     SENSITIVITY, SEARCH, SOLVE (the one driver: prove or discover)
60 CHAPTER 4 Synthesis and results
61     MONTE CARLO, FIGURES, __main__ (the invariant self-check suite)
62

```

```

63 A bold CHAPTER divider precedes each group, so as you scroll you always know
64 which chapter's machinery you are reading. Each section keeps its own banner.
65 """
66
67 # =====
68 # Hoisted imports
69 # -----
70 # All external dependencies, gathered into a single block.  ``from __future__
71 # import annotations`` must be (and is) the very first statement so that
72 # string-form type hints work uniformly.  Everything in this program lives in one
73 # namespace, so there are no intra-module imports to follow.
74 # =====
75 from __future__ import annotations
76
77 import copy
78 import json
79 import math
80 import pickle
81 import random
82 import shutil
83 import statistics
84 import subprocess
85 import time
86 import warnings
87 from collections import deque
88 from collections.abc import Iterable
89 from dataclasses import dataclass, field
90 from itertools import combinations, combinations_with_replacement, permutations,
    ↪ product
91 from pathlib import Path
92 from typing import Callable, Iterator
93
94 import concurrent.futures
95
96 import numpy as np
97 from scipy.sparse import csr_matrix as _csr
98 from scipy.sparse.csgraph import maximum_flow as _csgraph_maxflow
99
100 # networkx is OPTIONAL.  Every connectivity measure the thesis reports is computed
101 # through scipy's integer max-flow on a capacity matrix, so the core checker,
102 # search, provers, and enumeration run on numpy + scipy alone.  networkx is needed
103 # only for the Gomory-Hu tree (one figure/analysis helper), which raises a clear
104 # message if called without it installed.
105 try:
106     import networkx as nx
107     NETWORKX_AVAILABLE = True
108 except ImportError:
109     nx = None
110     NETWORKX_AVAILABLE = False
111
112
113 def _require_networkx(feature: str):
114     """Raise a clear error when an optional networkx-only feature is used headless
    ↪ ."""
115     if not NETWORKX_AVAILABLE:
116         raise ImportError(
117             f"{feature} needs networkx (pip install networkx). The core
    ↪ connectivity "

```

```

118         f"measures do not: they run on numpy + scipy alone."
119     )
120
121     # pulp is OPTIONAL. It backs the MILP certifier (prove_directed_multigraph and
122     # prove_integral_arc_bound): one solver-agnostic cut-counting model that CBC (pulp
123     ↪ 's
124     # bundled solver) runs by default and Gurobi runs by a one-line switch. The model
125     ↪ ,
126     # checker, search, enumeration, and sampling do not need it.
127     try:
128         import pulp
129         # pulp 3.x emits migration notices for its 4.0 API (LpVariable construction,
130         # the PULP_CBC_CMD name). We use the stable 3.x API on purpose and pin it in
131         # requirements.txt, so silence that forward-looking noise.
132         import warnings as _warnings
133         _warnings.filterwarnings("ignore", message=r".*PuLP 4\.0.*",
134                                 category=DeprecationWarning)
135         PULP_AVAILABLE = True
136     except ImportError:
137         pulp = None
138         PULP_AVAILABLE = False
139
140     def _require_pulp():
141         """Raise a clear error when the MILP certifier is used without pulp installed.
142         ↪ """
143         if not PULP_AVAILABLE:
144             raise ImportError(
145                 "the MILP certifier needs pulp (pip install pulp). The checker, search
146                 ↪ , "
147                 "and enumeration do not: they run on numpy + scipy alone."
148             )
149
150     import ctypes as _ct
151     _C = None
152     _so_path = Path(__file__).with_name("_erdos_fast.so")
153     if _so_path.exists():
154         _C = _ct.CDLL(str(_so_path))
155         _C.tiny_maxflow.restype = _ct.c_int
156         _C.tiny_maxflow.argtypes = [
157             _ct.POINTER(_ct.c_int), _ct.c_int, _ct.c_int, _ct.c_int, _ct.c_int,
158         ]
159         _C.max_connectivity_exceeds.restype = _ct.c_int
160         _C.max_connectivity_exceeds.argtypes = [
161             _ct.POINTER(_ct.c_int), _ct.c_int, _ct.c_int, _ct.c_int,
162         ]
163         _C.canonical_form_min.restype = None
164         _C.canonical_form_min.argtypes = [
165             _ct.POINTER(_ct.c_int), _ct.c_int, _ct.POINTER(_ct.c_int),
166         ]
167     C_EXTENSION_LOADED = _C is not None
168
169     # matplotlib is OPTIONAL: it is needed only to render the figures (the plot_*
170     # functions in the FIGURES section). It needs its backend chosen before pyplot is
171     # imported, so that the figure code runs head-less (writes PNG files, never opens
172     ↪ a
173     # window). When it is absent the model, checker, provers, search, and enumeration
174     # all still run; only figure generation is unavailable.

```

```

171 try:
172     import matplotlib
173     matplotlib.use("Agg")
174     import matplotlib.pyplot as plt # noqa: E402 (after backend selection)
175     from matplotlib.ticker import MaxNLocator # noqa: E402 (integer-only axis
        ↪ ticks)
176     from mpl_toolkits.mplot3d import Axes3D # noqa: F401 (registers the '3d'
        ↪ projection)
177     MATPLOTLIB_AVAILABLE = True
178 except ImportError:
179     plt = None
180     MaxNLocator = None
181     MATPLOTLIB_AVAILABLE = False

```

C.1.2 The objects: graphs, hypergraphs, and the values we already know

This part corresponds to [Chapter 1](#). It defines what a graph is for the purposes of this thesis, which is a single square table of numbers: entry (u, v) records how many parallel connections run from u to v . That one table covers all eight of the matrix variants at once, because a Variant record says whether the graph is directed and whether it is simple, and those two switches are the only difference between them. Hypergraphs need a different shape, so they are stored as a list of the sets of vertices each hyperedge joins.

After the objects come the closed-form values the thesis proves or conjectures, each written as a small function whose docstring records whether it is proved, conjectured, or a bound only. Then come the named constructions, the specific graphs that achieve those values, so that a claimed bound can always be met with a concrete example rather than an existence argument.

```

184 #####
185 ##
186 ## CHAPTER 1 -- THE PROBLEM AND ITS BASE CASES
187 ## Objects, variants, and the closed-form values we already know.
188 ##
189 #####
190
191 # --- VARIANT: mu[u,v] = multiplicity from u to v; simple = {0,1}; undirected =
    ↪ symmetric ---
192
193
194 @dataclass(frozen=True)
195 class Variant:
196     """Which kind of graph object we are working with.
197
198     Attributes:
199         directed: ‘True’ for digraphs (arcs), ‘False’ for graphs (edges).
200         simple: ‘True’ caps every multiplicity at one. ‘False’ allows
201             arbitrarily many parallel edges or arcs (a multigraph).
202         name: a short human-readable label, used in figures and reports.
203     """
204
205     directed: bool
206     simple: bool
207     # loops: bool
208     name: str = ""

```

```

209
210     def describe(self) -> str:
211         """Return a readable description such as "simple directed"."""
212         # Compose the label from the two boolean axes; used in plot titles.
213         words = ["simple" if self.simple else "multi",
214                 "directed" if self.directed else "undirected"]
215         return " ".join(words)
216
217
218     # The four matrix-representable variants, named for convenience. Random graphs
219     # reuse these (a sampled simple undirected graph is SIMPLE_UNDIRECTED).
220     SIMPLE_UNDIRECTED = Variant(directed=False, simple=True, name="simple undirected")
221     MULTI_UNDIRECTED = Variant(directed=False, simple=False, name="multi undirected")
222     SIMPLE_DIRECTED = Variant(directed=True, simple=True, name="simple directed")
223     MULTI_DIRECTED = Variant(directed=True, simple=False, name="multi directed")
224
225
226     class Graph:
227         """A graph or digraph stored as an integer multiplicity matrix.
228
229         The single source of truth is 'self.mu', an 'n by n' array of
230         non-negative integers with a zero diagonal (no self-loops). For an
231         undirected variant the matrix is kept symmetric at all times, so the
232         invariant 'mu[u, v] == mu[v, u]' may be relied upon everywhere.
233
234         All mutating methods respect the variant: they keep an undirected graph
235         symmetric and never let a simple graph exceed multiplicity one.
236         """
237
238         def __init__(self, num_vertices: int, variant: Variant):
239             if num_vertices < 0:
240                 raise ValueError("number of vertices must be non-negative")
241             self.variant = variant
242             # The whole graph is this one matrix; integer dtype keeps it exact.
243             self.mu = np.zeros((num_vertices, num_vertices), dtype=int)
244
245             # -- basic queries -----
246
247             @property
248             def num_vertices(self) -> int:
249                 """The number of vertices 'n'."""
250                 return self.mu.shape[0]
251
252             def vertices(self) -> range:
253                 """The vertex set '{0, 1, ..., n - 1}' as a range."""
254                 return range(self.num_vertices)
255
256             def multiplicity(self, u: int, v: int) -> int:
257                 """Number of parallel edges or arcs from 'u' to 'v'."""
258                 return int(self.mu[u, v])
259
260             def has_edge(self, u: int, v: int) -> bool:
261                 """Whether at least one edge or arc joins 'u' to 'v'."""
262                 return self.mu[u, v] > 0
263
264             def edge_count(self) -> int:
265                 """Total number of edges or arcs, counted with multiplicity.
266

```

```

267     For an undirected graph each edge is stored twice in the matrix, so we
268     count the strict upper triangle. For a digraph every ordered pair
269     counts once.
270     """
271     if self.variant.directed:
272         return int(self.mu.sum())
273     # Strict upper triangle (k=1) avoids double-counting the symmetric pair.
274     return int(np.triu(self.mu, k=1).sum())
275
276 def out_degree(self, v: int) -> int:
277     """Number of edges or arcs leaving ‘v’ (with multiplicity)."""
278     return int(self.mu[v, :].sum())
279
280 def in_degree(self, v: int) -> int:
281     """Number of edges or arcs entering ‘v’ (with multiplicity)."""
282     return int(self.mu[:, v].sum())
283
284 def degree(self, v: int) -> int:
285     """Total degree of ‘v’.”
286
287     For a digraph this is the sum of in- and out-degree. For an undirected
288     graph in- and out-degree coincide, so we report the out-degree.
289     """
290     if self.variant.directed:
291         return self.out_degree(v) + self.in_degree(v)
292     return self.out_degree(v)
293
294 def edges(self) -> Iterator[tuple[int, int, int]]:
295     """Yield ‘(u, v, multiplicity)’ for every present edge or arc.
296
297     For an undirected graph each edge is yielded once, with ‘u < v’.
298     """
299     n = self.num_vertices
300     for u in range(n):
301         # Undirected: only scan v > u so each edge is emitted once.
302         # Directed: scan all v so both arc directions are emitted.
303         start = u + 1 if not self.variant.directed else 0
304         for v in range(start, n):
305             if u != v and self.mu[u, v] > 0:
306                 yield u, v, int(self.mu[u, v])
307
308 # -- mutation -----
309
310 def add_edge(self, u: int, v: int, count: int = 1) -> None:
311     """Add ‘count’ parallel edges or arcs from ‘u’ to ‘v’.”
312
313     A simple graph silently saturates at multiplicity one. An undirected
314     graph updates both matrix entries to stay symmetric.
315     """
316     self._require_distinct(u, v)
317     new_value = self.mu[u, v] + count
318     if self.variant.simple:
319         new_value = min(new_value, 1) # enforce the 0/1 simple-graph cap
320     self._assign(u, v, new_value)
321
322 def remove_edge(self, u: int, v: int, count: int = 1) -> None:
323     """Remove ‘count’ parallel edges or arcs (never below zero)."""
324     self._require_distinct(u, v)

```

```

325         new_value = max(0, self.mu[u, v] - count)
326         self._assign(u, v, new_value)
327
328     def set_multiplicity(self, u: int, v: int, value: int) -> None:
329         """Set the multiplicity from 'u' to 'v' directly."""
330         self._require_distinct(u, v)
331         if value < 0:
332             raise ValueError("multiplicity must be non-negative")
333         if self.variant.simple and value > 1:
334             raise ValueError("a simple graph cannot have multiplicity above one")
335         self._assign(u, v, value)
336
337     def copy(self) -> "Graph":
338         """Return an independent copy with the same variant and edges."""
339         clone = Graph(self.num_vertices, self.variant)
340         clone.mu = self.mu.copy() # deep-copy the matrix so edits don't alias
341         return clone
342
343     # -- internal helpers -----
344
345     def _assign(self, u: int, v: int, value: int) -> None:
346         """Write 'value' into the matrix, mirroring it when undirected."""
347         self.mu[u, v] = value
348         # Maintain the symmetry invariant for undirected graphs in one place.
349         if not self.variant.directed:
350             self.mu[v, u] = value
351
352     def _require_distinct(self, u: int, v: int) -> None:
353         if u == v:
354             raise ValueError("self-loops are not allowed in this model")
355
356     def __repr__(self) -> str:
357         return (f"Graph(n={self.num_vertices}, variant={self.variant.describe()}!r
358                 ↪ }, "
359                 f"edges={self.edge_count()}")
360
361     # --- BOUNDS: closed-form extremal values; docstrings record proved/conjectured
362     ↪ status ---
363
364     def simple_undirected_edge(n: int, m: int) -> int:
365         """ $e_{ll\_m}(n) = \lfloor m(n-1) / 2 \rfloor$ . Proved (Mader, 1973)."""
366         return (m * (n - 1)) // 2
367
368
369     def multigraph_undirected_edge(n: int, m: int) -> int:
370         """ $L_m(n) = (m-1)(n-1)$ . Proved (multi-tree construction and bound)."""
371         return (m - 1) * (n - 1)
372
373
374     def simple_undirected_vertex_le4(n: int, m: int) -> int:
375         """ $k_m(n) = \lfloor m(n-1) / 2 \rfloor$  for  $m \leq 4$ . Cited (Leonard, 1972)."""
376         if m > 4:
377             raise ValueError("this equality is only known for  $m \leq 4$ ")
378         # For  $m \leq 4$  the vertex problem coincides with the edge problem.
379         return simple_undirected_edge(n, m)
380

```

```

381
382 def simple_undirected_vertex_m5(n: int) -> int:
383     """'k5(n) = floor(8n/3) - 3' for large 'n'. Cited (Sorensen-Thomassen).
        ↪ ""
384     return (8 * n) // 3 - 3
385
386
387 def directed_arc_m2(n: int) -> int:
388     """'ell2dir(n) = max(2(n-1), floor(n2/4))'. Proved (induction on n)."""
389     # The max picks whichever of the two competing constructions wins at this n:
390     # the linear hub (double star) versus the quadratic one-directional wall.
391     return max(2 * (n - 1), (n * n) // 4)
392
393
394 def directed_arc_lower_bound(n: int, m: int) -> int:
395     """Lower bound and conjectured value for 'ellmdir(n), 'm ≥ 2.".
396
397     'max(m(n-1), floor((n+m-2)2/4))'. The two branches are the hub
398     construction ('const:directed-hub') and the shifted-partition augmented
399     bipartite construction ('const:augmented-bipartite'). Proved as a lower
400     bound for all 'm'; conjectured to be tight for 'm ≥ 3
401     ('conj:dir-arc'), proved tight for 'm = 2.".
402     """
403     hub_branch = m * (n - 1)
404     bipartite_branch = (n + m - 2) ** 2 // 4
405     return max(hub_branch, bipartite_branch)
406
407
408 def hypergraph_edge(n: int, m: int, r: int) -> int:
409     """'floor((m-1)(n-1)/(r-1))' hyperedges for the 'r'-uniform edge problem.
410
411     Proved modulo the hypergraph Gomory-Hu theorem; the star hypertree attains
412     the 'm = 2' case and a sparse hypergraph attains the general bound.
413     """
414     return ((m - 1) * (n - 1)) // (r - 1)
415
416
417 def directed_multigraph_arc(n: int, m: int) -> int:
418     """'Lmdir(n) = 2(n-1)(m-1). Proved for 'n ≤ 6' (by counting cuts).
419
420     Conjectured to continue to hold while the double star dominates the
421     one-directional bipartite branch (small 'n'); for larger 'n' the
422     quadratic branch takes over.
423     """
424     double_star_branch = 2 * (n - 1) * (m - 1)
425     bipartite_branch = (m - 1) * ((n * n) // 4)
426     return max(double_star_branch, bipartite_branch)
427
428
429 # --- HYPERGRAPH: stored as incidence list; Berge-path connectivity via helper-
    ↪ node max-flow ---
430
431
432 class Hypergraph:
433     """A hypergraph stored by its incidence (a list of hyperedges).
434
435     An undirected hyperedge is a set of vertices, crossed in either direction. A
436     directed hyperedge ('directed=True') splits an 'r'-set into a non-empty

```

```

437 tail set 'T' and a non-empty head set 'H': a Berge route may enter it only
438 at a tail and leave only at a head. Two storage forms are accepted, and the
439 checker treats them identically:
440
441 * the legacy *forward* form '(tail, frozenset(heads))' with a single integer
442   tail and 'r - 1' heads ('|T| = 1'), the one-tail / many-head model the
443   rest of the thesis uses; and
444 * the *general* form '(frozenset(tails), frozenset(heads))' for any split,
445   which subsumes forward ('|T| = 1'), backward ('|H| = 1') and, from
446   'r >= 4', genuinely mixed ('|T|, |H| >= 2') hyperarcs.
447
448 For 'r = 2' every form collapses to a single arc, so a directed graph is the
449 two-vertex special case.
450 """
451
452 def __init__(self, num_vertices: int, hyperedges: Iterable = (),
453              *, directed: bool = False):
454     self.num_vertices = num_vertices
455     self.directed = directed
456     # Each entry is a frozenset, or a '(tail, frozenset(heads))' pair.
457     self.hyperedges: list = []
458     for edge in hyperedges:
459         self.add_hyperedge(edge)
460
461 def add_hyperedge(self, edge) -> None:
462     """Add an undirected vertex set, or a directed '(tail, heads)' pair.
463
464     A directed edge may be the forward form '(tail:int, heads)' or the
465     general form '(tails:frozenset, heads:frozenset)'; the general form is
466     kept verbatim, the forward form is kept verbatim too (no normalisation),
467     so existing forward results are unchanged byte-for-byte.
468     """
469     if self.directed:
470         first, raw_heads = edge
471         if isinstance(first, (set, frozenset)): # general (tails, heads)
472             tails, heads = frozenset(first), frozenset(raw_heads)
473             members = tails | heads
474             if (tails & heads) or not tails or not heads or len(members) < 2:
475                 raise ValueError("a directed hyperedge needs disjoint, non-
476                                     ↪ empty tail and head sets")
477             if any(v < 0 or v >= self.num_vertices for v in members):
478                 raise ValueError("hyperedge refers to a vertex outside the
479                                     ↪ graph")
480             self.hyperedges.append((tails, heads))
481             return
482         tail, heads = first, frozenset(raw_heads) # legacy forward (tail,
483             ↪ heads)
484         members = heads | {tail}
485         if tail in heads or len(members) < 2:
486             raise ValueError("a directed hyperedge needs a tail and a distinct
487                 ↪ head")
488         if any(v < 0 or v >= self.num_vertices for v in members):
489             raise ValueError("hyperedge refers to a vertex outside the graph")
490         self.hyperedges.append((tail, heads))
491     else:
492         vertices = frozenset(edge) # a hyperedge is a set of distinct
493             ↪ vertices
494         if len(vertices) < 2:

```

```

490         raise ValueError("a hyperedge must contain at least two vertices")
491     if any(v < 0 or v >= self.num_vertices for v in vertices):
492         raise ValueError("hyperedge refers to a vertex outside the graph")
493     self.hyperedges.append(vertices)
494
495     def members(self, edge) -> frozenset:
496         """The vertex set of a stored hyperedge, directed or not."""
497         if self.directed:
498             tails, heads = _dir_tails_heads(edge)
499             return tails | heads
500         return edge
501
502     def vertices(self) -> range:
503         return range(self.num_vertices)
504
505     def edge_count(self) -> int:
506         """Number of hyperedges."""
507         return len(self.hyperedges)
508
509     def incident_hyperedges(self, v: int) -> list[int]:
510         """Indices of the hyperedges containing vertex 'v'."""
511         return [i for i, edge in enumerate(self.hyperedges) if v in self.members(
512             ↪ edge)]
513
514     def _dir_tails_heads(edge) -> tuple[frozenset, frozenset]:
515         """Tail and head sets of a stored directed hyperedge, as frozensets.
516
517         Accepts the legacy forward form '(tail:int, heads)' and the general form
518         '(tails:frozenset, heads:frozenset)', so every consumer of a directed
519         hyperedge can read one '(tails, heads)' shape regardless of how it was
520         built.
521         """
522         first, heads = edge
523         if isinstance(first, (set, frozenset)):
524             return frozenset(first), frozenset(heads)
525         return frozenset((first,)), frozenset(heads)
526
527     def _hyper_capacity_matrix(hypergraph: Hypergraph, *, vertex_split: bool = False):
528         """Integer capacity matrix of the hypergraph flow network (one per variant).
529
530         Each hyperedge becomes an in/out gate joined by a capacity-1 arc, so at most
531         one disjoint Berge route may traverse it: this is the hyperedge-as-node trick.
532         With 'vertex_split' each ORIGINAL vertex is *also* split into an in/out pair
533         of capacity one, so the flow counts internally vertex-disjoint Berge routes
534         instead of edge-disjoint ones. For a directed hypergraph the gate is entered
535         only from a tail and left only toward a head (one tail for the forward model,
536         several for the general one); for an undirected one every member links to the
537         gate both ways. The hypergraph variants are thus the same construction with a
538         boolean or two flipped, not separate measures.
539
540         Vertices index '0..base-1' ('base = 2n' split, else 'n'); hyperedge gate
541         'i' indexes 'base+2i' (in) and 'base+2i+1' (out). 'leave(v)'/enter(
542             ↪ v)'
543         return the index a route leaves/enters vertex 'v' through, so the same scipy
544         max-flow that serves plain graphs serves hypergraphs too.
545         """

```

```

546     n = hypergraph.num_vertices
547     base = 2 * n if vertex_split else n
548     size = base + 2 * len(hypergraph.hyperedges)
549     cap = np.zeros((size, size), dtype=int)
550     leave = (lambda v: 2 * v + 1) if vertex_split else (lambda v: v)
551     enter = (lambda v: 2 * v) if vertex_split else (lambda v: v)
552     if vertex_split:
553         for v in range(n):
554             cap[2 * v, 2 * v + 1] = 1 # one route through each vertex
555     for index, edge in enumerate(hypergraph.hyperedges):
556         gate_in, gate_out = base + 2 * index, base + 2 * index + 1
557         cap[gate_in, gate_out] = 1 # one route through each
558             ↪ hyperedge
559     if hypergraph.directed:
560         tails, heads = _dir_tails_heads(edge)
561         for tail in tails: # enter the gate from any tail
562             cap[leave(tail), gate_in] = _UNBOUNDED
563         for head in heads: # leave the gate toward any head
564             cap[gate_out, enter(head)] = _UNBOUNDED
565     else:
566         for vertex in edge:
567             cap[leave(vertex), gate_in] = _UNBOUNDED
568             cap[gate_out, enter(vertex)] = _UNBOUNDED
569     return cap, size, leave, enter
570
571 def hyper_connectivity(hypergraph: Hypergraph, source: int, target: int,
572                        *, vertex_split: bool = False) -> int:
573     """Edge- or vertex-disjoint Berge routes from ‘source’ to ‘target’.
574     A single scipy integer max-flow on the matrix above. An isolated endpoint
575     (in no hyperedge) has no incident arcs, so the flow is then zero.
576     """
577     cap, _, leave, enter = _hyper_capacity_matrix(hypergraph, vertex_split=
578         ↪ vertex_split)
579     if vertex_split:
580         # Endpoints must not be the bottleneck: only interior vertices are capped.
581         cap[2 * source, 2 * source + 1] = _UNBOUNDED
582         cap[2 * target, 2 * target + 1] = _UNBOUNDED
583     return int(_csgraph_maxflow(_csr(cap, dtype=int), leave(source), enter(target)
584         ↪ ).flow_value)
585
586 def max_hyper_connectivity(hypergraph: Hypergraph, *, vertex_split: bool = False)
587     ↪ -> int:
588     """‘lambda^max’ (edge) or ‘kappa^max’ (vertex) over all pairs."""
589     return max((hyper_connectivity(hypergraph, source, target, vertex_split=
590         ↪ vertex_split)
591         for source, target in _pairs(hypergraph)), default=0)
592
593 def hyperedge_connectivity(hypergraph: Hypergraph, source: int, target: int) ->
594     ↪ int:
595     """Maximum number of edge-disjoint Berge ‘source’-‘target’ paths."""
596     return hyper_connectivity(hypergraph, source, target, vertex_split=False)
597
598 def max_hyperedge_connectivity(hypergraph: Hypergraph) -> int:

```

```

598     """ $\lambda^{\max}$  for the undirected edge hypergraph problem."""
599     return max_hyper_connectivity(hypergraph, vertex_split=False)
600
601
602 # --- CONSTRUCTIONS: named extremal graphs; each states its edge count and  $\lambda^{\max}$ 
603      $\hookrightarrow$  max ---
604
605 def double_star(n: int, m: int, directed: bool = True) -> Graph:
606     """The double star: one hub joined to every leaf with multiplicity  $m-1$ .
607
608     For a digraph the hub carries  $m-1$  arcs in both directions to each
609     leaf, attaining  $2(n-1)(m-1)$  arcs with  $\lambda^{\max} = m-1$ . For an
610     undirected multigraph it attains  $(m-1)(n-1)$  edges. This is the
611     small- $n$  extremiser of the multigraph problems.
612     """
613     variant = MULTI_DIRECTED if directed else MULTI_UNDIRECTED
614     graph = Graph(n, variant)
615     hub = 0 # vertex 0 is the central hub; everything else is a leaf
616     for leaf in range(1, n):
617         graph.set_multiplicity(hub, leaf, m - 1)
618         if directed:
619             graph.set_multiplicity(leaf, hub, m - 1) # arcs in both directions
620     return graph
621
622
623 def one_directional_bipartite(n: int) -> Graph:
624     """All arcs from one part to the other:  $\lfloor n^2/4 \rfloor$  arcs,  $\lambda^{\max} =$ 
625          $\hookrightarrow 1$ .
626
627     The vertices split into a smaller part  $A$  and a larger part  $B$  and
628     every arc runs  $A \rightarrow B$ . No two vertices have two arc-disjoint routes, so
629     this single-direction wall of arcs is feasible for  $m = 2$  while holding
630     quadratically many arcs: the phenomenon that has no undirected analogue.
631     """
632     graph = Graph(n, SIMPLE_DIRECTED)
633     size_a = n // 2 # smaller part; the  $n^2/4$  maximum is at the balanced split
634     part_a = range(size_a)
635     part_b = range(size_a, n)
636     for a in part_a:
637         for b in part_b:
638             graph.add_edge(a, b) # every arc runs  $A \rightarrow B$  only (one direction)
639     return graph
640
641 def augmented_bipartite(n: int, m: int) -> Graph:
642     """The conjectured directed extremiser for  $m \geq 2$ .
643
644     Set  $|B| = \lceil (n+m-2)/2 \rceil$  and  $|A| = n - |B|$ . Fill every arc
645      $A \rightarrow B$  (the one-directional wall), then give each  $B$ -vertex
646      $m - 2$  extra in-arcs from its circulant predecessors inside  $B$ .
647     Because no arc leaves  $B$  toward  $A$ , a route between two  $B$ 
648     vertices never escapes  $B$ , and a route from  $A$  to  $b$  uses the
649     direct arc plus one short detour through each of  $b$ 's  $m - 2$ 
650     in-arcs, giving  $\lambda^{\max} = m - 1$ . The arc count is
651      $\lfloor (n+m-2)^2/4 \rfloor$  (const:augmented-bipartite in the thesis).
652
653     Requires  $n \geq m$  so that  $|A| \geq 1$  and the circulant offsets stay

```

```

654     distinct. At ' $m = 2$ ' or ' $m = 3$ ' the partition equals the balanced
655     ' $(\lfloor n/2 \rfloor, \lceil n/2 \rceil)$ ' split and the formula reduces to
656     ' $\lfloor n^2/4 \rfloor + (m-2)\lceil n/2 \rceil$ '. At ' $m \geq 4$ ' the shifted partition
657     gives strictly more arcs. For ' $m = 3, n = 10$ ' this is the 30-arc
658     counterexample.
659     """
660     size_b = (n + m - 2 + 1) // 2 #  $\lceil (n+m-2)/2 \rceil$ 
661     size_a = n - size_b
662     if size_a < 1:
663         raise ValueError("augmented_bipartite needs  $n \geq m$ ")
664
665     graph = Graph(n, SIMPLE_DIRECTED)
666     part_a = range(size_a)
667     part_b = list(range(size_a, n))
668
669     # Complete wall of arcs from the small part to the large part.
670     for a in part_a:
671         for b in part_b:
672             graph.add_edge(a, b)
673
674     # Thicken B by a circulant giving every B-vertex in-degree exactly  $m - 2$ .
675     # Each offset adds a full cycle of arcs within B;  $m - 2$  offsets give each
676     # B-vertex  $m - 2$  extra in-arcs from other B-vertices (the short detours).
677     for offset in range(1, m - 1):
678         for i in range(size_b):
679             source = part_b[i]
680             target = part_b[(i + offset) % size_b]
681             graph.add_edge(source, target)
682     return graph
683
684
685 def clique_tree(clique_size: int, blocks_added: int) -> Graph:
686     """A tree of cliques (a ' $k$ '-tree): the chordal scaffold of the vertex case.
687
688     Begin with a complete graph ' $K_r$ ' on ' $r = \text{clique\_size}$ ' vertices, then
689     repeatedly attach a new vertex to the previous ' $r - 1$ ' vertices, which
690     always form a clique. The result is a maximal chordal graph of treewidth
691     ' $r - 1$ ' and *global* vertex connectivity ' $r - 1$ '.
692
693     A caution that the thesis makes explicit: the controlled quantity here is the
694     global connectivity, not ' $\kappa^{\max}$ '. Attaching a vertex to a triangle
695     raises the local connectivity of that triangle's vertices above ' $r - 1$ ', so
696     a ' $k$ '-tree is not itself feasible for the ' $\kappa^{\max}$ ' problem. It appears
697     in the thesis to illustrate the chordal structure underlying the
698     Sorensen-Thomassen analysis of the ' $m = 5$ ' divergence, not as a feasible
699     extremiser.
700     """
701     if clique_size < 2:
702         raise ValueError("clique_size must be at least two")
703     n = clique_size + blocks_added
704     graph = Graph(n, SIMPLE_UNDIRECTED)
705
706     # The seed clique on the first  $r$  vertices.
707     for u in range(clique_size):
708         for v in range(u + 1, clique_size):
709             graph.add_edge(u, v)
710
711     # Each new vertex joins the  $r - 1$  most recently added vertices.

```

```

712     # Those r - 1 always form a clique, which keeps the graph chordal (a k-tree).
713     for new_vertex in range(clique_size, n):
714         attach_to = range(new_vertex - (clique_size - 1), new_vertex)
715         for old_vertex in attach_to:
716             graph.add_edge(new_vertex, old_vertex)
717     return graph
718
719
720 def complete_uniform_hypergraph(n: int, r: int) -> Hypergraph:
721     """The complete 'r'-uniform hypergraph: every 'r'-subset is a hyperedge.
722
723     It is dense and highly connected; for example  $K_4^{(3)}$  has Berge
724     edge-connectivity three between every pair, exceeding the two-hyperedge count
725     of the pairs because longer Berge routes also contribute.
726     """
727     # One hyperedge per r-subset of the vertex set.
728     return Hypergraph(n, (set(subset) for subset in combinations(range(n), r)))
729
730
731 def star_hypertree(n: int, r: int) -> Hypergraph:
732     """A hub joined to disjoint blocks: the feasible extremiser at 'm = 2'.
733
734     Vertex '0' is the hub, and the remaining 'n - 1' vertices are split into
735     blocks of size 'r - 1'; each block together with the hub forms one
736     hyperedge. No two vertices share more than one hyperedge and the only routes
737     between blocks pass through the single hub, so the Berge edge-connectivity is
738     one everywhere, and the construction holds  $\lfloor (n-1)/(r-1) \rfloor$ 
739     hyperedges, the 'm = 2' case of the hypergraph edge bound.
740     """
741     if r < 2:
742         raise ValueError("hyperedge size r must be at least two")
743     hub = 0
744     others = list(range(1, n))
745     hypergraph = Hypergraph(n)
746     # Walk the non-hub vertices in blocks of r - 1; each block plus the hub is
747     # one hyperedge (a "star" of r-sets all sharing the single hub).
748     for start in range(0, len(others) - (r - 2), r - 1):
749         block = others[start:start + (r - 1)]
750         if len(block) == r - 1: # drop a trailing partial block
751             hypergraph.add_hyperedge([hub, *block])
752     return hypergraph

```

C.1.3 Measuring, and proving

This part corresponds to [Chapter 2](#). Measuring comes first. The question “how many independent routes join these two vertices” is answered exactly, by turning it into a maximum-flow problem and solving that, which is Menger’s theorem used as an algorithm. The same routine answers all four versions of the question, because whether routes must avoid shared edges or shared intermediate vertices is controlled by one flag that decides how the flow network is built, and hypergraphs are handled by giving each hyperedge a gate of its own.

Proving comes second and is a different kind of work. To show that no graph of a given size can beat a bound, it is not enough to fail to find one. The program instead writes the whole question as a single optimisation problem whose variables describe an arbitrary graph and whose

constraints say that every pair of vertices admits a small cut. If that problem has no solution, no such graph exists, and that is a proof rather than an absence of evidence.

```

755 #####
756 ##
757 CHAPTER 2 -- PROVING BOUNDS BY MACHINE
758 ## Measure connectivity exactly, then prove small-case upper bounds.
759 ##
760 #####
761
762 # --- CHECKER: exact max-flow connectivity; vertex_split=True gives kappa^max ---
763
764 # A capacity large enough to never be the bottleneck of any cut we build.
765 _UNBOUNDED = 10 ** 9
766
767
768 # -----
769 # One flow network and one measure, parameterised by edge vs vertex
770 # -----
771 # Every measure here is a single scipy integer max-flow on a capacity matrix
772 # (Menger). EDGE mode runs directly on the multiplicity matrix mu: each unit of
773 # capacity is one parallel edge, so a max-flow of value f is f edge-disjoint
774 # routes. VERTEX mode runs on the 2n x 2n split matrix (see
775     ↪ _split_capacity_matrix)
776 # where each vertex becomes a capacity-one in/out gate, so a route uses a vertex
777     ↪ at
778 # most once. Edge vs vertex is one boolean, not a second code path.
779
780 def local_connectivity(graph: Graph, source: int, target: int,
781                        *, vertex_split: bool = False) -> int:
782     """Disjoint 'source'-'target' routes by a single max-flow (Menger).
783
784     Edge-disjoint with 'vertex_split=False'; internally vertex-disjoint with
785     'vertex_split=True'. An isolated endpoint yields zero (no augmenting path).
786     """
787     if not vertex_split:
788         # scipy.sparse.csgraph.maximum_flow is a C implementation; faster than
789         # NetworkX for the small integer capacity matrices we use.
790         return int(_csgraph_maxflow(_csr(graph.mu, dtype=int), source, target).
791                    ↪ flow_value)
792
793     # Vertex mode: the same scipy max-flow on the split matrix. Uncapping the two
794     # endpoints' own in->out gates makes Menger count internally vertex-disjoint
795     # routes (only interior vertices stay capacity-capped at one).
796     cap, _ = _split_capacity_matrix(graph)
797     cap[2 * source, 2 * source + 1] = _UNBOUNDED
798     cap[2 * target, 2 * target + 1] = _UNBOUNDED
799     return int(_csgraph_maxflow(_csr(cap, dtype=int), 2 * source + 1, 2 * target).
800                ↪ flow_value)
801
802
803 def max_connectivity(graph: Graph, *, vertex_split: bool = False) -> int:
804     """The largest local connectivity over all pairs: 'lambda^max' (edge) or
805     'kappa^max' (vertex). For a digraph every ordered pair is examined, for an
806     undirected graph each unordered pair once.
807     """
808     if not vertex_split:
809         # Build the CSR matrix once and reuse it for every pair.
810         csr = _csr(graph.mu, dtype=int)

```

```

806         return max((int(_csgraph_maxflow(csr, s, t).flow_value)
807                     for s, t in _pairs(graph)), default=0)
808     return max((local_connectivity(graph, s, t, vertex_split=True)
809               for s, t in _pairs(graph)), default=0)
810
811
812 # -----
813 # Named measures: thin views on the single path above (so the rest of the
814 # program can pass ‘‘max_edge_connectivity’’ / ‘‘max_vertex_connectivity’’ around
815 # by name -- just as ‘‘hyperedge_connectivity’’ names one branch of
816 # ‘‘hyper_connectivity’’ in the hypergraph section).
817 # -----
818
819 def local_edge_connectivity(graph: Graph, source: int, target: int) -> int:
820     """Maximum number of edge-disjoint ‘‘source’’-‘‘target’’ paths."""
821     return local_connectivity(graph, source, target, vertex_split=False)
822
823
824 def max_edge_connectivity(graph: Graph) -> int:
825     """‘‘lambda^max(G)’’: the largest local edge-connectivity over all pairs."""
826     return max_connectivity(graph, vertex_split=False)
827
828
829 def local_vertex_connectivity(graph: Graph, source: int, target: int) -> int:
830     """Maximum number of internally vertex-disjoint ‘‘source’’-‘‘target’’ paths."""
831     ↪ "
832     return local_connectivity(graph, source, target, vertex_split=True)
833
834
835 def max_vertex_connectivity(graph: Graph) -> int:
836     """‘‘kappa^max(G)’’: the largest local vertex-connectivity over all pairs."""
837     return max_connectivity(graph, vertex_split=True)
838
839
840 def min_vertex_connectivity(graph: Graph) -> int:
841     """The global vertex connectivity ‘‘kappa(G)’’: the smallest over all pairs.
842
843     This is the classical connectivity of a graph (the least number of vertices
844     whose removal disconnects it). It is distinct from ‘‘kappa^max’’: a ‘‘k’’-
845     ↪ tree,
846     for instance, is globally ‘‘k’’-connected yet has pairs with higher local
847     connectivity.
848     """
849     pairs = list(_pairs(graph))
850     if not pairs:
851         return 0
852     # Global connectivity is the MINIMUM local connectivity over all pairs.
853     return min(local_vertex_connectivity(graph, source, target)
854               for source, target in pairs)
855
856 # -----
857 # Shared helpers
858 # -----
859
860 def _pairs(obj):
861     """Yield the vertex pairs to test: ordered if directed, unordered if not.

```

```

862     Works for both a ‘‘Graph’’ (directedness on ‘‘variant’’) and a ‘‘Hypergraph’’
863     (directedness on the object), so one pair sweep serves every variant.
864     """
865     directed = obj.directed if hasattr(obj, "directed") else obj.variant.directed
866     n = obj.num_vertices
867     for source in range(n):
868         # Directed objects need both (s, t) and (t, s); undirected need each
869         # unordered pair once, so we start the inner loop above the source.
870         start = 0 if directed else source + 1
871         for target in range(start, n):
872             if source != target:
873                 yield source, target
874
875
876 # -----
877 # Capped feasibility predicate (used inside the search, Section 7)
878 # -----
879 # The exact measures above answer "what is  $\lambda^{\max} / \kappa^{\max}$ ?". The search
880 # only ever needs the cheaper YES/NO question "is it greater than k?", because
881 # its energy depends on the value solely through the excess  $\max(0, \text{value} - (m-1))$ .
882 # A capped flow answers exactly that and stops the moment the answer is known, so
883 # this predicate is equivalent to ‘‘ $\max\_connectivity(\text{graph}, \dots) > k$ ’’ but usually
884 # far cheaper: each pair’s flow halts after k+1 augmenting paths, and the pair
885 # loop halts at the first violating pair. The exact value is recomputed only on
886 # the rare step where the search genuinely needs it (an infeasible proposal).
887 # DEPENDENCIES: _pairs (above), _tiny_maxflow (Section ENUMERATE), _UNBOUNDED.
888
889 def _split_capacity_matrix(graph: Graph,
890                           parallel_routes: bool = False) -> tuple[np.ndarray, int
891                               ↪ ]:
892
893     """The ‘‘2n x 2n’’ capacity matrix of the vertex-split network.
894
895     The vertex-mode network as a plain matrix, consumed by both the exact measure
896     (:func:‘local_connectivity’ via scipy max-flow) and the capped search
897     ↪ predicate
898     (:func:‘exceeds_bound’ via :func:‘_tiny_maxflow’). Vertex ‘‘v’’ becomes an
899     in-copy ‘‘2v’’ and an out-copy ‘‘2v+1’’ joined by an internal arc of capacity
900     one (so any route uses ‘‘v’’ at most once), and each adjacency ‘‘u -> v’’
901     becomes an arc ‘‘(2u+1) -> (2v)’’. The caller uncaps the two endpoints’ own
902     in->out gates so Menger counts internally disjoint routes (see
903     :func:‘exceeds_bound’).
904
905     ‘‘parallel_routes’’ selects the counting convention for parallel copies, the
906     choice discussed at length in the thesis (sec:parallel-convention). The
907     default False caps every adjacency at one, so a bundle of parallel edges is a
908     single direct route and a multigraph measures as its underlying simple graph:
909     that is the convention the twelve variants use. Setting it True gives the
910     adjacency capacity ‘‘ $\mu(u,v)$ ’’, so ‘‘q’’ parallel copies count as ‘‘q’’
911     internally disjoint routes, which is the alternative convention explored in
912     sec:multi-vertex-standard.
913     """
914     n = graph.num_vertices
915     size = 2 * n
916     cap = np.zeros((size, size), dtype=int)
917     for v in range(n):
918         cap[2 * v, 2 * v + 1] = 1 # internal gate: one route through v
919     for u in range(n):
920         for v in range(n):

```

```

918         if u != v and graph.mu[u, v] > 0:
919             cap[2 * u + 1, 2 * v] = (int(graph.mu[u, v]) if parallel_routes
920                                     else 1)
921     return cap, size
922
923
924 def max_multigraph_vertex_standard(n: int, m: int,
925                                   deadline: float | None = None) -> tuple[int,
926                                   ↪ Graph | None, bool]:
927
928     """Exhaustive maximum for the multigraph vertex problem, OTHER convention.
929
930     Maximises the total multiplicity of an undirected multigraph on ‘n’
931     vertices subject to ‘ $\kappa^{\max} \leq m - 1$ ’ when parallel copies are counted
932     as distinct internally vertex-disjoint routes. Every multiplicity is capped
933     at ‘m - 1’ because ‘m’ parallel copies already exceed the ceiling on
934     their own. Branch and bound over the pairs in a fixed order, trying the
935     largest multiplicity first, pruning on feasibility (which is monotone) and
936     on the incumbent. Returns ‘(best_total, witness, completed)’.”
937
938     pairs = list(combinations(range(n), 2))
939     cap = m - 1
940     graph = Graph(n, MULTI_UNDIRECTED)
941     best = [0, None]
942     timed_out = [False]
943
944     def feasible() -> bool:
945         return not exceeds_bound(graph, m - 1, separation="vertex",
946                                 parallel_routes=True)
947
948     def recurse(i: int, total: int) -> None:
949         if deadline is not None and time.time() > deadline:
950             timed_out[0] = True
951             return
952         if total + cap * (len(pairs) - i) <= best[0]:
953             return
954         if i == len(pairs):
955             if total > best[0]:
956                 best[0] = total
957                 best[1] = graph.copy()
958             return
959         u, v = pairs[i]
960         for q in range(cap, -1, -1):
961             graph.set_multiplicity(u, v, q)
962             if q == 0 or feasible():
963                 recurse(i + 1, total + q)
964             if timed_out[0]:
965                 break
966         graph.set_multiplicity(u, v, 0)
967
968     recurse(0, 0)
969     return best[0], best[1], not timed_out[0]
970
971 def exceeds_bound(graph: Graph, k: int, *, separation: str = "edge",
972                  parallel_routes: bool = False) -> bool:
973     """‘True’ iff ‘ $\lambda^{\max}(G) > k$ ’ (edge) or ‘ $\kappa^{\max}(G) > k$ ’ (vertex).
974
975     Equivalent to ‘max_connectivity(graph, vertex_split=(separation=='vertex')) >

```

```

    ↪ k''
975 but with two early exits: each pair's capped flow stops after 'k+1'
    ↪ augmenting
976 paths, and the pair loop stops at the first violating pair. On an infeasible
977 graph this typically returns after a single pair. The pair set is exactly the
978 one :func:'max_connectivity' iterates, so the predicate matches it pair for
    ↪ pair.

979
980 'parallel_routes' is passed through to :func:'_split_capacity_matrix' and
981 only affects vertex mode: it selects the convention in which parallel copies
982 are distinct routes, used by :func:'max_multigraph_vertex_standard'.
983 """
984 n = graph.num_vertices
985 if separation == "edge":
986     mu = graph.mu
987     if _C is not None and n <= 16:
988         flat, ptr = _c_flat(mu)
989         directed = int(graph.variant.directed)
990         return bool(_C.max_connectivity_exceeds(ptr, n, k, directed))
991     for s, t in _pairs(graph):
992         if _tiny_maxflow(mu, n, s, t, k):
993             return True
994     return False
995 if separation == "vertex":
996     cap, size = _split_capacity_matrix(graph, parallel_routes)
997     for s, t in _pairs(graph):
998         # Uncap the endpoints' own in->out gates so Menger counts INTERNAL
999         # routes (mirrors local_connectivity's vertex mode); flow leaves s's
1000         # out-copy (2s+1) and arrives at t's in-copy (2t).
1001         cap[2 * s, 2 * s + 1] = _UNBOUNDED
1002         cap[2 * t, 2 * t + 1] = _UNBOUNDED
1003         hit = _tiny_maxflow(cap, size, 2 * s + 1, 2 * t, k)
1004         cap[2 * s, 2 * s + 1] = 1 # restore the gates for the next
            ↪ pair
1005         cap[2 * t, 2 * t + 1] = 1
1006         if hit:
1007             return True
1008     return False
1009 raise ValueError("separation must be 'edge' or 'vertex'")
1010
1011
1012 # --- GOMORY-HU: encode all pairwise edge-connectivities in n-1 flows (undirected
    ↪ only) ---
1013
1014
1015 def _undirected_capacity_graph(graph: Graph) -> nx.Graph:
1016     """Build the undirected capacitated graph that Gomory-Hu consumes."""
1017     if graph.variant.directed:
1018         raise ValueError("Gomory-Hu trees are defined for undirected graphs only")
1019     capacity_graph = nx.Graph()
1020     capacity_graph.add_nodes_from(graph.vertices())
1021     # One undirected edge per adjacency, capacity = multiplicity (its edge count).
1022     for u, v, multiplicity in graph.edges():
1023         capacity_graph.add_edge(u, v, capacity=float(multiplicity))
1024     return capacity_graph
1025
1026
1027 def gomory_hu_tree(graph: Graph) -> nx.Graph:

```

```

1028     """Return a Gomory-Hu tree of ‘graph’ as a weighted ‘networkx’ tree.
1029
1030     Each tree edge carries a ‘weight’ equal to the capacity of the minimum cut
1031     it represents. This is the one place that genuinely needs networkx’s
1032     Gomory-Hu routine; it backs a figure, not any reported bound.
1033     """
1034     _require_networkx("gomory_hu_tree")
1035     return nx.gomory_hu_tree(_undirected_capacity_graph(graph), capacity="capacity
        ↳ ")
1036
1037
1038 def max_edge_connectivity_via_tree(graph: Graph) -> int:
1039     """‘lambda^max(G)’ read off the Gomory-Hu tree as its heaviest edge.
1040
1041     Returns ‘0’ for graphs with fewer than two vertices or no edges.
1042     """
1043     if graph.num_vertices < 2:
1044         return 0
1045     tree = gomory_hu_tree(graph)
1046     # READ-OFF: lambda(u, v) = lightest tree edge on the u-v path, so the LARGEST
1047     # such bottleneck over all pairs is simply the heaviest edge in the tree.
1048     weights = [data["weight"] for _, _, data in tree.edges(data=True)]
1049     if not weights:
1050         return 0
1051     return int(round(max(weights)))
1052
1053
1054 # --- PROVE: MILP for M*(n) via cut-counting (scipy/HiGHS or Gurobi backend) ---
1055 # L_m^dir(n) = (m-1)*M*(n) where M*(n) = max sum(w) s.t. maxflow(s,t;w)<=1 all
        ↳ pairs.
1056 # Flow constraint encoded exactly: choose a cut x in {0,1}^n per pair, cap
        ↳ crossing
1057 # weight via p; zero MIP gap = proof. Shared helpers build constraints for both
        ↳ backends.
1058
1059
1060 @dataclass
1061 class ProofResult:
1062     """The outcome of a proof run."""
1063
1064     n: int
1065     status: str # OPTIMAL, LIMIT, INFEASIBLE, UNBOUNDED, ...
1066     scaled_optimum: float # M*(n) = max total weight, exact if status OPTIMAL
1067     relative_gap_zero: bool # whether the solver closed the gap to zero
1068     solve_seconds: float
1069     weight_matrix: np.ndarray | None # a witnessing matrix w, if one was found
1070
1071     def value_for(self, m: int) -> int:
1072         """The proved directed multigraph value ‘L_m^dir(n) = (m-1) M*(n)’.””
1073         # Undo the (m-1) scaling: one proved M*(n) yields every m.
1074         return (m - 1) * int(round(self.scaled_optimum))
1075
1076     def is_proof(self) -> bool:
1077         """Whether this run is a genuine upper-bound proof (optimal, gap zero)."""
1078         # Only an OPTIMAL solve with a closed gap counts as a proof; a LIMIT
1079         # (timed out) result is merely a feasible lower bound, never a proof.
1080         return self.status == "OPTIMAL" and self.relative_gap_zero
1081

```

```

1082
1083 def _ordered_pairs(n: int) -> list[tuple[int, int]]:
1084     return [(u, v) for u in range(n) for v in range(n) if u != v]
1085
1086
1087 def _two_hop_triples(n: int) -> list[tuple[int, int, int]]:
1088     # All (source, middle, target) with three distinct vertices: the s -> x -> t
1089     # detours that the two-hop inequality reasons about.
1090     return [(s, x, t)
1091             for s in range(n) for t in range(n) if s != t
1092             for x in range(n) if x != s and x != t]
1093
1094
1095 # Proved values of M*(k) for the deletion cuts below (proved: M*(k) =
1096 # 2(k-1) for k <= 6, proved by this very optimisation with zero gap).
1097 _PROVEN_MSTAR = {2: 2, 3: 4, 4: 6, 5: 8, 6: 10}
1098
1099
1100 # -----
1101 # The cut-counting MILP: one PuLP model for both provers, any solver
1102 # -----
1103 # The two provers below are the SAME cut-counting optimisation over two weight
1104 # domains: the fractional one (prove_directed_multigraph, weights w in [0, 1], cut
1105 # cap 1) maximises the total weight M*(n); the integral one (prove_integral_arc_
1106 # bound, multiplicities mu in {0..m-1}, cut cap m-1) decides feasibility at a
1107 ↳ fixed
1108 # arc target. One builder serves both, and any MILP solver runs it: CBC (bundled
1109 # with pulp) by default, Gurobi by a one-line switch. Because the cut formulation
1110 ↳ proof.
1111
1112 _PULP_STATUS = {"Optimal": "OPTIMAL", "Infeasible": "INFEASIBLE",
1113                 "Unbounded": "UNBOUNDED"}
1114
1115 def _pick_solver(time_limit: float, show_log: bool, use_gurobi: bool | None):
1116     """Choose the MILP solver. 'gapRel=0' demands a closed gap, which is what
1117     makes an OPTIMAL verdict a proof rather than a heuristic.
1118
1119     'use_gurobi': 'True' forces Gurobi (error if it is not usable), 'False'
1120     forces CBC, 'None' uses Gurobi when its licence is active and CBC otherwise.
1121     Switching to Gurobi is the whole change needed to attack the open n=7 facts.
1122     """
1123     if use_gurobi is not False:
1124         gurobi = pulp.GUROBI_CMD(msg=show_log, timeLimit=time_limit,
1125                                  options=[("MIPGap", 0)])
1126         if gurobi.available():
1127             return gurobi
1128         if use_gurobi is True:
1129             raise RuntimeError(
1130                 "use_gurobi=True but no usable Gurobi was found. Install gurobipy
1131                 ↳ "
1132                 "and activate a licence, then retry."
1133             )
1134     return pulp.PULP_CBC_CMD(msg=show_log, timeLimit=time_limit, gapRel=0.0)
1135
1136 def _cut_counting_model(n: int, *, cap: float, integer: bool, two_hop: bool,

```

```

1137         symmetry: bool, deletion: bool, degree_pair: bool):
1138     """Build the shared cut-counting MILP as a PuLP model (no objective set yet).
1139
1140     For every ordered pair (s, t) the model CHOOSES one cut: the side indicators
1141     fix s on the source side (constant 1) and t on the sink side (constant 0), and
1142     every other vertex takes a binary side. The helper p picks up each crossing
1143     arc's weight through 'p >= w + cap*(x_u - x_v) - cap', which forces 'p >= w
1144     ↪ exactly on a crossing arc (x_u = 1, x_v = 0) and leaves p free at 0 otherwise.
1145     Capping the crossing total at 'cap' says maxflow(s, t) <= cap, and letting
1146     ↪ the
1147     optimisation pick the cut means it picks the MINIMUM cut, so this is exactly
1148     maxflow(s, t) <= cap, the true feasible region. 'cap=1' with continuous
1149     weights is the scaled prover; 'cap=m-1' with integer weights the arc one.
1150
1151     The optional families are all proved-valid inequalities that tighten the
1152     relaxation without ever moving the optimum: 'two_hop' (a two-arc-disjoint
1153     detour bound via z = min of the two hops), 'degree_pair' (a flow lower bound
1154     on each pair), 'deletion' (an induced-subgraph bound from proved smaller
1155     M*(k)), and 'symmetry' (a degree ordering that prunes relabelled duplicates)
1156     ↪ .
1157
1158     Returns '(prob, w)' with 'w' the weight variables keyed by ordered pair.
1159     """
1160     pairs = _ordered_pairs(n)
1161     prob = pulp.LpProblem("erdos915", pulp.LpMaximize)
1162     category = "Integer" if integer else "Continuous"
1163     w = {(u, v): pulp.LpVariable(f"w_{u}_{v}", 0, cap, cat=category)
1164          for (u, v) in pairs}
1165
1166     for (s, t) in pairs:
1167         # s is always on the source side, t always on the sink side: constants,
1168         ↪ not
1169         # variables (which also avoids pulp resetting a Binary's bounds to [0, 1])
1170         ↪ .
1171         side = {u: 1 if u == s else 0 if u == t
1172                 else pulp.LpVariable(f"x_{s}_{t}_{u}", cat="Binary")
1173                 for u in range(n)}
1174         crossing = []
1175         for (u, v) in pairs:
1176             p_uv = pulp.LpVariable(f"p_{s}_{t}_{u}_{v}", 0, cap)
1177             prob += p_uv >= w[(u, v)] + cap * (side[u] - side[v]) - cap
1178             crossing.append(p_uv)
1179         prob += pulp.lpSum(crossing) <= cap # crossing weight <= cap
1180
1181     if two_hop:
1182         z = {}
1183         for (s, x, t) in _two_hop_triples(n):
1184             z[(s, x, t)] = pulp.LpVariable(f"z_{s}_{x}_{t}", 0, cap)
1185             selector = pulp.LpVariable(f"b_{s}_{x}_{t}", cat="Binary")
1186             # z = min(w[s,x], w[x,t]): two upper bounds, and the selector forces z
1187             # up to whichever argument is the minimum (big-M with M = cap).
1188             prob += z[(s, x, t)] <= w[(s, x)]
1189             prob += z[(s, x, t)] <= w[(x, t)]
1190             prob += z[(s, x, t)] >= w[(s, x)] - cap * (1 - selector)
1191             prob += z[(s, x, t)] >= w[(x, t)] - cap * selector
1192
1193     for (s, t) in pairs:
1194         prob += w[(s, t)] + pulp.lpSum(
1195             z[(s, x, t)] for x in range(n) if x != s and x != t) <= cap

```

```

1190
1191 if degree_pair:
1192     #  $d(s) + d(t) - w[s,t] \leq (n-1)*cap$ . This is the two-hop bound
1193     # aggregated:  $w[s,t] + \sum_x \min(w[s,x], w[x,t]) \leq cap$ , plus each middle
1194     #  $\max(w[s,x], w[x,t]) \leq cap$ , sums to  $(n-1)*cap$ . It is a genuine cut, NOT
1195     # trivially satisfied: the box bound alone allows the LHS up to  $(2n-3)*cap$ 
1196     #  $\hookrightarrow$ ,
1197     # and a two-route witness reaches  $2(n-1)*cap$  (e.g.  $n=4$ :  $s \rightarrow x_1 \rightarrow t$ ,  $s \rightarrow x_2 \rightarrow t$ 
1198     # gives LHS 4 > 3). Dominated by two_hop when that family is on, but
1199     # standalone-useful when it is off; valid in both weight domains.
1200     for (s, t) in pairs:
1201         prob += (pulp.lpSum(w[(s, x)] for x in range(n) if x != s)
1202                 + pulp.lpSum(w[(x, t)] for x in range(n) if x != t)
1203                 - w[(s, t)]) <= (n - 1) * cap
1204
1205 if deletion:
1206     for k, holes in ((n - 1, 1), (n - 2, 2)):
1207         if k not in _PROVEN_MSTAR:
1208             continue
1209         bound = cap * _PROVEN_MSTAR[k] # induced subgraph on k vertices
1210         for gone in combinations(range(n), holes):
1211             prob += pulp.lpSum(w[(a, b)] for (a, b) in pairs
1212                               if a not in gone and b not in gone) <= bound
1213
1214 if symmetry:
1215     degree = [pulp.lpSum(w[(u, v)] for (u, v) in pairs if v0 in (u, v))
1216              for v0 in range(n)]
1217     for v0 in range(n - 1):
1218         prob += degree[v0] <= degree[v0 + 1]
1219
1220 return prob, w
1221
1222 def prove_directed_multigraph(
1223     n: int,
1224     *,
1225     time_limit: float = 1500.0,
1226     use_gurobi: bool | None = None,
1227     use_two_hop: bool = True,
1228     use_symmetry_breaking: bool = True,
1229     use_deletion_cuts: bool = False,
1230     show_solver_log: bool = False,
1231 ) -> ProofResult:
1232     """Prove ' $M*(n)$ ' for the directed multigraph arc problem.
1233
1234     Returns a :class:`ProofResult`. When its :meth:`is_proof` is true the value
1235     is a proven upper bound, and ' $value\_for(m)$ ' gives ' $L_m^{dir}(n)$ ' for every
1236     ' $m$ '. ' $use\_gurobi$ ' picks the solver (see :func:`_pick_solver`); the
1237      $\hookrightarrow$  default
1238     uses Gurobi when its licence is active and CBC otherwise.
1239     """
1240     _require_pulp()
1241     prob, w = _cut_counting_model(
1242         n, cap=1.0, integer=False, two_hop=use_two_hop,
1243         symmetry=use_symmetry_breaking, deletion=use_deletion_cuts,
1244         degree_pair=use_deletion_cuts,
1245     )
1246     prob += pulp.lpSum(w.values()) # maximise total weight =  $M*(n)$ 

```

```

1246
1247     start = time.time()
1248     prob.solve(_pick_solver(time_limit, show_solver_log, use_gurobi))
1249     elapsed = time.time() - start
1250     status = _PULP_STATUS.get(pulp.LpStatus[prob.status], "LIMIT")
1251
1252     weight_matrix = None
1253     if status == "OPTIMAL":
1254         weight_matrix = np.zeros((n, n))
1255         for (u, v), variable in w.items():
1256             weight_matrix[u, v] = variable.value() or 0.0
1257
1258     return ProofResult(
1259         n=n, status=status,
1260         scaled_optimum=(pulp.value(prob.objective) if status == "OPTIMAL"
1261                        else float("nan")),
1262         relative_gap_zero=(status == "OPTIMAL"),
1263         solve_seconds=elapsed, weight_matrix=weight_matrix,
1264     )

```

C.1.4 Searching

This part corresponds to [Chapter 3](#). Where proving is out of reach, the program looks for good examples instead. It starts from a graph, repeatedly makes a small random change, and keeps the change if it helps. Keeping only improvements gets stuck in the first decent shape it finds, so early on it also accepts some changes that make things worse, and it does this less and less as it goes, in the manner of hot metal cooling into an ordered state. A second engine, tabu search, walks the same landscape with a different rule: it always takes the best available move but refuses to revisit recent positions, which is what stops it circling.

Two ideas make this fast enough to be useful. The search is told which edges are load-bearing, so that it changes those rather than wasting moves on edges that carry nothing. And it almost never measures connectivity fully: adding one connection can raise the count by at most one and removing one can never raise it at all, so most steps can be settled by arithmetic on a bound the search already carries, with an exact measurement only when the answer is genuinely in doubt. This part ends with `solve`, the single entry point that either proves or searches depending on one argument.

```

1267 #####
1268 ##
1269 ##  CHAPTER 3 -- DISCOVERING BOUNDS BY SEARCH
1270 ##  The temperature search, and the one driver that proves or discovers.
1271 ##
1272 #####
1273
1274 # --- SENSITIVITY: sigma(e) = lambda^max(G) - lambda^max(G-e); dial for the
1275     ↪ cooling schedule ---
1276
1277 # A connectivity measure maps a graph to its lambda^max or kappa^max.
1278 ConnectivityMeasure = Callable[[Graph], int]

```

```

1279
1280 def edge_sensitivity(
1281     graph: Graph,
1282     u: int,
1283     v: int,
1284     connectivity: ConnectivityMeasure = max_edge_connectivity,
1285 ) -> int:
1286     """Return 'sigma(e)' for the edge or arc 'e = (u, v)'.
1287
1288     The edge is deleted in full (all parallel copies) before remeasuring, so the
1289     result reflects the structural role of the adjacency rather than of a single
1290     parallel copy. If the edge is absent the sensitivity is zero by convention.
1291     """
1292     if not graph.has_edge(u, v):
1293         return 0 # absent edge: nothing to remove, sensitivity is zero
1294     before = connectivity(graph)
1295     reduced = graph.copy()
1296     reduced.set_multiplicity(u, v, 0) # delete the WHOLE adjacency (all copies)
1297     after = connectivity(reduced)
1298     # How much  $\lambda^{\max}$  dropped: positive means e was load-bearing.
1299     return before - after
1300
1301
1302 def sensitivity_map(
1303     graph: Graph,
1304     connectivity: ConnectivityMeasure = max_edge_connectivity,
1305 ) -> dict[tuple[int, int], int]:
1306     """Return '{(u, v): sigma((u, v))}' over all present edges or arcs.
1307
1308     Useful for visualising an extremiser: high-sensitivity edges are the load
1309     bearers, low-sensitivity edges are slack.
1310
1311     The baseline connectivity 'before = connectivity(graph)' is the SAME for
1312     every edge, so it is measured once here rather than re-measured inside each
1313     :func:'edge_sensitivity' call. This changes no value (the result equals
1314     '{e: edge_sensitivity(graph, *e) for e in graph.edges()}'); it only avoids
1315     recomputing the shared baseline once per edge.
1316     """
1317     before = connectivity(graph)
1318     result: dict[tuple[int, int], int] = {}
1319     for u, v, _ in graph.edges():
1320         reduced = graph.copy()
1321         reduced.set_multiplicity(u, v, 0) # delete the WHOLE adjacency (all
            ↪ copies)
1322         result[(u, v)] = before - connectivity(reduced)
1323     return result
1324
1325
1326 # --- SEARCH: simulated annealing;  $E = -|E(G)| + \text{penalty} * \max(0, \lambda^{\max} - (m-1))$ 
            ↪ ---
1327 # Metropolis rule with geometric cooling; removal proposals biased by edge
            ↪ sensitivity.
1328
1329
1330 @dataclass
1331 class SearchStep:
1332     """One step of the search, recorded for plotting the temperature trace."""
1333

```

```

1334     step: int
1335     temperature: float
1336     energy: float
1337     edge_count: int
1338     connectivity: int
1339     accepted: bool
1340     seconds: float = 0.0          # wall-clock since the search began (for timed
        ↳ traces)
1341     best_feasible: int = 0        # densest feasible graph seen so far (for
        ↳ convergence)
1342
1343
1344 @dataclass
1345 class SearchResult:
1346     """The outcome of a search: the best feasible graph and the full trace."""
1347
1348     best_graph: Graph
1349     best_edge_count: int
1350     feasible_found: bool
1351     history: list[SearchStep] = field(default_factory=list)
1352
1353     def acceptance_rate(self) -> float:
1354         """Fraction of proposed moves that were accepted."""
1355         if not self.history:
1356             return 0.0
1357         return sum(record.accepted for record in self.history) / len(self.history)
1358
1359
1360 def _connectivity_measure(separation: str):
1361     """Return the connectivity function named by 'separation'."""
1362     if separation == "edge":
1363         return max_edge_connectivity
1364     if separation == "vertex":
1365         return max_vertex_connectivity
1366     raise ValueError("separation must be 'edge' or 'vertex'")
1367
1368
1369 def _energy(graph: Graph, m: int, measure, penalty: float) -> float:
1370     """The energy '-|E| + penalty * excess connectivity'."""
1371     # Excess is how far connectivity exceeds the allowed m-1; zero when feasible.
1372     excess = max(0, measure(graph) - (m - 1))
1373     # Lower energy is better: more edges (negative term) and no excess.
1374     return -graph.edge_count() + penalty * excess
1375
1376
1377 def _proposal_energy(
1378     proposal: Graph,
1379     *,
1380     is_add: bool,
1381     feasible_current: bool,
1382     lam_ub: int,
1383     m: int,
1384     penalty: float,
1385     measure_full,
1386     separation: str,
1387 ) -> tuple[float, bool]:
1388     """Return '(_energy(proposal), proposal_is_feasible)' doing the least work.
1389

```

```

1390 The energy '-|E| + penalty * max(0, connectivity - (m-1))' depends on the
1391 connectivity only through the excess, and by monotonicity (Facts M1/M2 in the
1392 thesis) that excess is provably zero on most steps, so no flow is needed:
1393
1394 - a removal from a feasible graph stays feasible (M1), excess 0;
1395 - an addition while 'lam_ub <= m-2' stays feasible (M2), excess 0.
1396
1397 Only when neither shortcut applies do we run one capped predicate, and only
    ↳ when
1398 that reports an infeasible proposal do we spend the exact 'measure_full' to
    ↳ get
1399 the penalty term right. The float returned is byte-for-byte the one '_energy
    ↳ '
1400 would return (the same arithmetic on the same excess), so the search
    ↳ trajectory
1401 is unchanged. The boolean is the EXACT feasibility of 'proposal'.
1402 """
1403 edges = proposal.edge_count()
1404 # Decide excess = max(0, measure(proposal) - (m-1)) with the least work.
1405 if (is_add and lam_ub <= m - 2) or (not is_add and feasible_current):
1406     excess = 0 # M2 (add) or M1 (remove): provably
    ↳ feasible
1407 elif not exceeds_bound(proposal, m - 1, separation=separation):
1408     excess = 0 # at the boundary but the capped flow
    ↳ clears it
1409 else:
1410     excess = measure_full(proposal) - (m - 1) # infeasible: exact penalty
    ↳ term
1411 # Identical arithmetic to _energy, so the returned float matches it exactly.
1412 return -edges + penalty * excess, excess == 0
1413
1414
1415 def _candidate_pairs(graph: Graph) -> list[tuple[int, int]]:
1416     """All ordered pairs (directed) or unordered pairs (undirected)."""
1417     n = graph.num_vertices
1418     pairs = []
1419     for u in range(n):
1420         start = 0 if graph.variant.directed else u + 1
1421         for v in range(start, n):
1422             if u != v:
1423                 pairs.append((u, v))
1424     return pairs
1425
1426
1427 def _propose_removal(
1428     graph: Graph,
1429     temperature: float,
1430     rng: random.Random,
1431     cached_sensitivity: dict[tuple[int, int], int] | None,
1432 ) -> tuple[int, int]:
1433     """Choose a present edge to thin out, biased toward free (low sigma) edges.
1434
1435     Without a sensitivity cache the choice is uniform. With one, edge 'e' is
1436     drawn with weight 'exp(-sigma(e) / T)': at high 'T' the weights flatten
1437     (uniform exploration), at low 'T' they concentrate on 'sigma = 0' edges.
1438     """
1439     present = [(u, v) for u, v, _ in graph.edges()]
1440     if cached_sensitivity is None:

```

```

1441         return rng.choice(present) # no guidance: uniform removal
1442     # exp(-sigma/T): big sigma (load-bearing) is suppressed, and the suppression
1443     # sharpens as T -> 0. max(T, eps) guards against division by zero.
1444     weights = [
1445         math.exp(-cached_sensitivity.get((u, v), 0) / max(temperature, 1e-9))
1446         for (u, v) in present
1447     ]
1448     return rng.choices(present, weights=weights, k=1)[0]
1449
1450
1451 def search_for_dense_graph(
1452     variant: Variant,
1453     n: int,
1454     m: int,
1455     *,
1456     separation: str = "edge",
1457     steps: int = 4000,
1458     initial_temperature: float = 3.0,
1459     cooling: float = 0.9985,
1460     penalty: float = 6.0,
1461     max_multiplicity: int | None = None,
1462     sensitivity_guided: bool = True,
1463     bias_refresh: int = 200,
1464     seed: int = 0,
1465     deadline: float | None = None,
1466     record_exact_connectivity: bool = False,
1467     reference_mode: bool = False,
1468 ) -> SearchResult:
1469     """Find the densest feasible graph by a guided random search (simulated
1470         ↪ annealing).
1471
1472     Args:
1473         variant: which graph model to search in.
1474         n: number of vertices.
1475         m: forbidden value. Feasibility means ‘connectivity <= m - 1’.
1476         separation: ‘“edge”’ for arc/edge-disjoint, ‘“vertex”’ for
1477             internally vertex-disjoint paths.
1478         steps: number of proposed moves.
1479         initial_temperature: starting temperature ‘T_0’.
1480         cooling: factor (< 1) the temperature is multiplied by each step.
1481         penalty: weight of the connectivity violation in the energy.
1482         max_multiplicity: cap on parallel edges (defaults to ‘m - 1’, a simple
1483             variant is always capped at one regardless).
1484         sensitivity_guided: bias removals toward free edges (the temperature
1485             dial described in the module docstring).
1486         bias_refresh: recompute the sensitivity cache every this many steps.
1487         seed: random seed, fixed so figures are reproducible.
1488         record_exact_connectivity: log the EXACT ‘lambda^max’ per step in
1489             ‘SearchStep.connectivity’ (the figure-trace needs it). When False,
1490             the cheaper maintained upper bound ‘lam_ub’ is logged instead, which
1491             does not affect the search at all (the energy never reads this field).
1492         reference_mode: INTERNAL, test only. Force the slow, exact per-step
1493             energy (call :func:‘_energy’ and :func:‘max_connectivity’ directly)
1494             instead of the capped/monotone fast path. Used by the equivalence
1495             test to prove the fast path reproduces this reference step for step.
1496
1497     Returns:
1498         An :class:‘SearchResult’ with the best feasible graph found and the

```

```

1498         full step-by-step history.
1499
1500     The fast path is *trajectory-preserving*: it computes the identical per-step
1501     energy as ‘reference_mode‘ (proven by ‘test_search.
        ↳ test_fast_path_matches_exact‘),
1502     so every accept/reject decision, the random-number draw order, and the
        ↳ returned
1503     witness are unchanged. It only avoids flows whose result the energy cannot
1504     depend on (Facts M1/M2: a removal from a feasible graph stays feasible, an
1505     addition while ‘lam_ub <= m-2‘ stays feasible), and resyncs the upper bound
1506     ‘lam_ub‘ to the exact value at the ‘bias_refresh‘ cadence.
1507     """
1508     rng = random.Random(seed)
1509     measure = _connectivity_measure(separation)
1510     # Multiplicity cap: simple graphs are always 0/1; multigraphs default to m-1.
1511     cap = 1 if variant.simple else (max_multiplicity if max_multiplicity else m -
        ↳ 1)
1512     pairs = None # built lazily once we know we need it
1513
1514     current = Graph(n, variant) # start from the empty graph: feasible, energy 0
1515     current_energy = _energy(current, m, measure, penalty)
1516
1517     best_graph = current.copy()
1518     best_edges = 0
1519     feasible_found = True # the empty graph is feasible
1520
1521     # State carried across steps so the energy can avoid recomputing connectivity
1522     # from scratch (see the trajectory-preserving note in the docstring):
1523     #   feasible -- is 'current' feasible (lambda^max <= m-1)? Exact at all times
        ↳ .
1524     #   lam_ub   -- a SOUND upper bound on lambda^max(current). Facts M1/M2 keep
        ↳ it
1525     #               sound: +1 on an accepted addition, unchanged on a removal.
1526     feasible = True # the empty graph is feasible
1527     lam_ub = 0 # lambda^max(empty graph) = 0
1528
1529     temperature = initial_temperature
1530     cached_sensitivity = None
1531     history: list[SearchStep] = []
1532     clock_start = time.perf_counter() # for the timed convergence trace only
1533
1534     for step in range(steps):
1535         # Periodically recompute the sensitivity map that guides removals. It
1536         # is expensive, so we cache it and refresh only every bias_refresh steps.
1537         if sensitivity_guided and step % bias_refresh == 0 and current.edge_count
            ↳ () > 0:
1538             cached_sensitivity = sensitivity_map(current, measure)
1539             # Resync the upper bound to the truth at the same cadence (the bound can
1540             # only drift upward over a run of removals, which costs extra capped
            ↳ checks
1541             # but never correctness). This step already pays for flows above, and the
1542             # resync touches neither the energy nor the RNG, so the trajectory is safe
            ↳ .
1543         if step % bias_refresh == 0:
1544             lam_ub = measure(current)
1545             feasible = lam_ub <= m - 1
1546
1547     proposal = current.copy()

```

```

1548     # Flip a coin: remove an edge (only if some exist) or add one.
1549     removing = current.edge_count() > 0 and rng.random() < 0.5
1550     if removing:
1551         # Sensitivity-guided removal: prefers free edges, especially cold.
1552         u, v = _propose_removal(current, temperature, rng, cached_sensitivity)
1553         proposal.remove_edge(u, v)
1554     else:
1555         if pairs is None:
1556             pairs = _candidate_pairs(current)
1557         # Only pairs that have room under the multiplicity cap can grow.
1558         addable = [(u, v) for (u, v) in pairs if proposal.multiplicity(u, v) <
1559                     ↪ cap]
1560         if not addable:
1561             continue # nothing to add; skip this step
1562         u, v = rng.choice(addable)
1563         proposal.add_edge(u, v)
1564     is_add = not removing
1565
1566     # The proposal's energy and exact feasibility. reference_mode is the slow
1567     ↪ ,
1568     # exact path the equivalence test compares against; the fast path returns
1569     # the IDENTICAL energy by Facts M1/M2 (see _proposal_energy).
1570     if reference_mode:
1571         proposal_energy = _energy(proposal, m, measure, penalty)
1572         proposal_feasible = measure(proposal) <= m - 1
1573     else:
1574         proposal_energy, proposal_feasible = _proposal_energy(
1575             proposal, is_add=is_add, feasible_current=feasible, lam_ub=lam_ub,
1576             m=m, penalty=penalty, measure_full=measure, separation=separation)
1577     change = proposal_energy - current_energy
1578     # ACCEPT OR NOT: always take an improvement (change <= 0); take a
1579     ↪ worsening
1580     # move only with probability exp(-change / T). As T falls this
1581     ↪ probability
1582     # shrinks, so the walk increasingly refuses to go uphill.
1583     accept = change <= 0 or rng.random() < math.exp(-change / max(temperature,
1584         ↪ 1e-9))
1585
1586     if accept:
1587         current, current_energy = proposal, proposal_energy
1588         # Carry the exact feasibility and update the bound by M2 (an accepted
1589         # addition can raise lambda^max by at most 1; a removal cannot raise
1590         ↪ it).
1591         feasible = proposal_feasible
1592         if is_add:
1593             lam_ub += 1
1594         # Track the best FEASIBLE graph found (within bound and denser than
1595         ↪ any
1596         # seen so far): this is the lower-bound witness. 'feasible' is exact
1597         # here, so the witness is certified with no extra flow.
1598         if feasible and current.edge_count() > best_edges:
1599             best_graph, best_edges = current.copy(), current.edge_count()
1600             feasible_found = True
1601
1602     # The connectivity field feeds fig:trace only. Log the exact value when
1603     ↪ the
1604     # figure asks for it (or in reference_mode), else the maintained upper
1605     ↪ bound.

```

```

1597         if record_exact_connectivity or reference_mode:
1598             recorded_connectivity = measure(current)
1599             # Cheap audit in the exact paths: the maintained bound must never
1600             # underestimate, and the feasibility flag must match the truth.
1601             assert lam_ub >= recorded_connectivity, (lam_ub, recorded_connectivity
1602             ↪ )
1603             assert feasible == (recorded_connectivity <= m - 1)
1604         else:
1605             recorded_connectivity = lam_ub
1606             history.append(SearchStep(
1607                 step=step,
1608                 temperature=temperature,
1609                 energy=current_energy,
1610                 edge_count=current.edge_count(),
1611                 connectivity=recorded_connectivity,
1612                 accepted=accept,
1613                 seconds=time.perf_counter() - clock_start,
1614                 best_feasible=best_edges,
1615             ))
1616             temperature *= cooling # cool down: T <- cooling * T each step
1617
1618             # Respect the deadline when supplied; check every 100 steps to amortise
1619             # the time.time() call.
1620             if deadline is not None and step % 100 == 99 and time.time() > deadline:
1621                 break
1622
1623         return SearchResult(best_graph, best_edges, feasible_found, history)
1624
1625 def _neighbour_moves(graph: Graph, cap: int) -> list[tuple[int, int, int]]:
1626     """Every single add ('+1') or remove ('-1') of one edge/arc under the cap.
1627
1628     The shared neighbourhood of both discoverers: one ordered pair for a digraph,
1629     one unordered pair for an undirected graph, addable while below the
1630     multiplicity 'cap' and removable while above zero.
1631     """
1632     n = graph.num_vertices
1633     directed = graph.variant.directed
1634     moves: list[tuple[int, int, int]] = []
1635     for u in range(n):
1636         for v in range(n):
1637             if u == v or (not directed and v < u):
1638                 continue
1639             mult = graph.multiplicity(u, v)
1640             if mult < cap:
1641                 moves.append((u, v, +1))
1642             if mult > 0:
1643                 moves.append((u, v, -1))
1644     return moves
1645
1646
1647 def tabu_search_for_dense_graph(
1648     variant: Variant,
1649     n: int,
1650     m: int,
1651     *,
1652     separation: str = "edge",
1653     steps: int = 4000,

```

```

1654     penalty: float = 6.0,
1655     tenure: int = 7,
1656     stall_limit: int = 40,
1657     max_multiplicity: int | None = None,
1658     seed: int = 0,
1659     deadline: float | None = None,
1660     record_exact_connectivity: bool = True,
1661 ) -> SearchResult:
1662     """Find the densest feasible graph by tabu search, the deterministic twin of
1663     :func:'search_for_dense_graph'.
1664
1665     Same energy '-|E| + penalty * max(0, connectivity - (m-1))' and the same
1666     neighbourhood (:func:'_neighbour_moves') as the annealer, but a different
1667     engine. Each step scans the whole neighbourhood and takes the best move whose
1668     pair is not on the tabu list, forbids that pair for 'tenure' steps, and
1669     perturbs from the incumbent once it has stalled for 'stall_limit' steps.
1670     Aspiration overrides the tabu flag for any move that beats the global best.
1671     There is no temperature and no accept-worse coin: the only randomness is the
1672     seed driving the perturbation kicks, so two runs from one seed are identical.
1673
1674     Returns the same :class:'SearchResult' as the annealer (the best feasible
1675     ↪ graph
1676     and the full timed history), so the two engines are interchangeable through
1677     ↪ the
1678     'method' argument of :func:'best_of_searches' and :func:'solve'.
1679     """
1680     rng = random.Random(seed)
1681     measure = _connectivity_measure(separation)
1682     cap = 1 if variant.simple else (max_multiplicity if max_multiplicity else m -
1683     ↪ 1)
1684
1685     current = Graph(n, variant) # start empty: feasible, energy 0
1686     best_graph, best_edges = current.copy(), 0
1687     tabu: dict[tuple[int, int], int] = {}
1688     stall = 0
1689     history: list[SearchStep] = []
1690     clock_start = time.perf_counter()
1691
1692     def trial_energy_and_feasibility(graph: Graph) -> tuple[float, bool]:
1693         """The annealer's energy and exact feasibility, by the same capped trick.
1694
1695         'exceeds_bound' decides feasibility with an early-exit capped flow, and
1696         the full 'measure' runs only on the rare infeasible trial that needs its
1697         penalty term, so most moves cost one cheap predicate instead of a full
1698         all-pairs flow. The returned float equals '_energy' exactly.
1699         """
1700         edges = graph.edge_count()
1701         if not exceeds_bound(graph, m - 1, separation=separation):
1702             return float(-edges), True # feasible: excess is zero
1703         return -edges + penalty * (measure(graph) - (m - 1)), False
1704
1705     for step in range(steps):
1706         if deadline is not None and time.time() > deadline:
1707             break
1708
1709         best_move: tuple[int, int, int] | None = None
1710         best_move_energy: float | None = None
1711         best_move_global = False
1712         for (u, v, d) in _neighbour_moves(current, cap):

```

```

1709         trial = current.copy()
1710         trial.add_edge(u, v) if d > 0 else trial.remove_edge(u, v)
1711         trial_energy, trial_feasible = trial_energy_and_feasibility(trial)
1712         improves_global = trial_feasible and trial.edge_count() > best_edges
1713         if tabu.get((u, v), 0) > step and not improves_global:
1714             continue # tabu, with no aspiration to override
1715         if (best_move_energy is None or trial_energy < best_move_energy
1716             or (improves_global and not best_move_global)):
1717             best_move, best_move_energy = (u, v, d), trial_energy
1718             best_move_global = improves_global
1719     if best_move is None: # every move tabu: clear the list,
1720         ↪ retry
1721         tabu.clear()
1722         continue
1723
1724     u, v, d = best_move
1725     current.add_edge(u, v) if d > 0 else current.remove_edge(u, v)
1726     tabu[(u, v)] = step + tenure
1727
1728     connectivity = measure(current) # computed anyway for best-tracking
1729     if connectivity <= m - 1 and current.edge_count() > best_edges:
1730         best_graph, best_edges = current.copy(), current.edge_count()
1731         stall = 0
1732     else:
1733         stall += 1
1734     history.append(SearchStep(
1735         step=step, temperature=0.0, energy=best_move_energy,
1736         edge_count=current.edge_count(),
1737         connectivity=connectivity if record_exact_connectivity else -1,
1738         accepted=True, # tabu always takes its best legal move
1739         seconds=time.perf_counter() - clock_start,
1740         best_feasible=best_edges,
1741     ))
1742
1743     if stall >= stall_limit: # perturb from the incumbent and reset
1744         current = best_graph.copy()
1745         for _ in range(rng.randint(2, 4)):
1746             kicks = _neighbour_moves(current, cap)
1747             if kicks:
1748                 a, b, dd = rng.choice(kicks)
1749                 current.add_edge(a, b) if dd > 0 else current.remove_edge(a, b)
1750                 ↪ )
1751         tabu.clear()
1752         stall = 0
1753
1754     return SearchResult(best_graph, best_edges, feasible_found=True, history=
1755         ↪ history)
1756
1757 def _run_one_search(args: tuple) -> SearchResult:
1758     """Module-level worker for parallel restarts (must be picklable for
1759     ↪ ProcessPoolExecutor)."""
1760     variant, n, m, seed, method, search_options = args
1761     engine = tabu_search_for_dense_graph if method == "tabu" else
1762         ↪ search_for_dense_graph
1763     return engine(variant, n, m, seed=seed, **search_options)

```

```

1762 def best_of_searches(
1763     variant: Variant,
1764     n: int,
1765     m: int,
1766     *,
1767     restarts: int = 6,
1768     seed: int = 0,
1769     parallel: bool = True,
1770     method: str = "sa",
1771     **search_options,
1772 ) -> SearchResult:
1773     """Run several independent discovery schedules and keep the densest result.
1774
1775     A single search can settle into a local optimum, so for reliable
1776     rediscovery we restart from a fresh random seed a handful of times and return
1777     the run that found the most edges. Any keyword understood by the chosen
1778     engine ('steps', 'penalty', 'separation', and so on) is forwarded
1779     unchanged.
1780
1781     Args:
1782         parallel: when 'True' (default), run the restarts concurrently using
1783             :class:`~concurrent.futures.ProcessPoolExecutor`. Each restart has
1784             a distinct seed so the runs are genuinely independent. Falls back
1785             to sequential execution if multiprocessing is unavailable.
1786         method: '"sa"' (default, simulated annealing,
1787             :func:`search_for_dense_graph`) or '"tabu"'
1788             (:func:`tabu_search_for_dense_graph`). Both optimise the same energy
1789             over the same neighbourhood and return the same result type.
1790     """
1791     if method not in ("sa", "tabu"):
1792         raise ValueError("method must be 'sa' or 'tabu'")
1793     task_args = [(variant, n, m, seed + r, method, search_options) for r in range(
1794         ↪ restarts)]
1795
1796     if parallel and restarts > 1:
1797         try:
1798             with concurrent.futures.ProcessPoolExecutor() as executor:
1799                 results = list(executor.map(_run_one_search, task_args))
1800             except (OSError, concurrent.futures.BrokenExecutor) as exc:
1801                 # Multiprocessing is genuinely unavailable here (a sandbox with no
1802                 # fork/spawn, or a worker that could not start). Fall back to a
1803                 # sequential run, but say so rather than masking the loss of cores.
1804                 # A bug inside the worker is NOT caught here, so it fails loudly.
1805                 warnings.warn(
1806                     f"parallel restarts unavailable ({exc!r}); running sequentially",
1807                     RuntimeWarning, stacklevel=2,
1808                 )
1809                 results = [_run_one_search(a) for a in task_args]
1810         else:
1811             results = [_run_one_search(a) for a in task_args]
1812     return max(results, key=lambda r: r.best_edge_count)
1813
1814 # --- SOLVE: unified driver; exhaustive=True proves, exhaustive=False discovers
1815 ↪ ---
1816
1817

```

```

1818 @dataclass
1819 class SolveResult:
1820     """The outcome of a solve, carrying an honest bound label."""
1821
1822     n: int
1823     m: int
1824     variant: str          # human label, e.g. "simple directed"
1825     separation: str       # "edge" or "vertex"
1826     value: int            # the extremal count we report
1827     bound: str            # "exact" / "lower" / "upper"
1828     method: str           # how the value was obtained
1829     seconds: float
1830     complete: bool        # an exhaustive run that actually finished
1831     witness: object | None # a Graph/Hypergraph attaining 'value', if any
1832     note: str = ""
1833
1834     @property
1835     def proven(self) -> bool:
1836         """Whether 'value' is a proved exact extremal value."""
1837         return self.bound == "exact"
1838
1839     def describe(self) -> str:
1840         """One readable line summarising the result."""
1841         relation = {"exact": "= ", "lower": ">=", "upper": "<="}[self.bound]
1842         line = (f"{self.variant} ({self.separation}), n={self.n}, m={self.m}: "
1843               f"value {relation}{self.value} "
1844               f"[{self.bound}, {self.method}, {self.seconds:.1f}s]")
1845         return line if not self.note else f"{line}\n    note: {self.note}"
1846
1847
1848     def _variant_for(directed: bool, simple: bool) -> Variant:
1849         """Pick the matching named Variant constant from the two booleans."""
1850         if directed:
1851             return SIMPLE_DIRECTED if simple else MULTI_DIRECTED
1852         return SIMPLE_UNDIRECTED if simple else MULTI_UNDIRECTED
1853
1854
1855     def _matrix_cells(n: int, directed: bool) -> list[tuple[int, int]]:
1856         """The off-diagonal matrix cells a graph of this kind may fill.
1857
1858         Directed graphs vary every ordered pair; undirected graphs vary only the
1859         upper triangle, since 'set_multiplicity' mirrors the lower half for us.
1860         """
1861         if directed:
1862             return [(u, v) for u in range(n) for v in range(n) if u != v]
1863         return [(u, v) for u in range(n) for v in range(n) if u < v]
1864
1865
1866     def _directed_witness(n: int, m: int, simple: bool) -> Graph | None:
1867         """The densest named directed construction that is feasible for '(n, m)'.
1868
1869         Used to exhibit a concrete graph attaining the proved value: the better
1870         of the double star and the (augmented) one-directional bipartite, or
1871         'None' if neither applies to this case.
1872         """
1873         candidates: list[Graph] = []
1874         if not (simple and m > 2):
1875             # a simple star needs m == 2
1876             candidates.append(double_star(n, m, directed=True)) # 2(n-1)(m-1) arcs

```

```

1876     if m == 2:
1877         candidates.append(one_directional_bipartite(n))           # floor(n^2/4) arcs
1878     elif (m - 2) < (n - n // 2):                                # augmented needs m-2 < ceil(n/2)
1879         candidates.append(augmented_bipartite(n, m))
1880     feasible = [g for g in candidates if max_edge_connectivity(g) <= m - 1]
1881     return max(feasible, key=lambda g: g.edge_count()) if feasible else None
1882
1883
1884 def _brute_force_matrix(
1885     variant: Variant, n: int, m: int, separation: str, deadline: float,
1886 ) -> tuple[int, Graph | None, bool]:
1887     """Exhaustively maximise edges over every graph of 'variant' on 'n'.
1888
1889     Returns '(best_edge_count, witness, completed)'. 'completed' is False if
1890     the budget ran out first, in which case the value is only a lower bound.
1891     A simple cell ranges over '{0, 1}'; a multigraph cell over '{0, ..., m-1}'
1892     because multiplicity 'm' already gives 'm' parallel disjoint routes.
1893     """
1894     measure = _connectivity_measure(separation)
1895     cells = _matrix_cells(n, variant.directed)
1896     span = 2 if variant.simple else m           # cell value lives in range(span)
1897     best_count, best_graph, completed = 0, None, True
1898     base = Graph(n, variant)
1899     for tick, values in enumerate(product(range(span), repeat=len(cells))):
1900         if tick % 256 == 0 and time.time() > deadline:
1901             completed = False                     # ran out before exhausting the space
1902             break
1903         candidate = base.copy()
1904         for (u, v), value in zip(cells, values):
1905             if value:
1906                 candidate.set_multiplicity(u, v, value)
1907             if measure(candidate) <= m - 1 and candidate.edge_count() > best_count:
1908                 best_count, best_graph = candidate.edge_count(), candidate.copy()
1909     return best_count, best_graph, completed
1910
1911
1912 def _search_within_budget(
1913     variant: Variant, n: int, m: int, separation: str,
1914     deadline: float, seed: int, method: str = "sa",
1915 ) -> SearchResult:
1916     """Repeated search restarts until the deadline; keep the densest run."""
1917     engine = tabu_search_for_dense_graph if method == "tabu" else
1918         ↪ search_for_dense_graph
1919     best: SearchResult | None = None
1920     restart = 0
1921     while True:
1922         # Pass the deadline through so a single long restart can be cut short.
1923         outcome = engine(variant, n, m, separation=separation,
1924                         steps=4000, seed=seed + restart, deadline=deadline)
1925         if best is None or outcome.best_edge_count > best.best_edge_count:
1926             best = outcome
1927         restart += 1
1928         if time.time() > deadline or restart >= 200:
1929             break
1930     assert best is not None
1931     return best
1932

```

```

1933 def _hyperedge_candidates(n: int, r: int, directed: bool, *, kind: str = "forward"
↳ ) -> list:
1934     """Every possible ‘r’-uniform hyperedge.
1935
1936     Undirected: every ‘r’-subset. Directed: the ‘kind’ selects the
1937     orientation model.
1938
1939     * “forward” (default): one tail, ‘r - 1’ heads, stored in the legacy
1940       ‘(tail:int, heads)’ form so existing forward results are untouched.
1941     * “backward”: ‘r - 1’ tails, one head (the arc-reversal dual of
1942       forward), stored in the general ‘(tails, heads)’ form.
1943     * “general”: every split of an ‘r’-subset into a non-empty tail set and
1944       a non-empty head set, i.e. forward, backward, and (from ‘r >= 4’) mixed.
1945     """
1946     if not directed:
1947         return [frozenset(s) for s in combinations(range(n), r)]
1948     if kind == "forward":
1949         # One tail and r-1 heads chosen from the rest (legacy form).
1950         return [(tail, frozenset(heads))
1951                 for tail in range(n)
1952                 for heads in combinations([v for v in range(n) if v != tail], r -
↳ 1)]
1953     if kind == "backward":
1954         # r-1 tails and one head: the reverse of every forward arc.
1955         return [(frozenset(tails), frozenset((head,)))
1956                 for head in range(n)
1957                 for tails in combinations([v for v in range(n) if v != head], r -
↳ 1)]
1958     if kind == "general":
1959         # Every ordered (tails, heads) split of each r-subset, both sides non-
↳ empty.
1960         out: list = []
1961         for subset in combinations(range(n), r):
1962             for mask in range(1, (1 << r) - 1): # exclude all-heads and all-
↳ tails
1963                 tails = frozenset(subset[i] for i in range(r) if mask & (1 << i))
1964                 heads = frozenset(subset[i] for i in range(r) if not (mask & (1 <<
↳ i)))
1965                 out.append((tails, heads))
1966         return out
1967     raise ValueError(f"unknown directed hyperedge kind {kind!r}")
1968
1969
1970 def _brute_force_hypergraph(
1971     n: int, r: int, m: int, deadline: float,
1972     *, directed: bool = False, vertex_split: bool = False, kind: str = "forward",
1973 ) -> tuple[int, Hypergraph | None, bool]:
1974     """Exhaustively maximise hyperedges over all ‘r’-uniform hypergraphs.
1975
1976     Each possible hyperedge is present or absent, so the search visits
1977     ‘2 ** (#candidates)’ hypergraphs; only tiny ‘(n, r)’ finish. The
1978     ‘directed’ and ‘vertex_split’ flags select which of the four hypergraph
1979     measures decides feasibility, exactly as ‘solve’ passes them through, and
1980     ‘kind’ selects the directed orientation model (forward/backward/general).
1981     """
1982     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
1983     best_count, best_h, completed = 0, None, True
1984     for tick, mask in enumerate(product((0, 1), repeat=len(candidates))):

```

```

1985         if tick % 256 == 0 and time.time() > deadline:
1986             completed = False
1987             break
1988         chosen = [candidates[i] for i, on in enumerate(mask) if on]
1989         hypergraph = Hypergraph(n, chosen, directed=directed)
1990         if (max_hyper_connectivity(hypergraph, vertex_split=vertex_split) <= m - 1
1991             and hypergraph.edge_count() > best_count):
1992             best_count, best_h = hypergraph.edge_count(), hypergraph
1993     return best_count, best_h, completed
1994
1995
1996 def _random_hypergraph_search(
1997     n: int, r: int, m: int, deadline: float, seed: int,
1998     *, directed: bool = False, vertex_split: bool = False, kind: str = "forward",
1999 ) -> tuple[int, Hypergraph | None]:
2000     """Greedy randomised growth: add random hyperedges while feasible, restart.
2001
2002     A discovery heuristic for the hypergraph model (search is matrix-only):
2003     each pass shuffles the candidate hyperedges and adds each one that keeps the
2004     Berge connectivity within ‘m - 1’; the densest pass within budget wins. The
2005     feasible hypergraph it returns is the easy construction behind a lower bound.
2006     ‘kind’ selects the directed orientation model.
2007     """
2008     rng = random.Random(seed)
2009     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
2010     best_count, best_h = 0, None
2011     while time.time() < deadline:
2012         order = candidates[:]
2013         rng.shuffle(order)
2014         hypergraph = Hypergraph(n, directed=directed)
2015         for edge in order:
2016             hypergraph.add_hyperedge(edge)
2017             if max_hyper_connectivity(hypergraph, vertex_split=vertex_split) > m -
2018                 ↪ 1:
2019                 hypergraph.hyperedges.pop() # this one broke feasibility; undo
2020             if hypergraph.edge_count() > best_count:
2021                 best_count, best_h = hypergraph.edge_count(), hypergraph
2022     return best_count, (best_h if best_h is not None else Hypergraph(n, directed=
2023         ↪ directed))
2024
2025
2026 def _arc_flow_at_least(out_adj: list[set[int]], n: int, s: int, t: int,
2027     k: int) -> bool:
2028     """True if there are at least ‘k’ arc-disjoint ‘s’-‘t’ paths.
2029
2030     A plain Ford--Fulkerson with unit capacities: find an augmenting path in the
2031     residual (forward arcs and the reverse arcs left by earlier augmentations),
2032     augment, and repeat at most ‘k’ times. Used as the inner feasibility test
2033     of the exhaustive digraph search, where the question is only whether the
2034     local connectivity has reached the forbidden value ‘m’ (so ‘k = m’).
2035     """
2036     cap: dict[tuple[int, int], int] = {}
2037     for a in range(n):
2038         for b in out_adj[a]:
2039             cap[(a, b)] = 1
2040     for _ in range(k):
2041         prev = {s: s}
2042         stack = [s] # DFS for any residual augmenting path

```

```

2041     while stack:
2042         x = stack.pop()
2043         if x == t:
2044             break
2045         for y in range(n):
2046             if y not in prev and cap.get((x, y), 0) > 0:
2047                 prev[y] = x
2048                 stack.append(y)
2049     if t not in prev:
2050         return False # no further path: fewer than k exist
2051     node = t # walk back, pushing one unit of flow
2052     while node != s:
2053         p = prev[node]
2054         cap[(p, node)] -= 1
2055         cap[(node, p)] = cap.get((node, p), 0) + 1
2056         node = p
2057     return True
2058
2059
2060 def _vertex_flow_at_least(out_adj: list[set[int]], n: int, s: int, t: int,
2061                          k: int) -> bool:
2062     """True if there are at least 'k' internally vertex-disjoint 's'-'t'
2063         ⇨ paths.
2064
2065     The vertex twin of :func:`_arc_flow_at_least`, and the same Ford--Fulkerson
2066     loop, run on the split network instead of the plain one: every vertex 'x'
2067     becomes an entry copy '2x' and an exit copy '2x+1' joined by a single
2068     unit-capacity arc, so a path may pass through 'x' at most once. The two
2069     endpoints keep an uncapped gate, since a route is allowed to start at 's'
2070     and finish at 't'. A direct arc 's -> t' therefore counts as one route
2071     with no interior, exactly as the thesis's separation axis intends.
2072     """
2073     cap: dict[tuple[int, int], int] = {}
2074     for x in range(n):
2075         cap[(2 * x, 2 * x + 1)] = k if (x == s or x == t) else 1
2076     for a in range(n):
2077         for b in out_adj[a]:
2078             cap[(2 * a + 1, 2 * b)] = 1
2079     source, sink = 2 * s + 1, 2 * t
2080     for _ in range(k):
2081         prev = {source: source}
2082         stack = [source]
2083         while stack:
2084             x = stack.pop()
2085             if x == sink:
2086                 break
2087             for y in range(2 * n):
2088                 if y not in prev and cap.get((x, y), 0) > 0:
2089                     prev[y] = x
2090                     stack.append(y)
2091         if sink not in prev:
2092             return False
2093     node = sink
2094     while node != source:
2095         p = prev[node]
2096         cap[(p, node)] -= 1
2097         cap[(node, p)] = cap.get((node, p), 0) + 1
2098         node = p

```

```

2098     return True
2099
2100
2101 def _exhaustive_directed(
2102     n: int, m: int, separation: str, deadline: float,
2103     stats: dict[str, int] | None = None,
2104 ) -> tuple[int, Graph | None, bool]:
2105     """Prove the simple directed maximum by a pruned exhaustive search.
2106
2107     This is the honest version of the thesis's base-case search: branch and bound
2108     over every simple digraph on 'n' vertices, keeping the densest one whose
2109     connectivity stays within 'm - 1'. Two prunes make it finish where naive
2110     enumeration cannot. Feasibility is monotone, so the include branch adds an
2111     arc only when the digraph is still feasible (an infeasible prefix can never be
2112     rescued by adding more arcs), and the bound prune drops a subtree once even
2113     taking every remaining arc could not beat the best found. Both separations
2114     use the same incremental feasibility test: a new arc can only lift the
2115     connectivity of a pair whose source reaches its tail and whose sink is
2116     reachable from its head, so only those pairs are re-measured, with
2117     :func: '_arc_flow_at_least' or :func: '_vertex_flow_at_least' according to the
2118     separation. Returns '(max_count, witness, completed)'; 'completed' is
2119     False if the time budget ran out first, leaving the value a lower bound.
2120     Pass 'stats' to receive the number of search-tree nodes visited.
2121     """
2122     pairs = [(u, v) for u in range(n) for v in range(n) if u != v]
2123     total = len(pairs)
2124     out: list[set[int]] = [set() for _ in range(n)]
2125     inc: list[set[int]] = [set() for _ in range(n)]
2126     # Seed the best at one below a known construction so the bound prune bites
2127     # immediately, and the search still records an actual witness when it ties it.
2128     # The same seed is valid in both separations: by Whitney's inequality
2129     #  $\kappa \leq \lambda$ , a digraph feasible for the arc problem is feasible for the
2130     # vertex problem too, so an arc construction is an honest vertex lower bound.
2131     seed = _directed_witness(n, m, simple=True)
2132     seed_value = seed.edge_count() if seed is not None else 0
2133     best_count = [max(0, seed_value - 1)]
2134     best_arcs: list[tuple[int, int]] = []
2135     timed_out = [False]
2136
2137     def reaches_to(target: int) -> set[int]:
2138         seen, stack = {target}, [target] # reverse reachability (can reach t)
2139         while stack:
2140             x = stack.pop()
2141             for y in inc[x]:
2142                 if y not in seen:
2143                     seen.add(y); stack.append(y)
2144         return seen
2145
2146     def reaches_from(source: int) -> set[int]:
2147         seen, stack = {source}, [source] # forward reachability (s can reach)
2148         while stack:
2149             x = stack.pop()
2150             for y in out[x]:
2151                 if y not in seen:
2152                     seen.add(y); stack.append(y)
2153         return seen
2154
2155     flow_at_least = (_arc_flow_at_least if separation == "edge"

```

```

2156         else _vertex_flow_at_least)
2157
2158     def feasible_after(u: int, v: int) -> bool:
2159         for s in reaches_to(u):
2160             for t in reaches_from(v):
2161                 if s != t and flow_at_least(out, n, s, t, m):
2162                     return False # this pair reached m disjoint paths
2163         return True
2164
2165     def recurse(idx: int, count: int) -> None:
2166         if stats is not None:
2167             stats["nodes"] = stats.get("nodes", 0) + 1
2168         if time.time() > deadline:
2169             timed_out[0] = True
2170             return
2171         if count + (total - idx) <= best_count[0]:
2172             return # cannot beat the incumbent: prune
2173         if idx == total:
2174             if count > best_count[0]:
2175                 best_count[0] = count
2176                 best_arcs[:] = [(a, b) for a in range(n) for b in out[a]]
2177             return
2178         u, v = pairs[idx]
2179         out[u].add(v); inc[v].add(u) # include the arc, if still feasible
2180         if feasible_after(u, v):
2181             recurse(idx + 1, count + 1)
2182         out[u].discard(v); inc[v].discard(u)
2183         if timed_out[0]:
2184             return
2185         recurse(idx + 1, count) # exclude the arc
2186
2187     recurse(0, 0)
2188     witness = None
2189     if best_arcs:
2190         witness = Graph(n, SIMPLE_DIRECTED)
2191         for (a, b) in best_arcs:
2192             witness.mu[a, b] = 1
2193     elif seed is not None:
2194         witness = seed # max equalled the seed construction
2195     # A completed search always reaches at least the seed's count, so the max
2196     # below only matters on a timeout before any recorded leaf: there it lifts
2197     # the reported lower bound from seed_value - 1 (the pruning incumbent) to
2198     # the seed witness's own arc count, keeping value and witness consistent.
2199     return max(best_count[0], seed_value), witness, not timed_out[0]
2200
2201
2202 def solve(
2203     n: int, m: int, *,
2204     directed: bool = False, simple: bool = True,
2205     hypergraph: bool = False, r: int = 3,
2206     exhaustive: bool = False, separation: str = "edge",
2207     max_seconds: float = 60.0, seed: int = 0, method: str = "tabu",
2208 ) -> SolveResult:
2209     """The single driver: prove the exact value, or discover a dense example.
2210
2211     Discovery defaults to 'method="tabu"' : tabu search is the stronger engine on
2212     the harder directed problems and the one used to solve the problems in this
2213     work. Simulated annealing ('method="sa"') is kept for the self-check and

```

```

2214     verification, where the lighter engine is enough.
2215
2216     Args:
2217         n, m: the vertices and the forbidden number of independent routes.
2218         directed, simple: the two matrix axes (ignored when 'hypergraph').
2219         hypergraph, r: switch to the 'r'-uniform hypergraph model instead.
2220         exhaustive: 'True' to PROVE the optimum, 'False' to DISCOVER one.
2221         separation: '"edge"' or '"vertex"' disjointness (matrix models).
2222         max_seconds: wall-clock budget; always respected.
2223         seed: random seed for the discovery searches.
2224
2225     Returns:
2226         A :class:'SolveResult' whose 'bound' is '"exact"' (proved),
2227         '"lower"' (a witness), or '"upper"' (proved, no matching witness).
2228     """
2229     start = time.time()
2230     deadline = start + max_seconds
2231
2232     # ----- the hypergraph model -----
2233     # Same driver, a different internal measure: the edge/vertex separation and
2234     # the direction pick which of the four Berge measures decides feasibility.
2235     if hypergraph:
2236         vertex_split = (separation == "vertex")
2237         label = f"{r}-uniform {'directed ' if directed else '}'hypergraph"
2238         if exhaustive:
2239             value, witness, done = _brute_force_hypergraph(
2240                 n, r, m, deadline, directed=directed, vertex_split=vertex_split)
2241             bound = "exact" if done else "lower"
2242             method = "brute-force enumeration"
2243             note = "" if done else "budget ran out; value is only a lower bound"
2244         else:
2245             value, witness = _random_hypergraph_search(
2246                 n, r, m, deadline, seed, directed=directed, vertex_split=
2247                 ↪ vertex_split)
2248             bound, done, method = "lower", True, "randomised greedy search"
2249             note = "discovery only ever yields a lower bound"
2250         return SolveResult(n, m, label, separation, value, bound, method,
2251                             time.time() - start, done, witness, note)
2252
2253     # ----- the matrix models -----
2254     variant = _variant_for(directed, simple)
2255     label = variant.describe()
2256
2257     # DISCOVER: search within the budget; the witness found is a lower bound.
2258     if not exhaustive:
2259         if method not in ("sa", "tabu"):
2260             raise ValueError("method must be 'sa' or 'tabu'")
2261         result = _search_within_budget(variant, n, m, separation, deadline, seed,
2262             ↪ method)
2263         method_label = "tabu search" if method == "tabu" else "random search"
2264         return SolveResult(
2265             n, m, label, separation, result.best_edge_count, "lower",
2266             method_label, time.time() - start, True,
2267             result.best_graph, "discovery only ever yields a lower bound")
2268
2269     # EXHAUSTIVE, simple directed: a pruned exhaustive digraph search is exact.
2270     # This is the prover for the m=2 base cases of the directed theorem, and it
2271     # reaches n = 6, 7 where the cut-counting cannot close the gap in any sane

```

```

2270         ↪ time.
2271     if directed and simple:
2272         value, witness, done = _exhaustive_directed(n, m, separation, deadline)
2273         bound = "exact" if done else "lower"
2274         method = "exhaustive digraph search (branch and bound)"
2275         note = " if done else "budget ran out; value is only a lower bound"
2276         return SolveResult(n, m, label, separation, value, bound, method,
2277                             time.time() - start, done, witness, note)
2278
2279     # EXHAUSTIVE, directed MULTIGRAPH arc problem: the cut-counting is the prover.
2280     if directed and separation == "edge":
2281         budget = max(1.0, deadline - time.time())
2282         proof = prove_directed_multigraph(n, time_limit=budget)
2283         if proof.is_proof():
2284             # We only reach here for a multigraph, where (m-1) M*(n) IS the value.
2285             value = proof.value_for(m) # (m-1) * M*(n)
2286             witness = _directed_witness(n, m, simple)
2287             return SolveResult(n, m, label, separation, value, "exact",
2288                               "cut-counting (gap zero)", time.time() - start,
2289                               True, witness, "")
2290         # Timed out: fall back to the best construction as a lower bound.
2291         witness = _directed_witness(n, m, simple)
2292         low = witness.edge_count() if witness else 0
2293         return SolveResult(
2294             n, m, label, separation, low, "lower",
2295             "cut-counting hit the time limit; reporting a construction",
2296             time.time() - start, False, witness,
2297             "raise max_seconds to let the cut-counting close the gap")
2298
2299     # EXHAUSTIVE otherwise (undirected, or vertex separation): brute force.
2300     value, witness, done = _brute_force_matrix(
2301         variant, n, m, separation, deadline)
2302     bound = "exact" if done else "lower"
2303     method = "brute-force enumeration"
2304     note = (" if done else "budget ran out; value is only a lower bound "
2305            "(no cut-counting exists for this case)")
2306     return SolveResult(n, m, label, separation, value, bound, method,
2307                       time.time() - start, done, witness, note)

```

C.1.5 The random model

Here begins the material behind [Chapter 4](#). This first piece builds random graphs and measures them, which is how the thesis looks at the typical case rather than the extreme one. Each of the six models gets its own sampler, and the multigraph samplers use a two-stage rule: decide whether a pair is connected at all, then decide how many parallel copies it gets, which recovers the ordinary random graph when the second stage is switched off.

```

2309 #####
2310 ##
2311 ## CHAPTER 4 -- SYNTHESIS AND RESULTS
2312 ## Sampling the random model, the thesis figures, and the self-check.
2313 ##
2314 #####
2315
2316 # --- MONTE CARLO: sample G(n,p) graphs, measure with exact checker, average ---

```

```

2317
2318
2319 def sample_random_graph(n: int, p: float, directed: bool, rng: random.Random) ->
    ↳ Graph:
2320     """Draw one sample of  $G(n, p)$  (or the random digraph  $D(n, p)$ ).
2321
2322     Each possible edge is included independently with probability ‘p’: unordered
2323     pairs for an undirected graph, ordered pairs for a digraph.
2324     """
2325     variant = SIMPLE_DIRECTED if directed else SIMPLE_UNDIRECTED
2326     graph = Graph(n, variant)
2327     for u in range(n):
2328         # Undirected: scan  $v > u$  (each pair once). Directed: scan all  $v$ .
2329         start = 0 if directed else u + 1
2330         for v in range(start, n):
2331             if u != v and rng.random() < p: # include this edge with prob  $p$ 
2332                 graph.add_edge(u, v)
2333     return graph
2334
2335
2336 def estimate_appearance_probability(
2337     n: int, p: float, m: int, *, trials: int = 200,
2338     separation: str = "edge", directed: bool = False, seed: int = 0,
2339 ) -> float:
2340     """Estimate the probability that a sample has  $\lambda^{\max} \geq m$ ."""
2341     rng = random.Random(seed)
2342     measure = _connectivity_measure(separation)
2343     # Count the fraction of samples that already exhibit  $m$  independent routes
2344     # somewhere ( $\lambda^{\max} \geq m$ ): the empirical appearance probability.
2345     appearances = sum(
2346         measure(sample_random_graph(n, p, directed, rng)) >= m
2347         for _ in range(trials)
2348     )
2349     return appearances / trials
2350
2351
2352 def connectivity_distribution(
2353     n: int, p: float, *, trials: int = 200,
2354     separation: str = "edge", directed: bool = False, seed: int = 0,
2355 ) -> list[int]:
2356     """Return the list of  $\lambda^{\max}$  (or  $\kappa^{\max}$ ) over samples.
2357
2358     Same arguments as the estimators above. The returned list has one entry per
2359     sample, so its histogram shows how the binding connectivity is distributed
2360     for that kind of graph. Calling this with the same ‘seed’ and ‘(n, p)’
2361     but different ‘separation’ reuses the very same random graphs, which is why
2362     the edge and vertex distributions can be compared sample by sample.
2363     """
2364     rng = random.Random(seed)
2365     measure = _connectivity_measure(separation)
2366     # Same seed + same (n, p) reproduces the identical sequence of graphs, so an
2367     # edge run and a vertex run can be compared sample by sample (Whitney check).
2368     return [measure(sample_random_graph(n, p, directed, rng)) for _ in range(
        ↳ trials)]
2369
2370
2371 @dataclass
2372 class ThresholdCurve:

```

```

2373     """The appearance probability sampled across a range of densities."""
2374
2375     n: int
2376     m: int
2377     probabilities: list[tuple[float, float]] # (p, estimated probability)
2378
2379     @property
2380     def predicted_threshold(self) -> float:
2381         """The proved location  $p^{\{*\}} = m/n$ ."""
2382         return self.m / self.n
2383
2384
2385     def threshold_curve(
2386         n: int, m: int, p_values: list[float], *, trials: int = 200,
2387         separation: str = "edge", directed: bool = False, seed: int = 0,
2388     ) -> ThresholdCurve:
2389         """Sweep 'p' and return the appearance probability at each value."""
2390         points = [
2391             (p, estimate_appearance_probability(
2392                 n, p, m, trials=trials, separation=separation, directed=directed, seed
2393                 ↪ =seed))
2394             for p in p_values
2395         ]
2396         return ThresholdCurve(n=n, m=m, probabilities=points)
2397
2398     # --- edge against vertex disjointness in the random model -----
2399     #
2400     # Whitney's inequality  $\kappa^{\max} \leq \lambda^{\max}$  leaves open whether the two
2401     # binding connectivities of a graph actually differ. For the EXTREMAL problem
2402     # they first part company at  $m = 5$  (Sorensen-Thomassen). The function below lets
2403     # the same split be watched in the random model: it draws samples of  $G(n, p)$  and
2404     # measures both maxima on each one. On small graphs, where the binding
2405     # connectivity stays at or below four, the two coincide on every sample; on
2406     # larger, denser graphs, where it climbs past five, a pair can carry more
2407     # edge-disjoint routes than internally vertex-disjoint ones, and  $\kappa^{\max}$  drops
2408     # below  $\lambda^{\max}$  on a growing fraction of samples.
2409
2410
2411     def edge_vertex_distribution(
2412         n: int, p: float, *, trials: int = 300, seed: int = 0,
2413     ) -> tuple[list[int], list[int]]:
2414         """Return paired  $(\lambda^{\max}, \kappa^{\max})$  over the same samples of  $G(n, p)$ .
2415
2416         Both lists are measured on the identical sequence of random graphs, because
2417         connectivity_distribution reuses the seed, so the two can be compared sample
2418         by sample: the fraction of entries with  $\kappa^{\max} < \lambda^{\max}$  is exactly how
2419         often vertex disjointness is strictly more demanding than edge disjointness
2420         at that size and density.
2421         """
2422         lambda_max = connectivity_distribution(
2423             n, p, trials=trials, separation="edge", seed=seed)
2424         kappa_max = connectivity_distribution(
2425             n, p, trials=trials, separation="vertex", seed=seed)
2426         return lambda_max, kappa_max
2427
2428

```

```

2429 # --- SAMPLING ALL VARIANTS: Bernoulli for simple/hypergraph, hurdle-geometric for
    ↪ multigraph ---
2430
2431
2432 def sample_random_multigraph(
2433     n: int, p: float, alpha: float, directed: bool, rng: random.Random,
2434     *, cap: int | None = None,
2435 ) -> Graph:
2436     """Random multigraph with hurdle-geometric edge multiplicities.
2437
2438     Each cell is empty with probability '1 - p'; otherwise it carries one copy
2439     plus a Geometric('alpha') number of further copies, so the multiplicity
2440     decays exponentially with rate 'alpha' and 'alpha = 0' recovers simple
2441     'G(n, p)'. With 'cap' set (use 'm - 1') a single fat edge cannot
2442     trivially break feasibility, so the panel stays a structural question.
2443     """
2444     variant = MULTI_DIRECTED if directed else MULTI_UNDIRECTED
2445     graph = Graph(n, variant)
2446     for u in range(n):
2447         start = 0 if directed else u + 1
2448         for v in range(start, n):
2449             if u == v or rng.random() >= p:
2450                 continue
2451             k = 1
2452             while rng.random() < alpha:
2453                 k += 1
2454             graph.set_multiplicity(u, v, k if cap is None else min(k, cap))
2455     return graph
2456
2457
2458 def sample_random_hypergraph(
2459     n: int, p: float, r: int, directed: bool, rng: random.Random,
2460 ) -> Hypergraph:
2461     """Random 'r'-uniform hypergraph: each candidate hyperedge included w.p. 'p'
    ↪ 'p'"""
2462     chosen = [c for c in _hyperedge_candidates(n, r, directed) if rng.random() < p
    ↪ ]
2463     return Hypergraph(n, chosen, directed=directed)
2464
2465
2466 def _sample_variant(n, p, *, directed, simple, hypergraph, r, alpha, cap, rng):
2467     """Draw one sample of whichever generative model the flags select."""
2468     if hypergraph:
2469         return sample_random_hypergraph(n, p, r, directed, rng)
2470     if simple:
2471         return sample_random_graph(n, p, directed, rng)
2472     return sample_random_multigraph(n, p, alpha, directed, rng, cap=cap)
2473
2474
2475 def _measure_variant(obj, *, separation, hypergraph):
2476     """Binding connectivity of a sampled object under the chosen separation."""
2477     if hypergraph:
2478         return max_hyper_connectivity(obj, vertex_split=(separation == "vertex"))
2479     return _connectivity_measure(separation)(obj)
2480
2481
2482 def _mean_binding_degree(obj, *, directed, hypergraph):

```

```

2483     """The degree that the degree bound caps connectivity by, averaged over
        ↪ vertices.
2484
2485     Undirected: total incident multiplicity. Directed: ‘‘min(d+, d-)’’ (the
2486     quantity bounding an arc-disjoint count). Hypergraph: the Berge degree, the
2487     number of hyperedges through a vertex.
2488     """
2489     n = obj.num_vertices
2490     if hypergraph:
2491         deg = [0] * n
2492         for edge in obj.hyperedges:
2493             for v in obj.members(edge):
2494                 deg[v] += 1
2495         return sum(deg) / n
2496     if directed:
2497         return sum(min(obj.out_degree(v), obj.in_degree(v)) for v in range(n)) / n
2498     return sum(obj.degree(v) for v in range(n)) / n
2499
2500
2501 def _p_for_target_degree(target_deg, n, *, directed, simple, hypergraph, r, alpha)
        ↪ :
2502     """Presence probability ‘‘p’’ that puts the expected degree near ‘‘target_deg
        ↪ ‘‘.
2503
2504     Only an aiming heuristic -- the figures plot the *measured* mean degree -- so
2505     the exact value of the cap or of the geometric tail does not need inverting.
2506     """
2507     if hypergraph:
2508         per_vertex = math.comb(n - 1, r - 1)
2509         if directed:
2510             per_vertex += (n - 1) * math.comb(n - 2, r - 2)
2511         return min(1.0, target_deg / max(per_vertex, 1))
2512     copies = 1.0 if simple else 1.0 / (1.0 - alpha)
2513     return min(1.0, target_deg / ((n - 1) * copies))
2514
2515
2516 # Twelve sampled panels, mirroring _VARIANT_ENUM_CONFIGS but at sampling sizes
2517 # well past the enumeration wall (n=6). ‘‘sample_n’’ is chosen per variant so
2518 # the binding-connectivity sweep stays a few minutes overall.
2519 _VARIANT_SAMPLE_CONFIGS: list[dict] = [
2520     dict(key="simple_undirected_edge", title="simple undirected edge",
2521         directed=False, simple=True, hypergraph=False, r=3, separation="edge",
2522         ↪ sample_n=26),
2523     dict(key="simple_undirected_vertex", title="simple undirected vertex",
2524         directed=False, simple=True, hypergraph=False, r=3, separation="vertex",
2525         ↪ sample_n=18),
2526     dict(key="simple_directed_edge", title="simple directed arc",
2527         directed=True, simple=True, hypergraph=False, r=3, separation="edge",
2528         ↪ sample_n=14),
2529     dict(key="simple_directed_vertex", title="simple directed vertex",
2530         directed=True, simple=True, hypergraph=False, r=3, separation="vertex",
2531         ↪ sample_n=14),
2532     dict(key="multi_undirected_edge", title="multigraph undirected edge",
2533         directed=False, simple=False, hypergraph=False, r=3, separation="edge",
2534         ↪ sample_n=22),
2535     dict(key="multi_undirected_vertex", title="multigraph undirected vertex",
2536         directed=False, simple=False, hypergraph=False, r=3, separation="vertex",
2537         ↪ sample_n=16),

```

```

2532     dict(key="multi_directed_edge",      title="multigraph directed arc",
2533           directed=True, simple=False, hypergraph=False, r=3, separation="edge",
           ↪ sample_n=12),
2534     dict(key="multi_directed_vertex",    title="multigraph directed vertex",
2535           directed=True, simple=False, hypergraph=False, r=3, separation="vertex",
           ↪ sample_n=12),
2536     dict(key="hyper_undirected_edge",    title="hypergraph undirected edge",
2537           directed=False, simple=True, hypergraph=True, r=3, separation="edge",
           ↪ sample_n=12),
2538     dict(key="hyper_undirected_vertex",  title="hypergraph undirected vertex",
2539           directed=False, simple=True, hypergraph=True, r=3, separation="vertex",
           ↪ sample_n=12),
2540     dict(key="hyper_directed_edge",      title="hypergraph directed arc",
2541           directed=True, simple=True, hypergraph=True, r=3, separation="edge",
           ↪ sample_n=9),
2542     dict(key="hyper_directed_vertex",    title="hypergraph directed vertex",
2543           directed=True, simple=True, hypergraph=True, r=3, separation="vertex",
           ↪ sample_n=9),
2544 ]

```

C.1.6 Drawing the figures

Every picture in the thesis is produced by the routines below, from the same data structures the rest of the program uses. They are collected here rather than in the figure script next door so that a plot and the computation behind it stay in one file.

```

2547 # --- FIGURES: headless matplotlib PNGs; each function takes a path and writes a
           ↪ file ---
2548
2549 _KUL_BLUE = "#1D8DB0"      # feasible / proved
2550 _KUL_DARK = "#1E6E87"     # row labels, known-max lines
2551 _KUL_LIGHT = "#52BDEC"    # named branches
2552 _WARM = "#DC8C28"         # feasibility boundary, certain interval
2553 _RED = "#E05050"          # infeasible bars
2554 _GREEN = "#3CA050"        # machine-checked exact markers
2555 _VIOLET = "#9B5DE5"       # search lower-bound markers
2556 _GUESS = "#E6B800"        # "guess" interpolation: a clear yellow/gold, kept well
2557                             # apart from the red conjecture curve (proved=blue,
2558                             # conjectured=red, guess=yellow)
2559
2560 # Four-level sensitivity palette: cool (sigma=0) -> hot (sigma=max).
2561 _SIGMA_PALETTE = ["#52BDEC", "#F9C74F", "#F4914B", "#DC8C28"] # 0,1,2,3+
2562
2563
2564 def plot_directed_crossover(m: int, max_n: int, path: str | Path) -> None:
2565     """Plot the two competing directed branches and their maximum versus 'n'".
2566
2567     Shows how the linear hub branch 'm(n-1)' is overtaken by the quadratic
2568     augmented-bipartite branch near 'n ~ 2m': the heart of the directed story.
2569     """
2570     ns = list(range(2, max_n + 1))
2571     hub = [m * (n - 1) for n in ns] # linear hub branch
2572     # Quadratic branch: the shifted-partition augmented bipartite count
2573     # floor((n+m-2)^2/4) of const: augmented-bipartite (at m <= 3 it equals the
2574     # balanced-partition count floor(n^2/4) + (m-2)ceil(n/2)).
2575     bipartite = [(n + m - 2) ** 2 // 4 for n in ns]

```

```

2576 envelope = [directed_arc_lower_bound(n, m) for n in ns] # their pointwise max
2577
2578 plt.figure(figsize=(7, 4.3))
2579 plt.plot(ns, hub, "--", color=_KUL_DARK, label=r"hub branch $m(n-1)$")
2580 plt.plot(ns, bipartite, "--", color=_WARM,
2581          label=r"bipartite branch $\lfloor (n+m-2)^2/4 \rfloor$")
2582 plt.plot(ns, envelope, "-", color=_KUL_BLUE, linewidth=2.4, label="their
    ↳ maximum")
2583
2584 # Mark the crossover: the first n at which the quadratic branch overtakes the
2585 # linear one, the order (linear in m) where the directed story changes hands.
2586 cross_n = next((n for n, h, b in zip(ns, hub, bipartite) if b > h), None)
2587 if cross_n is not None:
2588     y_cross = directed_arc_lower_bound(cross_n, m)
2589     plt.axvline(cross_n, color="gray", linestyle=":", linewidth=1.3, alpha
    ↳ =0.8)
2590     plt.scatter([cross_n], [y_cross], color=_KUL_BLUE, zorder=5,
2591                s=45, edgecolor="white", linewidth=0.8)
2592     plt.annotate(f"crossover at $n = {cross_n}$",
2593                xy=(cross_n, y_cross),
2594                xytext=(cross_n - 2.8, max(envelope) * 0.48),
2595                fontsize=9.5, color="black", ha="center",
2596                arrowprops=dict(arrowstyle="->", color="gray", lw=1.1))
2597
2598 plt.xlabel("number of vertices $n$")
2599 plt.ylabel("arcs")
2600 plt.title(f"Directed lower bound at $m = {m}$")
2601 plt.legend(loc="upper left")
2602 plt.grid(True, alpha=0.3)
2603 _save(path)
2604
2605
2606 def plot_edge_vertex_divergence(max_n: int, path: str | Path) -> None:
2607     """Show agreement through  $m \leq 4$  and divergence at  $m=5$ , all starting at the same
    ↳  $n$ .
2608
2609     All curves begin at  $n_{\text{start}}$  so the reader can compare directly.
2610     For  $m=2,3,4$  the two problems coincide (one line each, blue shades).
2611     At  $m=5$  they part: edge below, vertex above, with the gap shaded.
2612     """
2613     n_start = 4 # common start for all curves
2614     ns = list(range(n_start, max_n + 1))
2615
2616     plt.figure(figsize=(8, 5))
2617
2618     #  $m=2,3,4$ : edge == vertex -- one curve per  $m$ 
2619     agree_palette = ["#AED9EE", "#5CB4D9", _KUL_DARK]
2620     for m_val, color in zip([2, 3, 4], agree_palette):
2621         vals = [simple_undirected_edge(n, m_val) for n in ns]
2622         plt.plot(ns, vals, "-", color=color, linewidth=1.8, alpha=0.9,
2623                label=f"$m={m\_val}$: edge $=$ vertex")
2624
2625     #  $m=5$ : edge and vertex (same starting point, diverge from  $n=6$  onwards)
2626     edge5 = [simple_undirected_edge(n, 5) for n in ns]
2627     vert5 = [simple_undirected_vertex_m5(n) for n in ns]
2628     plt.plot(ns, edge5, "--", color=_KUL_BLUE, linewidth=2.4,
2629            label=r"$m=5$: edge $\ell_5(n)=\lfloor 5(n-1)/2 \rfloor$")
2630     plt.plot(ns, vert5, "-", color=_WARM, linewidth=2.4,

```

```

2631         label=r"$m=5$: vertex $k_5(n)=\lfloor 8n/3 \rfloor - 3$"
2632     # shade only where the gap is positive
2633     plt.fill_between(ns, edge5, vert5, where=[v > e for v, e in zip(vert5, edge5)
2634         ↪ ],
2635         alpha=0.15, color=_WARM)
2636
2637     plt.xlabel("number of vertices $n$")
2638     plt.ylabel("maximum edges")
2639     plt.title("Agreement through $m \leq 4$, first divergence at $m = 5$")
2640     plt.legend(fontsize=9, loc="upper left")
2641     plt.grid(True, alpha=0.3)
2642     _save(path)
2643
2644     # NOTE: plot_degree_threshold generated the 2-D random-sampling threshold figure
2645     # (the old Figure 4.2) removed from the thesis on 2026-06-20. Kept here so the
2646     # threshold phenomenon can be restored; see
2647     # research_notes/removed_threshold_phenomenon.md for the recipe. (Its 3-D
2648     # companion plot_conn_threshold_3d is still used, for the appendix.)
2649     def plot_degree_threshold(
2650         path: str | Path, *, n: int = 12, m: int = 4, alpha: float = 0.5,
2651         trials: int = 70, seed: int = 7,
2652         targets=(0.3, 0.5, 0.65, 0.8, 0.95, 1.1, 1.3, 1.6, 2.0),
2653     ) -> None:
2654         """Appearance probability against mean binding-degree, across every model.
2655
2656         Six random models -- simple, multigraph and  $r$ -uniform hypergraph, each
2657         undirected and directed -- are swept in density and plotted by their
2658         *measured* mean binding degree (in units of ' $m$ ') against the probability
2659         that a sample already contains a pair of (Berge) edge-connectivity at least
2660         ' $m$ '. The vertical line at degree ' $= m$ ' is the asymptotic threshold the
2661         degree argument of thm:gnp-threshold predicts; at the fixed small ' $m$ ' a
2662         computation can reach the curves are ORDERED rather than coincident -- a
2663         directed multigraph forces the pair at the least degree (heavy parallel edges
2664         concentrate connectivity), a directed hypergraph at the most (a hyperedge
2665         serves only one Berge route), with the simple cases between. Universality is
2666         the ' $m/\ln n \rightarrow \infty$ ' limit; the spread here is the finite- $m$  gap.
2667         """
2668         models = [
2669             ("simple, undirected", dict(directed=False, simple=True,
2670                 ↪ hypergraph=False, r=3), _KUL_LIGHT, "o", "-"),
2671             ("simple, directed", dict(directed=True, simple=True,
2672                 ↪ hypergraph=False, r=3), _KUL_BLUE, "s", "-"),
2673             ("multigraph, undirected", dict(directed=False, simple=False,
2674                 ↪ hypergraph=False, r=3), _WARM, "o", "--"),
2675             ("multigraph, directed", dict(directed=True, simple=False,
2676                 ↪ hypergraph=False, r=3), "#C0392B", "s", "--"),
2677             (" $r$ -uniform hyper, undir.", dict(directed=False, simple=True,
2678                 ↪ hypergraph=True, r=3), _VIOLET, "^", ":"),
2679             (" $r$ -uniform hyper, dir.", dict(directed=True, simple=True,
2680                 ↪ hypergraph=True, r=3), _GREEN, "^", ":"),
2681         ]
2682
2683     plt.figure(figsize=(7.8, 5.0))
2684     for label, spec, colour, mk, ls in models:
2685         xs, ys = [], []
2686         for t in targets:
2687             p = _p_for_target_degree(t * m, n, alpha=alpha, **spec)
2688             rng = random.Random(seed)

```

```

2682         degs, hits = [], 0
2683         for _ in range(trials):
2684             obj = _sample_variant(n, p, alpha=alpha, cap=m - 1, rng=rng, **
                ↪ spec)
2685             degs.append(_mean_binding_degree(
2686                 obj, directed=spec["directed"], hypergraph=spec["hypergraph"])
                ↪ )
2687             if _measure_variant(obj, separation="edge",
2688                 hypergraph=spec["hypergraph"]) >= m:
2689                 hits += 1
2690             xs.append(sum(degs) / len(degs) / m)
2691             ys.append(hits / trials)
2692             plt.plot(xs, ys, ls, marker=m, color=colour, markersize=4.5,
2693                 linewidth=1.8, label=label)
2694             plt.axvline(1.0, color="black", linestyle="--", linewidth=1.6,
2695                 label=r"asymptotic threshold (degree $= m$)")
2696             plt.axhline(0.5, color="gray", linestyle=":", linewidth=1.0, alpha=0.6)
2697             plt.xlabel(r"mean binding degree, in units of $m$")
2698             plt.ylabel(r"$\hat{P}\backslash, \backslash, \text{binding connectivity} \geq m\backslash, \backslash$")
2699             plt.title(f"Appearance vs expected degree, by model "
2700                 f"$n = {n}$, $m = {m}$, multiplicities capped at $m-1$")
2701             plt.ylim(-0.03, 1.03)
2702             plt.legend(fontsize=8.5, loc="lower right")
2703             plt.grid(True, alpha=0.3)
2704             _save(path)
2705
2706
2707 def plot_connectivity_distribution(series: dict, n: int, p: float, path: str |
    ↪ Path) -> None:
2708     """Plot overlaid histograms of the binding connectivity for several models.
2709
2710     ‘series’ maps a label to a list of connectivity values (one per random
2711     sample), as produced by :func:‘montecarlo.connectivity_distribution’. The
2712     integer-valued histograms are drawn on shared integer bins so the shapes can
2713     be compared directly across the kinds of graph.
2714
2715     # Shared integer bins across every series so the shapes line up exactly.
2716     all_values = [value for values in series.values() for value in values]
2717     low, high = min(all_values), max(all_values)
2718     levels = list(range(low, high + 1))
2719     colours = [_KUL_BLUE, _WARM, _KUL_DARK, _KUL_LIGHT]
2720     group_count = len(series)
2721     bar_width = 0.8 / group_count # split each integer slot among the series
2722
2723     plt.figure(figsize=(7, 4.3))
2724     for index, ((label, values), colour) in enumerate(zip(series.items(), colours)
        ↪ ):
2725         counts = [values.count(level) for level in levels]
2726         # Offset each series' bars so grouped bars sit side by side per level.
2727         offset = (index - (group_count - 1) / 2) * bar_width
2728         positions = [level + offset for level in levels]
2729         plt.bar(positions, counts, width=bar_width, color=colour, label=label,
2730             edgecolor="white", linewidth=0.4)
2731     plt.xticks(levels)
2732     plt.xlabel("binding connectivity")
2733     plt.ylabel("number of samples")
2734     plt.title(f"Connectivity over random graphs, $n = {n}$, $p = {p}$")
2735     plt.legend()

```

```

2736     plt.grid(True, axis="y", alpha=0.3)
2737     _save(path)
2738
2739
2740 _FRACTION_LABEL = {0.25: "1/4", 0.5: "1/2", 0.75: "3/4"}
2741
2742
2743 def plot_edge_vertex_histograms(
2744     data: dict[float, tuple[list[int], list[int]]],
2745     n: int,
2746     p_values: list[float],
2747     path: str | Path,
2748 ) -> None:
2749     """Six histograms of the binding connectivity in  $G(n, p)$ , at one size  $n$ .
2750
2751     'data' maps each density 'p' to a pair '(lambda_max_values,
2752     kappa_max_values)' measured on the same samples, from
2753     :func:'montecarlo.edge_vertex_distribution'. The grid is two rows by three
2754     columns: the top row is the edge-connectivity ' $\lambda^{\max}$ ' and the bottom
2755     row the vertex-connectivity ' $\kappa^{\max}$ ', one column per density. Reading a
2756     column top to bottom compares the two distributions at a fixed density, so
2757     Whitney's inequality shows up as the vertex histogram sitting at or to the
2758     left of the edge one. The fraction of samples with ' $\kappa^{\max} <
2759     \lambda^{\max}$ ' is printed on each vertex panel. All six share one axis range.
2760
2761     """
2762     rows = [(r"edge  $\lambda^{\max}$ ", 0, _KUL_BLUE),
2763             (r"vertex  $\kappa^{\max}$ ", 1, _WARM)]
2764     all_values = [v for p in p_values for series in data[p] for v in series]
2765     lo, hi = min(all_values), max(all_values)
2766     bins = range(lo, hi + 2) # one integer-wide bin per connectivity value
2767     trials = len(data[p_values[0]][0])
2768
2769     fig, axes = plt.subplots(2, len(p_values), figsize=(11, 6),
2770                             sharex=True, sharey=True)
2771     for col, p in enumerate(p_values):
2772         lam, kap = data[p]
2773         gap = 100 * sum(k < l for k, l in zip(kap, lam)) / len(lam)
2774         plabel = _FRACTION_LABEL.get(p, f"{p:.2f}")
2775         for label, ridx, colour in rows:
2776             axis = axes[ridx][col]
2777             values = lam if ridx == 0 else kap
2778             axis.hist(values, bins=bins, align="left", color=colour,
2779                     edgecolor="white", linewidth=0.4, rwidth=0.9)
2780             axis.axvline(statistics.mean(values), color="black", linestyle="--",
2781                         linewidth=1.2)
2782             if ridx == 0:
2783                 axis.set_title(f"$p = {plabel}$")
2784             else:
2785                 axis.set_xlabel("binding connectivity")
2786                 axis.text(0.03, 0.92, fr"$\kappa \!< \! \lambda$: {gap:.0f}%",
2787                         transform=axis.transAxes, ha="left", va="top",
2788                         fontsize=9, color=_WARM)
2789             if col == 0:
2790                 axis.set_ylabel(f"{label}\nsamples")
2791             axis.grid(True, axis="y", alpha=0.3)
2792
2793     fig.suptitle(fr"Distribution of the binding connectivity in  $G(\{n\}, p)$  "
2794                fr"over {trials} samples", fontsize=12)

```

```

2794     _save(path)
2795
2796
2797 def plot_search_trace(result: SearchResult, path: str | Path,
2798                      optimum: int | None = None,
2799                      ceiling: int | None = None,
2800                      *,
2801                      show_cooling: bool = True,
2802                      show_histogram: bool = True,
2803                      connectivity_label: str = r"$\lambda^{\max}$",
2804                      edge_label: str = "arcs",
2805                      title: str = "Cooling toward an extremal graph") -> None:
2806     """Plot the search trajectory; optionally include the cooling schedule.
2807
2808     Layout: [cooling schedule (optional)] | [scatter] | [visit histogram].
2809
2810     The scatter colours points by step number with a heavily nonlinear
2811     PowerNorm (gamma=0.2, RdYlBu colourmap) so that only the brief hot
2812     exploration phase (early steps) stands out in warm reds/yellows, while
2813     the long converged tail is uniformly blue. When 'ceiling' is given the
2814     feasibility boundary is marked and "feasible"/"infeasible" labels are
2815     placed just above the axes box using a blended data/axes transform, so
2816     they sit centred over their respective shaded regions regardless of zoom.
2817
2818     The rightmost panel is an optional horizontal density histogram showing how
2819     many search steps landed at each edge-count level; it shares the y-axis with
2820     the scatter so the bars align with the dots.
2821
2822     New keyword arguments (all optional, all have sensible defaults):
2823     - 'show_cooling': set False to omit the left cooling-schedule panel.
2824     - 'show_histogram': set False to omit the right visit-density panel.
2825     - 'connectivity_label': x-axis label for the scatter (default  $\lambda^{\max}$ ).
2826     - 'edge_label': y-axis label (default "arcs"; use "edges" for undirected).
2827     - 'title': figure supitle.
2828     """
2829     from collections import Counter
2830     from matplotlib.colors import PowerNorm
2831     from matplotlib.transforms import blended_transform_factory
2832     from matplotlib.gridspec import GridSpec
2833
2834     steps = [record.step for record in result.history]
2835     temperature = [record.temperature for record in result.history]
2836     edges = [record.edge_count for record in result.history]
2837     connectivity = [record.connectivity for record in result.history]
2838
2839     # Build layout: cooling (optional) | scatter | histogram (optional).
2840     # Histogram axes use no sharey -- that avoids a tight_layout warning, so the
2841     # y-limits are synced to the scatter manually after it is fully drawn.
2842     if show_cooling and show_histogram:
2843         fig = plt.figure(figsize=(13.5, 4.8))
2844         gs = GridSpec(1, 3, figure=fig, width_ratios=[1.0, 1.6, 0.28],
2845                      wspace=0.40)
2846         ax_cool = fig.add_subplot(gs[0])
2847         ax_scatter = fig.add_subplot(gs[1])
2848         ax_hist = fig.add_subplot(gs[2])
2849     elif show_cooling and not show_histogram:
2850         fig = plt.figure(figsize=(11.5, 4.8))
2851         gs = GridSpec(1, 2, figure=fig, width_ratios=[1.0, 1.6], wspace=0.40)

```

```

2852     ax_cool = fig.add_subplot(gs[0])
2853     ax_scatter = fig.add_subplot(gs[1])
2854     ax_hist = None
2855     elif (not show_cooling) and show_histogram:
2856         fig = plt.figure(figsize=(8.5, 4.8))
2857         gs = GridSpec(1, 2, figure=fig, width_ratios=[1.6, 0.28], wspace=0.12)
2858         ax_cool = None
2859         ax_scatter = fig.add_subplot(gs[0])
2860         ax_hist = fig.add_subplot(gs[1])
2861     else:
2862         fig = plt.figure(figsize=(6.6, 4.8))
2863         ax_cool = None
2864         ax_scatter = fig.add_subplot(1, 1, 1)
2865         ax_hist = None
2866
2867     # Left: cooling schedule.
2868     if ax_cool is not None:
2869         ax_cool.plot(steps, temperature, color=_WARM, linewidth=1.8)
2870         ax_cool.set_xlabel("step")
2871         ax_cool.set_ylabel("temperature")
2872         ax_cool.set_title("Cooling schedule")
2873         ax_cool.grid(True, alpha=0.3)
2874
2875     # Middle: scatter of (connectivity, edge_count), coloured by step.
2876     ax = ax_scatter
2877     lo = min(connectivity) - 0.5
2878     hi = max(connectivity) + 0.5
2879     if ceiling is not None:
2880         boundary = ceiling + 0.5
2881         ax.axvspan(lo, boundary, color=_GREEN, alpha=0.10)
2882         ax.axvspan(boundary, hi, color=_RED, alpha=0.10)
2883         ax.axvline(boundary, color=_WARM, linestyle="--", linewidth=1.4)
2884         ax.set_xlim(lo, hi)
2885         # Labels centred over each shaded region, placed just above the axes box.
2886         # blended_transform mixes data-x (tracks the region) with axes-fraction-y
2887         # (1.03 is always just above the top of the box regardless of zoom level).
2888         xt = blended_transform_factory(ax.transData, ax.transAxes)
2889         ax.text((lo + boundary) / 2, 1.03, "feasible",
2890                transform=xt, ha="center", va="bottom",
2891                fontsize=8.5, color=_GREEN, fontweight="bold", clip_on=False)
2892         ax.text((boundary + hi) / 2, 1.03, "infeasible",
2893                transform=xt, ha="center", va="bottom",
2894                fontsize=8.5, color="#C03030", fontweight="bold", clip_on=False)
2895
2896     # Nonlinear colour norm: gamma < 1 squashes large step values toward 1,
2897     # mapping the long converged tail (most points) to the BLUE end of RdYlBu
2898     # while only the first few hot steps appear in warm reds and yellows.
2899     max_step = max(steps) if steps else 1
2900     norm = PowerNorm(gamma=0.2, vmin=0, vmax=max_step)
2901     sc = ax.scatter(connectivity, edges, c=steps, cmap="RdYlBu",
2902                    norm=norm, s=16, alpha=0.75, linewidths=0)
2903
2904     if optimum is not None:
2905         ax.axhline(optimum, color=_KUL_DARK, linestyle="--", linewidth=1.4,
2906                   label=f"certified optimum = {optimum}")
2907     if optimum is not None and ceiling is not None:
2908         ax.plot([ceiling], [optimum], "*", color="#D11A2A", markersize=20,
2909                markeredgewidth=0.9, zorder=5,

```

```

2910         label="densest feasible graph")
2911 ax.set_xlabel(connectivity_label + " of the visited graph")
2912 ax.set_ylabel(edge_label)
2913 # No panel title: the supitle and the feasible/infeasible labels describe
2914 # the panel; a panel title would overlap with the labels above the axes box.
2915 ax.xaxis.set_major_locator(MaxNLocator(integer=True))
2916 ax.grid(True, alpha=0.3)
2917 handles, labels = ax.get_legend_handles_labels()
2918 if labels:
2919     ax.legend(loc="upper left", fontsize=8, framealpha=0.9)
2920 cbar = fig.colorbar(sc, ax=ax, shrink=0.92)
2921 cbar.set_label(r"step (cooling $\rightarrow$)")
2922 cbar.set_ticks([0, max_step // 4, max_step // 2,
2923                3 * max_step // 4, max_step])
2924
2925 # Right: horizontal histogram -- visit density per edge-count level.
2926 # Sync y-limits with the scatter manually (avoids tight_layout warning
2927 # that sharey axes cause inside _save's plt.tight_layout() call).
2928 if ax_hist is not None:
2929     visit_counts = Counter(edges)
2930     arc_vals = sorted(visit_counts)
2931     ax_hist.barh(arc_vals, [visit_counts[a] for a in arc_vals],
2932                 height=0.72, color=_KUL_BLUE, alpha=0.75, linewidth=0)
2933     ax_hist.set_ylim(ax.get_ylim())
2934     ax_hist.set_xlabel("visits", fontsize=8)
2935     ax_hist.set_title("Distribution", fontsize=9)
2936     ax_hist.tick_params(axis="y", left=False, labelleft=False)
2937     ax_hist.xaxis.set_major_locator(MaxNLocator(integer=True, nbins=3))
2938     ax_hist.tick_params(labelsize=7)
2939     ax_hist.grid(True, axis="x", alpha=0.3)
2940
2941 fig.suptitle(title, fontsize=12, y=1.04)
2942 _save(path)
2943
2944
2945 def draw_graph_with_sensitivity(
2946     graph: Graph,
2947     path: str | Path,
2948     layout: dict | None = None,
2949     show_sigma: bool = False,
2950     node_labels: dict | None = None,
2951 ) -> None:
2952     """Draw a directed multigraph with every parallel arc drawn, coloured by sigma
2953         ↪ .
2954
2955     Each parallel copy of an adjacency is rendered as its own curved arc, so the
2956     multiplicity ‘‘mu(u, v)’’ is read by *counting* arcs rather than from a label,
2957     and the whole adjacency is coloured by its sensitivity ‘‘sigma(e)’’ -- how far
2958     ‘‘lambda^max’’ falls when the adjacency is deleted -- on a four-level
2959     cool-to-hot palette. An over-provisioned adjacency (several parallel copies
2960     but a downstream bottleneck) therefore shows as a bundle of arcs in a cool
2961     colour: the visual point that multiplicity and load-bearing role differ.
2962
2963     Pass a ‘‘layout’’ dict mapping vertex id to (x, y) and a ‘‘node_labels’’ dict
2964     to override the numeric labels. ‘‘show_sigma’’ is accepted for backward
2965     compatibility and ignored (sigma is always shown).
2966     """
2967     from matplotlib.patches import FancyArrowPatch

```

```

2967
2968 sensitivities = sensitivity_map(graph)
2969 if layout is None:
2970     # Evenly spaced on the unit circle (the same picture nx.circular_layout
2971     # draws), computed directly so drawing needs no networkx.
2972     verts = list(graph.vertices())
2973     layout = {v: (math.cos(2 * math.pi * i / len(verts)),
2974                  math.sin(2 * math.pi * i / len(verts)))
2975               for i, v in enumerate(verts)}
2976 labels = (node_labels if node_labels is not None
2977           else {v: str(v) for v in graph.vertices()})
2978
2979 fig, ax = plt.subplots(figsize=(8.6, 6.2))
2980
2981 for (u, v), sigma in sensitivities.items():
2982     mu = graph.multiplicity(u, v)
2983     colour = _SIGMA_PALETTE[min(sigma, len(_SIGMA_PALETTE) - 1)]
2984     # Symmetric fan of curvatures, one per parallel copy.
2985     if mu <= 1:
2986         rads = [0.0]
2987     else:
2988         spread = 0.13 * (mu - 1)
2989         rads = [-spread + 2 * spread * k / (mu - 1) for k in range(mu)]
2990     for rad in rads:
2991         ax.add_patch(FancyArrowPatch(
2992             layout[u], layout[v], connectionstyle=f"arc3,rad={rad}",
2993             arrowstyle="->", mutation_scale=13, lw=2.3, color=colour,
2994             shrinkA=13, shrinkB=13, zorder=1))
2995         # One sigma label per adjacency, at the chord midpoint.
2996         (x0, y0), (x1, y1) = layout[u], layout[v]
2997         ax.text((x0 + x1) / 2, (y0 + y1) / 2, f"$\\sigma={sigma}$",
2998                fontsize=9, ha="center", va="center", zorder=3,
2999                bbox={"boxstyle": "round,pad=0.18", "fc": "white",
3000                     "ec": colour, "alpha": 0.9})
3001
3002 xs = [p[0] for p in layout.values()]
3003 ys = [p[1] for p in layout.values()]
3004 ax.scatter(xs, ys, s=620, c=_KUL_DARK, edgecolors="white", linewidths=1.2,
3005            zorder=2)
3006 for v, (x, y) in layout.items():
3007     ax.text(x, y, str(labels.get(v, v)), color="white", fontweight="bold",
3008            ha="center", va="center", zorder=4)
3009
3010 mx = (max(xs) - min(xs)) * 0.12 + 0.3
3011 my = (max(ys) - min(ys)) * 0.12 + 0.3
3012 ax.set_xlim(min(xs) - mx, max(xs) + mx)
3013 ax.set_ylim(min(ys) - my, max(ys) + my)
3014 ax.set_aspect("equal")
3015 ax.axis("off")
3016 ax.set_title("Parallel arcs drawn explicitly, coloured by sensitivity "
3017             "$\\sigma$: \ncool ($\\sigma=0$, free) $\\to$ warm (load-bearing)"
3018             "\n↔",
3019             fontsize=12)
3020
3021 _save(path)
3022
3023 def _running_best_trace(result: SearchResult) -> tuple[list[float], list[int]]:
3024     """Best feasible edge count so far against wall-clock seconds, from a history.

```

```

3024
3025 Reads the timed :class:'SearchStep' log (its 'best_feasible' field already
3026 carries the densest feasible graph seen up to each step) and returns the timed
3027 convergence curve both engines record in their native fast modes.
3028 """
3029 xs = [record.seconds for record in result.history]
3030 ys = [record.best_feasible for record in result.history]
3031 return xs, ys
3032
3033
3034 def _settle_time(xs: list[float], ys: list[int]) -> float:
3035     """Wall-clock second at which a trace first reached its own final best value.
3036
3037     The convergence curves plateau long before the time budget runs out, so this
3038     is what the x-axis should track: cropping to the budget would leave a long
3039     flat tail and the cropped view would barely change when the budget does.
3040     """
3041     if not xs:
3042         return 0.0
3043     final = ys[-1]
3044     return next((x for x, y in zip(xs, ys) if y == final), xs[-1])
3045
3046
3047 def plot_sa_vs_tabu_convergence(
3048     path: str | Path, *,
3049     cases: tuple[tuple[int, int], ...] = ((5, 3), (7, 3)),
3050     budget: float = 8.0,
3051     seed: int = 0,
3052     xmaxs: tuple[float, ...] | None = None,
3053     settle_margin: float = 1.3,
3054 ) -> None:
3055     """Best-feasible-value against wall-clock for annealing vs tabu search.
3056
3057     For each '(n, m)' it runs one simulated-annealing schedule
3058     (:func:'search_for_dense_graph') and one tabu schedule
3059     (:func:'tabu_search_for_dense_graph') on the directed multigraph at an equal
3060     wall-clock 'budget', and overlays their timed best-so-far traces against the
3061     conjectured optimum. The chosen cases are where the two engines diverge: tabu
3062     assembles the bipartite extremiser and reaches the optimum while annealing
3063     plateaus a few arcs short. The time axis is wall-clock, so this one figure is
3064     a representative timed run on the reference machine rather than a seed-exact
3065     reproduction like the others.
3066
3067     By default each panel's x-axis stops shortly after the slower engine settles
3068     ('settle_margin' times the later of the two plateau times), so the view
3069     tracks the actual run instead of a fixed window: cropping to a hard-coded
3070     second count leaves a long flat tail because both engines converge well
3071     inside one second. Pass 'xmaxs' to force fixed per-panel limits instead.
3072     """
3073     if not MATPLOTLIB_AVAILABLE:
3074         raise RuntimeError("matplotlib is required for figures")
3075     fig, axes = plt.subplots(1, len(cases), figsize=(11, 4.3))
3076     if len(cases) == 1:
3077         axes = [axes]
3078     for i, (ax, (n, m)) in enumerate(zip(axes, cases)):
3079         sa = search_for_dense_graph(MULTI_DIRECTED, n, m, seed=seed, steps=10**7,
3080                                     deadline=time.time() + budget)
3081         tb = tabu_search_for_dense_graph(MULTI_DIRECTED, n, m, seed=seed,

```

```

3082                                     steps=10**7, deadline=time.time() +
                                     ↪ budget)

3083     settle = 0.0
3084     for res, label, colour in ((sa, "simulated annealing", _KUL_BLUE),
3085                               (tb, "tabu search", _WARM)):
3086         xs, ys = _running_best_trace(res)
3087         ax.step(xs, ys, where="post", label=label, color=colour, linewidth
3088               ↪ =2.0)
3089         settle = max(settle, _settle_time(xs, ys))
3090     opt = directed_multigraph_arc(n, m)
3091     ax.axhline(opt, linestyle=":", color=_KUL_DARK, linewidth=1.6,
3092               label=f"optimum  $L_3^{\{\mathrm{dir}\}}(n) = \{opt\}$")
3093     ax.set_title(f"directed multigraph, $n = {n}$, $m = {m}$", fontsize=11)
3094     ax.set_xlabel("wall-clock seconds", fontsize=9.5)
3095     ax.set_ylabel("densest feasible arc count", fontsize=9.5)
3096     # Crop the long flat tail: stop just after the slower engine settles, so
3097     # the rise fills the panel instead of being squeezed against the y-axis.
3098     # A fixed second count cannot do this, since the settle time shifts run to
3099     # run; an explicit amaxes overrides for a reproducible fixed window.
3100     if xaxes is not None and i < len(xaxes):
3101         ax.set_xlim(0, xaxes[i])
3102     else:
3103         ax.set_xlim(0, max(settle * settle_margin, 0.5))
3104     ax.grid(True, alpha=0.3)
3105     ax.legend(fontsize=8.5, loc="lower right")
3106     # Fewer x-ticks declutters the time axis: the action concentrates early
3107     # and dense ticks add noise.
3108     if MaxNLocator is not None:
3109         ax.xaxis.set_major_locator(MaxNLocator(nbins=5))
3110     _save(path)
3111
3112 def _save(path: str | Path, *, tight: bool = True) -> None:
3113     """Tighten the layout, write the file, and close the figure.
3114
3115     ‘tight=False’ skips ‘tight_layout’ for the 3-D figures, where it
3116     misbehaves with the projected axes (they manage their own spacing).
3117     """
3118     import warnings
3119     Path(path).parent.mkdir(parents=True, exist_ok=True)
3120     if tight:
3121         # tight_layout can warn when figures contain colorbars or shared axes;
3122         # bbox_inches="tight" in savefig handles the actual bounding box so the
3123         # layout warning is cosmetic and safe to suppress.
3124         with warnings.catch_warnings():
3125             warnings.simplefilter("ignore", UserWarning)
3126             plt.tight_layout()
3127     plt.savefig(path, dpi=150, bbox_inches="tight")
3128     plt.close() # release the figure so head-less runs don't leak memory
3129
3130
3131 def plot_complexity_growth(path: str | Path) -> None:
3132     """The combinatorial explosion: number of graphs to enumerate, by direction.
3133
3134     Two panels, directed (left) and undirected (right), sharing one axis range.
3135     Each plots, on a log scale, how many graphs blind enumeration must visit
3136     against the vertex count ‘n’, for the simple model, two multigraph
3137     connectivity caps, and (dotted) the 3-uniform hypergraph, with a dashed line$ 
```

```

3138 at a generous budget of  $10^9$  candidates. The count does not depend on the
3139 edge-versus-vertex separation, because both search the very same set of
3140 graphs, so one pair of panels covers all of those variants at once.
3141 Direction doubles the number of cells to fill, which is the whole difference
3142 between the two panels.
3143 """
3144 ns = list(range(2, 13))
3145
3146 def log10_pow(base: float, exponent: float) -> float:
3147     return exponent * math.log10(base)
3148
3149 def curves(directed: bool):
3150     # Each model fills a number of off-diagonal cells; the candidate count is
3151     # (values per cell) ^ (number of cells). Labels stay plain -- the precise
3152     # exponential forms live in the caption, not the legend. Direction
3153     # multiplies the cell count: by 2 for graphs (ordered vs unordered pairs,
3154     #  $n(n-1)$  vs  $C(n,2)$ ), but by 3 for 3-uniform hypergraphs, since a directed
3155     # hyperedge is a tail plus 2 heads ( $n \cdot C(n-1,2) = 3 \cdot C(n,3)$  candidates).
3156     cells = (lambda n: n * (n - 1)) if directed else (lambda n: n * (n - 1) /
3157         ↪ 2)
3158     hyper_cells = ((lambda n: n * (n - 1) * (n - 2) / 2) if directed
3159         else (lambda n: n * (n - 1) * (n - 2) / 6))
3160     return [
3161         ("simple / random", _KUL_BLUE, "-",
3162          [log10_pow(2, cells(n)) for n in ns]),
3163         ("multigraph  $m=3$ ", _KUL_DARK, "-",
3164          [log10_pow(3, cells(n)) for n in ns]),
3165         ("multigraph  $m=4$ ", _WARM, "-",
3166          [log10_pow(4, cells(n)) for n in ns]),
3167         (" $3$ -uniform hypergraph", _VIOLET, ":",
3168          [log10_pow(2, hyper_cells(n)) for n in ns]),
3169     ]
3170
3171 fig, axes = plt.subplots(1, 2, figsize=(11, 4.6), sharey=True)
3172 for ax, directed, title in [(axes[0], True, "directed graphs"),
3173     (axes[1], False, "undirected graphs")]:
3174     for label, colour, style, vals in curves(directed):
3175         ax.plot(ns, vals, style, color=colour, marker="o", markersize=3.5,
3176             label=label)
3177     ax.axhline(9, color="gray", linestyle="--", linewidth=1,
3178         label=r" $10^9$  budget")
3179     ax.set_title(title)
3180     ax.set_xlabel("vertices  $n$ ")
3181     ax.grid(True, alpha=0.3)
3182     # Cap the shared vertical axis at 100 so the slowest curve (the directed
3183     # 3-uniform hypergraph) runs off the top instead of compressing every other
3184     # curve into a thin band near the bottom. sharey=True propagates the limit.
3185     axes[0].set_ylim(0, 100)
3186     axes[0].annotate(r" $3$ -uniform directed" + "\n" + r"continues off the top",
3187         xy=(9.55, 99.3), xytext=(7.5, 92), fontsize=8, color=_VIOLET,
3188         ha="center", va="center",
3189         arrowprops=dict(arrowstyle="→", color=_VIOLET, lw=1.0))
3190     axes[0].set_ylabel(r" $\log_{10}$  (number of graphs)")
3191     axes[0].legend(fontsize=8.5, loc="upper left", title="model")
3192     fig.suptitle("Number of graphs blind enumeration must visit",
3193         fontsize=12)
3194     _save(path)

```

```

3195
3196 def _circle_layout(n, hub=None):
3197     """Vertex positions for a panel: all on a unit circle, or the ‘hub‘ vertex
3198     at the centre and the rest on the circle, which draws stars and hub graphs
3199     far more clearly than a plain circle does."""
3200     if hub is None:
3201        angs = [math.pi / 2 - 2 * math.pi * k / n for k in range(n)]
3202         return [(math.cos(a), math.sin(a)) for a in angs]
3203     pos = [(0.0, 0.0) for _ in range(n)]
3204     others = [k for k in range(n) if k != hub]
3205     for idx, k in enumerate(others):
3206         a = math.pi / 2 - 2 * math.pi * idx / len(others)
3207         pos[k] = (math.cos(a), math.sin(a))
3208     return pos
3209
3210
3211 def _draw_extremal_panel(ax, matrix, *, directed: bool, hub=None) -> None:
3212     """Draw one extremal multiplicity matrix on ‘ax‘ as a graph.
3213
3214     Vertices sit on a circle (or, when ‘hub‘ is given, that vertex sits at the
3215     centre and the rest on the circle). An adjacency of multiplicity one is a
3216     single line (an arrowed line when ‘directed‘), and a higher multiplicity is
3217     shown by a small count label rather than parallel curves, so the picture
3218     stays legible. Opposite arcs of a bidirected pair are bent apart.
3219
3220     Every named family the gallery draws is *uniform*: all its adjacencies carry
3221     one and the same multiplicity. Repeating that count on every arc buries the
3222     structure under labels (worst on the bidirected ‘K_4‘, twelve labels over
3223     six crossing arcs), so a uniform multiplicity above one is stated once in a
3224     corner note, and the per-arc labels are kept only for the genuinely mixed
3225     case. ‘hub‘ may also be an explicit list of ‘(x, y)’ positions, used
3226     for the bipartite wall panel, where two columns read far better than a circle.
3227     """
3228     from matplotlib.patches import FancyArrowPatch
3229     n = len(matrix)
3230     pos = hub if isinstance(hub, list) else _circle_layout(n, hub)
3231     mults = {int(matrix[i][j]) for i in range(n) for j in range(n)
3232              if i != j and matrix[i][j] > 0}
3233     uniform = mults.pop() if len(mults) == 1 else None
3234     per_arc_labels = uniform is None
3235
3236     def edge(i, j, rad, arrow):
3237         ax.add_patch(FancyArrowPatch(
3238             pos[i], pos[j], connectionstyle=f"arc3,rad={rad}",
3239             arrowstyle="->" if arrow else "-"), mutation_scale=12,
3240             lw=1.7, color=_KUL_BLUE, shrinkA=10, shrinkB=10, zorder=1))
3241
3242     def mlabel(i, j, rad, mu):
3243         if mu <= 1:
3244             return
3245         mx, my = (pos[i][0] + pos[j][0]) / 2, (pos[i][1] + pos[j][1]) / 2
3246         dx, dy = pos[j][0] - pos[i][0], pos[j][1] - pos[i][1]
3247         length = math.hypot(dx, dy) or 1.0
3248         mx += (rad * 0.9 + 0.06) * (-dy) / length
3249         my += (rad * 0.9 + 0.06) * (dx) / length
3250         ax.text(mx, my, f"$\\times{mu}$", fontsize=8, ha="center", va="center",
3251                 color=_KUL_DARK, zorder=3,
3252                 bbox=dict(boxstyle="round,pad=0.08", fc="white", ec="none"))

```

```

3253
3254     if directed:
3255         for i in range(n):
3256             for j in range(n):
3257                 if i == j or matrix[i][j] == 0:
3258                     continue
3259                 rad = 0.16 if matrix[j][i] else 0.0
3260                 edge(i, j, rad, True)
3261                 if per_arc_labels:
3262                     mlabel(i, j, rad, matrix[i][j])
3263     else:
3264         for i in range(n):
3265             for j in range(i + 1, n):
3266                 if matrix[i][j] == 0:
3267                     continue
3268                 edge(i, j, 0.0, False)
3269                 if per_arc_labels:
3270                     mlabel(i, j, 0.0, matrix[i][j])
3271     for k, (x, y) in enumerate(pos):
3272         ax.scatter([x], [y], s=300, c="white", edgecolors=_KUL_BLUE,
3273                 linewidths=1.6, zorder=2)
3274         ax.text(x, y, str(k + 1), ha="center", va="center", fontsize=9,
3275                 zorder=3, color=_KUL_DARK)
3276     if uniform is not None and uniform > 1:
3277         word = "arcs" if directed else "edges"
3278         ax.text(0.0, -1.34, f"all {word}  $\times$ {uniform}", ha="center",
3279                 va="center", fontsize=9, color=_KUL_DARK)
3280     ax.set_xlim(-1.45, 1.45)
3281     ax.set_ylim(-1.45, 1.45)
3282
3283
3284     # metro-line palette for hypergraph panels: red, blue, green, orange, purple, grey
3285     _GALLERY_METRO = ["#C85050", "#1D8DB0", "#3CA050", "#DC8C28", "#8C50A0", "#787878"
3286                      ↪ ]
3287
3288     def _draw_hyper_panel(ax, hyperedges, n, *, directed: bool, hub=None) -> None:
3289         """Draw a hypergraph extremiser as a metro map: vertices on a circle (or a
3290         hub at the centre), each hyperedge a thick coloured line through its members
3291         when undirected, or a fan of coloured arrows from its tail when directed."""
3292         from matplotlib.patches import FancyArrowPatch
3293         pos = _circle_layout(n, hub)
3294         for idx, he in enumerate(hyperedges):
3295             col = _GALLERY_METRO[idx % len(_GALLERY_METRO)]
3296             if directed:
3297                 tail, heads = he
3298                 tx, ty = pos[tail]
3299                 for h in heads:
3300                     hx, hy = pos[h]
3301                     ax.add_patch(FancyArrowPatch(
3302                         (tx, ty), (hx, hy), arrowstyle="->", mutation_scale=11,
3303                         lw=2.4, color=col, alpha=0.85, shrinkA=11, shrinkB=11,
3304                         zorder=1, connectionstyle="arc3,rad=0.10"))
3305             else:
3306                 pts = [pos[v] for v in sorted(he)]
3307                 cx = sum(p[0] for p in pts) / len(pts)
3308                 cy = sum(p[1] for p in pts) / len(pts)
3309                 pts.sort(key=lambda p: math.atan2(p[1] - cy, p[0] - cx))

```

```

3310         ax.plot([p[0] for p in pts], [p[1] for p in pts], "-", color=col,
3311                 lw=4.2, solid_capstyle="round", solid_joinstyle="round",
3312                 alpha=0.85, zorder=1)
3313     for k, (x, y) in enumerate(pos):
3314         ax.scatter([x], [y], s=300, c="white", edgecolors="#333333",
3315                 linewidths=1.5, zorder=2)
3316         ax.text(x, y, str(k + 1), ha="center", va="center", fontsize=9,
3317                 zorder=3, color="#222222")
3318     ax.set_xlim(-1.45, 1.45)
3319     ax.set_ylim(-1.45, 1.45)
3320
3321
3322 def plot_extremal_gallery(path: str | Path, *,
3323                          gallery_json: str | Path | None = None) -> None:
3324     """A gallery of the named extremal graphs, drawn at a large representative
3325         ↪ size.
3326
3327     These are the structured extremisers the analysis identifies, not the
3328     trivial small trees: the directed double and bidirected stars, the doubled
3329     bipartite wall, the dense bidirected complete graph, the multigraph star at
3330     full multiplicity, and the star hypertrees (drawn as metro maps). The program
3331     ↪ both builds these and,
3332     at small 'n', proves them optimal by the enumeration of
3333     :func:'gallery_extremal_graphs'. The 'gallery_json' argument is accepted for
3334     backward compatibility and ignored.
3335     """
3336     if not MATPLOTLIB_AVAILABLE:
3337         raise RuntimeError("matplotlib is required for figures")
3338     # dense bidirected complete graphs, built directly
3339     bidir_k4 = Graph(4, SIMPLE_DIRECTED)
3340     for i in range(4):
3341         for j in range(4):
3342             if i != j:
3343                 bidir_k4.set_multiplicity(i, j, 1)
3344     # 2.B_{3,4}: the doubled one-directional bipartite wall, one of the three
3345     # proved 24-arc extremisers at n = 7, m = 3 (rem:odd-step-roadmap, fact (b)).
3346     # It ties the double star at the crossover, so the gallery shows both branches
3347     ↪ .
3348     bip_2b34 = Graph(7, MULTI_DIRECTED)
3349     for a in range(3):
3350         for b in range(3, 7):
3351             bip_2b34.set_multiplicity(a, b, 2)
3352     bip_pos = [(-0.85, 0.85), (-0.85, 0.0), (-0.85, -0.85),
3353               (0.85, 1.05), (0.85, 0.35), (0.85, -0.35), (0.85, -1.05)]
3354     multi_star = Graph(9, MULTI_UNDIRECTED) # undirected star, mult 2
3355     for i in range(1, 9):
3356         multi_star.set_multiplicity(0, i, 2)
3357     dir_hypertree = [(0, frozenset({1, 2})), (0, frozenset({3, 4})),
3358                     (0, frozenset({5, 6}))] # directed star hypertree on 7
3359         ↪ vertices
3360
3361     # ("g"/"h"/"hd", object, directed, hub, title)
3362     panels = [
3363         ("g", double_star(7, 3, directed=True), True, 0,
3364          "multigraph directed, arc\n$m=3$, $n=7$: double star ($24$ arcs)"),
3365         ("g", double_star(7, 2, directed=True), True, 0,
3366          "simple directed, arc\n$m=2$, $n=7$: bidirected star ($12$ arcs)"),
3367         ("g", multi_star, False, 0,

```

```

3364         "multigraph undirected, edge\n$m=3$, $n=9$: star at multiplicity $2$ (
           ↪ $16$ edges)"),
3365     ("g", bidir_k4, True, None,
3366      "simple directed, arc\n$m=4$, $n=4$: bidirected $K_4$ ($12$ arcs)"),
3367     ("g", bip_2b34, True, bip_pos,
3368      "multigraph directed, arc\n$m=3$, $n=7$: bipartite wall $2 \cdot B_{\{3,4\}}$
           ↪ $24$ arcs)"),
3369     ("h", star_hypertree(10, 3), False, 0,
3370      "hypergraph, $r=3$\n$m=2$, $n=10$: star hypertree ($4$ hyperedges)"),
3371     ("h", star_hypertree(13, 4), False, 0,
3372      "hypergraph, $r=4$\n$m=2$, $n=13$: star hypertree ($4$ hyperedges)"),
3373     ("h", star_hypertree(16, 3), False, 0,
3374      "hypergraph, $r=3$\n$m=2$, $n=16$: star hypertree ($7$ hyperedges)"),
3375     ("hd", (dir_hypertree, 7), True, 0,
3376      "directed hypergraph, $r=3$\n$m=2$, $n=7$: tail-to-heads star"),
3377 ]
3378 fig, axes = plt.subplots(3, 3, figsize=(12, 12))
3379 for ax, (kind, obj, directed, hub, title) in zip(axes.flat, panels):
3380     if kind == "g":
3381         _draw_extremal_panel(ax, obj.mu.tolist(), directed=directed, hub=hub)
3382     elif kind == "h":
3383         _draw_hyper_panel(ax, list(obj.hyperedges), obj.num_vertices,
3384                          directed=False, hub=hub)
3385     else: # "hd": (hyperedges, n)
3386         hes, nn = obj
3387         _draw_hyper_panel(ax, hes, nn, directed=True, hub=hub)
3388     ax.set_title(title, fontsize=9.5)
3389     ax.set_aspect("equal")
3390     ax.axis("off")
3391 fig.suptitle("Extremal graphs of the named families, drawn large", fontsize
           ↪ =14)
3392 _save(path)
3393
3394
3395 _GUESS_STYLE = (0, (1, 1)) # dense dots: "we are very much guessing"
3396
3397
3398 def plot_variant_grid(panels: list[dict], path: str | Path,
3399                      m: int | None = None) -> None:
3400     """All twelve variants on one grid, in the proved/conjectured/guessed language
           ↪ .
3401
3402     ‘panels’ is a row-major list of twelve dictionaries (rows = model simple,
3403     multi, hyper; columns = undirected edge, undirected vertex, directed arc,
3404     directed vertex). Each panel may carry any of:
3405
3406     ‘proved’ ‘(xs, ys)’ solid blue line, a theorem holding for all ‘n’;
3407     ‘conj’ ‘(xs, ys)’ solid red line, a conjecture formula;
3408     ‘guess’ ‘(xs, ys)’ solid yellow line, a bare extrapolation of the
3409     machine points where no formula is known;
3410     ‘band’ ‘(xs, lo, hi)’ the certain interval, an easy construction below
3411     and the trivial maximum edge count above;
3412     ‘exact’ ‘(xs, ys)’ filled squares, sizes the machine proved;
3413     ‘search’ ‘(xs, ys)’ open circles, the search lower bounds.
3414
3415     The point is one honest picture, distinguished by colour rather than dash
3416     pattern: a blue line is settled, a red line is conjectured, a yellow line is
3417     a guess, and the shaded band is the interval we are certain the truth lies in.

```

```

3418     """
3419     def draw_panel(ax, panel):
3420         band = panel.get("band")
3421         if band is not None:
3422             xs, lo, hi = band
3423             ax.fill_between(xs, lo, hi, color=_WARM, alpha=0.12,
3424                             label="certain interval")
3425             ax.plot(xs, lo, "-", color=_WARM, linewidth=0.8, alpha=0.6)
3426             ax.plot(xs, hi, "-", color=_WARM, linewidth=0.8, alpha=0.6)
3427         for branch in panel.get("branches", []):
3428             bxs, bys, blabel = branch
3429             ax.plot(bxs, bys, ":", color=_KUL_LIGHT, linewidth=1.3, label=blabel)
3430         if panel.get("proved") is not None:
3431             xs, ys = panel["proved"]
3432             ax.plot(xs, ys, "-", color=_KUL_BLUE, linewidth=2.3, label="proved")
3433         if panel.get("conj") is not None:
3434             xs, ys = panel["conj"]
3435             ax.plot(xs, ys, "-", color=_RED, linewidth=2.0, label="conjectured")
3436         if panel.get("guess") is not None:
3437             xs, ys = panel["guess"]
3438             ax.plot(xs, ys, "-", color=_GUESS, linewidth=2.0,
3439                     label="guess (interpolated)")
3440         if panel.get("exact") is not None:
3441             xs, ys = panel["exact"]
3442             if len(xs):
3443                 ax.plot(xs, ys, "s", color=_GREEN, markersize=7,
3444                         label="machine-checked (exact)")
3445         if panel.get("search") is not None:
3446             xs, ys = panel["search"]
3447             if len(xs):
3448                 ax.plot(xs, ys, "o", mfc="none", mec=_VIOLET, mew=1.8,
3449                         markersize=8, label="search (lower bound)")
3450         ax.set_title(panel["title"], fontsize=9.5)
3451         ax.set_xlabel("vertices $n$", fontsize=8.5)
3452         ax.set_ylabel(panel.get("ylabel", "edges"), fontsize=8.5)
3453         ax.tick_params(labelsize=8)
3454         ax.grid(True, alpha=0.3)
3455         ax.legend(fontsize=6.8, loc="upper left", framealpha=0.85)
3456
3457         # The line styles (solid / dashed / dotted) are named in the per-panel
3458         # legends and spelled out in the caption, so the supitle stays short and
3459         # just records the variant family and the threshold m.
3460         supitle = "Erdős 915 across the twelve variants"
3461         if m is not None:
3462             supitle += fr"    $m = {m}$"
3463         _variant_panel_grid(draw_panel, configs=panels, supitle=supitle, path=path,
3464                             supitle_fontsize=13)

```

C.1.7 Walking the whole space

At small sizes it is possible to visit every graph there is and record what it looks like, which is what these routines do. The result is the distribution behind the landscape figures, and, more importantly, a way to confirm a claimed maximum by exhaustion rather than by search. The three-dimensional surfaces are built on the same idea, run across a grid of sizes and route limits and cached so that the sweep is paid for once.

```

3467 # --- ENUMERATION LANDSCAPES: visit every labeled graph, collect (edges, lambda^
    ↪ max) ---
3468
3469 # Row-major table of all twelve variants, matching gather_variant_grid.
3470 # "enum_n" is the vertex count used for full enumeration.
3471 # "max_mult" caps per-cell multiplicity for multigraph variants.
3472 _VARIANT_ENUM_CONFIGS: list[dict] = [
3473     # row 1 -- simple
3474     dict(key="simple_undirected_edge", title="simple undirected edge",
3475           directed=False, simple=True, hypergraph=False, r=3, separation="edge",
3476           enum_n=6, max_mult=1),
3477     dict(key="simple_undirected_vertex", title="simple undirected vertex",
3478           directed=False, simple=True, hypergraph=False, r=3, separation="vertex",
3479           enum_n=6, max_mult=1),
3480     dict(key="simple_directed_edge", title="simple directed arc",
3481           directed=True, simple=True, hypergraph=False, r=3, separation="edge",
3482           enum_n=4, max_mult=1),
3483     dict(key="simple_directed_vertex", title="simple directed vertex",
3484           directed=True, simple=True, hypergraph=False, r=3, separation="vertex",
3485           enum_n=4, max_mult=1),
3486     # row 2 -- multigraph
3487     dict(key="multi_undirected_edge", title="multigraph undirected edge",
3488           directed=False, simple=False, hypergraph=False, r=3, separation="edge",
3489           enum_n=5, max_mult=2),
3490     dict(key="multi_undirected_vertex", title="multigraph undirected vertex",
3491           directed=False, simple=False, hypergraph=False, r=3, separation="vertex",
3492           enum_n=5, max_mult=2),
3493     dict(key="multi_directed_edge", title="multigraph directed arc",
3494           directed=True, simple=False, hypergraph=False, r=3, separation="edge",
3495           enum_n=3, max_mult=5),
3496     dict(key="multi_directed_vertex", title="multigraph directed vertex",
3497           directed=True, simple=False, hypergraph=False, r=3, separation="vertex",
3498           enum_n=3, max_mult=5),
3499     # row 3 -- hypergraph r=3
3500     dict(key="hyper_undirected_edge", title="hypergraph undirected edge",
3501           directed=False, simple=True, hypergraph=True, r=3, separation="edge",
3502           enum_n=5, max_mult=1),
3503     dict(key="hyper_undirected_vertex", title="hypergraph undirected vertex",
3504           directed=False, simple=True, hypergraph=True, r=3, separation="vertex",
3505           enum_n=5, max_mult=1),
3506     dict(key="hyper_directed_edge", title="hypergraph directed arc",
3507           directed=True, simple=True, hypergraph=True, r=3, separation="edge",
3508           enum_n=4, max_mult=1),
3509     dict(key="hyper_directed_vertex", title="hypergraph directed vertex",
3510           directed=True, simple=True, hypergraph=True, r=3, separation="vertex",
3511           enum_n=4, max_mult=1),
3512 ]
3513
3514
3515 def enumerate_all_graphs(
3516     n: int, *,
3517     directed: bool,
3518     simple: bool,
3519     hypergraph: bool = False,
3520     r: int = 3,
3521     max_mult: int = 3,
3522     separation: str = "edge",
3523 ) -> list[tuple[int, int]]:

```

```

3524     """Return '(edge_count, lambda_max)' for every labeled graph of this type on
        ↳ 'n' vertices.
3525
3526     For simple graphs each cell is 0/1; for multigraphs each cell ranges over
3527     '{0,...,max_mult}'. For hypergraphs each candidate hyperedge is present or
3528     absent. The list covers the full labeled graph space (isomorphic copies
3529     included), which is what we want for the distribution over the search space.
3530     """
3531     results: list[tuple[int, int]] = []
3532
3533     if hypergraph:
3534         vertex_split = (separation == "vertex")
3535         candidates = _hyperedge_candidates(n, r, directed)
3536         for mask in product((0, 1), repeat=len(candidates)):
3537             chosen = [candidates[i] for i, flag in enumerate(mask) if flag]
3538             h = Hypergraph(n, chosen, directed=directed)
3539             conn = max_hyper_connectivity(h, vertex_split=vertex_split)
3540             results.append((h.edge_count(), conn))
3541         return results
3542
3543     variant = _variant_for(directed, simple)
3544     measure = _connectivity_measure(separation)
3545     cells = _matrix_cells(n, directed)
3546     span = 2 if simple else (max_mult + 1)
3547     base = Graph(n, variant)
3548     for values in product(range(span), repeat=len(cells)):
3549         candidate = base.copy()
3550         for (u, v), value in zip(cells, values):
3551             if value:
3552                 candidate.set_multiplicity(u, v, value)
3553             conn = measure(candidate)
3554             results.append((candidate.edge_count(), conn))
3555     return results
3556
3557
3558 def _pair_connectivities(obj, *, separation: str, hypergraph: bool) -> list[int]:
3559     """Local connectivity of every vertex pair of one graph or hypergraph.
3560
3561     The same pair sweep 'max_connectivity' runs, but keeping each value rather
3562     than only the maximum. For the edge case the capacity matrix is built once
3563     and reused across pairs (as 'max_connectivity' does); vertex and hypergraph
3564     cases reuse their named per-pair routines.
3565     """
3566     vsplit = (separation == "vertex")
3567     if hypergraph:
3568         return [hyper_connectivity(obj, s, t, vertex_split=vsplit)
3569                 for s, t in _pairs(obj)]
3570     if not vsplit:
3571         csr = _csr(obj.mu, dtype=int)
3572         return [int(_csgraph_maxflow(csr, s, t).flow_value) for s, t in _pairs(obj)
3573                 ↳ ]
3574     return [local_connectivity(obj, s, t, vertex_split=True) for s, t in _pairs(
3575         ↳ obj)]
3576
3577 def enumerate_pair_connectivities(
3578     n: int, *,
3579     directed: bool,

```

```

3579     simple: bool,
3580     hypergraph: bool = False,
3581     r: int = 3,
3582     max_mult: int = 3,
3583     separation: str = "edge",
3584 ) -> dict[tuple[int, int], int]:
3585     """Pooled 2-D histogram '{(lambda_max, pair_conn): count}' over all graphs.
3586
3587     For every labeled graph on 'n' vertices and every vertex pair, record the
3588     pair's local connectivity tagged with the graph's own 'lambda^max'. The
3589     result is a small threshold-independent table: at plot time a feasibility
3590     threshold 'T' splits each pair-connectivity column into observations from
3591     graphs that stay at 'lambda^max <= T' and those that exceed it. Much
3592     smaller than the raw observation list (a few dozen entries per variant), so
3593     it pickles compactly even though the sweep itself enumerates every graph.
3594     """
3595     from collections import Counter
3596     table: Counter = Counter()
3597
3598     def record(obj):
3599         pcs = _pair_connectivities(obj, separation=separation, hypergraph=
3600             ↪ hypergraph)
3601         lmax = max(pcs, default=0)
3602         for c in pcs:
3603             table[(lmax, c)] += 1
3604
3605     if hypergraph:
3606         candidates = _hyperedge_candidates(n, r, directed)
3607         for mask in product((0, 1), repeat=len(candidates)):
3608             chosen = [candidates[i] for i, flag in enumerate(mask) if flag]
3609             record(Hypergraph(n, chosen, directed=directed))
3610         return dict(table)
3611
3612     variant = _variant_for(directed, simple)
3613     cells = _matrix_cells(n, directed)
3614     span = 2 if simple else (max_mult + 1)
3615     base = Graph(n, variant)
3616     for values in product(range(span), repeat=len(cells)):
3617         candidate = base.copy()
3618         for (u, v), value in zip(cells, values):
3619             if value:
3620                 candidate.set_multiplicity(u, v, value)
3621         record(candidate)
3622     return dict(table)
3623
3624 def compute_pair_enumeration_cache(
3625     cache_path: str | Path = "figures/pair_enumeration_cache.pkl",
3626     configs: list[dict] | None = None,
3627 ) -> dict[str, dict[tuple[int, int], int]]:
3628     """Per-variant pooled pair-connectivity tables, loading from cache if present.
3629
3630     The pooled analogue of :func:'compute_enumeration_cache': one
3631     '{(lambda_max, pair_conn): count}' table per variant. Slow on first run
3632     (it re-sweeps the enumeration measuring every pair), cached thereafter.
3633     """
3634     cache_path = Path(cache_path)
3635     if configs is None:

```

```

3636         configs = _VARIANT_ENUM_CONFIGS
3637
3638     if cache_path.exists():
3639         with open(cache_path, "rb") as f:
3640             cached = pickle.load(f)
3641     else:
3642         cached = {}
3643
3644     changed = False
3645     for cfg in configs:
3646         key = cfg["key"]
3647         if key in cached:
3648             continue
3649         print(f"    pooling pairs for {cfg['title']} (n={cfg['enum_n']})...", flush=
↳ True)
3650         cached[key] = enumerate_pair_connectivities(
3651             cfg["enum_n"],
3652             directed=cfg["directed"], simple=cfg["simple"],
3653             hypergraph=cfg["hypergraph"], r=cfg["r"],
3654             max_mult=cfg["max_mult"], separation=cfg["separation"])
3655         changed = True
3656
3657     if changed:
3658         cache_path.parent.mkdir(parents=True, exist_ok=True)
3659         with open(cache_path, "wb") as f:
3660             pickle.dump(cached, f)
3661
3662     return cached
3663
3664
3665 def compute_enumeration_cache(
3666     cache_path: str | Path = "figures/enumeration_cache.pkl",
3667     configs: list[dict] | None = None,
3668 ) -> dict[str, list[tuple[int, int]]]:
3669     """Return the full enumeration for every variant, loading from cache if
↳ available.
3670
3671     On first run this takes several minutes; subsequent runs load the pickle
3672     instantly. The cache lives at 'cache_path' relative to the caller's
3673     working directory.
3674     """
3675     cache_path = Path(cache_path)
3676     if configs is None:
3677         configs = _VARIANT_ENUM_CONFIGS
3678
3679     if cache_path.exists():
3680         with open(cache_path, "rb") as f:
3681             cached = pickle.load(f)
3682     else:
3683         cached = {}
3684
3685     changed = False
3686     for cfg in configs:
3687         key = cfg["key"]
3688         if key in cached:
3689             continue
3690         print(f"    enumerating {cfg['title']} (n={cfg['enum_n']})...", flush=True)
3691         data = enumerate_all_graphs(

```

```

3692         cfg["enum_n"],
3693         directed=cfg["directed"],
3694         simple=cfg["simple"],
3695         hypergraph=cfg["hypergraph"],
3696         r=cfg["r"],
3697         max_mult=cfg["max_mult"],
3698         separation=cfg["separation"],
3699     )
3700     cached[key] = data
3701     changed = True
3702
3703     if changed:
3704         cache_path.parent.mkdir(parents=True, exist_ok=True)
3705         with open(cache_path, "wb") as f:
3706             pickle.dump(cached, f)
3707
3708     return cached
3709
3710
3711 def plot_scatter_lambda_edges(
3712     enum_data: dict[str, list[tuple[int, int]]],
3713     path: str | Path,
3714 ) -> None:
3715     """Extremal envelope of edge count against binding connectivity, all twelve
3716         ↪ variants.
3717
3718     For each variant the full enumeration is drawn as a faint grey cloud (one
3719     point per labeled graph) with the binding connectivity " $\lambda^{\max}$ " on the
3720     horizontal axis and the edge count on the vertical, matching the
3721     edges-against-parameter layout of the all-variant grid. The extremal
3722     envelope -- the densest feasible graph available at each connectivity level
3723     -- is overlaid as an integer staircase with one dot per level, so every
3724     connectivity value carries an exact extremal edge count and the frontier is
3725     read off without any interpolation between levels.
3726
3727     """
3728     def draw_panel(ax, cfg):
3729         key = cfg["key"]
3730         data = enum_data.get(key, [])
3731         if not data:
3732             ax.set_title(cfg["title"], fontsize=9.5)
3733             return
3734
3735         edges_arr = [e for e, _ in data]
3736         conn_arr = [c for _, c in data]
3737
3738         # Faint cloud of every graph: connectivity (x) against edge count (y).
3739         ax.scatter(conn_arr, edges_arr, s=1.5, c="grey", alpha=0.25,
3740                 linewidths=0, rasterized=True)
3741
3742         # Extremal envelope  $e(\lambda)$  = densest feasible graph with  $\lambda^{\max}$ 
3743         # at most this level: a non-decreasing integer staircase. Reading it at
3744         #  $\lambda = m-1$  gives the extremal value of Problem 915 at this n.
3745         levels = sorted(set(conn_arr))
3746         running, env_edges = 0, []
3747         for lv in levels:
3748             running = max(running, max(e for e, c in data if c == lv))
3749             env_edges.append(running)
3750         ax.step(levels, env_edges, where="mid", color=_KUL_BLUE, linewidth=1.8,

```

```

3749         zorder=3)
3750     ax.plot(levels, env_edges, "o", color=_KUL_BLUE, markersize=6,
3751             zorder=4, label="extremal envelope")
3752
3753     ax.set_title(cfg["title"], fontsize=9.5)
3754     ax.set_xlabel(r"$\lambda^{\{\max\}}$", fontsize=8.5)
3755     ax.set_ylabel("edge count", fontsize=8.5)
3756     ax.tick_params(labelsize=8)
3757     # Both axes count discrete objects (independent routes, edges).
3758     ax.xaxis.set_major_locator(MaxNLocator(integer=True))
3759     ax.yaxis.set_major_locator(MaxNLocator(integer=True))
3760     ax.grid(True, alpha=0.3)
3761     ax.text(0.02, 0.98, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3762            ha="left", va="top", fontsize=8, color="grey")
3763
3764     _variant_panel_grid(
3765         draw_panel, path=path,
3766         suptitle=("Extremal envelope: edge count against binding connectivity "
3767                  "across all twelve variants (full enumeration)"),
3768         suptitle_fontsize=13)
3769
3770
3771 def _variant_panel_grid(draw_panel, *, suptitle: str, path: str | Path,
3772                        configs=None, suptitle_fontsize: float = 12.0,
3773                        row_label_fontsize: float = 12.0) -> None:
3774     """Shared scaffold for every twelve-panel variant grid (three model rows by
3775     four columns): the distribution grids, the proved/conjectured bound grid, the
3776     sampled grid, and the extremal-envelope scatter. Builds the axes, calls
3777     ‘draw_panel(ax, cfg)’ on each panel config in turn, adds the per-row model
3778     labels and the suptitle, and saves. Only the per-panel drawing and the panel
3779     configs differ between the grids, so the caller supplies both; everything else
3780     (layout, row labels, save) lives here once. ‘configs’ defaults to the
3781     twelve enumeration variants but may be any 12-item list (e.g. the sampled
3782     configs or a precomputed ‘panels’ list).
3783     """
3784     if configs is None:
3785         configs = _VARIANT_ENUM_CONFIGS
3786     fig, axes = plt.subplots(3, 4, figsize=(16, 11))
3787     for cfg, ax in zip(configs, axes.flat):
3788         draw_panel(ax, cfg)
3789     for row, name in enumerate(("simple", "multigraph", "hypergraph $r=3$")):
3790         axes[row, 0].annotate(name, xy=(-0.32, 0.5), xycoords="axes fraction",
3791                               rotation=90, ha="center", va="center",
3792                               fontsize=row_label_fontsize, fontweight="bold",
3793                               color=_KUL_DARK)
3794     fig.suptitle(suptitle, fontsize=suptitle_fontsize)
3795     fig.tight_layout(rect=(0.02, 0.0, 1.0, 0.97))
3796     _save(path)
3797
3798
3799 def plot_conn_dist_grid(
3800     enum_data: dict[str, list[tuple[int, int]]],
3801     m: int,
3802     path: str | Path,
3803     known_maxima: dict[str, int] | None = None,
3804 ) -> None:
3805     """12-panel histogram of lambda_max distribution, all variants, fixed m.
3806

```

```

3807     Bars to the left of the feasibility boundary  $m-1$  are coloured blue (the
3808     graphs we study); bars at or above  $m$  are coloured red (infeasible for this
3809      $m$ ). A vertical dashed line marks the boundary, and a dotted line marks
3810     the known or conjectured maximum if supplied in ‘‘known_maxima‘‘.
3811     """
3812     threshold = m - 1 # feasible means  $\lambda_{\max} \leq \text{threshold}$ 
3813
3814     def draw_panel(ax, cfg):
3815         key = cfg["key"]
3816         data = enum_data.get(key, [])
3817         if not data:
3818             ax.set_title(cfg["title"], fontsize=9)
3819             return
3820
3821         conn_vals = [c for _, c in data]
3822         lo, hi = min(conn_vals), max(conn_vals)
3823         levels = list(range(lo, hi + 1))
3824         counts = [conn_vals.count(lv) for lv in levels]
3825         bar_colours = [_KUL_BLUE if lv <= threshold else _RED for lv in levels]
3826         ax.bar(levels, counts, color=bar_colours, edgecolor="white", linewidth
3827             ↳ =0.4)
3828
3829         ax.axvline(threshold + 0.5, color=_WARM, linestyle="--", linewidth=1.8)
3830         if known_maxima and key in known_maxima:
3831             ax.axvline(known_maxima[key], color=_KUL_DARK, linestyle=":",
3832                 linewidth=1.6, label=f"known max = {known_maxima[key]}")
3833
3834         ax.set_title(cfg["title"], fontsize=9.5)
3835         ax.set_xlabel(r"$\lambda^{\{\max\}}$", fontsize=8.5)
3836         ax.set_ylabel("graphs", fontsize=8.5)
3837         ax.tick_params(labelsize=8)
3838         # Connectivity is integer-valued: force integer ticks so the directed
3839         # panels stop showing spurious half-integer labels (0.5, 1.5, ...).
3840         ax.set_xticks(levels)
3841         ax.grid(True, axis="y", alpha=0.3)
3842         # The feasibility boundary is identical in all twelve panels, so it is
3843         # explained once in the title and caption rather than repeated as a
3844         # legend in every panel. A legend is drawn only when a per-panel known
3845         # maximum line is present (a value that genuinely differs by panel).
3846         if known_maxima and key in known_maxima:
3847             ax.legend(fontsize=6.8, loc="upper right", framealpha=0.85)
3848             ax.text(0.02, 0.98, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3849                 ha="left", va="top", fontsize=8, color="grey")
3850
3851     suptitle = (fr"Connectivity distribution across all twelve variants,  $m = \{m\}$ 
3852         ↳ "
3853         fr"(blue = feasible  $\lambda^{\{\max\}} \leq \{\text{threshold}\}$ , red =
3854             ↳ infeasible)")
3855     _variant_panel_grid(draw_panel, suptitle=suptitle, path=path,
3856         row_label_fontsize=12)
3857
3858     def _midrange_lambda_threshold(hi: int) -> int:
3859         """A per-variant connectivity boundary at the midpoint of the achievable range
3860             ↳ .
3861
3862         The distribution figures split graphs by ‘‘ $\lambda^{\max}$ ‘‘ into a blue (low) and

```

```

3860     a red (high) population. A single global boundary collapses some panels to
           ↳ one
3861     colour (every graph below it, or almost none), because the twelve variants
3862     reach very different connectivity ranges at the sizes enumeration allows.
3863     Splitting instead at ‘‘round(hi/2)’’ -- the middle of what each enumeration
           ↳ can
3864     reach -- keeps both populations visible in every panel, scaled to what is
3865     possible there. Floored at one so the boundary is never zero, which would
3866     push every graph above it and leave the blue (low) side empty.
3867     """
3868     return max(1, round(hi / 2))
3869
3870
3871 def plot_pair_conn_dist_grid(
3872     pair_data: dict[str, dict[tuple[int, int], int]],
3873     path: str | Path,
3874 ) -> None:
3875     """12-panel histogram of per-PAIR connectivity, pooled over the full
           ↳ enumeration.
3876
3877     Where :func:‘plot_conn_dist_grid‘ plots one ‘‘lambda~max’’ per graph, this
3878     pools every vertex pair of every graph: one observation per (graph, pair).
3879     Each bar is split and STACKED by the connectivity of the graph the pair came
3880     from -- blue (bottom) for pairs from graphs whose ‘‘lambda~max’’ stays at or
3881     below the per-panel boundary ‘‘T’’, red (top) for pairs from graphs above it
3882     -- so the bar’s full height is the true number of observations at that level.
3883
3884     ‘‘T’’ is set per variant by :func:‘_midrange_lambda_threshold’, not by a
           ↳ single
3885     global ‘‘m’’: at a fixed boundary some panels are entirely one colour, while
           ↳ the
3886     mid-range split keeps both populations visible everywhere. Pairs whose own
3887     connectivity exceeds ‘‘T’’ can only come from graphs above ‘‘T’’, so those
           ↳ bars
3888     are wholly red; below ‘‘T’’ the colours mix, since a high-connectivity graph
3889     still has many low-connectivity pairs.
3890     """
3891     def draw_panel(ax, cfg):
3892         table = pair_data.get(cfg["key"], {})
3893         if not table:
3894             ax.set_title(cfg["title"], fontsize=9)
3895             return
3896
3897         hi = max(lmax for (lmax, _pc) in table)
3898         T = _midrange_lambda_threshold(hi)
3899         levels = sorted({pc for (_lmax, pc) in table})
3900         blue = [sum(c for (lmax, pc), c in table.items()
3901                     if pc == lv and lmax <= T) for lv in levels]
3902         red = [sum(c for (lmax, pc), c in table.items()
3903                   if pc == lv and lmax > T) for lv in levels]
3904
3905         ax.bar(levels, blue, color=_KUL_BLUE, edgecolor="white", linewidth=0.4,
3906               label=fr"from $\lambda^{\{\max\}}\!\!\leq\!\{T\}$")
3907         ax.bar(levels, red, bottom=blue, color=_RED, edgecolor="white",
3908               linewidth=0.4, label=fr"from $\lambda^{\{\max\}}\!\!>\!\{T\}$")
3909         ax.axvline(T + 0.5, color=_WARM, linestyle="--", linewidth=1.8)
3910
3911         ax.set_title(cfg["title"], fontsize=9.5)

```

```

3912     ax.set_xlabel("pair connectivity", fontsize=8.5)
3913     ax.set_ylabel("pair observations", fontsize=8.5)
3914     ax.tick_params(labelsize=8)
3915     ax.set_xticks(levels)
3916     ax.grid(True, axis="y", alpha=0.3)
3917     ax.legend(fontsize=6.8, loc="upper right", framealpha=0.85)
3918     ax.text(0.02, 0.98, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3919            ha="left", va="top", fontsize=8, color="grey")
3920
3921     suptitle = ("Pair-connectivity distribution across all twelve variants "
3922                "(every vertex pair of every graph pooled, split at each variant's
3923                 ↪ ")
3924
3925     r"mid-range  $\lambda^{\max}$ : blue from low-connectivity graphs,
3926         ↪ red from high")
3927
3928     _variant_panel_grid(draw_panel, suptitle=suptitle, path=path,
3929                        row_label_fontsize=12)
3930
3931 def plot_edge_dist_grid(
3932     enum_data: dict[str, list[tuple[int, int]]],
3933     path: str | Path,
3934 ) -> None:
3935     """12-panel histogram of edge count distribution, all variants.
3936
3937     Each bar is split and STACKED by graph connectivity: blue (bottom) for graphs
3938     whose ' $\lambda^{\max}$ ' stays at or below the per-panel boundary ' $T$ ', red (top)
3939     for graphs above it, so the full bar height is the true number of graphs at
3940     that edge count. ' $T$ ' is the midpoint of each variant's achievable
3941     connectivity range (:func: '_midrange_lambda_threshold'); a single global
3942     boundary leaves some panels all one colour, the mid-range split keeps both
3943     populations visible. A dotted line marks the densest graph at or below ' $T$ '.
3944
3945     """
3946     def draw_panel(ax, cfg):
3947         key = cfg["key"]
3948         data = enum_data.get(key, [])
3949         if not data:
3950             ax.set_title(cfg["title"], fontsize=9)
3951             return
3952
3953         hi = max(c for _e, c in data)
3954         T = _midrange_lambda_threshold(hi)
3955         max_e = max(e for e, _ in data)
3956         levels = list(range(0, max_e + 1))
3957         blue = [0] * (max_e + 1)
3958         red = [0] * (max_e + 1)
3959         for e, c in data:
3960             (blue if c <= T else red)[e] += 1
3961
3962         # Stacked, not overlaid: low-connectivity graphs (blue) on the bottom,
3963         # high-connectivity (red) on top, so the full height is the true count.
3964         ax.bar(levels, blue, color=_KUL_BLUE, edgecolor="white", linewidth=0.3,
3965                label=fr"$\lambda^{\{\max\}}\!\!\leq\!\{T\}$")
3966         ax.bar(levels, red, bottom=blue, color=_RED, edgecolor="white",
3967                linewidth=0.3, label=fr"$\lambda^{\{\max\}}\!\!>\!\{T\}$")
3968
3969         # The densest graph in the lower-connectivity population: read straight
3970         # off the data, so it lands on the right edge of the blue mass.
3971         if any(blue):

```

```

3968         kmax = max(e for e in levels if blue[e])
3969         ax.axvline(kmax, color=_KUL_DARK, linestyle=":", linewidth=1.8,
3970                  label=fr"densest  $\leq \{T\}$ : {kmax}")
3971
3972         ax.set_title(cfg["title"], fontsize=8.5)
3973         ax.set_xlabel("edge count", fontsize=7.5)
3974         ax.set_ylabel("graphs", fontsize=7.5)
3975         ax.tick_params(labels=7)
3976         # Edge counts are integers: keep the tick labels integer-valued.
3977         ax.xaxis.set_major_locator(MaxNLocator(integer=True))
3978         ax.grid(True, axis="y", alpha=0.25)
3979         ax.legend(fontsize=6.2, loc="upper right", framealpha=0.8)
3980         ax.text(0.98, 0.02, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3981                ha="right", va="bottom", fontsize=7, color="grey")
3982
3983         suptitle = ("Edge-count distribution across all twelve variants "
3984                    r"(stacked by connectivity, split at each variant's mid-range "
3985                    r"$\lambda^{\max}$: blue low-connectivity graphs, red high)")
3986         _variant_panel_grid(draw_panel, suptitle=suptitle, path=path,
3987                             row_label_fontsize=11)
3988
3989
3990 # --- 3-D BOUND SURFACE: cache solve() over (variant, n, m) grid;
3991     ↳ plot_variant_3d_surfaces draws it ---
3992
3992 # The two multigraph vertex problems reduce exactly to the simple ones: vertex
3993 # mode is blind to parallel copies (see _split_capacity_matrix), so a multigraph
3994 # and its underlying simple graph have the same vertex connectivity. We mirror
3995 # the simple-vertex surface into these keys rather than searching multigraphs,
3996 # whose degenerate free-parallel fills would report a lower bound far above the
3997 # true (simple) value and spike the plot.
3998 _SURFACE_ALIASED_VERTEX = {
3999     "multi_undirected_vertex": "simple_undirected_vertex",
4000     "multi_directed_vertex": "simple_directed_vertex",
4001 }
4002
4003 # 12 variant configs for the surface computation.
4004 _SURFACE_VARIANT_CONFIGS: list[dict] = [
4005     dict(key="simple_undirected_edge", title="simple undirected edge",
4006          directed=False, simple=True, hypergraph=False, r=3, separation="edge"),
4007     dict(key="simple_undirected_vertex", title="simple undirected vertex",
4008          directed=False, simple=True, hypergraph=False, r=3, separation="vertex")
4009     ↳ ,
4010     dict(key="simple_directed_edge", title="simple directed arc",
4011          directed=True, simple=True, hypergraph=False, r=3, separation="edge"),
4012     dict(key="simple_directed_vertex", title="simple directed vertex",
4013          directed=True, simple=True, hypergraph=False, r=3, separation="vertex")
4014     ↳ ,
4015     dict(key="multi_undirected_edge", title="multigraph undirected edge",
4016          directed=False, simple=False, hypergraph=False, r=3, separation="edge"),
4017     dict(key="multi_undirected_vertex", title="multigraph undirected vertex",
4018          directed=False, simple=False, hypergraph=False, r=3, separation="vertex")
4019     ↳ ,
4020     dict(key="multi_directed_edge", title="multigraph directed arc",
4021          directed=True, simple=False, hypergraph=False, r=3, separation="edge"),
4022     dict(key="multi_directed_vertex", title="multigraph directed vertex",
4023          directed=True, simple=False, hypergraph=False, r=3, separation="vertex")
4024     ↳ ,

```

```

4021     dict(key="hyper_undirected_edge",    title="hypergraph undirected edge",
4022           directed=False, simple=True, hypergraph=True, r=3, separation="edge"),
4023     dict(key="hyper_undirected_vertex",  title="hypergraph undirected vertex",
4024           directed=False, simple=True, hypergraph=True, r=3, separation="vertex")
4025     ↪ ,
4026     dict(key="hyper_directed_edge",      title="hypergraph directed arc",
4027           directed=True, simple=True, hypergraph=True, r=3, separation="edge"),
4028     dict(key="hyper_directed_vertex",    title="hypergraph directed vertex",
4029           directed=True, simple=True, hypergraph=True, r=3, separation="vertex")
4030     ↪ ,
4031 ]
4032 def _surface_known_value(vkey: str, n: int, m: int) -> int | None:
4033     """The proved exact value at '(n, m)' for 'vkey', or 'None' if open here
4034     ↪ .
4035
4036     Returns the closed-form extremal value precisely in the regime where the
4037     thesis proves it, so the surface can colour those bars as exact (blue)
4038     rather than leaving every bar a search lower bound. Outside the proved
4039     regime it returns 'None' and the caller falls back to a discovery search
4040     (a purple lower bound). Each value is capped at the trivial maximum for
4041     its model, since the closed form overshoots in the small-'n' large-'m'
4042     corner where no graph that dense exists.
4043     """
4044     tri_simple = n * (n - 1) // 2
4045     tri_dir = n * (n - 1)
4046     tri_hyp = math.comb(n, 3)
4047     if vkey == "simple_undirected_edge":      # Mader, all m
4048         return min(simple_undirected_edge(n, m), tri_simple)
4049     if vkey == "simple_undirected_vertex":    # Leonard, m<=4
4050         return min(simple_undirected_edge(n, m), tri_simple) if m <= 4 else None
4051     if vkey == "simple_directed_edge":        # exact only at m=2
4052         return min(directed_arc_lower_bound(n, 2), tri_dir) if m == 2 else None
4053     if vkey == "simple_directed_vertex":      # exact only at m=2
4054         return min(directed_arc_lower_bound(n, 2), tri_dir) if m == 2 else None
4055     if vkey == "multi_undirected_edge":      # multi-tree bound, all m
4056         return min(multigraph_undirected_edge(n, m), (m - 1) * tri_simple)
4057     if vkey == "multi_undirected_vertex":    # = simple vertex, m<=4
4058         return min(simple_undirected_edge(n, m), tri_simple) if m <= 4 else None
4059     if vkey == "multi_directed_edge":        # cut-counting proves n<=6
4060         return min(directed_multigraph_arc(n, m), (m - 1) * tri_dir) if n <= 6
4061         ↪ else None
4062     if vkey == "multi_directed_vertex":      # = simple digraph, exact m=2
4063         return min(directed_arc_lower_bound(n, 2), tri_dir) if m == 2 else None
4064     if vkey == "hyper_undirected_edge":      # incidence-rank, all m
4065         return min(hypergraph_edge(n, m, 3), tri_hyp)
4066     if vkey == "hyper_undirected_vertex":    # incidence-rank lemma, m<=3
4067         return min(hypergraph_edge(n, m, 3), tri_hyp) if m <= 3 else None
4068     # hyper_directed_edge, hyper_directed_vertex: open (new model), no formula.
4069     return None
4070 def compute_surface_cache(
4071     n_range: tuple[int, int] = (3, 9),
4072     m_range: tuple[int, int] = (2, 6),
4073     max_seconds: float = 20.0,
4074     cache_path: str | Path = "figures/surface_cache.json",

```

```

4075 ) -> dict:
4076     """Compute the optimal bound at every (variant, n, m) triple, save to a JSON
        ↳ cache.

4077
4078     Returns the cache dict: '{variant_key: {str(n): {str(m): {value, bound}}}}'.
4079     Where the value is proved (see :func: '_surface_known_value') the bound is
4080     recorded as '"exact"' using the closed form; otherwise a discovery search
4081     supplies a '"lower"' bound. Missing entries are skipped on later runs.
4082     """
4083     cache_path = Path(cache_path)
4084     if cache_path.exists():
4085         with open(cache_path) as f:
4086             cache = json.load(f)
4087     else:
4088         cache = {}
4089
4090     changed = False
4091     ns = list(range(n_range[0], n_range[1] + 1))
4092     ms = list(range(m_range[0], m_range[1] + 1))
4093
4094     for cfg in _SURFACE_VARIANT_CONFIGS:
4095         vkey = cfg["key"]
4096         if vkey in _SURFACE_ALIASED_VERTEX:
4097             continue # filled by mirroring its simple counterpart, below
4098         if vkey not in cache:
4099             cache[vkey] = {}
4100         for n in ns:
4101             sn = str(n)
4102             if sn not in cache[vkey]:
4103                 cache[vkey][sn] = {}
4104             for m in ms:
4105                 sm = str(m)
4106                 known = _surface_known_value(vkey, n, m)
4107                 if known is not None:
4108                     # Proved regime: use the closed form, mark it exact (blue).
4109                     # Always (re)write, so an older cache whose every cell was
4110                     # tagged "lower" is corrected here without recomputation.
4111                     entry = {"value": known, "bound": "exact"}
4112                     if cache[vkey][sn].get(sm) != entry:
4113                         cache[vkey][sn][sm] = entry
4114                         changed = True
4115                     continue
4116                 if sm in cache[vkey][sn]:
4117                     continue # open cell already searched; keep the lower bound
4118                 print(f" surface: {cfg['title']} n={n} m={m}", flush=True)
4119                 res = solve(
4120                     n, m,
4121                     directed=cfg["directed"],
4122                     simple=cfg["simple"],
4123                     hypergraph=cfg["hypergraph"],
4124                     r=cfg["r"],
4125                     exhaustive=False,
4126                     separation=cfg["separation"],
4127                     max_seconds=max_seconds,
4128                     seed=0,
4129                 )
4130                 cache[vkey][sn][sm] = {"value": res.value, "bound": res.bound}
4131                 changed = True

```

```

4132
4133     # Mirror the simple-vertex surfaces into their multigraph aliases, so the two
4134     # provably-equal problems show identical bars instead of two independent
4135     # (and possibly degenerate) searches.
4136     for multi_key, simple_key in _SURFACE_ALIASED_VERTEX.items():
4137         if simple_key in cache and cache.get(multi_key) != cache[simple_key]:
4138             cache[multi_key] = copy.deepcopy(cache[simple_key])
4139             changed = True
4140
4141     if changed:
4142         cache_path.parent.mkdir(parents=True, exist_ok=True)
4143         with open(cache_path, "w") as f:
4144             json.dump(cache, f)
4145
4146     return cache
4147
4148
4149 def plot_variant_3d_surfaces(
4150     cache_path: str | Path,
4151     path: str | Path,
4152 ) -> None:
4153     """3-D bar chart of the optimal bound over the (n, m) grid, all twelve
4154         ↪ variants.
4155
4156     Each bar's height is the bound value; exact points are blue, lower-bound
4157     points are purple. The 3x4 layout matches the flat grid figure.
4158     """
4159     with open(cache_path) as f:
4160         cache = json.load(f)
4161
4162     fig = plt.figure(figsize=(20, 13))
4163     row_names = ("simple", "multigraph", "hypergraph $r=3$")
4164
4165     for idx, cfg in enumerate(_SURFACE_VARIANT_CONFIGS):
4166         vkey = cfg["key"]
4167         ax = fig.add_subplot(3, 4, idx + 1, projection="3d")
4168
4169         variant_data = cache.get(vkey, {})
4170         ns_exact, ms_exact, vs_exact = [], [], []
4171         ns_lower, ms_lower, vs_lower = [], [], []
4172
4173         for sn, m_dict in sorted(variant_data.items(), key=lambda x: int(x[0])):
4174             n = int(sn)
4175             for sm, entry in sorted(m_dict.items(), key=lambda x: int(x[0])):
4176                 m_val = int(sm)
4177                 v = entry["value"]
4178                 if entry["bound"] == "exact":
4179                     ns_exact.append(n)
4180                     ms_exact.append(m_val)
4181                     vs_exact.append(v)
4182                 else:
4183                     ns_lower.append(n)
4184                     ms_lower.append(m_val)
4185                     vs_lower.append(v)
4186
4187         dx = dy = 0.6
4188         if ns_lower:
4189             ax.bar3d(

```

```

4189         [x - dx / 2 for x in ns_lower],
4190         [y - dy / 2 for y in ms_lower],
4191         [0] * len(vs_lower),
4192         dx, dy, vs_lower,
4193         color=_VIOLET, alpha=0.7, shade=True,
4194         edgecolor="white", linewidth=0.25,
4195     )
4196     if ns_exact:
4197         ax.bar3d(
4198             [x - dx / 2 for x in ns_exact],
4199             [y - dy / 2 for y in ms_exact],
4200             [0] * len(vs_exact),
4201             dx, dy, vs_exact,
4202             color=_KUL_BLUE, alpha=0.9, shade=True,
4203             edgecolor="white", linewidth=0.25,
4204         )
4205
4206     # Same camera for every panel so the twelve are read side by side, and a
4207     # z-axis anchored at 0 so bar heights are comparable within a panel.
4208     ax.view_init(elev=24, azim=-58)
4209     peak = max(vs_exact + vs_lower, default=1)
4210     ax.set_zlim(0, peak * 1.05)
4211     for pane in (ax.xaxis, ax.yaxis, ax.zaxis):
4212         pane.pane.set_alpha(0.04)
4213
4214     ax.set_title(cfg["title"], fontsize=7.5, pad=2)
4215     ax.set_xlabel("$n$", fontsize=7, labelpad=2)
4216     ax.set_ylabel("$m$", fontsize=7, labelpad=2)
4217     ax.set_zlabel("bound", fontsize=7, labelpad=2)
4218     ax.tick_params(labelsize=6)
4219
4220     # Row labels
4221     for row, name in enumerate(row_names):
4222         row_ax = fig.axes[row * 4]
4223         row_ax.text2D(-0.28, 0.5, name, transform=row_ax.transAxes,
4224                      rotation=90, ha="center", va="center",
4225                      fontsize=11, fontweight="bold", color=_KUL_DARK)
4226
4227     # Manual legend patches
4228     from matplotlib.patches import Patch
4229     legend_elems = [
4230         Patch(facecolor=_KUL_BLUE, alpha=0.85, label="exact (proved)"),
4231         Patch(facecolor=_VIOLET, alpha=0.65, label="lower bound (search)"),
4232     ]
4233     fig.legend(handles=legend_elems, loc="lower center", ncol=2,
4234              fontsize=10, framealpha=0.9)
4235
4236     fig.suptitle("Erdős 915: optimal bound over $(n, m)$, "
4237                "blue where proved, purple where only a search lower bound is  

4238                ↔ known",
4239                fontsize=13, y=0.99)
4239     _save(path, tight=False)
4240
4241
4242     # --- 3-D THRESHOLD HISTOGRAM: lambda^max distribution vs p for three variants ---
4243
4244     def plot_conn_threshold_3d(

```

```

4246     path: str | Path,
4247     n: int = 30,
4248     m: int = 3,
4249     p_values: list[float] | None = None,
4250     samples: int = 400,
4251     seed: int = 7,
4252 ) -> None:
4253     """3-D bar chart of the lambda_max distribution over density, for three
        ↳ variants.

4254
4255     x = density in units of that model's OWN threshold, y = lambda_max value,
4256     z = fraction of samples. Each panel is swept in units of its own p*, and the
4257     threshold line sits at 1 in all three, because the models do NOT share a
4258     density scale. The threshold is where a vertex's expected degree reaches m,
4259     which for a graph is  $p*(n-1) = m$ , giving  $p* = m/n$ , but for an r-uniform
4260     binomial hypergraph is  $p*C(n-1, r-1) = m$ , giving  $p* = m/C(n-1, r-1) =$ 
4261      $\Theta(m/n^{(r-1)})$ . Plotting the hypergraph panel against  $m/n$  would put the
4262     line in the wrong place by a factor of order  $n^{(r-2)}$ . Only the graph panels
4263     are covered by thm:gnp-threshold; the hypergraph panel is an observation.
4264
4265     Three panels: undirected edge (uses 'n'), directed arc (uses 'n'),
4266     hypergraph edge (uses  $\min(n, 8)$  to keep flow computation fast).
4267     """
4268     # Sweep in units of the model's own threshold, so all three panels are
4269     # directly comparable and each transition is visible where it actually is.
4270     ratios = ([i / 5.0 for i in range(1, 20)] if p_values is None else None)
4271
4272     rng = random.Random(seed)
4273     # Hypergraph flow computation scales poorly: cap n at 8 for that panel.
4274     n_hyper = min(n, 8)
4275
4276     variants_3d = [
4277         dict(label="undirected edge", directed=False, separation="edge",
4278             hypergraph=False, panel_n=n, p_star=m / n,
4279             star_text=f"$m/n = {m}/{n}$"),
4280         dict(label="directed arc", directed=True, separation="edge",
4281             hypergraph=False, panel_n=n, p_star=m / n,
4282             star_text=f"$m/n = {m}/{n}$"),
4283         dict(label=f"hypergraph edge (<math>r=3</math>, <math>n={n\_hyper}</math>)", directed=False,
4284             separation="edge", hypergraph=True, panel_n=n_hyper,
4285             p_star=m / math.comb(n_hyper - 1, 2),
4286             star_text=(fr"$m/\text{binom}{{{n\_hyper}-1}}{{2}} = "
4287                       fr"{m}/{math.comb(n\_hyper - 1, 2)}$")),
4288     ]
4289
4290     fig = plt.figure(figsize=(18, 6))
4291
4292     for panel_idx, vdesc in enumerate(variants_3d):
4293         ax = fig.add_subplot(1, 3, panel_idx + 1, projection="3d")
4294         panel_n = vdesc["panel_n"]
4295         p_star = vdesc["p_star"]
4296         panel_ps = (p_values if p_values is not None
4297                     else [min(1.0, r * p_star) for r in ratios])
4298
4299         all_conn_vals: list[int] = []
4300         dist_by_p: list[tuple[float, list[int]]] = []
4301
4302         for p in panel_ps:

```

```

4303     conn_list: list[int] = []
4304     for _ in range(samples):
4305         if vdesc["hypergraph"]:
4306             cand = _hyperedge_candidates(panel_n, 3, vdesc["directed"])
4307             chosen = [e for e in cand if rng.random() < p]
4308             h = Hypergraph(panel_n, chosen, directed=vdesc["directed"])
4309             c = max_hyper_connectivity(h, vertex_split=False)
4310         else:
4311             g = sample_random_graph(panel_n, p, vdesc["directed"], rng)
4312             if vdesc["separation"] == "edge":
4313                 c = max_edge_connectivity(g)
4314             else:
4315                 c = max_vertex_connectivity(g)
4316             conn_list.append(c)
4317         dist_by_p.append((p, conn_list))
4318         all_conn_vals.extend(conn_list)
4319
4320     if not all_conn_vals:
4321         continue
4322
4323     lo, hi = 0, max(all_conn_vals)
4324     levels = list(range(lo, hi + 1))
4325
4326     # x is the density in units of this panel's own threshold.
4327     xs = [p / p_star for p, _ in dist_by_p]
4328     width = 0.9 * min((b - a) for a, b in zip(xs, xs[1:])) if len(xs) > 1 else
4329         ↪ 0.1
4330     for x, (_, conn_list) in zip(xs, dist_by_p):
4331         total = len(conn_list)
4332         for lv in levels:
4333             frac = conn_list.count(lv) / total
4334             if frac > 0:
4335                 ax.bar3d(x - width / 2, lv - 0.4, 0, width, 0.8, frac,
4336                         color=_KUL_BLUE, alpha=0.7, shade=True)
4337
4338     # The threshold sits at 1 in these units, by construction.
4339     for lv in levels:
4340         ax.plot([1.0, 1.0], [lv - 0.4, lv + 0.4], [0, 0],
4341                 color=_WARM, linewidth=0.5, alpha=0.4)
4342     zmax = max(conn_list.count(lv) / len(conn_list)
4343                for _, conn_list in dist_by_p
4344                for lv in levels
4345                if conn_list.count(lv) > 0)
4346     ax.plot([1.0, 1.0], [lo, hi], [zmax * 0.9, zmax * 0.9],
4347            color=_WARM, linewidth=2.5, linestyle="--", label="$p / p^* = 1$")
4348
4349     ax.set_title(f"{vdesc['label']}\n$p^* = $ {vdesc['star_text']}", fontsize
4350         ↪ =9)
4351     ax.set_xlabel("$p / p^*$", fontsize=9)
4352     ax.set_ylabel(r"$\lambda^{\{\max\}}$", fontsize=9)
4353     ax.set_zlabel("fraction", fontsize=9)
4354     ax.tick_params(labelsize=7)
4355
4356     fig.suptitle(
4357         fr"Distribution of $\lambda^{\{\max\}}$ against density, $m = \{m\}$, each
4358         ↪ model "
4359         fr"in units of its own threshold $p^*$ (the density at which a vertex's "
4360         fr"expected degree reaches $m$)",

```

```

4358         fontsize=11,
4359     )
4360     _save(path, tight=False)

```

C.1.8 The open variants

These routines exist to attack the questions the thesis leaves open. They measure the hypergraph vertex problem, work with fractional weightings where the integer problem is out of reach, and provide the exhaustive maximisers used to test conjectures at the sizes where testing is still possible. Several results in [Appendix A](#) were found, or in two cases refuted, by running exactly these functions.

```

4363 # --- OPEN-VARIANT EXPLORATION: hypergraph vertex connectivity, fractional search
4364     ↪ tools ---
4365
4366 def max_hyper_vertex_connectivity(hypergraph: Hypergraph) -> int:
4367     """‘kappa^max’ for the hypergraph vertex problem (readability wrapper)."""
4368     return max_hyper_connectivity(hypergraph, vertex_split=True)
4369
4370
4371 def _hyper_codegree_ok(edges, bound: int) -> bool:
4372     """Necessary feasibility filter: two vertices in > ‘bound’ common
4373     hyperedges already carry that many one-step disjoint Berge routes."""
4374     from collections import Counter
4375
4376     codegree: Counter = Counter()
4377     for edge in edges:
4378         for pair in combinations(sorted(edge), 2):
4379             codegree[pair] += 1
4380             if codegree[pair] > bound:
4381                 return False
4382     return True
4383
4384
4385 def hyper_vertex_feasible_exists(n: int, r: int, m: int, q: int,
4386                                 multi: bool = True) -> bool:
4387     """Exhaustively decide whether some ‘r’-uniform (multi-)hypergraph on
4388     ‘n’ vertices with ‘q’ hyperedges has ‘kappa^max ≤ m - 1’.
4389
4390     Repeated hyperedges are allowed when ‘multi’ is set (they are forced
4391     infeasible at m = 2 by the codegree filter, so m = 2 needs no repeats).
4392     Practical up to roughly C(C(n,r), q) ~ a few hundred thousand candidates.
4393     """
4394     all_edges = list(combinations(range(n), r))
4395     chooser = combinations_with_replacement if multi else combinations
4396     for sub in chooser(all_edges, q):
4397         if not _hyper_codegree_ok(sub, m - 1):
4398             continue
4399         candidate = Hypergraph(n, [frozenset(edge) for edge in sub])
4400         feasible = True
4401         for s in range(n):
4402             for t in range(s + 1, n):
4403                 if hyper_connectivity(candidate, s, t, vertex_split=True) > m - 1:
4404                     feasible = False

```

```

4405         break
4406     if not feasible:
4407         break
4408     if feasible:
4409         return True
4410 return False
4411
4412
4413 def verify_hyper_vertex_value(n: int, r: int, m: int) -> bool:
4414     """Confirm exhaustively that the ‘r’-uniform vertex value at ‘(n, m)’
4415     equals ‘floor((m-1)(n-1)/(r-1))’: the bound’s witness is feasible
4416     ( $\kappa_{\max} \leq m-1$ ) and one more hyperedge never is."""
4417     target = ((m - 1) * (n - 1)) // (r - 1)
4418     if m == 2:
4419         witness = star_hypertree(n, r)
4420         attained = (len(witness.hyperedges) == target
4421                     and max_hyper_vertex_connectivity(witness) <= 1)
4422     else:
4423         attained = hyper_vertex_feasible_exists(n, r, m, target)
4424     return attained and not hyper_vertex_feasible_exists(n, r, m, target + 1)
4425
4426
4427 def max_feasible_hyperedges(
4428     n: int, r: int, m: int,
4429     *, directed: bool = False, vertex_split: bool = False, kind: str = "forward",
4430     time_limit: float = 20.0, seed_lb: int = 0,
4431 ) -> tuple[int, bool]:
4432     """Largest number of ‘r’-uniform hyperedges with Berge connectivity ‘ $\leq m$ 
4433      $\hookrightarrow -1$ ’.
4434
4435     Returns ‘(value, exact)’. ‘exact’ is ‘True’ when the branch and bound
4436     below proved optimality within ‘time_limit’ (the value is then the true
4437     extremal number for that variant); otherwise ‘value’ is the best feasible
4438     construction found, a lower bound. ‘kind’ selects the directed orientation
4439     model (forward / backward / general); for the undirected case it is ignored.
4440
4441     The search is a depth-first include/exclude over the candidate hyperedges with
4442     two prunings. Feasibility is monotone (adding a hyperedge never lowers any
4443     Berge connectivity), so once the active set is infeasible no superset can be
4444     feasible and the include branch is dropped; and a branch is cut once
4445     ‘active + remaining’ cannot beat the best feasible count already seen. A
4446     short randomised warm start raises that incumbent first, so the bound prune
4447     bites early. ‘seed_lb’ is an externally known feasible lower bound (e.g.
4448      $\hookrightarrow$  the
4449     forward value when maximising over the general model, which contains every
4450     forward hypergraph) used to start the incumbent; it never changes the proven
4451     optimum, only the pruning.
4452     """
4453     deadline = time.time() + time_limit
4454     # Warm start: a short greedy randomised lower bound sharpens the bound prune;
4455     # kept brief so the branch and bound, which does the actual proving, keeps
4456     # the budget.
4457     warm_share = min(0.3 * time_limit, 1.5)
4458     lb, _ = _random_hypergraph_search(
4459         n, r, m, time.time() + warm_share, seed=0,
4460         directed=directed, vertex_split=vertex_split, kind=kind)
4461     lb = max(lb, seed_lb)

```

```

4461     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
4462     total = len(candidates)
4463     best = [lb]
4464     timed_out = [False]
4465     active = Hypergraph(n, directed=directed)
4466
4467     def dfs(pos: int, count: int) -> None:
4468         if timed_out[0]:
4469             return
4470         if count + (total - pos) <= best[0]:
4471             return # cannot beat the incumbent
4472         if time.time() > deadline:
4473             timed_out[0] = True
4474             return
4475         if pos == total:
4476             if count > best[0]:
4477                 best[0] = count
4478             return
4479         # Include candidates[pos], but only if the set stays feasible.
4480         active.add_hyperedge(candidates[pos])
4481         if max_hyper_connectivity(active, vertex_split=vertex_split) <= m - 1:
4482             dfs(pos + 1, count + 1)
4483         active.hyperedges.pop()
4484         # Exclude candidates[pos].
4485         dfs(pos + 1, count)
4486
4487     dfs(0, 0)
4488     return best[0], (not timed_out[0])
4489
4490
4491 def _c_flat(mu: np.ndarray) -> tuple[np.ndarray, "_ct.POINTER[_ct.c_int]"]:
4492     """Return a contiguous int32 copy of ‘mu’ and a ctypes c_int pointer into it
4493     ↪ ."""
4494     flat = np.ascontiguousarray(mu, dtype=np.int32)
4495     ptr = flat.ctypes.data_as(_ct.POINTER(_ct.c_int))
4496     return flat, ptr # caller must keep flat alive while ptr is in use
4497
4498 def _tiny_maxflow(mu: np.ndarray, n: int, s: int, t: int, cap: int) -> bool:
4499     """Return True iff the integer max-flow from s to t under capacities ‘mu’
4500     exceeds ‘cap’. Plain DFS augmentation (Ford-Fulkerson with a stack, not
4501     BFS/Edmonds-Karp); built for n <= 7, where it beats both the networkx call
4502     and scipy’s C ‘maximum_flow’ by orders of magnitude inside the hot
4503     enumeration/search loops (measured ~16x faster than scipy at n=7: the
4504     library calls’ per-call sparse-matrix construction and dispatch overhead
4505     dominates at this size, while the ‘cap’ lets this routine stop after
4506     cap+1 augmenting paths). Correct for integer capacities: each augmentation
4507     increases flow by at least 1, so the loop terminates in at most
4508     max-flow-value iterations."""
4509     if _C is not None and n <= 7:
4510         flat, ptr = _c_flat(mu)
4511         return bool(_C.tiny_maxflow(ptr, n, s, t, cap))
4512     residual = mu.astype(int) # astype already returns a fresh, mutable
4513     ↪ copy
4514     flow = 0
4515     while flow <= cap:
4516         parent = [-1] * n
4517         parent[s] = s

```

```

4517     queue = [s]
4518     while queue and parent[t] == -1:
4519         u = queue.pop()
4520         for v in range(n):
4521             if parent[v] == -1 and residual[u, v] > 0:
4522                 parent[v] = u
4523                 queue.append(v)
4524     if parent[t] == -1:
4525         return False # flow is maximal and <= cap
4526     bottleneck = 10 ** 9
4527     v = t
4528     while v != s:
4529         bottleneck = min(bottleneck, residual[parent[v], v])
4530         v = parent[v]
4531     v = t
4532     while v != s:
4533         residual[parent[v], v] -= bottleneck
4534         residual[v, parent[v]] += bottleneck
4535         v = parent[v]
4536     flow += bottleneck
4537     return True
4538
4539
4540 def _canonical_form(mu: np.ndarray) -> bytes:
4541     """A canonical key for the isomorphism class of a directed multigraph.
4542
4543     The key is the lexicographically smallest flattened multiplicity matrix over
4544     all vertex relabellings: two multiplicity matrices have the same key iff one
4545     is a vertex permutation of the other. This is the SOURCE OF TRUTH for
4546     isomorphism here, with no dependency on any external library, so the
4547     enumeration's correctness never rests on an outside binary. For 'n=7' it is
4548     5040 permutations per graph, which is cheap on CPU; storing one key per class
4549     is what bounds the enumeration's memory to the number of classes rather than
4550     the number of labelled copies.
4551     """
4552     n = mu.shape[0]
4553     if _C is not None and n <= 7:
4554         flat, ptr = _c_flat(mu)
4555         out = np.empty(n * n, dtype=np.int32)
4556         out_ptr = out.ctypes.data_as(_ct.POINTER(_ct.c_int))
4557         _C.canonical_form_min(ptr, n, out_ptr)
4558         return out.tobytes()
4559     best: bytes | None = None
4560     for perm in permutations(range(n)):
4561         p = list(perm)
4562         cand = mu[np.ix_(p, p)].tobytes()
4563         if best is None or cand < best:
4564             best = cand
4565     assert best is not None # n >= 1, so at least the identity permutation exists
4566     return best
4567
4568
4569 def enumerate_extremal_directed_multigraphs(
4570     n: int, m: int, target_arcs: int, max_degree: int | None = None,
4571     up_to_iso: bool = True,
4572 ) -> list[np.ndarray]:
4573     """Exhaustively list the directed multigraphs on 'n' vertices with
4574     multiplicities in {0..m-1}, 'lambda^max <= m-1', exactly 'target_arcs'

```

```

4575 arcs and (optionally) maximum total degree at most 'max_degree'.
4576
4577 This is the finite-base tool for the m = 3 chain (see claude.md and the
4578 commented rem:odd-step-roadmap): the characterisations at n = 4, 5 feed
4579 the recursion behind statement (b) at n = 7. DFS in vertex-block order
4580 with three prunings: multiplicity capacity, the induced-subgraph arc bound
4581 on every completed prefix, and exact flow checks on completed prefixes
4582 (induced flows only grow, so a violation is final).
4583
4584 SOUNDNESS: the induced-subgraph bound is a PROVED upper bound for j <= 6
4585 (from the cut-counting MILP; see 'known' dict below). For j >= 7 it
4586 falls back to the CONJECTURED bipartite bound floor(j^2/4), which is not
4587 yet proved. Therefore this function is a sound complete search only for
4588 n <= 6. For n >= 7 it is a heuristic lower-bound search: it may miss
4589 extremal graphs whose j=7 prefix exceeds the bipartite bound (if the
4590 conjecture is false) or whose search branch was pruned by it.
4591 Returns multiplicity matrices, deduplicated up to vertex permutation when
4592 'up_to_iso' is set. Deduplication is STREAMED through a canonical form
4593 (:func: '_canonical_form') as graphs are found, so peak memory is bounded by
4594 the number of isomorphism classes rather than the number of labelled copies
4595 (the previous buffer-everything approach exhausted RAM on n=7). The returned
4596 list, and its order, are unchanged by this.
4597 """
4598 # Ordered pairs grouped so that all pairs inside {0..j} precede vertex j+1.
4599 order: list[tuple[int, int]] = []
4600 for j in range(1, n):
4601     for i in range(j):
4602         order.append((i, j))
4603         order.append((j, i))
4604 block_end = {j: 2 * ((j + 1) * j // 2) for j in range(1, n)} # prefix length
4605 # SOUNDNESS NOTE: for j <= 6 the prefix cap (m-1)*M*(j) is a PROVED upper
4606 # bound, so pruning at that cap is safe. For j >= 7, _PROVEN_MSTAR.get(j, 0)
4607     ↳ == 0
4608 # and we fall back to (m-1)*floor(j^2/4), which equals the conjectured
4609 # bipartite bound and is NOT yet proved. Any call with n >= 7 therefore
4610 # uses an unverified pruning at the j=7 boundary: the enumeration is sound
4611 # as a lower-bound search but cannot certify completeness for n >= 7.
4612 # To produce a proof for n=7 the j=7 pruning must be disabled or replaced
4613 # by a proved bound (e.g. from the MILP once M*(7) is confirmed).
4614 prefix_cap = {j: (m - 1) * max(_PROVEN_MSTAR.get(j, 0), (j * j) // 4)
4615               for j in range(2, n + 1)}
4616
4617 mu = np.zeros((n, n), dtype=int)
4618 total_pairs = len(order)
4619 # Stream results straight into the deduplicated output instead of buffering
4620 # every labelled feasible matrix first. Memory is then bounded by the number
4621 # of ISOMORPHISM CLASSES (one canonical key in 'seen', one representative in
4622 # the list), not by the number of labelled copies, which is what previously
4623 # blew up to tens of gigabytes on n=7. The set of matrices returned, and
4624 # their order, are identical to the old buffer-then-dedup code.
4625 seen: set[bytes] = set()
4626 representatives: list[np.ndarray] = []
4627
4628 def _record(matrix: np.ndarray) -> None:
4629     # The DFS only checked prefixes {0..j} for j < n, so confirm the full
4630     # n-vertex graph is feasible before keeping it (cheap, and done first so a
4631     # rejected graph never pays for the canonical form).
4632     if any(_tiny_maxflow(matrix, n, s, t, m - 1)

```

```

4632         for s in range(n) for t in range(n) if s != t):
4633             return
4634     if up_to_iso:
4635         canon = _canonical_form(matrix)
4636         if canon in seen:
4637             return
4638         seen.add(canon)
4639     representatives.append(matrix.copy())
4640
4641 def dfs(pos: int, arcs: int) -> None:
4642     if arcs + (m - 1) * (total_pairs - pos) < target_arcs or arcs >
4643         ↪ target_arcs:
4644         return
4645     if pos == total_pairs:
4646         if arcs == target_arcs:
4647             _record(mu)
4648             return
4649     completed = next((j for j in range(1, n) if block_end[j] == pos), None)
4650     if completed is not None:
4651         j = completed + 1 # prefix {0..completed} has j vertices
4652         if arcs > prefix_cap.get(j, 10 ** 9):
4653             return
4654         for s in range(j):
4655             for t in range(j):
4656                 if s != t and _tiny_maxflow(mu, n, s, t, m - 1):
4657                     return
4658     u, v = order[pos]
4659     for value in range(m):
4660         mu[u, v] = value
4661         if max_degree is None or (mu[u, :].sum() + mu[:, u].sum() <=
4662             ↪ max_degree
4663             and mu[v, :].sum() + mu[:, v].sum() <=
4664             ↪ max_degree):
4665             dfs(pos + 1, arcs + value)
4666     mu[u, v] = 0
4667
4668 dfs(0, 0)
4669 return representatives
4670
4671 def _geng_support_graphs(
4672     n: int, min_edges: int, max_edges: int, geng_path: str = "geng",
4673 ) -> Iterator[list[tuple[int, int]]]:
4674     """Yield one edge list per isomorphism class of simple graph on 'n'
4675     vertices with 'min_edges' to 'max_edges' edges, via nauty's 'geng'.
4676
4677     Each edge is an ordered tuple '(u, v)' with 'u < v'; isolated vertices
4678     are kept ('geng' always emits all 'n' vertices). 'geng' writes the
4679     non-isomorphic graphs in graph6 to stdout and we parse each with _graph6_edges
4680     ↪ .
4681
4682     Raises 'RuntimeError' if 'geng' is not on PATH.
4683     """
4684     exe = shutil.which(geng_path)
4685     if exe is None:
4686         raise RuntimeError(
4687             f"nauty's '{geng_path}' was not found on PATH. Install nauty "
4688             "(e.g. the 'nauty' package) or pass geng_path=...; the "
4689             "generation enumerator needs it."

```

```

4686         )
4687     if max_edges < min_edges:
4688         return
4689     proc = subprocess.run(
4690         [exe, str(n), f"{min_edges}:{max_edges}"],
4691         capture_output=True, check=True,
4692     )
4693     for line in proc.stdout.splitlines():
4694         line = line.strip()
4695         if not line:
4696             continue
4697         yield _graph6_edges(line)
4698
4699
4700 def _graph6_edges(data: bytes) -> list[tuple[int, int]]:
4701     """Decode one graph6 line (geng's output format) into its '(i, j)' edges, '
4702         ↳ i < j'.
4703
4704     graph6 stores 'n' in the first byte (offset 63), then the upper-triangle
4705     adjacency bits column by column (for 'j' from 1, all 'i < j'), six bits
4706         ↳ per
4707     subsequent byte, high bit first. We only ever generate 'n <= 12' supports,
4708         ↳ so
4709     the single-byte header case is all that is needed and the format needs no
4710     networkx to read.
4711     """
4712     values = [byte - 63 for byte in data]
4713     n = values[0]
4714     bits = [(value >> shift) & 1 for value in values[1:] for shift in range(5, -1,
4715         ↳ -1)]
4716     index = 0
4717     edges: list[tuple[int, int]] = []
4718     for j in range(1, n):
4719         for i in range(j):
4720             if bits[index]:
4721                 edges.append((i, j))
4722             index += 1
4723     return edges
4724
4725
4726 def _decorate_support_worker(task: tuple) -> list[np.ndarray]:
4727     """Every feasible extremal decoration of ONE undirected support graph.
4728
4729     Module-level and picklable, so :func:
4730         ↳ enumerate_extremal_directed_multigraphs_via_generation'
4731     can fan the supports out across processes (each support is independent, and
4732     two non-isomorphic supports can never yield the same multigraph). Returns one
4733     multiplicity matrix per decoration, deduplicated within this support when
4734     'up_to_iso'. The decoration logic and its pruning are identical to the
4735     sequential generator; only the ownership of the scratch arrays moves here.
4736     """
4737     n, m, target_arcs, max_degree, up_to_iso, edges = task
4738     per_edge_max = 2 * (m - 1)
4739     # Per-edge states (a, b) = (mu[u,v], mu[v,u]) excluding (0,0): a support edge
4740     # carries at least one arc in some direction.
4741     states = [(a, b) for a in range(m) for b in range(m) if (a, b) != (0, 0)]
4742     mu = np.zeros((n, n), dtype=int)
4743     deg = np.zeros(n, dtype=int)

```

```

4739 out_deg = np.zeros(n, dtype=int)
4740 in_deg = np.zeros(n, dtype=int)
4741 seen: set[bytes] = set()
4742 reps: list[np.ndarray] = []
4743
4744 def feasible_prefix(j: int) -> bool:
4745     cap = m - 1
4746     for s in range(j):
4747         if out_deg[s] <= cap:
4748             continue
4749         for t in range(j):
4750             if s != t and in_deg[t] > cap and _tiny_maxflow(mu, n, s, t, cap):
4751                 return False
4752     return True
4753
4754 # Decorate in vertex-block order: pairs sorted by larger endpoint, then
4755 # smaller, so after the last pair whose larger endpoint is w the induced
4756 # subgraph on {0..w} is complete and the prefix bound for size w+1 applies.
4757 sps = sorted(edges, key=lambda e: (e[1], e[0]))
4758 bound: list[int | None] = [None] * len(sps)
4759 for i, (_, w) in enumerate(sps):
4760     if i + 1 == len(sps) or sps[i + 1][1] > w:
4761         bound[i] = w + 1
4762
4763 def decorate(idx: int, arcs: int) -> None:
4764     remaining = len(sps) - idx
4765     if arcs + remaining > target_arcs:
4766         return
4767     if arcs + remaining * per_edge_max < target_arcs:
4768         return
4769     if idx == len(sps): # every support edge decorated
4770         if arcs == target_arcs and feasible_prefix(n):
4771             if up_to_iso:
4772                 canon = _canonical_form(mu)
4773                 if canon in seen:
4774                     return
4775                 seen.add(canon)
4776                 reps.append(mu.copy())
4777             return
4778     u, v = sps[idx]
4779     j = bound[idx] # prefix size to verify after this edge, or None
4780     for a, b in states:
4781         s = a + b
4782         if arcs + s > target_arcs:
4783             continue
4784         if max_degree is not None and (deg[u] + s > max_degree
4785                                         or deg[v] + s > max_degree):
4786             continue
4787         mu[u, v] = a
4788         mu[v, u] = b
4789         deg[u] += s
4790         deg[v] += s
4791         out_deg[u] += a; in_deg[v] += a
4792         out_deg[v] += b; in_deg[u] += b
4793         ok = True
4794         if j is not None: # block boundary: prefix {0..j-1} is complete
4795             cap = _PROVEN_MSTAR.get(j) # j == prefix size
4796             if cap is not None and arcs + s > (m - 1) * cap:

```

```

4797         ok = False                                     # PROVED induced-arc bound
4798         elif not feasible_prefix(j):
4799             ok = False
4800         if ok:
4801             decorate(idx + 1, arcs + s)
4802             deg[u] -= s
4803             deg[v] -= s
4804             out_deg[u] -= a; in_deg[v] -= a
4805             out_deg[v] -= b; in_deg[u] -= b
4806             mu[u, v] = 0
4807             mu[v, u] = 0
4808
4809     decorate(0, 0)
4810     return reps
4811
4812
4813 def enumerate_extremal_directed_multigraphs_via_generation(
4814     n: int, m: int, target_arcs: int, max_degree: int | None = None,
4815     up_to_iso: bool = True, geng_path: str = "geng", parallel: bool = True,
4816 ) -> list[np.ndarray]:
4817     """Sound, geng-seeded twin of :func:'enumerate_extremal_directed_multigraphs'.
4818
4819     Lists every directed multigraph on 'n' vertices with multiplicities in
4820     {0..m-1}, 'lambda^max <= m-1', exactly 'target_arcs' arcs and
4821     (optionally) maximum total degree '<= max_degree', deduplicated up to
4822     isomorphism. Same object and same return format (multiplicity matrices) as
4823     the DFS enumerator.
4824
4825     Following J. Goedgebeur's suggestion, the underlying undirected SUPPORT
4826     graph (the pairs carrying at least one arc) is generated once per
4827     isomorphism class by nauty's 'geng'; each support is then decorated, in
4828     vertex-block order, with a directed multiplicity pair '(mu[u,v], mu[v,u])'
4829     in {0..m-1}^2 minus (0,0) per support edge. Pruning is the running arc
4830     total, the degree cap, and -- at each completed prefix on 'j' vertices --
4831     the PROVED induced-arc bound 'arcs(prefix) <= (m-1) M*(j)' together with a
4832     prefix feasibility check (induced max-flows only grow, so a prefix that
4833     already exceeds the cap is dead).
4834
4835     Soundness for every 'n'. Two directed multigraphs with non-isomorphic
4836     supports are non-isomorphic, and every isomorphism class with a given
4837     support class has a labelling whose support equals geng's representative, so
4838     decorating every representative and canonical-deduplicating returns exactly
4839     one matrix per isomorphism class. Crucially the induced-arc bound is used
4840     ONLY for 'j <= 6', where 'M*(j)' is proved; for 'j >= 7' no arc bound
4841     is applied (only the global 'target_arcs' and feasibility). This is the
4842     difference from :func:'enumerate_extremal_directed_multigraphs', which falls
4843     back to the CONJECTURED 'floor(j^2/4)' at 'j >= 7' and is therefore a
4844     complete search only for 'n <= 6'. This generator is complete for all
4845     'n' (it never assumes an unproved value), so it can certify the finite
4846     'n = 7' classification once it finishes.
4847
4848     Requires nauty's 'geng' on PATH. For 'n <= 6' it returns the same set
4849     of isomorphism classes as :func:'enumerate_extremal_directed_multigraphs';
4850     'tests/test_solve' checks that equality.
4851
4852     With 'parallel' (default) the supports are decorated concurrently across
4853     processor cores via a :class:'~concurrent.futures.ProcessPoolExecutor', since
4854     each support is independent. This is where the 'n = 7' classification

```

```

4855     becomes practical on a multi-core machine: the work scales with the number of
4856     supports, which is exactly what fans out. It falls back to a sequential run
4857     (with a warning) where multiprocessing is unavailable, and the result is
4858     identical either way.
4859     """
4860     if m < 1:
4861         raise ValueError("m must be >= 1")
4862     per_edge_max = 2 * (m - 1)          # most arcs one undirected pair can hold
4863     if per_edge_max == 0:               # m == 1: only the empty graph is
        ↪ feasible
4864         return [np.zeros((n, n), dtype=int)] if target_arcs == 0 else []
4865     min_edges = (target_arcs + per_edge_max - 1) // per_edge_max
4866     max_edges = min(target_arcs, n * (n - 1) // 2)
4867
4868     # geng emits each undirected support iso-class once, and decorating one
4869     # support is independent of the others (two non-isomorphic supports can never
4870     # yield the same multigraph), so the supports fan out across processes, each
4871     # with its own scratch arrays inside _decorate_support_worker. The PROVED
4872     # M*(j) bound is applied there only for j <= 6; j >= 7 gets no arc bound, so
4873     # the search stays sound at every n.
4874     tasks = [(n, m, target_arcs, max_degree, up_to_iso, edges)
4875              for edges in _geng_support_graphs(n, min_edges, max_edges, geng_path)
        ↪ ]
4876
4877     representatives: list[np.ndarray] = []
4878     if parallel and len(tasks) > 1:
4879         try:
4880             with concurrent.futures.ProcessPoolExecutor() as executor:
4881                 # chunksize=1: supports vary wildly in cost, so hand them out one
4882                 # at a time to keep every core busy to the end.
4883                 for reps in executor.map(_decorate_support_worker, tasks,
4884                                         chunksize=1):
4885                     representatives.extend(reps)
4886             except (OSError, concurrent.futures.BrokenExecutor) as exc:
4887                 warnings.warn(
4888                     f"parallel support enumeration unavailable ({exc!r}); running "
4889                     "sequentially", RuntimeWarning, stacklevel=2,
4890                 )
4891                 for task in tasks:
4892                     representatives.extend(_decorate_support_worker(task))
4893         else:
4894             for task in tasks:
4895                 representatives.extend(_decorate_support_worker(task))
4896
4897     if up_to_iso:
4898         # Distinct geng supports give non-isomorphic multigraphs, so this final
4899         # canonical pass is only a safety net over the process-local per-support
4900         # dedup; it keeps exactly one representative per isomorphism class.
4901         seen: set[bytes] = set()
4902         unique: list[np.ndarray] = []
4903         for matrix in representatives:
4904             key = _canonical_form(matrix)
4905             if key not in seen:
4906                 seen.add(key)
4907                 unique.append(matrix)
4908         representatives = unique
4909     return representatives

```

C.1.9 The gallery

Given a size and a route limit, these routines find every extremal graph there is, up to relabelling, and count the symmetries of each. Deciding when two graphs are the same graph drawn differently is done by trying all relabellings and keeping the smallest resulting table, which is slow in principle but perfectly quick at the sizes involved.

```
4912 # --- GALLERY: all non-isomorphic extremal graphs for small (n, m); entry point
      ↪ gallery_extremal_graphs ---
4913
4914
4915 def _graph_from_mu(mu: np.ndarray, variant: Variant) -> Graph:
4916     """Wrap a multiplicity matrix in a fresh Graph (zero-copy)."""
4917     g = Graph(mu.shape[0], variant)
4918     g.mu[:] = mu
4919     return g
4920
4921
4922 def _aut_count_matrix(mu: np.ndarray) -> int:
4923     """Count vertex permutations that preserve ‘mu’ (equals |Aut(G)|)."""
4924     n = mu.shape[0]
4925     canon = _canonical_form(mu)
4926     return sum(1 for perm in permutations(range(n))
4927                if mu[np.ix_(list(perm), list(perm))].tobytes() == canon)
4928
4929
4930 def _dir_relabel_key(edge, p: list) -> tuple:
4931     """Permutation-relabelled key of one directed hyperedge.
4932
4933     Keeps the legacy forward shape ‘(int, tuple_of_heads)’ for forward edges
4934     (so their canonical keys and dedup counts are unchanged) and uses
4935     ‘(tuple_of_tails, tuple_of_heads)’ for general edges.
4936     """
4937     first, heads = edge
4938     if isinstance(first, (set, frozenset)):
4939         return (tuple(sorted(p[t] for t in first)), tuple(sorted(p[h] for h in
4940                        ↪ heads)))
4941     return (p[first], tuple(sorted(p[h] for h in heads)))
4942
4943
4944 def _hyper_canonical(hyperedges: list, n: int, directed: bool) -> tuple:
4945     """Canonical key for a hyperedge collection under vertex permutation.
4946
4947     Each edge is a frozenset (undirected) or a directed hyperedge in either the
4948     forward ‘(tail, heads)’ or general ‘(tails, heads)’ form. Returns the
4949     lex-minimum over all n! relabellings.
4950     """
4951     best: tuple | None = None
4952     for perm in permutations(range(n)):
4953         p = list(perm)
4954         if directed:
4955             relabeled: tuple = tuple(sorted(_dir_relabel_key(edge, p) for edge in
4956                    ↪ hyperedges))
4957         else:
4958             relabeled = tuple(sorted(
4959                 tuple(sorted(p[v] for v in edge)) for edge in hyperedges
4960             ))
```

```

4959         if best is None or relabeled < best:
4960             best = relabeled
4961     return best # type: ignore[return-value]
4962
4963
4964 def _aut_count_hyper(hyperedges: list, n: int, directed: bool) -> int:
4965     """Count vertex permutations that preserve a hyperedge collection."""
4966     canon = _hyper_canonical(hyperedges, n, directed)
4967
4968     def _relabel(p: list) -> tuple:
4969         if directed:
4970             return tuple(sorted(_dir_relabel_key(edge, p) for edge in hyperedges))
4971         return tuple(sorted(
4972             tuple(sorted(p[v] for v in edge)) for edge in hyperedges
4973         ))
4974
4975     return sum(1 for perm in permutations(range(n))
4976               if _relabel(list(perm)) == canon)
4977
4978
4979 def _hyper_to_lists(hyperedges: list, directed: bool) -> list:
4980     """Convert hyperedges to JSON-serialisable sorted integer lists.
4981
4982     Forward edges keep their "[tail, [heads]]" shape; general edges use
4983     "[[tails], [heads]]".
4984     """
4985     if directed:
4986         out = []
4987         for edge in hyperedges:
4988             first, heads = edge
4989             if isinstance(first, (set, frozenset)):
4990                 out.append([sorted(first), sorted(heads)])
4991             else:
4992                 out.append([first, sorted(heads)])
4993         return out
4994     return [sorted(edge) for edge in hyperedges]
4995
4996
4997 def _enum_matrix_extremals(
4998     variant: Variant,
4999     n: int,
5000     m: int,
5001     separation: str,
5002     target: int,
5003     deadline: float,
5004 ) -> tuple[list[np.ndarray], bool]:
5005     """DFS over all (variant, n, m) multiplicity matrices achieving exactly '
5006         ↳ target' edges.
5007
5008     Generalises :func:'enumerate_extremal_directed_multigraphs' to all four
5009     matrix variants (simple/multi × directed/undirected) and to vertex as well
5010     as edge separation. Returns "(reps, timed_out)" where "reps" are
5011     deduplicated representatives (one per isomorphism class).
5012
5013     Pruning strategy: arc-count reachability at every step; prefix edge-
5014     connectivity check at each completed vertex block (edge separation only;
5015     vertex separation checks only the final graph because the per-node cost is
5016     too high for DFS).

```

```

5016 """
5017 max_mult = 1 if variant.simple else (m - 1)
5018 is_directed = variant.directed
5019
5020 # Vertex-block order: all pairs involving vertex j before j+1 appears.
5021 order: list[tuple[int, int]] = []
5022 for j in range(1, n):
5023     for i in range(j):
5024         order.append((i, j))
5025         if is_directed:
5026             order.append((j, i))
5027 total = len(order)
5028
5029 # Position at which the subgraph on {0..j} is fully decided.
5030 # Directed: j*(j+1) positions; undirected: j*(j+1)//2 positions.
5031 step_per_vertex = 2 if is_directed else 1
5032 block_end: dict[int, int] = {
5033     j: step_per_vertex * j * (j + 1) // 2 for j in range(1, n)
5034 }
5035
5036 mu = np.zeros((n, n), dtype=int)
5037 seen: set[bytes] = set()
5038 reps: list[np.ndarray] = []
5039 timed_out = [False]
5040
5041 def _feasible_full() -> bool:
5042     g = _graph_from_mu(mu, variant)
5043     if separation == "edge":
5044         return max_edge_connectivity(g) <= m - 1
5045     return max_vertex_connectivity(g) <= m - 1
5046
5047 def dfs(pos: int, edges: int) -> None:
5048     if timed_out[0]:
5049         return
5050     if time.time() > deadline:
5051         timed_out[0] = True
5052         return
5053     if edges + max_mult * (total - pos) < target or edges > target:
5054         return
5055     if pos == total:
5056         if edges == target and _feasible_full():
5057             canon = _canonical_form(mu)
5058             if canon not in seen:
5059                 seen.add(canon)
5060                 reps.append(mu.copy())
5061         return
5062     # At each block boundary: prune if the prefix subgraph is already
5063     # infeasible (edge connectivity only -- cheap with _tiny_maxflow).
5064     completed_j = next(
5065         (j for j in range(1, n) if block_end.get(j) == pos), None
5066     )
5067     if completed_j is not None and separation == "edge":
5068         sub_n = completed_j + 1
5069         if any(_tiny_maxflow(mu, n, s, t, m - 1)
5070             for s in range(sub_n) for t in range(sub_n) if s != t):
5071             return
5072     u, v = order[pos]
5073     for val in range(max_mult + 1):

```

```

5074         mu[u, v] = val
5075         if not is_directed:
5076             mu[v, u] = val
5077         dfs(pos + 1, edges + val)
5078     mu[u, v] = 0
5079     if not is_directed:
5080         mu[v, u] = 0
5081
5082     dfs(0, 0)
5083     return reps, timed_out[0]
5084
5085
5086 def _enum_hyper_extremals(
5087     directed: bool,
5088     vertex_split: bool,
5089     r: int,
5090     n: int,
5091     m: int,
5092     target: int,
5093     deadline: float,
5094     *,
5095     kind: str = "forward",
5096 ) -> tuple[list[list], bool]:
5097     """DFS over all r-uniform hypergraphs on n vertices with exactly ‘target‘
5098         ↪ edges.
5099
5100     Iterates candidate hyperedges in order (include / exclude) and prunes
5101     immediately when adding a hyperedge pushes the connectivity above m-1
5102     (monotonicity: adding edges never decreases connectivity, so no superset
5103     can be feasible at that branch either). ‘kind‘ selects the directed
5104     orientation model. Returns ‘(reps, timed_out)’ where each representative
5105     in ‘reps‘ is a list of JSON-serialisable sorted vertex lists.
5106     """
5107     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
5108     total = len(candidates)
5109     active: list = []
5110     seen: set[tuple] = set()
5111     reps: list[list] = []
5112     timed_out = [False]
5113
5114     def dfs(pos: int, count: int) -> None:
5115         if timed_out[0]:
5116             return
5117         if time.time() > deadline:
5118             timed_out[0] = True
5119             return
5120         if count + (total - pos) < target or count > target:
5121             return
5122         if pos == total:
5123             if count == target:
5124                 canon = _hyper_canonical(active, n, directed)
5125                 if canon not in seen:
5126                     seen.add(canon)
5127                     reps.append(_hyper_to_lists(active, directed))
5128             return
5129         # Include candidates[pos] and recurse only if still feasible.
5130         active.append(candidates[pos])
5131         h = Hypergraph(n, active, directed=directed)

```

```

5131         if max_hyper_connectivity(h, vertex_split=vertex_split) <= m - 1:
5132             dfs(pos + 1, count + 1)
5133         active.pop()
5134         # Exclude candidates[pos].
5135         dfs(pos + 1, count)
5136
5137     dfs(0, 0)
5138     return reps, timed_out[0]
5139
5140
5141 def gallery_extremal_graphs(
5142     max_n: int = 7,
5143     max_m: int = 4,
5144     r: int = 3,
5145     time_per_case: float = 3.0,
5146 ) -> dict:
5147     """Collect every non-isomorphic extremal graph for all 12 variants at small n,
5148         ↪ m.
5149
5149     For each (variant, n, m) triple where the enumeration finishes within
5150     'time_per_case' seconds, the function:
5151
5152     1. runs a short repeated search to discover the extremal edge count;
5153     2. exhaustively enumerates every graph achieving that count;
5154     3. deduplicates by isomorphism class and records 'n! / |Aut(G)|' (the
5155         number of labelled copies) per class.
5156
5157     Returns a JSON-serialisable dict with structure::
5158
5159         result[variant_key]["n={n}_m={m}"] = {
5160             "extremal_value": int,
5161             "classes": [{"repr": <matrix or edge list>, "labelled_count": int}],
5162             "total_labelled": int,      # sum of labelled_count over all classes
5163             "complete": bool,          # False when the deadline was hit
5164         }
5165
5166     For matrix variants 'repr' is the multiplicity matrix (list of lists of
5167     int). For hypergraph variants it is a list of hyperedges: each edge is a
5168     sorted list of vertex indices (undirected) or '[tail, [head, ...]]'
5169     (directed).
5170
5171     The twelve variant keys are:
5172     'simple_undirected_{edge,vertex}',
5173     'simple_directed_{edge,vertex}',
5174     'multi_undirected_{edge,vertex}',
5175     'multi_directed_{edge,vertex}',
5176     'hyper_undirected_r{r}_{edge,vertex}',
5177     'hyper_directed_r{r}_{edge,vertex}'.
5178
5179     The two 'multi*_vertex' keys report the reduced SIMPLE problem, because
5180     the thesis's vertex measure is blind to parallel copies and the variant
5181     collapses onto the simple one (see the config comment below).
5182     """
5183     # The two multi-vertex rows run on the SIMPLE variant: the checker's vertex
5184     # mode gives every adjacency capacity one, so parallel copies never change
5185     # kappa and a raw multigraph search would just fill every cell to m-1 for
5186     # free (it once reported a "36-arc extremal K_4 at multiplicity 3"). The
5187     # thesis reduces these variants to the simple ones (tab:summary), and the

```

```

5188     # gallery must report the reduced problem, exactly as the variant grids do.
5189     matrix_configs: list[tuple[str, Variant, str]] = [
5190         ("simple_undirected_edge",    SIMPLE_UNDIRECTED, "edge"),
5191         ("simple_undirected_vertex",  SIMPLE_UNDIRECTED, "vertex"),
5192         ("simple_directed_edge",      SIMPLE_DIRECTED,   "edge"),
5193         ("simple_directed_vertex",    SIMPLE_DIRECTED,   "vertex"),
5194         ("multi_undirected_edge",    MULTI_UNDIRECTED,  "edge"),
5195         ("multi_undirected_vertex",  SIMPLE_UNDIRECTED, "vertex"),
5196         ("multi_directed_edge",      MULTI_DIRECTED,    "edge"),
5197         ("multi_directed_vertex",    SIMPLE_DIRECTED,   "vertex"),
5198     ]
5199     hyper_configs: list[tuple[str, bool, bool]] = [
5200         (f"hyper_undirected_r{r}_edge",    False, False),
5201         (f"hyper_undirected_r{r}_vertex",  False, True),
5202         (f"hyper_directed_r{r}_edge",      True,  False),
5203         (f"hyper_directed_r{r}_vertex",    True,  True),
5204     ]
5205
5206     result: dict = {}
5207
5208     for key, variant, separation in matrix_configs:
5209         result[key] = {}
5210         for n in range(2, max_n + 1):
5211             for m in range(2, max_m + 1):
5212                 case_deadline = time.time() + time_per_case
5213                 # Step 1: discover the extremal value via repeated search.
5214                 # 2000 steps/restart is enough to converge for n <= 6;
5215                 # we run at least 3 seeds before the deadline to be robust
5216                 # against unlucky single-seed runs.
5217                 best_val = 0
5218                 seed = 0
5219                 while time.time() < case_deadline - 2.0 or seed < 3:
5220                     if time.time() > case_deadline - 0.5:
5221                         break
5222                     res = search_for_dense_graph(
5223                         variant, n, m, separation=separation,
5224                         steps=2000, seed=seed)
5225                     best_val = max(best_val, res.best_edge_count)
5226                     seed += 1
5227                 # Step 2: enumerate all graphs at that value.
5228                 enum_deadline = min(case_deadline, time.time() + 2.0)
5229                 reps, timed_out = _enum_matrix_extremals(
5230                     variant, n, m, separation, best_val, enum_deadline)
5231                 # Step 3: count automorphisms and labelled copies.
5232                 fn = math.factorial(n)
5233                 classes = [
5234                     {"repr": mu.tolist(),
5235                      "labelled_count": fn // _aut_count_matrix(mu)}
5236                     for mu in reps
5237                 ]
5238                 result[key][f"n={n}_m={m}"] = {
5239                     "extremal_value": best_val,
5240                     "classes": classes,
5241                     "total_labelled": sum(c["labelled_count"] for c in classes),
5242                     "complete": not timed_out,
5243                 }
5244
5245     for key, directed, vertex_split in hyper_configs:

```

```

5246     result[key] = {}
5247     for n in range(r, max_n + 1):
5248         for m in range(2, max_m + 1):
5249             case_deadline = time.time() + time_per_case
5250             # Brute-force search (small n): finds the extremal value.
5251             best_val, _, completed = _brute_force_hypergraph(
5252                 n, r, m, case_deadline,
5253                 directed=directed, vertex_split=vertex_split)
5254             if not completed:
5255                 result[key][f"n={n}_m={m}"] = {
5256                     "extremal_value": best_val, "classes": [],
5257                     "total_labelled": 0, "complete": False,
5258                 }
5259                 continue
5260             # Enumerate all achieving best_val.
5261             enum_deadline = min(case_deadline, time.time() + 2.0)
5262             hyper_reps, timed_out = _enum_hyper_extremals(
5263                 directed, vertex_split, r, n, m, best_val, enum_deadline)
5264             fn = math.factorial(n)
5265             classes = []
5266             for edge_lists in hyper_reps:
5267                 raw = [(row[0], frozenset(row[1])) for row in edge_lists]
5268                 if directed else [frozenset(e) for e in edge_lists]
5269                 classes.append({
5270                     "repr": edge_lists,
5271                     "labelled_count": fn // _aut_count_hyper(raw, n, directed)
5272                     ↪ ,
5273                 })
5274             result[key][f"n={n}_m={m}"] = {
5275                 "extremal_value": best_val,
5276                 "classes": classes,
5277                 "total_labelled": sum(c["labelled_count"] for c in classes),
5278                 "complete": not timed_out,
5279             }
5280     return result
5281
5282
5283 def save_gallery_json(gallery: dict, path: str | Path) -> None:
5284     """Write the gallery dict to a JSON file (pretty-printed)."""
5285     with open(path, "w") as f:
5286         json.dump(gallery, f, indent=2)
5287
5288
5289 def prove_integral_arc_bound(n: int, m: int, target: int, *,
5290                             time_limit: float = 3000.0,
5291                             use_gurobi: bool | None = None,
5292                             show_solver_log: bool = False) -> str:
5293     """Decide by MILP whether some directed multigraph on ‘n’ vertices with
5294     multiplicities in {0..m-1} and ‘lambda^max <= m-1’ has >= ‘target’ arcs.
5295
5296     Returns "INFEASIBLE" (proving  $L_m^{\text{dir}}(n) < \text{target}$ ), "FEASIBLE", or
5297     "LIMIT". This is the integral companion of the fractional prover: the
5298     m = 3 chain (rem:odd-step-roadmap) needs only  $L_3^{\text{dir}}(7) = 24$ , i.e.
5299     INFEASIBLE at target 25, which is a far friendlier MILP than  $M(7)$  because the
5300     weights themselves are integers. Same exact :func:‘_cut_counting_model’ with
5301     cap = m-1 and integer multiplicities, all its proved-valid families on, plus
5302     ↪ the

```

```

5302     one constraint that the multiplicities total at least ‘‘target‘‘.
5303     """
5304     _require_pulp()
5305     prob, w = _cut_counting_model(
5306         n, cap=float(m - 1), integer=True, two_hop=True,
5307         symmetry=True, deletion=True, degree_pair=True,
5308     )
5309     prob += pulp.lpSum(w.values()) >= target           # the arc target
5310     prob += 0                                           # feasibility, no objective
5311
5312     prob.solve(_pick_solver(time_limit, show_solver_log, use_gurobi))
5313     status = pulp.LpStatus[prob.status]
5314     if status == "Infeasible":
5315         return "INFEASIBLE"
5316     if status == "Optimal":
5317         return "FEASIBLE"
5318     return "LIMIT"
5319
5320
5321 def _float_maxflow_value(capacity: np.ndarray, source: int, target: int,
5322                         eps: float = 1e-12) -> float:
5323     """Max-flow value for FLOAT capacities (Edmonds-Karp, shortest augmenting path
5324         ↪ ).
5325
5326     scipy's csgraph max-flow is integer-only, so the fractional check below needs
5327     its own tiny routine. BFS augmenting paths give a polynomial bound
5328     ↪ independent
5329     of the capacity magnitudes, so the loop terminates even on irrational-looking
5330     floats; ‘‘eps‘‘ treats a near-zero residual as saturated.
5331     """
5332     n = capacity.shape[0]
5333     residual = capacity.astype(float).copy()
5334     flow = 0.0
5335     while True:
5336         parent = [-1] * n
5337         parent[source] = source
5338         queue = deque([source])
5339         while queue and parent[target] == -1:
5340             u = queue.popleft()
5341             # FIFO: shortest augmenting
5342             ↪ path
5343             for v in range(n):
5344                 if parent[v] == -1 and residual[u, v] > eps:
5345                     parent[v] = u
5346                     queue.append(v)
5347             if parent[target] == -1:
5348                 return flow
5349             bottleneck = float("inf")
5350             v = target
5351             while v != source:
5352                 bottleneck = min(bottleneck, residual[parent[v], v])
5353                 v = parent[v]
5354             v = target
5355             while v != source:
5356                 residual[parent[v], v] -= bottleneck
5357                 residual[v, parent[v]] += bottleneck
5358                 v = parent[v]
5359             flow += bottleneck
5360

```

```

5357
5358 def fractional_flows_feasible(weights: np.ndarray, tol: float = 1e-9) -> bool:
5359     """Check the scaled multigraph constraint: all pairwise max-flows <= 1.
5360
5361     'weights' is an 'n x n' matrix with entries in [0,1] (zero diagonal),
5362     the scaled multiplicity matrix mu/(m-1) of lem:scaling-reduction.
5363     """
5364     n = weights.shape[0]
5365     capacity = np.where(weights > 1e-12, weights, 0.0).astype(float)
5366     np.fill_diagonal(capacity, 0.0)
5367     for s in range(n):
5368         for t in range(n):
5369             if t != s and _float_maxflow_value(capacity, s, t) > 1 + tol:
5370                 return False
5371     return True
5372
5373
5374 def fractional_search(n: int, objective: str = "min_degree", steps: int = 6000,
5375                      seed: int = 0, start: str = "bipartite") -> tuple[np.ndarray
5376                               ↪ , float]:
5377     """Hunt for a fractional counterexample to the odd-n directed multigraph
5378     statements (conj:min-degree and the total-weight bound).
5379
5380     'objective': "min_degree" maximises the minimum weighted total degree
5381     (conjecture: cannot exceed  $k = (n-1)/2$  for odd  $n$ ); "total" maximises the
5382     total weight (conjecture: cannot exceed  $\max(2(n-1), \lfloor n^2/4 \rfloor)$ ).
5383     'start': "bipartite" (the conjectured extremiser), "zero", or "random".
5384     Returns the best weighting found and its objective value. A value
5385     exceeding the conjectured ceiling would refute the conjecture for every
5386      $m - 1$  divisible by the weight denominators; none was found at  $n = 7, 9$ .
5387     """
5388     rng = random.Random(seed)
5389     weights = np.zeros((n, n))
5390     if start == "bipartite":
5391         for a in range(n // 2):
5392             for b in range(n // 2, n):
5393                 weights[a, b] = 1.0
5394     elif start == "random":
5395         weights = np.array([[rng.random() if i != j else 0.0 for j in range(n)]
5396                             for i in range(n)])
5397         while not fractional_flows_feasible(weights):
5398             weights *= 0.85
5399
5400     def score(w: np.ndarray) -> float:
5401         if objective == "total":
5402             return float(w.sum())
5403         degrees = w.sum(axis=0) + w.sum(axis=1)
5404         return float(degrees.min()) + 1e-3 * float(w.sum())
5405
5406     current = score(weights)
5407     best, best_w = current, weights.copy()
5408     for it in range(steps):
5409         temperature = 0.08 * (1 - it / steps) + 1e-4
5410         u, v = rng.randrange(n), rng.randrange(n)
5411         if u == v:
5412             continue
5413         old = weights[u, v]
5414         weights[u, v] = min(1.0, max(0.0, old + (rng.random() * 2 - 1)

```

```

5414                                     * max(0.02, temperature)))
5415         if weights[u, v] == old:
5416             continue
5417         value = score(weights)
5418         accept = value > current or rng.random() < math.exp((value - current)
5419                                                         / temperature)
5420         if accept and fractional_flows_feasible(weights):
5421             current = value
5422             if value > best:
5423                 best, best_w = value, weights.copy()
5424         else:
5425             weights[u, v] = old
5426     return best_w, best
5427
5428
5429 # --- SELF-CHECK: python erdos915_unified.py runs every invariant; exits 0 on all-
5430 ↪ PASS ---
5431
5432 _failures = 0
5433
5434 def check(label: str, condition: bool) -> None:
5435     """Print a PASS/FAIL line and tally failures."""
5436     global _failures
5437     print(f" {'PASS' if condition else 'FAIL'} {label}")
5438     if not condition:
5439         _failures += 1
5440
5441
5442 def section(title: str) -> None:
5443     """Print a section header."""
5444     print(f"\n{title}")

```

C.1.10 The self-check

The file ends with a suite of checks that runs when the program is executed directly. It re-derives known values, tests each construction against its predicted size and connectivity, confirms the fast routines agree with the slow ones they stand in for, and verifies the bounds proved in [Appendix A](#) on sampled graphs. It prints a line per check and fails loudly rather than quietly if any of them stops holding, which is what makes it useful as a regression test after an edit.

```

5447 def _run_checks() -> int:
5448     """Run every invariant check; return 1 on any failure, else 0.
5449
5450     This is the program checking itself end to end, using the bare top-level
5451     names defined above in this single file.
5452     """
5453     section("Checker: the directed m=2 values ell_2^dir(n) = 2, 4, 6, 8")
5454     for n, expected in [(2, 2), (3, 4), (4, 6), (5, 8)]:
5455         graph = double_star(n, m=2, directed=True)
5456         check(f"double star n={n}: {graph.edge_count()} arcs, lambda^max={
5457             ↪ max_edge_connectivity(graph)}",
5458             graph.edge_count() == expected and max_edge_connectivity(graph) ==
5459             ↪ 1)
5458

```

```

5459     section("Checker: one-directional bipartite is quadratic and feasible")
5460     for n in range(2, 8):
5461         graph = one_directional_bipartite(n)
5462         check(f"bipartite n={n}: {graph.edge_count()} arcs (floor(n^2/4)={ (n*n)
                    ↳ //4 })",
5463               graph.edge_count() == (n * n) // 4 and max_edge_connectivity(graph)
                    ↳ == 1)
5464
5465     section("Checker: the undirected cycle is the infeasible witness for m=2")
5466     cycle = Graph(6, SIMPLE_UNDIRECTED)
5467     for v in range(6):
5468         cycle.add_edge(v, (v + 1) % 6)
5469     check(f"C_6 has lambda^max={max_edge_connectivity(cycle)}",
           ↳ max_edge_connectivity(cycle) == 2)
5470
5471     if NETWORKX_AVAILABLE:
5472         section("Gomory-Hu shortcut agrees with brute-force lambda^max")
5473         tree = clique_tree(3, 4)
5474         check(f"GH={max_edge_connectivity_via_tree(tree)} vs direct={
                    ↳ max_edge_connectivity(tree)}",
5475               max_edge_connectivity_via_tree(tree) == max_edge_connectivity(tree))
5476     else:
5477         section("Gomory-Hu shortcut skipped (optional networkx not installed)")
5478
5479     section("Vertex connectivity: K_4-trees are globally 3-connected")
5480     for blocks in range(0, 5):
5481         ktree = clique_tree(4, blocks)
5482         check(f"K_4-tree on {ktree.num_vertices} vertices: kappa={
                    ↳ min_vertex_connectivity(ktree)}",
5483               min_vertex_connectivity(ktree) == 3)
5484
5485     section("Constructions: the augmented bipartite conjecture values")
5486     counterexample = augmented_bipartite(10, 3)
5487     check(f"m=3,n=10: {counterexample.edge_count()} arcs, lambda^max={
                    ↳ max_edge_connectivity(counterexample)}",
5488           counterexample.edge_count() == 30 and max_edge_connectivity(
                    ↳ counterexample) == 2)
5489     for n, m in [(8, 3), (10, 4), (12, 4), (10, 5)]:
5490         graph = augmented_bipartite(n, m)
5491         expected = (n + m - 2) ** 2 // 4
5492         check(f"n={n},m={m}: {graph.edge_count()} arcs (expected {expected}),
                    ↳ lambda^max={max_edge_connectivity(graph)}",
5493               graph.edge_count() == expected and max_edge_connectivity(graph) == m
                    ↳ - 1)
5494
5495     section("Hypergraphs: Berge connectivity, exact and generalising the graph
           ↳ case")
5496     k43 = complete_uniform_hypergraph(4, 3)
5497     check(f"complete 3-uniform K_4^(3): Berge lambda^max = {
                    ↳ max_hyperedge_connectivity(k43)} (expect 3)",
5498           max_hyperedge_connectivity(k43) == 3)
5499     for n in [3, 4, 5]:
5500         complete = complete_uniform_hypergraph(n, 2)
5501         check(f"2-uniform K_{n} reduces to edge-connectivity {
                    ↳ max_hyperedge_connectivity(complete)} (expect {n-1})",
5502               max_hyperedge_connectivity(complete) == n - 1)
5503     for n, r in [(7, 3), (10, 4), (13, 4)]:
5504         star = star_hypertree(n, r)

```

```

5505     check(f"star hypertree n={n}, r={r}: {star.edge_count()} edges (expect {(n
        ↳ -1)/(r-1)}), "
5506         f"lambda^max={max_hyperedge_connectivity(star)}",
5507         star.edge_count() == (n - 1) // (r - 1) and
        ↳ max_hyperedge_connectivity(star) == 1)
5508
5509     section("Hypergraphs: directed orientation models (forward / backward /
        ↳ general)")
5510     # The general gate admits a mixed arc {0,1} -> {2,3}: a route enters at a tail
5511     # and leaves at a head, so 0->2 carries one route and 2->0 carries none.
5512     mixed = Hypergraph(4, [(frozenset({0, 1}), frozenset({2, 3}))], directed=True)
5513     check("mixed arc {0,1}->{2,3}: lambda(0,2)=1 and lambda(2,0)=0",
5514         hyper_connectivity(mixed, 0, 2) == 1 and hyper_connectivity(mixed, 2, 0)
        ↳ == 0)
5515     # Backward is the arc-reversal dual of forward, so their extremal numbers
        ↳ agree.
5516     fwd, _ = max_feasible_hyperedges(4, 3, 3, directed=True, kind="forward",
        ↳ time_limit=1.5)
5517     bwd, _ = max_feasible_hyperedges(4, 3, 3, directed=True, kind="backward",
        ↳ time_limit=1.5)
5518     check(f"forward == backward at n=4, r=3, m=3 ({fwd} == {bwd})", fwd == bwd)
5519     # General contains forward and strictly beats it where vertices are scarce
5520     # (n = r); both values here are proved exact by the branch and bound.
5521     g3, e3 = max_feasible_hyperedges(3, 3, 3, directed=True, kind="general",
        ↳ time_limit=1.5)
5522     f3, _ = max_feasible_hyperedges(3, 3, 3, directed=True, kind="forward",
        ↳ time_limit=1.5)
5523     check(f"general 4 > forward 3 at n=3, r=3, m=3 (got {g3} vs {f3}, exact={e3})"
        ↳ ,
5524         g3 == 4 and f3 == 3 and e3)
5525     g4, e4 = max_feasible_hyperedges(4, 4, 4, directed=True, kind="general",
        ↳ time_limit=10.0)
5526     f4, _ = max_feasible_hyperedges(4, 4, 4, directed=True, kind="forward",
        ↳ time_limit=3.0)
5527     check(f"general 8 doubles forward 4 at n = r = 4, m = 4 ({g4} vs {f4}, exact={
        ↳ e4})",
5528         g4 == 8 and f4 == 4 and e4)
5529
5530     section("Monte Carlo: the appearance probability crosses the m/n scale")
5531     n, m = 30, 3
5532     below = estimate_appearance_probability(n, 0.3 * m / n, m, trials=120, seed=1)
5533     above = estimate_appearance_probability(n, 1.8 * m / n, m, trials=120, seed=1)
5534     check(f"P[appears] rises from {below:.2f} (well below m/n) to {above:.2f} (
        ↳ well above)",
5535         below < 0.5 < above)
5536
5537     section("Monte Carlo: vertex-connectivity never exceeds edge-connectivity (
        ↳ Whitney)")
5538     edge_dist = connectivity_distribution(20, 0.25, trials=40, separation="edge",
        ↳ seed=3)
5539     vertex_dist = connectivity_distribution(20, 0.25, trials=40, separation="
        ↳ vertex", seed=3)
5540     check("kappa^max <= lambda^max on every one of the 40 identical samples",
5541         all(kappa <= lam for kappa, lam in zip(vertex_dist, edge_dist)))
5542
5543     section("Edge vs vertex in G(n,p): they agree at small m, part company at
        ↳ large m")
5544     lam_small, kap_small = edge_vertex_distribution(5, 0.5, trials=120, seed=7)

```

```

5545 small_gap = sum(k < 1 for k, l in zip(kap_small, lam_small))
5546 check("n=5: kappa^max = lambda^max on every sample (no gap)",
5547       small_gap == 0 and all(k <= 1 for k, l in zip(kap_small, lam_small)))
5548 lam_large, kap_large = edge_vertex_distribution(16, 0.25, trials=200, seed=7)
5549 large_gap = sum(k < 1 for k, l in zip(kap_large, lam_large))
5550 check(f"n=16: Whitney holds and kappa^max < lambda^max on {large_gap} of 200
    ↪ samples",
5551       large_gap > 0 and all(k <= 1 for k, l in zip(kap_large, lam_large)))
5552
5553 if not PULP_AVAILABLE:
5554     section("Prover sections skipped (optional pulp not installed)")
5555 else:
5556     section("Prover: cut-counting proves  $M^*(n) = 2(n-1)$  optimal for small  $n$ ")
5557     # The thesis (ch2) claims  $L_m^{\text{dir}}(n) = 2(n-1)(m-1)$  is proved for ALL  $n \leq$ 
    ↪ 6
5558     # via the cut-counting MILP (thm:dir-multi-small).  $n=3,4,5$  all finish in
5559     # well under 60 s and are verified here.  $n=6$  takes ~1315 s (~22 min) on a
5560     # typical laptop and is NOT included in this fast loop -- it must be
5561     # verified separately:
5562     #
5563     # result = prove_directed_multigraph(6, time_limit=2000.0)
5564     # assert result.is_proof() and round(result.scaled_optimum) == 10
5565     #
5566     # The confirmed run (2026-06-13) is logged in program/logs/selftest_check.
    ↪ log
5567     # (search "n=6 OPTIMAL  $M^*(6)=10$  in 1315s"). Omitting  $n=6$  from this loop
    ↪ is
5568     # not an error in the proof -- the proof ran -- but it means this self-
    ↪ test
5569     # alone does not reproduce the full  $n \leq 6$  certification.
5570     for n in [3, 4, 5]:
5571         result = prove_directed_multigraph(n, time_limit=300.0)
5572         check(f"n={n}: status={result.status},  $M^*={result.scaled\_optimum:.1f}$ ,
    ↪ proof={result.is_proof()}",
5573              result.is_proof() and round(result.scaled_optimum) == 2 * (n -
    ↪ 1))
5574
5575     section("Prover: the valid inequalities sharpen but never move the optimum
    ↪ ")
5576     # Turning the two-hop and symmetry-breaking rows off leaves the BARE exact
5577     # cut formulation, which must still prove the same  $M^*(3) = 4$ . If any row
    ↪ we
5578     # call a "valid inequality" were in fact invalid, it would cut a feasible
5579     # point and shift this optimum -- so this is the regression tripwire that
5580     # guards the prover's soundness, not just a re-run of the value.
5581     bare = prove_directed_multigraph(3, time_limit=120.0,
5582                                     use_two_hop=False, use_symmetry_breaking=
    ↪ False)
5583     check(f" $M^*(3)$  with the tightenings off = {bare.scaled_optimum:.1f} (expect
    ↪ 4), proof={bare.is_proof()}",
5584          bare.is_proof() and round(bare.scaled_optimum) == 4)
5585
5586     section("Prover (integral): an INFEASIBLE verdict is a genuine upper-bound
    ↪ proof")
5587     #  $L_3^{\text{dir}}(4) = 12$ : a 12-arc multigraph exists, no 13-arc one does. This
5588     # exercises the exact mechanism behind the thesis's  $L_3(7) = 24$  result
5589     # (INFEASIBLE one above the optimum) on a value small enough to re-run
    ↪ here,

```

```

5590     # so a regression in the integral encoding cannot pass unnoticed.
5591     feasible_at_12 = prove_integral_arc_bound(4, 3, 12, time_limit=120.0)
5592     infeasible_at_13 = prove_integral_arc_bound(4, 3, 13, time_limit=120.0)
5593     check(f"target 12 -> {feasible_at_12} (expect FEASIBLE), "
5594           f"target 13 -> {infeasible_at_13} (expect INFEASIBLE)",
5595           feasible_at_12 == "FEASIBLE" and infeasible_at_13 == "INFEASIBLE")
5596
5597     section("Search: rediscovering  $\text{ell}_2^{\text{dir}}(4) = 6$  by random search")
5598     result = search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2, steps=6000, seed=0)
5599     check(f"best found = {result.best_edge_count} arcs, feasible  $\lambda^{\text{max}} = \{$ 
5600            $\hookrightarrow$  max_edge_connectivity(result.best_graph)}",
5601           result.best_edge_count == 6 and max_edge_connectivity(result.best_graph)
5602            $\hookrightarrow$  <= 1)
5603
5604     section("Driver: one solve() call handles each kind of question")
5605     # Exhaustive directed: the cut-counting proves  $\text{ell}_2^{\text{dir}}(4) = 6$  exactly.
5606     proved = solve(4, 2, directed=True, simple=True, exhaustive=True,
5607                   max_seconds=120.0)
5608     check(f"solve exhaustive simple-directed n=4,m=2: {proved.value} ({proved.
5609            $\hookrightarrow$  bound})",
5610           proved.proven and proved.value == 6)
5611     # Discovery finds the same 6 arcs as a lower bound within a short budget.
5612     # The checks use sa (lighter); tabu is the default for actually solving.
5613     found = solve(4, 2, directed=True, simple=True, exhaustive=False,
5614                  max_seconds=3.0, method="sa")
5615     check(f"solve discovery simple-directed n=4,m=2: {found.value} ({found.bound})
5616            $\hookrightarrow$  ",
5617           found.bound == "lower" and found.value == 6)
5618     # The VERTEX separation is a separate question, not a corollary of the arc
5619     # one: Whitney's  $\kappa \leq \lambda$  makes the vertex-feasible family the LARGER
5620     # of the two, so an arc upper bound does not restrict it. thm:dir-vertex-m2
5621     # needs its own base cases, and these are the first three of them.
5622     for base_n, base_value in ((3, 4), (4, 6), (5, 8)):
5623         v_value, _, v_done = _exhaustive_directed(base_n, 2, "vertex",
5624                                                    time.time() + 120.0)
5625         check(f"exhaustive simple-directed VERTEX n={base_n},m=2: {v_value}",
5626               v_done and v_value == base_value)
5627     # The vertex flow helper must agree with the exact checker it stands in for.
5628     _vg = Graph(4, SIMPLE_DIRECTED)
5629     for _a, _b in ((0, 1), (0, 2), (1, 3), (2, 3)):
5630         _vg.mu[_a, _b] = 1
5631     _vout = [{_b for _b in range(4) if _vg.mu[_a, _b]} for _a in range(4)]
5632     check("vertex flow helper agrees with the exact checker on a theta digraph",
5633           _vertex_flow_at_least(_vout, 4, 0, 3, 2)
5634           and not _vertex_flow_at_least(_vout, 4, 0, 3, 3)
5635           and local_connectivity(_vg, 0, 3, vertex_split=True) == 2)
5636     # prop:dir-arc-stability, the unconditional quadratic bound. Both the
5637     # load-bearing counting step (sum of  $d^+$   $d^-$  capped by  $m n(n-1)$ ) and the bound
5638     # it yields are checked on sampled feasible digraphs.
5639     _rng = random.Random(3)
5640     _worst_slack = None
5641     _count_ok = True
5642     for _ in range(120):
5643         _sn, _sm = _rng.randint(3, 7), _rng.randint(2, 4)
5644         _sg = Graph(_sn, SIMPLE_DIRECTED)
5645         for _a in range(_sn):
5646             for _b in range(_sn):
5647                 if _a != _b and _rng.random() < 0.6:

```

```

5644         _sg.mu[_a, _b] = 1
5645     while max_edge_connectivity(_sg) > _sm - 1:  # prune down to feasible
5646         _present = [(a, b) for a in range(_sn) for b in range(_sn) if _sg.mu[a
                    ↪ , b]]
5647         _a, _b = _rng.choice(_present)
5648         _sg.mu[_a, _b] = 0
5649         _dout = [int(_sg.mu[x].sum()) for x in range(_sn)]
5650         _din = [int(_sg.mu[:, x].sum()) for x in range(_sn)]
5651         # The sharp form of the counting step: (m-1) n(n-1), not m n(n-1).
5652         if sum(_dout[x] * _din[x] for x in range(_sn)) > (_sm - 1) * _sn * (_sn -
                    ↪ 1):
5653             _count_ok = False
5654             _slack = ((_sn * _sn) // 4 + math.sqrt(_sm) * _sn ** 1.5
5655                      - int(_sg.mu.sum()))
5656             _worst_slack = _slack if _worst_slack is None else min(_worst_slack,
                    ↪ _slack)
5657             # thm:dir-arc-linear-error, the 0_m(n) bound.
5658             if int(_sg.mu.sum()) > (_sn * _sn) // 4 + 4 * (_sm - 1) * (_sn - 1):
5659                 _count_ok = False
5660             # Case 2 of its proof: min total degree >= n/2 forces the linear bound
5661             # on the sum of the smaller half-degrees.
5662             if min(_dout[x] + _din[x] for x in range(_sn)) >= _sn / 2:
5663                 if (sum(min(_dout[x], _din[x]) for x in range(_sn))
5664                     > 4 * (_sm - 1) * (_sn - 1)):
5665                     _count_ok = False
5666         check("prop:dir-arc-stability and thm:dir-arc-linear-error hold on 120 "
5667              f"sampled feasible digraphs (tightest slack {_worst_slack:.1f})",
5668              _count_ok and _worst_slack >= 0)
5669         # sec:multi-vertex-standard: the OTHER counting convention is a different
5670         # problem, and the theta construction beats the thickened tree at m=5, n=4.
5671         _mv4, _, _mv4_done = max_multigraph_vertex_standard(4, 3)
5672         _mv5, _, _mv5_done = max_multigraph_vertex_standard(4, 5)
5673         check(f"multigraph vertex, other convention: K_3(4) = {_mv4} = (m-1)(n-1)",
5674              _mv4_done and _mv4 == 6)
5675         check(f"multigraph vertex, other convention: K_5(4) = {_mv5} > 12, the "
5676              "multigraph edge value, so the two problems differ",
5677              _mv5_done and _mv5 == 14)
5678         # Exhaustive undirected by brute force: ell_2(5) = n-1 = 4 (a spanning tree).
5679         tree = solve(5, 2, directed=False, simple=True, exhaustive=True,
5680                     max_seconds=30.0)
5681         check(f"solve exhaustive simple-undirected n=5,m=2: {tree.value} ({tree.bound
                    ↪ })",
5682              tree.proven and tree.value == 4)
5683         # Hypergraph discovery returns a feasible lower bound for the r=3, m=2 case.
5684         hyper = solve(7, 2, hypergraph=True, r=3, exhaustive=False, max_seconds=3.0,
5685                     method="sa")
5686         check(f"solve discovery 3-uniform hypergraph n=7,m=2: {hyper.value} ({hyper.
                    ↪ bound})",
5687              hyper.bound == "lower" and hyper.value == 3)
5688
5689         # --- open-variant exploration coverage -----
5690         # These three functions are the program's only handle on the OPEN parts of
5691         # the problem (hypergraph vertex value, the fractional relaxation behind the
5692         # odd-n conjectures). The first author-reviewed draft of this block ran
5693         # exhaustive n>=7 sweeps and 6000-step searches, which added minutes; the
5694         # checks below cover the same code paths at n<=5 so they finish in ~2s.
5695         section("Open variants: hypergraph vertex value (kappa^max) at small n")
5696         # m=2 uses the incidence-rank/forest witness, m=3 the brute-force

```

```

5697     # feasibility sweep; both confirm value = floor((m-1)(n-1)/(r-1)).
5698     for n, r, m in [(4, 3, 2), (5, 3, 2), (4, 3, 3), (5, 3, 3)]:
5699         target = ((m - 1) * (n - 1)) // (r - 1)
5700         check(f"hypergraph vertex value at n={n}, r={r}, m={m} = {target}",
5701              verify_hyper_vertex_value(n, r, m))
5702
5703     section("Open variants: the fractional [0,1]-weight checker is exact")
5704     # A directed path keeps every max-flow at 1 (feasible); two internally
5705     # disjoint s->t routes push the s->t flow to 2 (infeasible). This is the
5706     # relaxation that fractional_search optimises over.
5707     path_w = np.zeros((4, 4))
5708     path_w[0, 1] = path_w[1, 2] = path_w[2, 3] = 1.0
5709     two_route = np.zeros((4, 4))
5710     two_route[0, 1] = two_route[0, 2] = two_route[1, 3] = two_route[2, 3] = 1.0
5711     check("path weighting feasible, two-route weighting infeasible",
5712          fractional_flows_feasible(path_w)
5713          and not fractional_flows_feasible(two_route))
5714
5715     section("Open variants: fractional search respects the odd-n ceilings")
5716     # Short n=5 runs (the conjectured bipartite point is rigid). The
5717     # min_degree score carries a 1e-3 * total-weight tie-breaker, so its
5718     # comparison allows that slack; a genuine refutation would clear the
5719     # ceiling by a full unit, far above it.
5720     _, total_best = fractional_search(5, "total", steps=1500, seed=1)
5721     check(f"fractional total weight at n=5 stays <= 8: {total_best:.3f}",
5722          total_best <= 8 + 1e-6)
5723     _, deg_best = fractional_search(5, "min_degree", steps=1500, seed=1)
5724     check(f"fractional min degree at n=5 stays <= k=2 (+tie-break slack): {
5725         ↪ deg_best:.3f}",
5726          deg_best <= 2 + 1e-3 * 5 * 5)
5727
5728     print(f"\n{'ALL CHECKS PASSED' if _failures == 0 else f'{'_failures'} CHECK(S)
5729         ↪ FAILED'}")
5730
5731     return 1 if _failures else 0
5732
5733 if __name__ == "__main__":
5734     print(f"C extension: {'LOADED (_erdos_fast.so)' if C_EXTENSION_LOADED else '
5735         ↪ not found (pure Python)'}")
5736     raise SystemExit(_run_checks())

```

C.2 The figure script

This is the only script that writes image files. It imports the main program and calls its plotting routines, so it holds no mathematics of its own. Keeping it separate means the figures can be regenerated in one command without running anything else, and every random choice it makes is seeded, so running it twice produces identical files.

```

1  """
2  Regenerate every figure the thesis uses, from the one program next door.
3
4  This is the only figure script. It imports the single program
5  ‘‘erdos915_unified.py’’ and calls its plotting routines, so the figures can never
6  drift from the code that produces the numbers. Run it from this ‘‘program/‘‘
7  directory:

```

```

8
9     python make_figures.py
10
11 Each figure is written into ‘../figures/’. Every randomised search uses a
12 fixed seed, so the output is reproducible.
13 """
14
15 from __future__ import annotations
16
17 import math
18 import sys
19 from pathlib import Path
20
21 import numpy as np
22
23 sys.path.insert(0, str(Path(__file__).resolve().parent))
24
25 from erdos915_unified import ( # noqa: E402
26     MULTI_DIRECTED,
27     MULTI_UNDIRECTED,
28     SIMPLE_DIRECTED,
29     SIMPLE_UNDIRECTED,
30     Graph,
31     _VARIANT_ENUM_CONFIGS,
32     search_for_dense_graph,
33     compute_enumeration_cache,
34     compute_pair_enumeration_cache,
35     compute_surface_cache,
36     connectivity_distribution,
37     gallery_extremal_graphs,
38     directed_arc_lower_bound,
39     directed_multigraph_arc,
40     draw_graph_with_sensitivity,
41     edge_vertex_distribution,
42     hypergraph_edge,
43     multigraph_undirected_edge,
44     plot_complexity_growth,
45     plot_extremal_gallery,
46     plot_conn_dist_grid,
47     plot_conn_threshold_3d,
48     plot_connectivity_distribution,
49     plot_pair_conn_dist_grid,
50     plot_directed_crossover,
51     plot_edge_vertex_divergence,
52     plot_edge_vertex_histograms,
53     plot_edge_dist_grid,
54     plot_scatter_lambda_edges,
55     plot_search_trace,
56     plot_sa_vs_tabu_convergence,
57     plot_variant_3d_surfaces,
58     plot_variant_grid,
59     save_gallery_json,
60     simple_undirected_edge,
61     solve,
62 )
63
64 FIGURES = Path(__file__).resolve().parent.parent / "figures"
65

```

```

66
67 # -----
68 # The all-variant grid: proved / conjectured / guessed, with machine points.
69 # Every number below comes from one driver, "solve": exhaustive for an exact
70 # point, discovery for a search lower bound. The formula curves come from the
71 # closed forms; the guess is a fit; the band is [construction LB, trivial max].
72 # -----
73
74 def _exact_points(ns, m, budget, **kw):
75     """Machine-proved sizes: increasing n until the exhaustion no longer finishes.
76     ↳ """
77     xs, ys = [], []
78     for n in ns:
79         res = solve(n, m, exhaustive=True, max_seconds=budget, **kw)
80         if res.bound == "exact":
81             xs.append(n)
82             ys.append(res.value)
83         else:
84             break # once it cannot finish, larger n cannot either
85     return xs, ys
86
87 def _search_points(ns, m, budget, **kw):
88     """Discovery lower bounds: a concrete feasible graph at each n (the LB
89     ↳ construction)."""
90     xs, ys = [], []
91     for n in ns:
92         res = solve(n, m, exhaustive=False, max_seconds=budget, seed=0, **kw)
93         xs.append(n)
94         ys.append(res.value)
95     return xs, ys
96
97 def _band(search, trivial_max, ns):
98     """The certain interval: an easy construction below, the trivial maximum above
99     ↳ .
100
101     The lower edge is the densest feasible graph the search exhibited (a real
102     lower bound); the upper edge is the most edges any graph on n vertices can
103     hold at all (a loose but certain upper bound). The truth lies in between.
104     """
105     found = dict(zip(search[0], search[1]))
106     lower = [min(found.get(n, 0), trivial_max(n)) for n in ns]
107     upper = [trivial_max(n) for n in ns]
108     return list(ns), lower, upper
109
110 def _extend_lower_bounds(search, lb_fn, ns):
111     """Raise the search lower bounds to a known verified construction where it
112     ↳ wins.
113
114     Past the size the quick search reaches, an explicit construction the checker
115     confirms is a better (still honest) lower bound, so we report the best of the
116     two at each n, exactly as the rediscovery section does by hand.
117     """
118     found = dict(zip(search[0], search[1]))
119     for n in ns:
120         found[n] = max(found.get(n, 0), lb_fn(n))

```

```

120     xs = sorted(found)
121     return xs, [found[n] for n in xs]
122
123
124 def _reconcile_panel(panel: dict) -> dict:
125     """Make a panel internally consistent before plotting.
126
127     Three honest invariants, enforced here once for every panel rather than
128     scattered through the construction code:
129
130     1. Nothing exceeds a proved optimum. A machine-checked exact value is
131     the true maximum at that 'n'; no proved/conjectured curve, no
132     named-construction lower bound, and no search circle may sit above it.
133     (This is what produced the green square below its own curve: a closed
134     form -- e.g. the hypergraph bound capped at the complete hypergraph -- can
135     overshoot what is actually achievable at small 'n'.)
136     2. Search lower bounds rise with 'n'. A feasible graph on 'n'
137     vertices stays feasible on 'n+1' (add an isolated vertex), so the best
138     feasible edge count can never drop as 'n' grows. We take a running
139     maximum, which is exactly that extend-by-an-isolated-vertex bound.
140     3. No certain-interval band, and the guess interpolates rather than
141     overshoots. The loose [construction, trivial-max] band dwarfed the
142     data and is dropped (axes rescale to the data). On an open panel the
143     dotted guess is the piecewise-linear interpolation through the search
144     points -- never a fitted curve riding above points it should pass through.
145     """
146     panel.pop("band", None) # invariant 3a: no shaded certain interval
147
148     exact = dict(zip(*panel["exact"])) if panel.get("exact") else {}
149
150     def cap_to_exact(curve):
151         if curve is None:
152             return None
153         xs, ys = curve
154         return xs, [min(y, exact[x]) if x in exact else y for x, y in zip(xs, ys)]
155
156     # invariant 1 applied to every formula line and to the named branches
157     for key in ("proved", "conj"):
158         if panel.get(key) is not None:
159             panel[key] = cap_to_exact(panel[key])
160     if panel.get("branches"):
161         panel["branches"] = [(cap_to_exact((bx, by)), lbl)
162                               for bx, by, lbl in panel["branches"]]
163
164     # invariants 1 then 2 applied to the search circles
165     if panel.get("search") is not None:
166         sx, sy = panel["search"]
167         sy = [min(y, exact[x]) if x in exact else y for x, y in zip(sx, sy)]
168         running, mono = -1, []
169         for y in sy:
170             running = max(running, y)
171             mono.append(running)
172         panel["search"] = (sx, mono)
173
174     # invariant 3b: the open-panel guess is the interpolation through those
175     # (monotone, exact-capped) points, so it can never ride above them.
176     if panel.get("guess") == "search":
177         panel["guess"] = panel.get("search")

```

```

178
179     return panel
180
181
182 def gather_variant_grid(m=3, exact_budget=4.0, search_budget=0.4,
183     ↪ open_search_budget=4.0):
184     """Build the twelve panels of the all-variant grid (row-major, model x col).
185
186     'm' is the forbidden connectivity value shown in every panel.
187
188     Two uniformity rules make the twelve panels directly comparable:
189
190     1. Every matrix panel plots its search lower bounds over the *same* vertex
191         range 'matrix_ns' and every hypergraph panel over 'hyper_ns', so each
192         panel carries the same number of data points -- no panel trails off early.
193     2. For every panel where a construction is known (proved or conjectured) the
194         search lower bound is raised to that named construction with
195         :func: '_extend_lower_bounds', exactly as the caption of the figure and
196         the rediscovery section state: the reported circle at each 'n' is the
197         better of the cooled search and the verified construction. This is the
198         honest "lands on the curve" rule, applied to *all* panels rather than
199         only the two directed ones it used to cover, so the proved curves pass
200         through their circles instead of floating above a jagged search tail.
201
202     Open panels also get a verified construction planted, not only the proved
203     ones: the open undirected vertex panels get the proved EDGE value (Whitney,
204     kappa <= lambda, so the edge extremiser is vertex-feasible) and the two
205     directed-hypergraph panels get the proved bipartite family of
206     prop:dir-hyper-first. This matters because at 'search_budget=0.4' the
207     vertex-mode search on a large matrix graph barely completes one move, so
208     the RAW result actively *degrades* with growing 'n' (measured:
209     n=6/10/14/16 gave 15/12/5/4 at m=6), and the monotone running-max in
210     :func: '_reconcile_panel' then freezes the plotted curve at its early peak,
211     a flat line that looks like a finding but is really budget starvation.
212     'open_search_budget' (default 4s, ~10x 'search_budget') still gives
213     those panels a stronger walk, since the search can genuinely beat the
214     planted construction where the open problem is richer (it does at m=6,
215     n=9 on the undirected vertex panel).
216     """
217     panels = []
218     # Uniform x-ranges for the search / construction curves. The machine-PROVED
219     # (exact) points are genuinely limited by enumeration cost and stop early; the
220     # search lower bounds and the formula curves are cheap, so they run well past
221     # that, to show each variant's trend and let the search discover dense
222     # ("maybe extremal") graphs at sizes exhaustion cannot reach.
223     matrix_ns = list(range(2, 17)) # all matrix panels, n = 2..16
224     hyper_ns = list(range(2, 13)) # all hypergraph panels, n = 2..12
225
226     def tri_undirected(n):
227         return n * (n - 1) // 2
228
229     def tri_hyper(n, r=3):
230         return math.comb(n, r)
231
232     def tri_dir_hyper(n, r=3):
233         return n * math.comb(n - 1, r - 1)
234
235     # Construction lower bounds (verified, proved-or-conjectured values), each

```

```

235 # capped at the trivial maximum for its model. The cap matters at small n:
236 # a closed form like Mader's floor(m(n-1)/2) is the value only for n >= m;
237 # for n < m the complete graph K_n already has lambda^max = n-1 <= m-1, so it
238 # is feasible and denser, and the true value is comb(n,2). Without the cap
239 # the plotted curve and its lower-bound circles float ABOVE the exact squares,
240 # which is impossible (a lower bound can never exceed the proved optimum).
241 def lb_simple_edge(n):
242     return min(simple_undirected_edge(n, m), n * (n - 1) // 2)
243
244 def lb_multi_edge(n):
245     return min(multigraph_undirected_edge(n, m), (m - 1) * (n * (n - 1) // 2))
246
247 def lb_dir(n):
248     return min(directed_arc_lower_bound(n, m), n * (n - 1))
249
250 def lb_multi_dir(n):
251     return min(directed_multigraph_arc(n, m), (m - 1) * n * (n - 1))
252
253 def lb_hyper_edge(n):
254     return min(hypergraph_edge(n, m, 3), math.comb(n, 3))
255
256 def lb_dir_hyper(n):
257     # The PROVED bipartite family of prop:dir-hyper-first at r = 3:
258     # alpha tails, and on the other n - alpha vertices a shared near-regular
259     # head graph of max degree m - 1 (realisable iff m - 1 <= n - alpha - 1),
260     # giving alpha * floor((m-1)(n-alpha)/2) single-step hyperarcs with
261     # lambda^max = kappa^max = m - 1. Sound for BOTH separations, since
262     # every route in it is a single tail -> head step.
263     best = 0
264     for alpha in range(1, n - m + 1):
265         best = max(best, alpha * ((m - 1) * (n - alpha) // 2))
266     return min(best, n * math.comb(n - 1, 2))
267
268 # Undirected vertex is proved for m<=4 (Leonard/Whitney), open for m>=5.
269 vert_proved = (m <= 4)
270 # Hypergraph vertex is proved for m<=3 (incidence-rank lemma, thm:hyper-vertex
271 #   -> -m3).
272 hyper_vert_proved = (m <= 3)
273
274 def searched(ns, lb_fn, **solve_kw):
275     """Search over the uniform range, then raise to the known construction."""
276     se = _search_points(ns, m, search_budget, **solve_kw)
277     if lb_fn is not None:
278         se = _extend_lower_bounds(se, lb_fn, ns)
279     return se
280
281 # ----- row 1: simple -----
282 # (1) simple undirected edge -- proved (Mader) for all m.
283 ex = _exact_points(range(2, 9), m, exact_budget,
284                    directed=False, simple=True, separation="edge")
285 se = searched(matrix_ns, lb_simple_edge,
286               directed=False, simple=True, separation="edge")
287 panels.append(dict(
288     title=f"undirected edge, $m={m}$ (proved)", ylabel="edges",
289     proved=(matrix_ns, [lb_simple_edge(n) for n in matrix_ns]),
290     exact=ex, search=se))
291
292 # (2) simple undirected vertex -- proved for m<=4, open for m>=5.

```

```

292 ex2 = _exact_points(range(2, 8), m, exact_budget,
293                     directed=False, simple=True, separation="vertex")
294 if vert_proved:
295     se2 = searched(matrix_ns, lb_simple_edge,
296                   directed=False, simple=True, separation="vertex")
297     panels.append(dict(
298         title=f"undirected vertex, $m={m}$ (proved)", ylabel="edges",
299         proved=(matrix_ns, [lb_simple_edge(n) for n in matrix_ns]),
300         exact=ex2, search=se2))
301 else:
302     se2 = _search_points(matrix_ns, m, open_search_budget,
303                         directed=False, simple=True, separation="vertex")
304     # Whitney: kappa <= lambda, so Mader's edge extremiser is vertex-feasible
305     # and the proved edge value is an honest lower bound on the open vertex
306     # panel too. Without it the starved vertex-mode search freezes into a
307     # flat plateau at large n (the "cut off" look); the search may still beat
308     # the edge value where the vertex problem is genuinely richer.
309     se2 = _extend_lower_bounds(se2, lb_simple_edge, matrix_ns)
310     band2 = _band(se2, tri_undirected, matrix_ns)
311     panels.append(dict(
312         title=f"undirected vertex, $m={m}$ (open)", ylabel="edges",
313         guess="search",
314         band=band2, exact=ex2, search=se2))
315
316 # (3) simple directed arc -- conjectured.
317 ex3 = _exact_points(range(2, 8), m, exact_budget,
318                     directed=True, simple=True, separation="edge")
319 se3 = searched(matrix_ns, lb_dir,
320               directed=True, simple=True, separation="edge")
321 panels.append(dict(
322     title=f"directed arc, $m={m}$ (conjectured)", ylabel="arcs",
323     conj=(matrix_ns, [lb_dir(n) for n in matrix_ns]),
324     branches=[(matrix_ns, [min(m * (n - 1), n * (n - 1)) for n in matrix_ns],
325                  "hub $m(n{-}1)$"),
326               # Shifted-partition augmented bipartite floor((n+m-2)^2/4)
327               # (the corrected conj:dir-arc branch; the old balanced count
328               # floor(n^2/4)+(m-2)ceil(n/2) agrees at m<=3 but is smaller
329               # at m>=4 and would sit below the conjecture curve at m=6).
330               (matrix_ns,
331                [min((n + m - 2) ** 2 // 4, n * (n - 1))
332                  for n in matrix_ns], "bipartite")],
333     exact=ex3, search=se3))
334
335 # (4) simple directed vertex -- conjectured (= arc).
336 ex4 = _exact_points(range(2, 8), m, exact_budget,
337                     directed=True, simple=True, separation="vertex")
338 se4 = searched(matrix_ns, lb_dir,
339               directed=True, simple=True, separation="vertex")
340 panels.append(dict(
341     title=f"directed vertex, $m={m}$ (conjectured)", ylabel="arcs",
342     conj=(matrix_ns, [lb_dir(n) for n in matrix_ns]),
343     exact=ex4, search=se4))
344
345 # ----- row 2: multigraph -----
346 # (5) multigraph undirected edge -- proved for all m.
347 ex5 = _exact_points(range(2, 7), m, exact_budget,
348                     directed=False, simple=False, separation="edge")
349 se5 = searched(matrix_ns, lb_multi_edge,

```

```

350         directed=False, simple=False, separation="edge")
351 panels.append(dict(
352     title=f"undirected edge, $m={m}$ (proved)", ylabel="edges",
353     proved=(matrix_ns, [lb_multi_edge(n) for n in matrix_ns]),
354     exact=ex5, search=se5))
355
356 # (6) multigraph undirected vertex -- equals simple (parallel edges irrelevant
    ↳ for vertex cuts).
357 multi_vert_label = ("proved, $=$ simple" if vert_proved else "open, $=$ simple
    ↳ ")
358 if vert_proved:
359     panels.append(dict(
360         title=f"undirected vertex, $m={m}$ ({multi_vert_label})", ylabel="
            ↳ edges",
361         proved=(matrix_ns, [lb_simple_edge(n) for n in matrix_ns]),
362         exact=ex2, search=se2))
363 else:
364     panels.append(dict(
365         title=f"undirected vertex, $m={m}$ ({multi_vert_label})", ylabel="
            ↳ edges",
366         guess="search",
367         band=_band(se2, tri_undirected, matrix_ns), exact=ex2, search=se2))
368
369 # (7) multigraph directed arc -- certified (cut-counting) then conjectured.
370 ex7 = _exact_points(range(3, 6), m, 10.0,
371     directed=True, simple=False, separation="edge")
372 se7 = searched(matrix_ns, lb_multi_dir,
373     directed=True, simple=False, separation="edge")
374 panels.append(dict(
375     title=f"directed arc, $m={m}$ (multigraph, certified)", ylabel="arcs",
376     conj=(matrix_ns, [lb_multi_dir(n) for n in matrix_ns]),
377     exact=ex7, search=se7))
378
379 # (8) multigraph directed vertex -- conjectured (reduces to simple digraph).
380 panels.append(dict(
381     title=f"directed vertex, $m={m}$ (conjectured, $=$ simple)", ylabel="arcs
        ↳ ",
382     conj=(matrix_ns, [lb_dir(n) for n in matrix_ns]),
383     exact=ex4, search=se4))
384
385 # ----- row 3: hypergraph (r=3) -----
386 # (9) hypergraph undirected edge -- proved for all m.
387 ex9 = _exact_points(range(2, 8), m, exact_budget,
388     hypergraph=True, r=3, directed=False, separation="edge")
389 se9 = searched(hyper_ns, lb_hyper_edge,
390     hypergraph=True, r=3, directed=False, separation="edge")
391 panels.append(dict(
392     title=f"undirected edge, $m={m}$ (proved)", ylabel="hyperedges",
393     proved=(hyper_ns, [lb_hyper_edge(n) for n in hyper_ns]),
394     exact=ex9, search=se9))
395
396 # (10) hypergraph undirected vertex -- PROVED for m<=3 (incidence-rank lemma),
397 # open for m>=4.
398 ex10 = _exact_points(range(2, 7), m, exact_budget,
399     hypergraph=True, r=3, directed=False, separation="vertex"
        ↳ )
400 if hyper_vert_proved:
401     se10 = searched(hyper_ns, lb_hyper_edge,

```

```

402             hypergraph=True, r=3, directed=False, separation="vertex")
403     panels.append(dict(
404         title=f"undirected vertex, $m={m}$ (proved)", ylabel="hyperedges",
405         proved=(hyper_ns, [lb_hyper_edge(n) for n in hyper_ns]),
406         exact=ex10, search=se10))
407     else:
408         se10 = _search_points(hyper_ns, m, open_search_budget,
409                             hypergraph=True, r=3, directed=False, separation="
410                                 ↪ vertex")
411         # Whitney again: the proved hyperedge value is a lower bound for the
412         # vertex separation (its extremiser is vertex-feasible). The exact cap
413         # in _reconcile_panel corrects the small-n corner where the simple
414         # attainment condition  $m-1 \leq n-2$  has not kicked in yet.
415         se10 = _extend_lower_bounds(se10, lb_hyper_edge, hyper_ns)
416         panels.append(dict(
417             title=f"undirected vertex, $m={m}$ (open)", ylabel="hyperedges",
418             guess="search",
419             band=_band(se10, tri_hyper, hyper_ns), exact=ex10, search=se10))
420
421     # (11) hypergraph directed arc -- OPEN (new directed Berge model).
422     ex11 = _exact_points(range(2, 6), m, exact_budget,
423                         hypergraph=True, r=3, directed=True, separation="edge")
424     se11 = _search_points(hyper_ns, m, open_search_budget,
425                         hypergraph=True, r=3, directed=True, separation="edge")
426     # The proved bipartite construction of prop:dir-hyper-first is the named
427     # lower bound here (the search alone slips below the quadratic at larger n).
428     se11 = _extend_lower_bounds(se11, lb_dir_hyper, hyper_ns)
429     panels.append(dict(
430         title=f"directed arc, $m={m}$ (open, new model)", ylabel="hyperarcs",
431         guess="search",
432         band=_band(se11, tri_dir_hyper, hyper_ns), exact=ex11, search=se11))
433
434     # (12) hypergraph directed vertex -- OPEN (new directed Berge model).
435     ex12 = _exact_points(range(2, 6), m, exact_budget,
436                         hypergraph=True, r=3, directed=True, separation="vertex")
437     se12 = _search_points(hyper_ns, m, open_search_budget,
438                         hypergraph=True, r=3, directed=True, separation="vertex
439                             ↪ )
440     # Same construction: all its routes are single tail -> head steps, so it is
441     # feasible for the vertex separation at the same value.
442     se12 = _extend_lower_bounds(se12, lb_dir_hyper, hyper_ns)
443     panels.append(dict(
444         title=f"directed vertex, $m={m}$ (open, new model)", ylabel="hyperarcs",
445         guess="search",
446         band=_band(se12, tri_dir_hyper, hyper_ns), exact=ex12, search=se12))
447
448     return [_reconcile_panel(p) for p in panels]
449
450 def main() -> None:
451     FIGURES.mkdir(parents=True, exist_ok=True)
452     plot_directed_crossover(m=3, max_n=24, path=FIGURES / "directed_crossover.png"
453                             ↪ )
454     print("wrote directed_crossover.png")
455     plot_edge_vertex_divergence(max_n=35, path=FIGURES / "edge_vertex_divergence.
456                             ↪ png")

```

```

456     print("wrote edge_vertex_divergence.png")
457
458     # Figure 3.2: cooling trace for the directed multigraph n=4, m=3 case.
459     # record_exact_connectivity logs the true  $\lambda^{\max}$  per step (more expensive
460     # but needed for an exact scatter; the search itself uses the cheaper upper
461     #  $\hookrightarrow$  bound).
462
463     run = search_for_dense_graph(MULTI_DIRECTED, n=4, m=3, steps=3000, seed=1,
464                                record_exact_connectivity=True)
465     plot_search_trace(run, path=FIGURES / "temperature_trace.png",
466                      optimum=12, ceiling=2, show_histogram=False)
467     print(f"wrote temperature_trace.png (search reached {run.best_edge_count} arcs
468            $\hookrightarrow$  of 12)")
469
470     # Additional scatter-only traces for other variants (no cooling panel needed
471     # since the cooling schedule is the same across all cases).
472     trace_cases = [
473         # (variant, n, m, filename_stem, ceiling, connectivity_label, edge_label,
474         #  $\hookrightarrow$  title)
475         (SIMPLE_UNDIRECTED, 7, 3, "trace_simple_undirected_n7_m3", 2,
476          r"$\lambda^{\{\max\}}$", "edges",
477          r"Search trace: simple undirected, $n=7$, $m=3$"),
478         (SIMPLE_DIRECTED, 5, 3, "trace_simple_directed_n5_m3", 2,
479          r"$\lambda^{\{\max\}}$", "arcs",
480          r"Search trace: simple directed, $n=5$, $m=3$"),
481         (MULTI_UNDIRECTED, 5, 3, "trace_multi_undirected_n5_m3", 2,
482          r"$\lambda^{\{\max\}}$", "edges",
483          r"Search trace: multi undirected, $n=5$, $m=3$"),
484         (MULTI_DIRECTED, 5, 3, "trace_multi_directed_n5_m3", 2,
485          r"$\lambda^{\{\max\}}$", "arcs",
486          r"Search trace: multi directed, $n=5$, $m=3$"),
487         (MULTI_DIRECTED, 4, 3, "trace_multi_directed_n4_m3_vertex", 2,
488          r"$\kappa^{\{\max\}}$", "arcs",
489          r"Search trace: multi directed vertex-sep, $n=4$, $m=3$"),
490     ]
491
492     for variant, n, m, stem, ceil_val, clabel, elabel, ttitle in trace_cases:
493         sep = "vertex" if "vertex" in stem else "edge"
494         tr = search_for_dense_graph(variant, n=n, m=m, separation=sep,
495                                    steps=3000, seed=1,
496                                    record_exact_connectivity=True)
497         plot_search_trace(tr, path=FIGURES / f"{stem}.png",
498                          ceiling=ceil_val, show_cooling=False,
499                          show_histogram=False,
500                          connectivity_label=clabel, edge_label=elabel,
501                          title=ttitle)
502         print(f"wrote {stem}.png (best={tr.best_edge_count})")
503
504     # A larger directed multigraph: four s->t routes of differing capacity plus a
505     # dead end. Parallel arcs are drawn explicitly, so multiplicity is read by
506     # counting. The s->a route is over-provisioned ( $\mu=3$  into a but a single arc
507     # a->t out), so it appears as three arcs in a COOL colour ( $\sigma=1$ ): the
508     # visual point that multiplicity and load-bearing role are not the same.
509     sens = Graph(8, MULTI_DIRECTED) # 0=s 1=t 2=a 3=b 4=c 5=d 6=e 7=f
510     sens.set_multiplicity(0, 2, 3); sens.set_multiplicity(2, 1, 1) # s->a->t,
511     #  $\hookrightarrow$  over-provisioned in
512     sens.set_multiplicity(0, 3, 3); sens.set_multiplicity(3, 1, 3) # s->b->t, cap
513     #  $\hookrightarrow$  3 (load-bearing)
514     sens.set_multiplicity(0, 4, 2); sens.set_multiplicity(4, 1, 2) # s->c->t, cap
515     #  $\hookrightarrow$  2

```

```

508     sens.set_multiplicity(0, 6, 1); sens.set_multiplicity(6, 1, 1)    # s->e->t, cap
        ↳ 1
509     sens.set_multiplicity(0, 7, 2); sens.set_multiplicity(7, 1, 2)    # s->f->t, cap
        ↳ 2
510     sens.set_multiplicity(0, 5, 2)                                     # s->d, dead
        ↳ end
511     sens_layout = {0: (-3.2, 0.0), 1: (3.2, 0.0),
512                   2: (0.0, 2.6), 3: (0.0, 1.3), 4: (0.0, 0.0),
513                   6: (0.0, -1.3), 7: (0.0, -2.6), 5: (-3.2, -2.4)}
514     sens_labels = {0: "s", 1: "t", 2: "a", 3: "b", 4: "c", 5: "d", 6: "e", 7: "f"}
515     draw_graph_with_sensitivity(
516         sens, path=FIGURES / "sensitivity_mixed.png",
517         layout=sens_layout, node_labels=sens_labels)
518     print("wrote sensitivity_mixed.png")
519
520     # Annealing vs tabu convergence (Ch.3): wall-clock timed, so a representative
521     # run rather than a seed-exact one (see the figure caption).
522     plot_sa_vs_tabu_convergence(FIGURES / "sa_vs_tabu_convergence.pdf",
523                                cases=((5, 3), (7, 3)), budget=8.0, seed=0)
524     print("wrote sa_vs_tabu_convergence.pdf")
525
526     plot_complexity_growth(path=FIGURES / "complexity_growth.png")
527     print("wrote complexity_growth.png")
528
529     # Gallery of machine-found extremal graphs (reads extremal_gallery.json).
530     plot_extremal_gallery(FIGURES / "extremal_graphs_gallery.png")
531     print("wrote extremal_graphs_gallery.png")
532
533     # Bounds grids: one for m=3 and one for m=6 (more purple dots).
534     for m_val in (3, 6):
535         print(f"gathering all-variant grid m={m_val} (runs solve many times)...")
536         panels = gather_variant_grid(m=m_val)
537         out = FIGURES / f"variant_bounds_m{m_val}.png"
538         plot_variant_grid(panels, path=out, m=m_val)
539         print(f"wrote {out.name}")
540
541     # -----
542     # Full-enumeration scatter and distribution figures.
543     # The enumeration cache is written once to figures/enumeration_cache.pkl.
544     # -----
545     print("building enumeration cache (slow on first run, cached thereafter)...")
546     enum_cache_path = FIGURES / "enumeration_cache.pkl"
547     enum_data = compute_enumeration_cache(cache_path=enum_cache_path)
548     print("enumeration cache ready")
549
550     plot_scatter_lambda_edges(enum_data, path=FIGURES / "scatter_lambda_edges.png"
        ↳ )
551     print("wrote scatter_lambda_edges.png")
552
553     # Pooled per-pair connectivity: every vertex pair of every enumerated graph,
554     # tagged with its graph's lambda^max. Slow first run, cached thereafter.
555     print("building pair-connectivity cache (slow on first run, cached thereafter)
        ↳ ...)")
556     pair_cache_path = FIGURES / "pair_enumeration_cache.pkl"
557     pair_data = compute_pair_enumeration_cache(cache_path=pair_cache_path)
558     print("pair-connectivity cache ready")
559
560     # Pooled per-pair connectivity histograms (Ch.4): each panel stacks blue/red

```

```

561     # at its own mid-range connectivity boundary, so the split is meaningful in
562     # every variant rather than collapsing to one colour at a fixed m.
563     plot_pair_conn_dist_grid(pair_data, path=FIGURES / "pair_conn_dist.png")
564     print("wrote pair_conn_dist.png")
565
566     # Edge-count histograms (Ch.4): same mid-range stacked colouring.
567     plot_edge_dist_grid(enum_data, path=FIGURES / "edges_dist.png")
568     print("wrote edges_dist.png")
569
570     # Per-graph connectivity distribution at the fixed forbidden threshold m=6
571     # (App. B), showing the m=6 feasibility split; m=3 body version removed
572     #   ↳ 2026-06-20.
573     plot_conn_dist_grid(enum_data, 6, path=FIGURES / "conn_dist_m6.png")
574     print("wrote conn_dist_m6.png")
575
576     # -----
577     # 3-D bound surface over the (n, m) grid.
578     # Uses a JSON cache (slow first run, instant thereafter).
579     # -----
580     surface_cache_path = FIGURES / "surface_cache.json"
581     print("building surface cache (slow on first run, cached thereafter)...")
582     compute_surface_cache(
583         n_range=(3, 9), m_range=(2, 6),
584         max_seconds=20, cache_path=surface_cache_path)
585     plot_variant_3d_surfaces(surface_cache_path, path=FIGURES / "
586         ↳ variant_surface_3d.png")
587     print("wrote variant_surface_3d.png")
588
589     # -----
590     # 3-D threshold histogram (appendix): how the lambda_max distribution shifts
591     # with density p, for three representative variants. Kept as a standalone
592     # appendix figure (fig:threshold-3d) after the Figure 4.2 threshold story was
593     # removed from the body on 2026-06-20.
594     # -----
595     # No p-values: each panel sweeps in units of its own threshold, since the
596     # graph and hypergraph models do not share a density scale.
597     plot_conn_threshold_3d(
598         path=FIGURES / "threshold_3d.png",
599         n=12, m=3, samples=150, seed=7)
600     print("wrote threshold_3d.png")
601
602     # NOTE: the 2-D random-sampling threshold figures (degree_threshold,
603     # sampled_variant_grid) were removed from the thesis on 2026-06-20. Their
604     # generators are kept in erdos915_unified.py and the restore recipe is in
605     # research_notes/removed_threshold_phenomenon.md.
606
607     # Edge vs vertex disjointness in  $G(16, p)$  at three densities (Chapter 1).
608     p_values = [0.25, 0.5, 0.75]
609     data = {pp: edge_vertex_distribution(16, pp, trials=400, seed=7)
610             for pp in p_values}
611     plot_edge_vertex_histograms(data, 16, p_values,
612                                FIGURES / "edge_vertex_sampling.png")
613     print("wrote edge_vertex_sampling.png")
614
615     # Gallery of extremal graphs: all 12 variants  $\times$  small (n, m), ~3s per case.
616     print("building extremal graph gallery (this takes a few minutes)...")
617     gallery_path = FIGURES / "extremal_gallery.json"
618     gal = gallery_extremal_graphs(max_n=7, max_m=4, r=3, time_per_case=3.0)

```

```

617     save_gallery_json(gal, gallery_path)
618     total_classes = sum(
619         len(cases.get("classes", []))
620         for variant_data in gal.values()
621         for cases in variant_data.values()
622     )
623     print(f"wrote {gallery_path.name} ({total_classes} iso-classes across all
        ↳ variants)")
624
625
626 if __name__ == "__main__":
627     main()

```

C.3 The C accelerator

Two operations dominate the running time at the sizes the searches reach: deciding whether a pair of vertices already has too many independent routes, and reducing a graph to a standard form so that two drawings of the same graph can be recognised as equal. Both are rewritten here in C, which is compiled separately and loaded if it is present.

Nothing depends on it. Every function below has a pure Python counterpart in the main program, the two are tested against each other, and if the compiled library is missing the program simply uses the Python version. The size limits in the code are deliberate: the fixed-size buffers are safe only up to the stated number of vertices, and above it the program falls back rather than risking a buffer it cannot hold.

```

1  /*
2  * _erdos_fast.c - hot-path C helpers for erdos915_unified.py
3  *
4  * Compile: gcc -O3 -march=native -shared -fPIC -o _erdos_fast.so _erdos_fast.c
5  *
6  * All functions take a flat row-major int32 array 'mu' of size n*n.
7  * Access: mu[u*n + v] is the capacity (multiplicity) of arc u->v.
8  */
9
10 #include <stdint.h>
11 #include <string.h>
12 #include <stdlib.h>
13
14 /* ----- */
15 /* tiny_maxflow */
16 /* DFS Ford-Fulkerson; returns 1 iff max-flow(s,t) > cap. */
17 /* Stops as soon as it finds cap+1 augmenting paths (early exit). */
18 /* n <= 7 in practice; residual lives on the stack. */
19 /* ----- */
20 int tiny_maxflow(const int *mu, int n, int s, int t, int cap)
21 {
22     /* local mutable residual; sized for n up to 16 (vertex-split at n=8) */
23     int res[256];
24     memcpy(res, mu, (size_t)(n * n) * sizeof(int));
25
26     int flow = 0;
27     while (flow <= cap) {

```

```

28     /* DFS to find an augmenting path */
29     int parent[16];
30     for (int i = 0; i < n; i++) parent[i] = -1;
31     parent[s] = s;
32
33     /* stack-based DFS (max depth = n vertices) */
34     int stack[16];
35     int sp = 0;
36     stack[sp++] = s;
37
38     while (sp > 0 && parent[t] == -1) {
39         int u = stack[--sp];
40         for (int v = 0; v < n; v++) {
41             if (parent[v] == -1 && res[u * n + v] > 0) {
42                 parent[v] = u;
43                 stack[sp++] = v;
44             }
45         }
46     }
47
48     if (parent[t] == -1)
49         return 0;    /* max-flow <= cap */
50
51     /* find bottleneck */
52     int bottleneck = 1 << 30;
53     for (int v = t; v != s; ) {
54         int u = parent[v];
55         int cap_uv = res[u * n + v];
56         if (cap_uv < bottleneck) bottleneck = cap_uv;
57         v = u;
58     }
59
60     /* augment */
61     for (int v = t; v != s; ) {
62         int u = parent[v];
63         res[u * n + v] -= bottleneck;
64         res[v * n + u] += bottleneck;
65         v = u;
66     }
67
68     flow += bottleneck;
69 }
70 return 1;    /* max-flow > cap */
71 }
72
73 /* ----- */
74 /* max_connectivity_exceeds */
75 /* Sweeps all pairs; returns 1 iff any pair has flow > k. */
76 /* directed=1: all ordered pairs; directed=0: unordered pairs. */
77 /* ----- */
78 int max_connectivity_exceeds(const int *mu, int n, int k, int directed)
79 {
80     for (int s = 0; s < n; s++) {
81         int start = directed ? 0 : s + 1;
82         for (int t = start; t < n; t++) {
83             if (s == t) continue;
84             if (tiny_maxflow(mu, n, s, t, k))
85                 return 1;

```

```

86     }
87 }
88     return 0;
89 }
90
91 /* ----- */
92 /* canonical_form_min */
93 /* Writes the lex-min row-major permutation of the n×n matrix into */
94 /* out (n*n int32 values). Uses Heap's algorithm to enumerate n! */
95 /* permutations. At n=7: 5040 iterations × 49 ints each. */
96 /* ----- */
97 void canonical_form_min(const int *mu, int n, int *out)
98 {
99     int perm[7], best[49], cand[49];
100     int c[7];    /* Heap's algorithm counters */
101     int nn = n * n;
102
103     for (int i = 0; i < n; i++) { perm[i] = i; c[i] = 0; }
104
105     /* identity permutation is the first candidate */
106     for (int r = 0; r < n; r++)
107         for (int col = 0; col < n; col++)
108             best[r * n + col] = mu[perm[r] * n + perm[col]];
109
110     /* Heap's algorithm */
111     int i = 1;
112     while (i < n) {
113         if (c[i] < i) {
114             /* swap */
115             if (i % 2 == 0) {
116                 int tmp = perm[0]; perm[0] = perm[i]; perm[i] = tmp;
117             } else {
118                 int tmp = perm[c[i]]; perm[c[i]] = perm[i]; perm[i] = tmp;
119             }
120
121             /* build candidate permuted matrix */
122             for (int r = 0; r < n; r++)
123                 for (int col = 0; col < n; col++)
124                     cand[r * n + col] = mu[perm[r] * n + perm[col]];
125
126             /* update best if cand is lex-smaller */
127             if (memcmp(cand, best, (size_t)nn * sizeof(int)) < 0)
128                 memcpy(best, cand, (size_t)nn * sizeof(int));
129
130             c[i]++;
131             i = 1;
132         } else {
133             c[i] = 0;
134             i++;
135         }
136     }
137
138     memcpy(out, best, (size_t)nn * sizeof(int));
139 }

```

C.4 The test suite

The suite checks the program against facts established independently of it: closed forms proved by hand, constructions whose size and connectivity are known in advance, and small cases where an exhaustive walk gives the answer outright. Several tests compare two routines that are supposed to agree, such as a fast path against the slow one it replaces, which is what catches an optimisation that changes an answer. Tests needing an optional library skip themselves when it is absent, so the suite passes on a minimal installation rather than reporting failures for tools it was never given.

C.4.1 Shared setup

```
1 """Test suite for the unified Erdos-915 program.
2
3 The tests import 'erdos915_unified' from the parent directory, so we put that
4 directory on the path here, once, for every test module. The whole suite runs
5 with the standard library alone:
6
7     cd program
8     python -m unittest discover -s tests
9
10 (no pytest required, though pytest will also discover and run these if present).
11 """
12
13 import sys
14 from pathlib import Path
15
16 sys.path.insert(0, str(Path(__file__).resolve().parent.parent))
```

C.4.2 The graph model

```
1 """The graph model: the multiplicity matrix and every operation on it."""
2
3 import unittest
4
5 from erdos915_unified import (
6     Graph,
7     MULTI_DIRECTED,
8     MULTI_UNDIRECTED,
9     SIMPLE_DIRECTED,
10    SIMPLE_UNDIRECTED,
11 )
12
13
14 class VariantBooleans(unittest.TestCase):
15     def test_the_two_booleans(self):
16         self.assertEqual((SIMPLE_UNDIRECTED.directed, SIMPLE_UNDIRECTED.simple), (
17             ↪ False, True))
18         self.assertEqual((MULTI_UNDIRECTED.directed, MULTI_UNDIRECTED.simple), (
19             ↪ False, False))
20         self.assertEqual((SIMPLE_DIRECTED.directed, SIMPLE_DIRECTED.simple), (True
21             ↪ , True))
22         self.assertEqual((MULTI_DIRECTED.directed, MULTI_DIRECTED.simple), (True,
```

```

        ↪ False))
20
21     def test_describe_is_nonempty(self):
22         for variant in (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED, SIMPLE_DIRECTED,
23             ↪ MULTI_DIRECTED):
24             self.assertTrue(variant.describe())
25
26 class EmptyGraph(unittest.TestCase):
27     def test_fresh_graph_is_empty(self):
28         g = Graph(5, SIMPLE_UNDIRECTED)
29         self.assertEqual(g.num_vertices, 5)
30         self.assertEqual(list(g.vertices()), list(range(5)))
31         self.assertEqual(g.edge_count(), 0)
32         self.assertEqual(list(g.edges()), [])
33         self.assertFalse(g.has_edge(0, 1))
34
35
36 class UndirectedModel(unittest.TestCase):
37     def test_add_edge_is_symmetric(self):
38         g = Graph(3, SIMPLE_UNDIRECTED)
39         g.add_edge(0, 1)
40         self.assertTrue(g.has_edge(0, 1))
41         self.assertTrue(g.has_edge(1, 0))
42         self.assertEqual(g.multiplicity(0, 1), 1)
43         self.assertEqual(g.multiplicity(1, 0), 1)
44         self.assertEqual(g.edge_count(), 1)
45         self.assertEqual(list(g.edges()), [(0, 1, 1)])
46
47     def test_simple_graph_saturates_at_one(self):
48         g = Graph(3, SIMPLE_UNDIRECTED)
49         g.add_edge(0, 1)
50         g.add_edge(0, 1)
51         self.assertEqual(g.multiplicity(0, 1), 1)
52         self.assertEqual(g.edge_count(), 1)
53
54     def test_multigraph_keeps_parallel_edges(self):
55         g = Graph(3, MULTI_UNDIRECTED)
56         g.add_edge(0, 1, 3)
57         self.assertEqual(g.multiplicity(0, 1), 3)
58         self.assertEqual(g.multiplicity(1, 0), 3)
59         self.assertEqual(g.edge_count(), 3)
60         self.assertEqual(g.degree(0), 3)
61         self.assertEqual(g.degree(1), 3)
62
63     def test_undirected_degree_counts_incident_multiplicity(self):
64         g = Graph(3, MULTI_UNDIRECTED)
65         g.add_edge(0, 1, 2)
66         g.add_edge(0, 2, 1)
67         self.assertEqual(g.degree(0), 3)
68         self.assertEqual(g.edge_count(), 3)
69
70
71 class DirectedModel(unittest.TestCase):
72     def test_arc_is_one_directional(self):
73         g = Graph(3, SIMPLE_DIRECTED)
74         g.add_edge(0, 1)
75         self.assertTrue(g.has_edge(0, 1))

```

```

76         self.assertFalse(g.has_edge(1, 0))
77         self.assertEqual(g.edge_count(), 1)
78         self.assertEqual(g.out_degree(0), 1)
79         self.assertEqual(g.in_degree(1), 1)
80         self.assertEqual(g.in_degree(0), 0)
81         self.assertEqual(g.degree(0), 1)
82         self.assertEqual(g.degree(1), 1)
83
84     def test_directed_multiplicities_are_independent(self):
85         g = Graph(3, MULTI_DIRECTED)
86         g.add_edge(0, 1, 2)
87         g.add_edge(1, 0, 1)
88         self.assertEqual(g.multiplicity(0, 1), 2)
89         self.assertEqual(g.multiplicity(1, 0), 1)
90         self.assertEqual(g.edge_count(), 3)
91         self.assertEqual(g.out_degree(0), 2)
92         self.assertEqual(g.in_degree(0), 1)
93         self.assertEqual(g.degree(0), 3)
94
95     def test_edges_yields_both_arc_directions(self):
96         g = Graph(3, SIMPLE_DIRECTED)
97         g.add_edge(0, 1)
98         g.add_edge(1, 0)
99         self.assertEqual(sorted(g.edges()), [(0, 1, 1), (1, 0, 1)])
100
101
102 class Mutation(unittest.TestCase):
103     def test_remove_never_goes_below_zero(self):
104         g = Graph(3, MULTI_UNDIRECTED)
105         g.add_edge(0, 1, 3)
106         g.remove_edge(0, 1, 5)
107         self.assertEqual(g.multiplicity(0, 1), 0)
108         self.assertFalse(g.has_edge(0, 1))
109
110     def test_set_multiplicity_directly(self):
111         g = Graph(3, MULTI_DIRECTED)
112         g.set_multiplicity(0, 1, 4)
113         self.assertEqual(g.multiplicity(0, 1), 4)
114         self.assertEqual(g.multiplicity(1, 0), 0)
115
116     def test_simple_rejects_multiplicity_above_one(self):
117         g = Graph(3, SIMPLE_UNDIRECTED)
118         with self.assertRaises(ValueError):
119             g.set_multiplicity(0, 1, 2)
120
121     def test_negative_multiplicity_rejected(self):
122         g = Graph(3, MULTI_UNDIRECTED)
123         with self.assertRaises(ValueError):
124             g.set_multiplicity(0, 1, -1)
125
126     def test_self_loops_rejected(self):
127         g = Graph(3, MULTI_UNDIRECTED)
128         with self.assertRaises(ValueError):
129             g.add_edge(1, 1)
130
131     def test_copy_is_independent(self):
132         g = Graph(3, MULTI_UNDIRECTED)
133         g.add_edge(0, 1, 2)

```

```

134         clone = g.copy()
135         clone.add_edge(0, 2, 1)
136         self.assertEqual(g.edge_count(), 2)
137         self.assertEqual(clone.edge_count(), 3)
138
139
140 if __name__ == "__main__":
141     unittest.main()

```

C.4.3 The closed-form bounds

```

1  """The closed-form bounds library: every formula, on a range of inputs."""
2
3  import unittest
4
5  from erdos915_unified import (
6      directed_arc_lower_bound,
7      directed_arc_m2,
8      directed_multigraph_arc,
9      hypergraph_edge,
10     multigraph_undirected_edge,
11     simple_undirected_edge,
12     simple_undirected_vertex_le4,
13     simple_undirected_vertex_m5,
14 )
15
16
17 class Bounds(unittest.TestCase):
18     def test_simple_undirected_edge(self):
19         self.assertEqual(simple_undirected_edge(5, 3), 6)
20         for n in range(2, 12):
21             for m in range(2, 6):
22                 self.assertEqual(simple_undirected_edge(n, m), (m * (n - 1)) // 2)
23
24     def test_multigraph_undirected_edge(self):
25         for n in range(2, 12):
26             for m in range(2, 6):
27                 self.assertEqual(multigraph_undirected_edge(n, m), (m - 1) * (n -
28                                     ↪ 1))
29
30     def test_vertex_le4_matches_edge_and_caps_at_four(self):
31         for m in range(2, 5):
32             self.assertEqual(simple_undirected_vertex_le4(10, m),
33                             ↪ simple_undirected_edge(10, m))
34         with self.assertRaises(ValueError):
35             simple_undirected_vertex_le4(10, 5)
36
37     def test_vertex_m5(self):
38         for n in range(6, 20):
39             self.assertEqual(simple_undirected_vertex_m5(n), (8 * n) // 3 - 3)
40
41     def test_directed_arc_m2_branches_and_crossover(self):
42         self.assertEqual(directed_arc_m2(4), 6)      # linear hub branch wins
43         self.assertEqual(directed_arc_m2(7), 12)     # the two branches meet
44         self.assertEqual(directed_arc_m2(8), 16)     # quadratic branch wins
45         for n in range(2, 20):
46             self.assertEqual(directed_arc_m2(n), max(2 * (n - 1), (n * n) // 4))

```

```

45
46     def test_directed_arc_lower_bound(self):
47         self.assertEqual(directed_arc_lower_bound(10, 3), 30) # the 30-arc graph
48         self.assertEqual(directed_arc_lower_bound(4, 2), 6)
49         for n in range(3, 14):
50             for m in range(2, 5):
51                 expected = max(m * (n - 1), (n + m - 2) ** 2 // 4)
52                 self.assertEqual(directed_arc_lower_bound(n, m), expected)
53
54     def test_hypergraph_edge(self):
55         self.assertEqual(hypergraph_edge(7, 2, 3), 3)
56         self.assertEqual(hypergraph_edge(13, 2, 4), 4)
57         for n in range(3, 14):
58             for m in range(2, 5):
59                 for r in range(2, 5):
60                     self.assertEqual(hypergraph_edge(n, m, r), ((m - 1) * (n - 1))
61                                     ↪ // (r - 1))
62
63     def test_directed_multigraph_arc(self):
64         self.assertEqual(directed_multigraph_arc(4, 3), 12)
65         self.assertEqual(directed_multigraph_arc(6, 2), 10)
66         for n in range(2, 12):
67             for m in range(2, 6):
68                 expected = max(2 * (n - 1) * (m - 1), (m - 1) * ((n * n) // 4))
69                 self.assertEqual(directed_multigraph_arc(n, m), expected)
70
71 if __name__ == "__main__":
72     unittest.main()

```

C.4.4 The named constructions

```

1  """The named extremal constructions: their exact sizes and connectivities."""
2
3  import unittest
4
5  from erdos915_unified import (
6      augmented_bipartite,
7      clique_tree,
8      double_star,
9      max_edge_connectivity,
10     min_vertex_connectivity,
11     one_directional_bipartite,
12 )
13
14
15 class Constructions(unittest.TestCase):
16     def test_double_star_directed(self):
17         for n in range(2, 7):
18             for m in range(2, 5):
19                 g = double_star(n, m, directed=True)
20                 self.assertEqual(g.edge_count(), 2 * (n - 1) * (m - 1))
21                 self.assertEqual(max_edge_connectivity(g), m - 1)
22
23     def test_double_star_undirected(self):
24         for n in range(2, 7):
25             for m in range(2, 5):

```

```

26         g = double_star(n, m, directed=False)
27         self.assertEqual(g.edge_count(), (m - 1) * (n - 1))
28         self.assertEqual(max_edge_connectivity(g), m - 1)
29
30     def test_one_directional_bipartite_is_quadratic_and_feasible(self):
31         for n in range(2, 8):
32             g = one_directional_bipartite(n)
33             self.assertEqual(g.edge_count(), (n * n) // 4)
34             self.assertEqual(max_edge_connectivity(g), 1)
35
36     def test_augmented_bipartite_conjecture_values(self):
37         self.assertEqual(augmented_bipartite(10, 3).edge_count(), 30)
38         for n, m in [(8, 3), (10, 4), (12, 4), (10, 5)]:
39             g = augmented_bipartite(n, m)
40             self.assertEqual(g.edge_count(), (n + m - 2) ** 2 // 4)
41             self.assertEqual(max_edge_connectivity(g), m - 1)
42
43     def test_augmented_bipartite_guards_its_precondition(self):
44         with self.assertRaises(ValueError):
45             augmented_bipartite(3, 5) # needs n >= m
46
47     def test_clique_tree_is_globally_connected(self):
48         for blocks in range(0, 5):
49             self.assertEqual(min_vertex_connectivity(clique_tree(4, blocks)), 3)
50
51
52 if __name__ == "__main__":
53     unittest.main()

```

C.4.5 The connectivity checker

```

1  """The connectivity checker (edge and vertex) and the Gomory-Hu shortcut."""
2
3  import random
4  import unittest
5
6  from erdos915_unified import (
7      Graph,
8      MULTI_DIRECTED,
9      MULTI_UNDIRECTED,
10     NETWORKXX_AVAILABLE,
11     SIMPLE_DIRECTED,
12     SIMPLE_UNDIRECTED,
13     _vertex_flow_at_least,
14     exceeds_bound,
15     local_edge_connectivity,
16     local_vertex_connectivity,
17     max_connectivity,
18     max_edge_connectivity,
19     max_edge_connectivity_via_tree,
20     max_vertex_connectivity,
21     sample_random_graph,
22 )
23
24
25 def random_multigraph(variant, n, rng, max_mult=2):
26     """A random graph of ‘variant’ on ‘n’ vertices for predicate testing.

```

```

27
28     Simple variants take multiplicities in '{0, 1}'; multigraph variants in
29     '{0, ..., max_mult}'. Directed variants fill every ordered pair, undirected
30     variants the upper triangle ('set_multiplicity' mirrors the lower half).
31     """
32     g = Graph(n, variant)
33     top = 1 if variant.simple else max_mult
34     for u in range(n):
35         start = 0 if variant.directed else u + 1
36         for v in range(start, n):
37             if u != v:
38                 g.set_multiplicity(u, v, rng.randint(0, top))
39     return g
40
41
42 def complete_graph(n, variant=SIMPLE_UNDIRECTED):
43     g = Graph(n, variant)
44     for u in range(n):
45         for v in range(u + 1, n):
46             g.add_edge(u, v)
47     return g
48
49
50 def star(n):
51     g = Graph(n, SIMPLE_UNDIRECTED)
52     for leaf in range(1, n):
53         g.add_edge(0, leaf)
54     return g
55
56
57 def cycle(n):
58     g = Graph(n, SIMPLE_UNDIRECTED)
59     for v in range(n):
60         g.add_edge(v, (v + 1) % n)
61     return g
62
63
64 class EdgeConnectivity(unittest.TestCase):
65     def test_tree_is_one(self):
66         self.assertEqual(max_edge_connectivity(star(6)), 1)
67
68     def test_cycle_is_two(self):
69         self.assertEqual(max_edge_connectivity(cycle(6)), 2)
70
71     def test_complete_graph(self):
72         for n in (3, 4, 5):
73             self.assertEqual(max_edge_connectivity(complete_graph(n)), n - 1)
74             self.assertEqual(local_edge_connectivity(complete_graph(n), 0, 1), n -
75                 ↪ 1)
76
77     def test_parallel_edges_count_as_routes(self):
78         g = Graph(3, MULTI_UNDIRECTED)
79         g.add_edge(0, 1, 4)
80         self.assertEqual(local_edge_connectivity(g, 0, 1), 4)
81         self.assertEqual(max_edge_connectivity(g), 4)
82
83     def test_directed_arc_disjoint_routes(self):
84         g = Graph(3, SIMPLE_DIRECTED)

```

```

84         g.add_edge(0, 1) # direct route
85         g.add_edge(0, 2) # detour 0 -> 2 -> 1
86         g.add_edge(2, 1)
87         self.assertEqual(local_edge_connectivity(g, 0, 1), 2)
88
89     def test_removing_an_edge_never_raises_lambda(self):
90         rng = random.Random(11)
91         for _ in range(20):
92             g = sample_random_graph(8, 0.4, False, rng)
93             before = max_edge_connectivity(g)
94             for u, v, _count in list(g.edges()):
95                 reduced = g.copy()
96                 reduced.set_multiplicity(u, v, 0)
97                 self.assertLessEqual(max_edge_connectivity(reduced), before)
98
99
100 class VertexConnectivity(unittest.TestCase):
101     def test_complete_graph(self):
102         for n in (3, 4, 5):
103             self.assertEqual(max_vertex_connectivity(complete_graph(n)), n - 1)
104             self.assertEqual(local_vertex_connectivity(complete_graph(n), 0, 1), n
105                 ↪ - 1)
106
107     def test_whitney_inequality_on_random_graphs(self):
108         rng = random.Random(0)
109         for _ in range(40):
110             g = sample_random_graph(12, 0.3, False, rng)
111             self.assertLessEqual(max_vertex_connectivity(g), max_edge_connectivity
112                 ↪ (g))
113
114 class VertexFlowPredicate(unittest.TestCase):
115     """‘_vertex_flow_at_least’ is the exhaustive search’s inner vertex test.
116
117     It must agree with the exact checker it replaces on every (pair, k), or the
118     base cases of the directed vertex theorem would rest on a faster wrong test.
119     """
120     def test_matches_the_exact_checker(self):
121         rng = random.Random(17)
122         for _ in range(150):
123             n = rng.randint(2, 6)
124             g = Graph(n, SIMPLE_DIRECTED)
125             out = [set() for _ in range(n)]
126             for a in range(n):
127                 for b in range(n):
128                     if a != b and rng.random() < 0.35:
129                         g.mu[a, b] = 1
130                         out[a].add(b)
131             for s in range(n):
132                 for t in range(n):
133                     if s == t:
134                         continue
135                     exact = local_vertex_connectivity(g, s, t)
136                     for k in range(1, 5):
137                         self.assertEqual(_vertex_flow_at_least(out, n, s, t, k),
138                             exact >= k)
139

```

```

140     def test_a_direct_arc_is_one_route_with_no_interior(self):
141         # s -> t plus the detour s -> x -> t: two internally disjoint routes.
142         out = [{1, 2}, set(), {1}, set()]
143         self.assertTrue(_vertex_flow_at_least(out, 4, 0, 1, 2))
144         self.assertFalse(_vertex_flow_at_least(out, 4, 0, 1, 3))
145
146
147 @unittest.skipUnless(NETWORKX_AVAILABLE, "Gomory-Hu trees need the optional
    ↳ networkx")
148 class GomoryHu(unittest.TestCase):
149     def test_tree_matches_direct_sweep(self):
150         for g in (cycle(6), complete_graph(5), star(7)):
151             self.assertEqual(max_edge_connectivity_via_tree(g),
    ↳ max_edge_connectivity(g))
152
153     def test_multigraph_matches(self):
154         g = Graph(4, MULTI_UNDIRECTED)
155         g.add_edge(0, 1, 3)
156         g.add_edge(1, 2, 2)
157         g.add_edge(2, 3, 1)
158         g.add_edge(0, 3, 2)
159         self.assertEqual(max_edge_connectivity_via_tree(g), max_edge_connectivity(
    ↳ g))
160
161
162 class CappedPredicate(unittest.TestCase):
163     """'exceeds_bound' must agree with the exact checker, pair set for pair set.
164
165     This pins the cheap capped predicate the search relies on to the trusted
166     exact 'max_connectivity' over every variant, separation and threshold.
167     """
168
169     def test_predicate_matches_exact_value(self):
170         variants = (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED,
171                     SIMPLE_DIRECTED, MULTI_DIRECTED)
172         rng = random.Random(2024)
173         for variant in variants:
174             for n in (3, 4, 5):
175                 for _ in range(8):
176                     g = random_multigraph(variant, n, rng)
177                     for sep in ("edge", "vertex"):
178                         split = sep == "vertex"
179                         exact = max_connectivity(g, vertex_split=split)
180                         for k in range(-1, exact + 2):
181                             self.assertEqual(
182                                 exceeds_bound(g, k, separation=sep),
183                                 exact > k,
184                                 msg=f"{variant.name} n={n} sep={sep} k={k} "
185                                     f"exact={exact}",
186                             )
187
188     def test_predicate_does_not_mutate_the_graph(self):
189         rng = random.Random(7)
190         g = random_multigraph(MULTI_DIRECTED, 5, rng)
191         before = g.mu.copy()
192         exceeds_bound(g, 1, separation="vertex")
193         exceeds_bound(g, 1, separation="edge")
194         self.assertTrue((g.mu == before).all())

```

```

195
196
197 if __name__ == "__main__":
198     unittest.main()

```

C.4.6 Hypergraphs

```

1  """The hypergraph model and Berge connectivity by the helper-point network."""
2
3  import unittest
4
5  from erdos915_unified import (
6      Hypergraph,
7      complete_uniform_hypergraph,
8      hyper_connectivity,
9      hyperedge_connectivity,
10     max_feasible_hyperedges,
11     max_hyperedge_connectivity,
12     star_hypertree,
13     _hyperedge_candidates,
14 )
15
16
17 class BergeConnectivity(unittest.TestCase):
18     def test_single_hyperedge_carries_one_route(self):
19         h = Hypergraph(3, [[0, 1, 2]])
20         self.assertEqual(h.edge_count(), 1)
21         self.assertEqual(hyperedge_connectivity(h, 0, 1), 1)
22         self.assertEqual(hyperedge_connectivity(h, 0, 2), 1)
23         self.assertEqual(hyperedge_connectivity(h, 1, 2), 1)
24
25     def test_two_hyperedges_give_two_routes(self):
26         h = Hypergraph(4, [[0, 1, 2], [0, 1, 3]])
27         self.assertEqual(h.edge_count(), 2)
28         self.assertEqual(hyperedge_connectivity(h, 0, 1), 2)
29         self.assertEqual(len(h.incident_hyperedges(0)), 2)
30         self.assertEqual(len(h.incident_hyperedges(2)), 1)
31
32     def test_complete_three_uniform_k4(self):
33         self.assertEqual(max_hyperedge_connectivity(complete_uniform_hypergraph(4,
34             ↪ 3)), 3)
35
36     def test_two_uniform_reduces_to_edge_connectivity(self):
37         for n in (3, 4, 5):
38             self.assertEqual(max_hyperedge_connectivity(
39                 ↪ complete_uniform_hypergraph(n, 2)), n - 1)
40
41     def test_star_hypertree(self):
42         for n, r in [(7, 3), (10, 4), (13, 4)]:
43             h = star_hypertree(n, r)
44             self.assertEqual(h.edge_count(), (n - 1) // (r - 1))
45             self.assertEqual(max_hyperedge_connectivity(h), 1)
46
47     def test_bad_hyperedge_size_rejected(self):
48         with self.assertRaises(ValueError):
49             star_hypertree(7, 1)

```

```

49
50 class DirectedOrientationModels(unittest.TestCase):
51     """Forward / backward / general directed hyperedge models."""
52
53     def test_general_gate_is_one_way(self):
54         # A mixed arc {0,1} -> {2,3}: a route enters at a tail, leaves at a head.
55         h = Hypergraph(4, [(frozenset({0, 1}), frozenset({2, 3}))], directed=True)
56         self.assertEqual(sorted(h.members(h.hyperedges[0])), [0, 1, 2, 3])
57         self.assertEqual(hyper_connectivity(h, 0, 2), 1) # tail -> head
58         self.assertEqual(hyper_connectivity(h, 2, 0), 0) # head -> tail: none
59
60     def test_forward_legacy_equals_general_form(self):
61         legacy = Hypergraph(3, [(0, frozenset({1, 2}))], directed=True)
62         general = Hypergraph(3, [(frozenset({0}), frozenset({1, 2}))], directed=
        ↪ True)
63         for s, t in [(0, 1), (0, 2), (1, 2)]:
64             self.assertEqual(hyper_connectivity(legacy, s, t),
65                             hyper_connectivity(general, s, t))
66
67     def test_candidate_counts_split_at_r3(self):
68         # At r = 3 every split has a singleton side, so general = forward +
69         ↪ backward.
70         fwd = _hyperedge_candidates(4, 3, True, kind="forward")
71         bwd = _hyperedge_candidates(4, 3, True, kind="backward")
72         gen = _hyperedge_candidates(4, 3, True, kind="general")
73         self.assertEqual(len(fwd), 12)
74         self.assertEqual(len(bwd), 12)
75         self.assertEqual(len(gen), 24)
76
77     def test_forward_backward_duality(self):
78         # Arc reversal makes backward the dual of forward: same extremal numbers.
79         for n, m in [(3, 3), (4, 3), (4, 2)]:
80             f, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="forward",
81             ↪ time_limit=2.0)
82             b, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="backward"
83             ↪ , time_limit=2.0)
84             self.assertEqual(f, b)
85
86     def test_general_beats_forward_when_vertices_scarce(self):
87         # n = r is the regime where the richer orientation set packs more arcs.
88         g3, e3 = max_feasible_hyperedges(3, 3, 3, directed=True, kind="general",
89         ↪ time_limit=2.0)
90         f3, _ = max_feasible_hyperedges(3, 3, 3, directed=True, kind="forward",
91         ↪ time_limit=2.0)
92         self.assertTrue(e3)
93         self.assertEqual((g3, f3), (4, 3))
94
95     def test_general_contains_forward(self):
96         # General can never be worse than forward (forward hypergraphs are general
97         ↪ ).
98         for n, m in [(4, 3), (5, 2)]:
99             f, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="forward",
100             ↪ time_limit=2.0)
101             g, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="general",
102             ↪ time_limit=3.0, seed_lb=f)
103             self.assertGreaterEqual(g, f)
104

```

```

99 if __name__ == "__main__":
100     unittest.main()

```

C.4.7 Edge sensitivity

```

1  """Edge sensitivity: the load-bearing measure that guides the search."""
2
3  import random
4  import unittest
5
6  from erdos915_unified import (
7      Graph,
8      MULTI_DIRECTED,
9      MULTI_UNDIRECTED,
10     SIMPLE_DIRECTED,
11     SIMPLE_UNDIRECTED,
12     double_star,
13     edge_sensitivity,
14     max_edge_connectivity,
15     max_vertex_connectivity,
16     one_directional_bipartite,
17     sensitivity_map,
18 )
19
20
21 def _random_graph(variant, n, rng, top):
22     g = Graph(n, variant)
23     cap = 1 if variant.simple else top
24     for u in range(n):
25         start = 0 if variant.directed else u + 1
26         for v in range(start, n):
27             if u != v:
28                 g.set_multiplicity(u, v, rng.randint(0, cap))
29     return g
30
31
32 class Sensitivity(unittest.TestCase):
33     def test_absent_edge_has_zero_sensitivity(self):
34         g = Graph(3, SIMPLE_UNDIRECTED)
35         self.assertEqual(edge_sensitivity(g, 0, 1), 0)
36
37     def test_sensitivity_matches_its_definition(self):
38         g = one_directional_bipartite(6)
39         before = max_edge_connectivity(g)
40         for (u, v), sigma in sensitivity_map(g).items():
41             reduced = g.copy()
42             reduced.set_multiplicity(u, v, 0)
43             self.assertEqual(sigma, before - max_edge_connectivity(reduced))
44             self.assertGreaterEqual(sigma, 0) # removing an edge never raises
45                                             ↪ lambda
46
47     def test_map_covers_exactly_the_present_edges(self):
48         g = double_star(5, 3, directed=True)
49         self.assertEqual(len(sensitivity_map(g)), sum(1 for _ in g.edges()))
50
51     def test_map_equals_per_edge_sensitivity(self):
52         # The before-once refactor must return the IDENTICAL dict as the old

```

```

52     # per-edge definition, across variants and both connectivity measures.
53     variants = (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED,
54                 SIMPLE_DIRECTED, MULTI_DIRECTED)
55     rng = random.Random(99)
56     for variant in variants:
57         for measure in (max_edge_connectivity, max_vertex_connectivity):
58             for _ in range(6):
59                 g = _random_graph(variant, 5, rng, top=2)
60                 expected = {(u, v): edge_sensitivity(g, u, v, measure)
61                             for u, v, _ in g.edges()}
62                 self.assertEqual(sensitivity_map(g, measure), expected)
63
64
65 if __name__ == "__main__":
66     unittest.main()

```

C.4.8 The search

```

1  """The temperature search: it rediscovers the known small extremal values.
2  The search is seeded, so these outcomes are deterministic and reproducible.
3  """
4
5
6  import unittest
7
8  from erdos915_unified import (
9      MULTI_DIRECTED,
10     MULTI_UNDIRECTED,
11     SIMPLE_DIRECTED,
12     SIMPLE_UNDIRECTED,
13     best_of_searches,
14     max_connectivity,
15     max_edge_connectivity,
16     search_for_dense_graph,
17     tabu_search_for_dense_graph,
18 )
19
20
21 class Search(unittest.TestCase):
22     def test_search_rediscovers_directed_m2(self):
23         result = search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2, steps=6000,
24                                         ↪ seed=0)
25         self.assertEqual(result.best_edge_count, 6) # ell_2^dir(4) = 6
26         self.assertTrue(result.feasible_found)
27         self.assertLessEqual(max_edge_connectivity(result.best_graph), 1)
28
29     def test_acceptance_rate_is_a_fraction(self):
30         result = search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2, steps=6000,
31                                         ↪ seed=0)
32         self.assertTrue(0.0 <= result.acceptance_rate() <= 1.0)
33         self.assertEqual(len(result.history), 6000)
34
35     def test_search_rediscovers_undirected_m3(self):
36         # ell_3(n) = floor(3(n-1)/2): n=4 -> 4, n=5 -> 6.
37         for n, expected in [(4, 4), (5, 6)]:
38             result = best_of_searches(SIMPLE_UNDIRECTED, n, 3, restarts=6, steps
39                                     ↪ =3000, seed=0)

```

```

37         self.assertEqual(result.best_edge_count, expected)
38         self.assertLessEqual(max_edge_connectivity(result.best_graph), 2)
39
40
41 class TabuSearch(unittest.TestCase):
42     """Tabu search is the deterministic twin of the annealer: same energy and
43     neighbourhood, so it rediscovers the same small extremal values.
44     """
45
46     def test_tabu_rediscovers_directed_m2(self):
47         result = tabu_search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2,
48                                             steps=150, seed=0)
49         self.assertEqual(result.best_edge_count, 6) #  $ell_2^{dir}(4) = 6$ 
50         self.assertTrue(result.feasible_found)
51         self.assertLessEqual(max_edge_connectivity(result.best_graph), 1)
52
53     def test_tabu_rediscovers_undirected_m3(self):
54         result = tabu_search_for_dense_graph(SIMPLE_UNDIRECTED, n=4, m=3,
55                                             steps=150, seed=0)
56         self.assertEqual(result.best_edge_count, 4) #  $ell_3(4) = 4$ 
57         self.assertLessEqual(max_edge_connectivity(result.best_graph), 2)
58
59     def test_best_of_searches_accepts_tabu_method(self):
60         result = best_of_searches(SIMPLE_DIRECTED, 4, 2, restarts=2,
61                                 method="tabu", parallel=False, steps=120)
62         self.assertEqual(result.best_edge_count, 6)
63
64     def test_unknown_method_rejected(self):
65         with self.assertRaises(ValueError):
66             best_of_searches(SIMPLE_UNDIRECTED, 4, 3, method="genetic")
67
68
69 class FastPathEquivalence(unittest.TestCase):
70     """The fast (capped + monotone) search must reproduce the slow exact search
71     step for step. This is the "nothing under the rug" guarantee: the speedup
72     changes no reported value because it changes no single step of the walk.
73     """
74
75     def test_fast_path_matches_exact(self):
76         variants = (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED,
77                   SIMPLE_DIRECTED, MULTI_DIRECTED)
78         for variant in variants:
79             for n in (3, 4, 5):
80                 for m in (2, 3):
81                     for sep in ("edge", "vertex"):
82                         for seed in range(2):
83                             opts = dict(separation=sep, steps=300, seed=seed)
84                             ref = search_for_dense_graph(
85                                 variant, n, m, reference_mode=True, **opts)
86                             fast = search_for_dense_graph(variant, n, m, **opts)
87                             tag = (f"{variant.name} n={n} m={m} sep={sep} "
88                                   f"seed={seed}")
89                             # Bit-for-bit: same energies, same accept/reject
90                             # decisions, same density at every step.
91                             self.assertEqual(
92                                 [s.energy for s in fast.history],
93                                 [s.energy for s in ref.history], tag)
94                             self.assertEqual(

```

```

95         [s.accepted for s in fast.history],
96         [s.accepted for s in ref.history], tag)
97     self.assertEqual(
98         [s.edge_count for s in fast.history],
99         [s.edge_count for s in ref.history], tag)
100    self.assertEqual(
101        fast.best_edge_count, ref.best_edge_count, tag)
102    self.assertTrue(
103        (fast.best_graph.mu == ref.best_graph.mu).all(),
104        tag)
105    # The returned witness is genuinely feasible by the
106    # exact checker (certified, not just flagged).
107    split = sep == "vertex"
108    self.assertLessEqual(
109        max_connectivity(fast.best_graph,
110                          vertex_split=split),
111        m - 1, tag)
112
113
114    if __name__ == "__main__":
115        unittest.main()

```

C.4.9 The prover

```

1  """The cut-counting prover: a zero-gap solve is a genuine upper-bound proof."""
2
3  import os
4  import unittest
5
6  import numpy as np
7
8  from erdos915_unified import (
9      MULTI_DIRECTED,
10     PULP_AVAILABLE,
11     Graph,
12     _canonical_form,
13     enumerate_extremal_directed_multigraphs,
14     max_edge_connectivity,
15     prove_directed_multigraph,
16 )
17
18
19 @unittest.skipUnless(PULP_AVAILABLE, "the MILP certifier needs the optional pulp")
20 class Prover(unittest.TestCase):
21     def test_small_optima_are_proved(self):
22         # M*(n) = 2(n-1), optimal with zero gap, for the reachable sizes.
23         for n in (3, 4, 5):
24             result = prove_directed_multigraph(n, time_limit=120.0)
25             self.assertTrue(result.is_proof())
26             self.assertEqual(result.status, "OPTIMAL")
27             self.assertEqual(round(result.scaled_optimum), 2 * (n - 1))
28
29     def test_one_solve_settles_every_m(self):
30         result = prove_directed_multigraph(4, time_limit=120.0)
31         self.assertTrue(result.is_proof())
32         for m in (2, 3, 4, 5):
33             # L_m^dir(4) = (m-1) * M*(4) = (m-1) * 6.

```

```

34         self.assertEqual(result.value_for(m), (m - 1) * 2 * (4 - 1))
35
36
37 class EnumerationDedup(unittest.TestCase):
38     """The streamed canonical-form dedup keeps one representative per iso-class
39     and returns exactly the known small extremal sets.
40
41     The active path stays at n in {3, 4} (sub-second). The n=5 enumeration is
42     minutes long (the DFS, not the dedup), so its known count is checked only when
43     'RUN_SLOW_ENUM' is set; the dedup logic it exercises is identical to n=4.
44     """
45
46     # m=3 directed multigraph, target = 4(n-1) arcs: the doubled bidirected
47     # spanning trees. Known counts (claude.md): n=3 -> 1 (path), n=4 -> 2
48     # (star, path), n=5 -> 3 (star, broom, path).
49     FAST_COUNTS = {3: 1, 4: 2}
50
51     def test_known_iso_class_counts(self):
52         for n, expected in self.FAST_COUNTS.items():
53             reps = enumerate_extremal_directed_multigraphs(n, 3, 4 * (n - 1))
54             self.assertEqual(len(reps), expected, f"n={n}")
55
56     def test_representatives_are_distinct_and_feasible(self):
57         for n in self.FAST_COUNTS:
58             reps = enumerate_extremal_directed_multigraphs(n, 3, 4 * (n - 1))
59             keys = [_canonical_form(mu) for mu in reps]
60             # One representative per class: all canonical keys distinct.
61             self.assertEqual(len(set(keys)), len(keys), f"n={n}")
62             for mu in reps:
63                 g = Graph(n, MULTI_DIRECTED)
64                 g.mu = mu.copy()
65                 self.assertEqual(g.edge_count(), 4 * (n - 1))
66                 self.assertLessEqual(max_edge_connectivity(g), 2, f"n={n}")
67
68     def test_dedup_preserves_the_iso_classes(self):
69         # The deduped run must cover exactly the iso-classes of the full labelled
70         # run: no class invented, none dropped.
71         for n in self.FAST_COUNTS:
72             labelled = enumerate_extremal_directed_multigraphs(
73                 n, 3, 4 * (n - 1), up_to_iso=False)
74             deduped = enumerate_extremal_directed_multigraphs(n, 3, 4 * (n - 1))
75             self.assertEqual(
76                 {_canonical_form(mu) for mu in labelled},
77                 {_canonical_form(mu) for mu in deduped},
78                 f"n={n}")
79             # Every labelled matrix is itself one of the kept iso-classes.
80             kept = {_canonical_form(mu) for mu in deduped}
81             for mu in labelled:
82                 self.assertIn(_canonical_form(mu), kept, f"n={n}")
83
84     def test_canonical_form_is_permutation_invariant(self):
85         rng = np.random.default_rng(0)
86         for n in (3, 4, 5, 6):
87             mu = rng.integers(0, 3, size=(n, n))
88             np.fill_diagonal(mu, 0)
89             key = _canonical_form(mu)
90             for _ in range(5):
91                 p = rng.permutation(n)

```

```

92         self.assertEqual(_canonical_form(mu[np.ix_(p, p)]), key, f"n={n}")
93
94     @unittest.skipUnless(os.environ.get("RUN_SLOW_ENUM"),
95                          "n=5 enumeration is minutes long; set RUN_SLOW_ENUM=1")
96     def test_n5_known_count(self):
97         reps = enumerate_extremal_directed_multigraphs(5, 3, 16)
98         self.assertEqual(len(reps), 3)  # star, broom, path
99
100
101 if __name__ == "__main__":
102     unittest.main()

```

C.4.10 The random model

```

1  """The random model: the appearance threshold and the Whitney relation.
2
3  All sampling is seeded, so these outcomes are deterministic.
4  """
5
6  import unittest
7
8  from erdos915_unified import (
9      connectivity_distribution,
10     edge_vertex_distribution,
11     estimate_appearance_probability,
12     threshold_curve,
13 )
14
15
16 class MonteCarlo(unittest.TestCase):
17     def test_appearance_probability_crosses_m_over_n(self):
18         n, m = 30, 3
19         below = estimate_appearance_probability(n, 0.3 * m / n, m, trials=120,
20         ↪ seed=1)
21         above = estimate_appearance_probability(n, 1.8 * m / n, m, trials=120,
22         ↪ seed=1)
23         self.assertLess(below, 0.5)
24         self.assertGreater(above, 0.5)
25
26     def test_whitney_holds_on_every_sample(self):
27         edge = connectivity_distribution(20, 0.25, trials=40, separation="edge",
28         ↪ seed=3)
29         vertex = connectivity_distribution(20, 0.25, trials=40, separation="vertex
30         ↪ ", seed=3)
31         self.assertEqual(len(edge), 40)
32         for kappa, lam in zip(vertex, edge):
33             self.assertLessEqual(kappa, lam)
34
35     def test_edge_and_vertex_agree_at_small_n(self):
36         lam, kap = edge_vertex_distribution(5, 0.5, trials=120, seed=7)
37         self.assertEqual(sum(k < 1 for k, l in zip(kap, lam)), 0)  # no gap at n =
38         ↪ 5
39         self.assertTrue(all(k <= 1 for k, l in zip(kap, lam)))  # Whitney
40
41     def test_edge_and_vertex_part_company_at_larger_n(self):
42         lam, kap = edge_vertex_distribution(16, 0.25, trials=200, seed=7)
43         gap = sum(k < 1 for k, l in zip(kap, lam))

```

```

39         self.assertGreater(gap, 0)
40         self.assertTrue(all(k <= 1 for k, l in zip(kap, lam)))
41
42     def test_threshold_curve_reports_the_predicted_threshold(self):
43         curve = threshold_curve(20, 2, [0.05, 0.2], trials=20, seed=0)
44         self.assertAlmostEqual(curve.predicted_threshold, 2 / 20)
45         self.assertEqual(len(curve.probabilities), 2)
46
47
48 if __name__ == "__main__":
49     unittest.main()

```

C.4.11 The solver

```

1  """The single solver: one solve() call handles every kind of question."""
2
3  import shutil
4  import unittest
5
6  import itertools
7
8  import numpy as np
9
10 from erdos915_unified import (
11     solve,
12     enumerate_extremal_directed_multigraphs as _dfs_enum,
13     enumerate_extremal_directed_multigraphs_via_generation as _gen_enum,
14     _decorate_support_worker,
15     _canonical_form,
16     PULP_AVAILABLE,
17 )
18
19
20 class Solve(unittest.TestCase):
21     def test_exhaustive_simple_directed_is_exact(self):
22         r = solve(4, 2, directed=True, simple=True, exhaustive=True, max_seconds
23             ↳ =120.0)
24         self.assertTrue(r.proven)
25         self.assertEqual(r.bound, "exact")
26         self.assertEqual(r.value, 6)
27
28     def test_discovery_returns_a_lower_bound(self):
29         # verification stays on sa; tabu is the default for solving the problems
30         r = solve(4, 2, directed=True, simple=True, exhaustive=False,
31             max_seconds=3.0, method="sa")
32         self.assertEqual(r.bound, "lower")
33         self.assertEqual(r.value, 6)
34
35     def test_exhaustive_undirected_finds_the_spanning_tree(self):
36         r = solve(5, 2, directed=False, simple=True, exhaustive=True, max_seconds
37             ↳ =30.0)
38         self.assertTrue(r.proven)
39         self.assertEqual(r.value, 4) # a spanning tree on 5 vertices
40
41     @unittest.skipUnless(PULP_AVAILABLE,
42         "solve() proves this one through the MILP certifier, "
43         "which needs the optional pulp")

```

```

42     def test_directed_multigraph_is_proved(self):
43         r = solve(4, 3, directed=True, simple=False, exhaustive=True, max_seconds
           ↪ =120.0)
44         self.assertTrue(r.proven)
45         self.assertEqual(r.value, 12) #  $L_3^{dir}(4) = 2(n-1)(m-1) = 12$ 
46
47     def test_vertex_separation_undirected(self):
48         #  $k_2(n) = n - 1$ : no cycle is allowed, so a spanning tree is extremal.
49         r = solve(5, 2, directed=False, simple=True, separation="vertex",
           exhaustive=True, max_seconds=30.0)
50         self.assertTrue(r.proven)
51         self.assertEqual(r.value, 4)
52
53
54     def test_hypergraph_discovery(self):
55         r = solve(7, 2, hypergraph=True, r=3, exhaustive=False, max_seconds=3.0,
           method="sa")
56         self.assertEqual(r.bound, "lower")
57         self.assertEqual(r.value, 3)
58
59
60     def test_result_describe_is_readable(self):
61         r = solve(4, 2, directed=True, simple=True, exhaustive=True, max_seconds
           ↪ =120.0)
62         self.assertIn("value", r.describe())
63
64
65 @unittest.skipIf(shutil.which("geng") is None, "nauty's geng is not installed")
66 class GengGeneration(unittest.TestCase):
67     """The geng-seeded enumerator must return exactly the same isomorphism
68     classes as the DFS enumerator wherever the DFS one is a sound complete
69     search ( $n \leq 6$ ). This pins down the new generation pipeline against the
70     method whose values are already in the thesis."""
71
72     @staticmethod
73     def _classes(reps):
74         return {_canonical_form(mu) for mu in reps}
75
76     def test_matches_dfs_at_n4(self):
77         #  $n = 4, m = 3$ : full target range and a degree cap. Fast, and the DFS
78         # enumerator is provably complete here, so equality is the gold standard.
79         for target, cap in [(12, None), (10, None), (8, None), (8, 6), (6, None)]:
80             with self.subTest(target=target, cap=cap):
81                 a = self._classes(_dfs_enum(4, 3, target, max_degree=cap))
82                 b = self._classes(_gen_enum(4, 3, target, max_degree=cap))
83                 self.assertEqual(a, b)
84
85     def test_extremal_is_the_doubled_spanning_trees(self):
86         # At  $n \leq 6$  the extremal directed multigraphs are exactly the doubled
87         # bidirected spanning trees, one per unlabelled tree: 2 on 4 vertices.
88         reps = _gen_enum(4, 3, 12) #  $L_3^{dir}(4) = 2(n-1)(m-1) = 12$ 
89         self.assertEqual(len(self._classes(reps)), 2)
90
91
92 class SupportWorker(unittest.TestCase):
93     """The per-support decorator is the unit the generation enumerator fans out
94     across processes. Over every support it must reproduce the DFS classes, so
95     the parallel refactor is covered here even without geng on PATH."""
96
97     @staticmethod

```

```

98     def _all_supports(n, min_e, max_e):
99         # Every non-isomorphic simple graph on n vertices in the edge range, the
100         # output geng would feed the enumerator, by brute force + canonical dedup.
101         pairs = [(i, j) for i in range(n) for j in range(i + 1, n)]
102         seen, out = set(), []
103         for mask in itertools.product((0, 1), repeat=len(pairs)):
104             if not (min_e <= sum(mask) <= max_e):
105                 continue
106             mu = np.zeros((n, n), dtype=int)
107             edges = []
108             for (i, j), bit in zip(pairs, mask):
109                 if bit:
110                     mu[i, j] = mu[j, i] = 1
111                     edges.append((i, j))
112             key = _canonical_form(mu)
113             if key not in seen:
114                 seen.add(key)
115                 out.append(edges)
116         return out
117
118     def test_worker_union_matches_dfs_at_n4(self):
119         n, m, target = 4, 3, 12
120         supports = self._all_supports(n, (target + 3) // 4, min(target, n * (n -
121             ↪ 1) // 2))
122         reps = []
123         for edges in supports:
124             reps.extend(_decorate_support_worker((n, m, target, None, True, edges)
125             ↪ ))
126         worker = {_canonical_form(mu) for mu in reps}
127         dfs = {_canonical_form(mu) for mu in _dfs_enum(n, m, target)}
128         self.assertEqual(worker, dfs)
129
130 if __name__ == "__main__":
131     unittest.main()

```

Bibliography

- [1] Béla Bollobás and Pál Erdős. Extremal problems in graph theory. *Matematikai Lapok*, 13:143–152, 1962. Original statement of Problem 915.
- [2] Karl Menger. Zur allgemeinen Kurventheorie. *Fundamenta Mathematicae*, 10:96–115, 1927. The original statement of Menger’s theorem.
- [3] Pál Erdős. Erdős problems. <https://www.erdosproblems.com/915>. Problem 915.
- [4] Wolfgang Mader. Ein Extremalproblem des Zusammenhangs von Graphen. *Mathematische Zeitschrift*, 131:223–231, 1973. Solves Erdős Problem 915 for edge-disjoint paths.
- [5] Hassler Whitney. Congruent graphs and the connectivity of graphs. *American Journal of Mathematics*, 54(1):150–168, 1932.
- [6] John L. Leonard. On graphs with at most four line-disjoint paths connecting any two vertices. *Journal of Combinatorial Theory, Series B*, 13:242–250, 1972.
- [7] B. A. Sørensen and Carsten Thomassen. On k -rails in graphs. *Journal of Combinatorial Theory, Series B*, 17(2):143–159, 1974. Determines $k_5(n) = \lfloor 8n/3 \rfloor - 3$.
- [8] John L. Leonard. On a conjecture of Bollobás and Erdős. *Periodica Mathematica Hungarica*, 3(3–4):281–284, 1973. Proves $\ell_m(n) = k_m(n)$ for $m \leq 4$.
- [9] Kristóf Bérczi, Karthekeyan Chandrasekaran, Tamás Király, and Shubhang Kulkarni. Splitting-off in hypergraphs. In *Proceedings of the 51st International Colloquium on Automata, Languages, and Programming (ICALP 2024)*, Leibniz International Proceedings in Informatics (LIPIcs), 2024. arXiv:2307.08555. A journal version has since appeared, which should replace this entry once the volume and pages are to hand.
- [10] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014. Describes the nauty/Traces canonical-labelling tools underlying geng.
- [11] Brendan D. McKay and Stanisław P. Radziszowski. $R(4, 5) = 25$. *Journal of Graph Theory*, 19(3):309–322, 1995. Exhaustive-generation determination of a Ramsey number.
- [12] Marijn J. H. Heule, Oliver Kullmann, and Victor W. Marek. Solving and verifying the Boolean Pythagorean triples problem via cube-and-conquer. In *Theory and Applications of Satisfiability Testing (SAT 2016)*, volume 9710 of *Lecture Notes in Computer Science*, pages 228–245. Springer, 2016. The certified SAT proof is roughly 200 terabytes in DRAT format.

- [13] Marijn J. H. Heule. Schur number five. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI-18)*, pages 6598–6606. AAAI Press, 2018.
- [14] Alexander A. Razborov. Flag algebras. *The Journal of Symbolic Logic*, 72(4):1239–1282, 2007.
- [15] Ralph E. Gomory and Tien Chung Hu. Multi-terminal network flows. *Journal of the Society for Industrial and Applied Mathematics*, 9(4):551–570, 1961. Introduces the Gomory-Hu tree structure.
- [16] Zsolt Baranyai. On the factorization of the complete uniform hypergraph. In *Infinite and Finite Sets (Colloq., Keszthely, 1973), Vol. I*, volume 10 of *Colloq. Math. Soc. Janos Bolyai*, pages 91–108. North-Holland, 1975.
- [17] W. T. Tutte. *Connectivity in Graphs*. University of Toronto Press, Toronto, 1966.
- [18] John E. Hopcroft and Robert E. Tarjan. Dividing a graph into triconnected components. *SIAM Journal on Computing*, 2(3):135–158, 1973.
- [19] Wolfgang Mulzer. Five proofs of Chernoff’s bound with applications. *Bulletin of the EATCS*, 124, 2018. Accessible survey of Chernoff bound variants.
- [20] Béla Bollobás. *Random Graphs*, volume 73 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2 edition, 2001. Chapter 7 gives $\kappa(G(n, p)) = \delta(G(n, p))$ with high probability in this regime.
- [21] Béla Bollobás and Andrew Thomason. Random graphs of small order. In *Random Graphs ’83*, volume 28 of *Annals of Discrete Mathematics*, pages 47–97. North-Holland, 1985. Hitting-time form of the connectivity result.
- [22] Alan Frieze and Michał Karoński. *Introduction to Random Graphs*. Cambridge University Press, 2016. Chapter 7 covers connectivity in $G(n, p)$ and $D(n, p)$; the directed analogue of Bollobás’s $\kappa = \delta$ result appears in Section 7.3.

